Libertinus and diacritics

To do

Long term: add mark-to-mark positioning (stacking) for the common above-marks in all four styles: ring, dieresis, macron, acute, grave, circumflex, caron, tilde, breve. The designer's stacked precomposed glyphs (ȅ, ȍ, ã, …) could supply the mark-to-mark anchors, as the precomposed capitals do for stage 4.

Long term: audit the SFD sources for consistent component referencing, across the four styles, and perhaps redraw or redefine components for consistency.

- Copies that could be references: glyphs that copy another glyph's splines (a derived letter drawn as a copy of its base, a precomposed letter as a copy of base and mark), so that a correction to the base reaches every glyph built on it, and the derived and precomposed stages need not rediscover the base in the outline.
- Dots, in letters and marks. The precomposed dot-below letters reference the dot above, moved below the letter (`uni0307` in regular and semibold, the spacing `dotaccent` in italic and semibold italic), except ṃ (and regular ṇ), which reference the combining dot below `dotbelowcomb`, so the dot of a precomposed letter and of a composed one differ in size: regular 100 against 108, italic 98 against 110, semibold 124 against 109, semibold italic 120 against 122. The shapes differ too: four-point circles in regular and semibold; in italic, `uni0307` a circle but `dotaccent` and `dotbelowcomb` skewed ovals; in semibold italic `dotaccent` a four-point skewed oval and `dotbelowcomb` a six-point one. The dot of semibold j is 14 units left of its stem (regular -3), a flaw in the drawing of j; ĵ and ǰ put the accent on the stem ($+2$). Options: move the dot onto the stem, or leave it and put j's above anchor over the dot ("Recheck all curated anchors A–Z and a–z"). Regular ṇ references both dots and n twice, so its two dots overlap. The diaeresis below U+0324 is two `dotbelowcomb` in regular and italic, but the precomposed ṳ is built from a diaeresis (`uni0308` in regular, semibold and semibold italic, the spacing `dieresis` in italic), with smaller dots in regular and italic.
- Commas and horns, in letters and marks (2026-11-11). In semibold and semibold italic the combining commas are drawn at the lighter style's weight: U+0312–U+0315 have 0.98 to 1.05 times the ink of regular's (italic's), and semibold's comma below U+0326 0.91, where the designer's own semibold comma and quotes have 1.21 to 1.27. The horn U+031B is regular weight in semibold (1.00). The koronis U+0343 is a reference to `uni0313` in regular and italic, but a reference to the horn in semibold (which had no mark anchor) and a horn-sized outline of its own in semibold italic; the patch curates both anchors ("Greek koronis U+0343" under "Comma-style marks"). The spacing Greek and modifier signs are bolded only in part: U+02BC, U+02BB, psili U+1FBF and koronis U+1FBD about 1.11 in semibold, U+1FBF and U+1FBD 1.00 in semibold italic; reversed comma U+02BD, dasia U+1FFE and varia U+1FEF 1.00 (varia is regular's outline). The precomposed Greek letters reference neither combining mark: ὰ ἐ Ἀ ῥ reference U+02BC (U+1FBD in semibold italic ὰ), and italic draws its own. After a base letter HarfBuzz shapes U+0343 as U+0313, its canonical equivalent, so a koronis on a letter shows the style's comma above: its weight matters more than the koronis glyph. Options: embolden the combining commas (as stage 3's `embolden()` does for the marks it copies) or derive them from the style's own comma; make U+0343 the style's U+0313.

Create the glyphs that are missing in the non-regular styles because the base groups now reach further than the designer's coverage (Ɐ R Ɤ ɤ ƥ ҁ Ɱ ʊ ʏ exist only in regular, Ɉ f f not in every style, and Ɋ Ⅼ Ƌ before them), as stage 2 already does for a few.

Revise how stage 8 moves Y away from an added part (the hook, tail or stroke of a derived letter). Now the anchor moves by how much further the added part reaches than the base letter within 120 units of the mark's axis, and a tail at the bottom takes the candidate's Y, from the letter's lowest point; both use the extreme point, not where the marks actually come close, so most anchors get more clearance than they need and a few too little (ɑ̈ above overlaps one mark). Alternatives measured on 2026-11-11 with every IPA mark: (A) the tightest mark on the derived letter gets the base letter's tightest gap, which removes over-clearance on 83 of 148 anchors but adds clearance on 43; (B) the same, but only toward the glyph, with a floor of about 20, so that only over-clearance and near-collisions change.

Write general instructions for using CGJ U+034F to block unwanted normalization. A below-mark typed after a precomposed letter that already has an above-mark is reordered and recomposed by normalization: ì + tilde below becomes ḭ + grave, which brings back the dot of i, and Ắ + dot below becomes Ạ + ring + acute (130 cases among the precomposed letters). That is a decomposition problem, to be avoided with a CGJ (or a dotless base) whether or not the font has mark-to-mark positioning; the sections "Lowercase i with tilde below and grave above, in italic and semibold italic" and following explore it, and "Floating tones" uses CGJ for a similar reason. The instructions should say when CGJ is needed, where it goes, and its cost (a precomposed glyph before CGJ has no below-anchor, so the below-mark falls back to a default position).

Recheck all curated anchors A–Z and a–z in all styles (2026-11-12). Curated anchors were made style by style and at different times, and some disagree between styles about where an anchor belongs: regular p has its below anchor $(120, -319)$ under the stem, but the other styles do not. For each letter, compare the four styles' curated anchors (and the scripted ones

they overrule) against the anchor model and the test sheets, and make them agree, or note why they differ. Part of it is curation that the stem-axis rule leaves to a human ("Stem-axis rule" under "Patch j regular, above-anchor"): in italic and semibold italic, the below anchors of i, ɪ and l and the above anchor of m, where a curl moves the optical centre off the stem (the curled foot of i ɪ l, right; the arches of m). None of the existing values (the designer's i, ɪ and m, the curated l) is known to be reliable. Semibold j: the stem-axis rule puts its above anchor on the stem, because the dot rules apply only in the italic styles, but its dot is 14 units left of the stem ("Dots, in letters and marks"); an above anchor of dotted j should be over the dot, as in the italic styles, unless the dot is moved. ȷ, and so ĵ and ǰ, keep the stem. Evidence from the precomposed glyphs, composed minus precomposed with the anchor on the stem: italic ị -14, ḷ $+4$ (median of the other dot-below letters $+11$); semibold italic ị -33, ḷ -22 (median -19, spread about ± 30); italic ṁ $+7$, semibold italic -21. The ink-area centroid of the bottom zone (baseline to 100 above the stem's straight part) came within 4 units of ị and ḷ in three of four cases. Test sheets `tex/stem_axis_test_sheet.pdf`, `tex/stem_angle_test_sheet.pdf`.

Audit the above anchors of the vowels against their precomposed glyphs, which are many and good evidence (survey of 2026-11-07). Where the patched anchor differs from the median mark centre of the precomposed glyphs by more than 15: italic o $+20$, O $+22$, E -20, Y -17; semibold y -19, Æ -16; semibold italic y $+29$ (regular's anchor, copied), Y -58, a $+18$, A -15, Æ -19. Regular O is within 12, but its marks land 16–19 left of the centre line of O. Two cautions: italic and semibold italic Y and U were just curated so that shaped marks land on their axes, and their precomposed glyphs disagree by 14–58; and the semibold italic precomposed glyphs run 30–40 right of the anchor model for g and E too. The vowel base anchors have not changed since; with the symmetric marks on their axes (2026-11-09), the median offset of composed marks from the precomposed glyphs per letter confirms the lowercase figures (italic o $+19$, semibold y -18, semibold italic y $+30$, a $+22$), and adds i: marks on ɪ land 12–18 right of ì í î ï in every style (regular $+16$, italic $+18$, semibold $+18$, semibold italic $+12$). The capitals were not remeasured. O and o lean to the left, unlike most of the other humanist letters, so their optical centre at the top may lie right of the bbox midpoint: a reason to move their above-anchors right, to weigh in the audit.

Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

Every font has five vertical reference lines (with respective Y coordinate values) – ascender line, capline, x-height or meanline, baseline, descender line – within which each glyph lives. These vertical reference lines are the gray vertical lines in Figures 1, 2, 3 and 4. Each glyph has a (tight) bounding box for real minimum and maximum X and Y coordinate values, as well as a so-called ideal bounding box where the glyph seems to live, but accounting for overshoots of bowls and serifs. The entire design goal is that the top and bottom of the ideal bounding box are two vertical reference lines, except for glyphs that are non-full ascenders or non-full descenders.

By clearance, I mean the (vertical) distance between a base and a mark, measured between the ideal bounding boxes of the base and the mark. For some designed fonts, the clearance is consistent for each vertical reference line, so that all glyphs that live at that reference line are separated from all marks with the same clearance; e.g. dots should not dance below the baseline for different letters that sit on the baseline, and acutes and graves sit at about the same height for different letters with similar top vertical reference line.

In roman styles of Libertinus, the clearance is encoded in the base anchor. The Y coordinate of an above-anchor is the top of the base's ideal bounding box plus the clearance of the nearest vertical reference line, and the Y coordinate of a below-anchor is the bottom of the ideal bounding box minus that clearance. For most glyphs that edge of the ideal bounding box is the reference line itself. For a non-full ascender or non-full descender it is not, but the clearance is the same; see the section on vertical reference lines below. The X coordinate of the anchor is at some horizontal optical center of the base glyph, but the meaning of the optical center depends on the glyph letterform, or whatever the designer thinks looks best. For a mostly symmetric letterform in the roman styles, a candidate for this X coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle

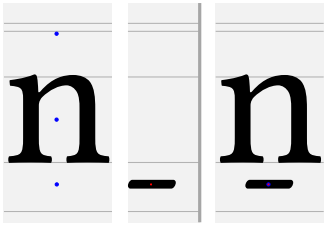


Figure 1: Libertinus n and macron with anchors.

panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark anchor, and in Libertinus it is encoded in the base anchor.

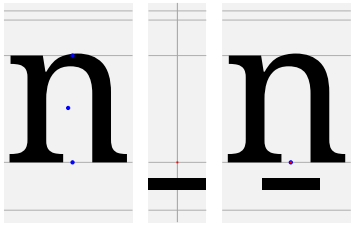


Figure 2: Noto n and macron with anchors.

Italic anchor model

In italic, the Libertinus anchor model is similar to the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle θ (12°). The midline of the italic lane is the line $X = W/2 + Y \tan\theta$, where W is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

The Y coordinate of an above- or below-anchor is set exactly as in the roman styles: the top or bottom of the glyph's ideal bounding box plus or minus the clearance of the nearest vertical reference line, measured vertically. So marks sit at the same height in roman and italic. The X coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$.

The height in this formula is the height of the anchor itself, that is, the edge of the ideal bounding box plus or minus the clearance, not the height of the reference line or of the ideal bounding box. So the Y coordinate of the anchor must be determined before the X coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the Y coordinate moves the X coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where Y is the height of the anchor, as in the roman styles, and C is where the axis of the mark meets the baseline: the horizontal center of the mark outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor X coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

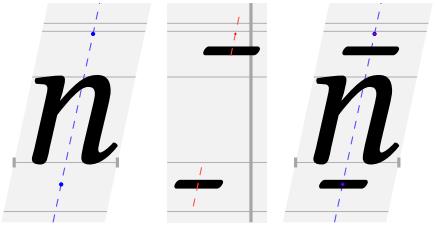


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The outlines are slanted (the italic angle θ is $11.5°$), but the anchors are, almost all, the anchors of regular, copied unchanged: 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have exactly the coordinates of regular.

So the anchors do not follow the slant. A slanted outline moves further right the higher it goes, but an anchor copied from regular stays where it was. An above-anchor therefore sits left of where the italic anchor model would put it, by roughly the amount the slant has moved the outline at that height. This is true of the base and of the mark alike: the base's above-anchor is left of the midline of its italic lane, and the mark's anchor is left of the mark's own slanted axis, by about the same amount. When the two anchors are superimposed, the two errors cancel, and the axis of the mark lands near the midline of the base, as it would in italic. See Figure 4, where the filled symbols are the designer's anchors and the hollow symbols are the positions given by the italic anchor model: the above-anchor of n and the anchor of the macron are both far left of the hollow symbols, but by nearly the same distance.

Each anchor in the font has its own shift k: how far it lies left of the position the italic anchor model gives it. For a base anchor, $k = W/2 + Y\tan\theta - X$. For a mark anchor, $k = C + Y\tan\theta - (X - d)$, where C is where the mark's axis meets the baseline and d is the mark's offset from its axis in regular ($d = 0$ for a symmetric mark; for the acute it is -32). In Figure 4, the above-anchor of n has $k = 170$ and the macron has $k = 141$. A mark lands $k_{\text{mark}} - k_{\text{base}}$ units right of the base's midline (left, if negative), so the macron lands $141 - 170 = -29$, that is, 29 units left of the midline of n.

The shifts differ from anchor to anchor, because the designer copied regular's X coordinates, which carry no slant. So an anchor's shift is about $Y\tan\theta$ at its own height, plus a small amount that depends on how the slanted outline was drawn. A higher anchor has a larger shift: comma above U+0313, with its anchor at $Y = 605$, has $k = 125$; hook above U+0309, at $Y = 765$, has $k = 168$. Only the difference between a mark's shift and a base's shift moves the mark off the midline, so any single value works, provided every base anchor and every mark anchor uses it.

The semibold italic anchor model therefore fixes one value of K for each anchor class and gives every anchor of that class the same shift: the base anchor is at $X = W/2 + Y\tan\theta - K$ and the mark anchor is at $X = C + Y\tan\theta + d - K$, with $\tan\theta = 0.2035$ and the same Y coordinates as in the other styles.

- Above-anchors: $K = 150$, for every above-mark and every base's above-anchor. Above-anchors are high, where the slant has moved the outline far to the right, so their shift is large. 150 is the median shift of the designer's above-marks; most are between 132 and 168.
- Below-anchors: $K = -12$, for every below-mark and every base's below-anchor. Below-anchors are just under the baseline, where the slant has moved the outline a little to the left, so their shift is small and negative. -12 is the median shift of the designer's below-marks; most are between -23 and -4.

The values are taken from the marks rather than the bases so that most of the designer's mark anchors already conform. The designer's base anchors have median shifts of 139 above and -1 below on lowercase Latin and IPA letters, and 173 above and -5 below on capitals. An anchor conforms if its own shift k is close to K. The italic anchor model is the case $K = 0$.

The candidates for a base are $ax = W/2 + ay\tan\theta - 150$ and $bx = W/2 + by\tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because the designer's anchors do not follow the slant at all, whereas the model's anchors do, apart from the constant K. A mark's shift is fixed by the height of its own anchor: for a mark, $k = Y\tan\theta$ plus a small amount, where Y is the height of the mark anchor. But the designer's base anchors are near $W/2 - 9$ at every height,

so they do not move right for a taller base, such as a capital or an ascender. The error of a mark on a base therefore grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital.

The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and corrects only the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: a base anchor without the shift would put every above-mark about 150 units too far right, because the marks carry the shift.

In the original font, several groups of anchors do not conform to this model. The patch corrects them:

- The capital forms of the above-marks (`grave.cap`, …) lacked K and landed 160 to 190 units left of the middle of a capital; they are set from the precomposed capitals, with K (stage 4).
- The above-anchors of a, s, h and J were set with the slant, as in italic, so that marks landed 130 to 190 units too far right; they are deleted and replaced by candidates (stages 1 and 6).
- Marks whose own shift k was far from K (the dieresis, tilde, inverted breve, hook above, koronis, right tack below and tilde below) are on the model, by the mark-axis rule or curated anchors ("Mark anchor model").
- Many capitals and other letters whose anchors were far from the candidate now have candidates, curated or stem-axis anchors, or anchors inherited from their base letter; the transposed anchors of ɥ and ʮ are replaced by candidates.

Still open:

- The bowl capitals and a few others keep the designer's above-anchor, 40 to 60 units left of the candidate: D, B, W, R, Q; A and X about 30 left, P 32 right. To be curated one by one.
- Designer below-anchors far from the candidate: Æ ($+123$), q ($+115$), r (-77, inherited by ɾ), j (-60), C and D (± 55), d (-47), A (-41), t (-40), and Ɑ̩; above, nᷠ (-63) and c ($+39$).
- Letters that overhang their advance width: the above-anchor of f is still the candidate and the below-anchor of j the designer's, in both italic styles; both need checking by eye.
- The Hebrew ligatures U+05F0–U+05F2 have above- and below-anchors in this lookup in semibold italic only; harmless, left as they are.

The base tables ("Font metrics per glyph") show which anchors are still the designer's.

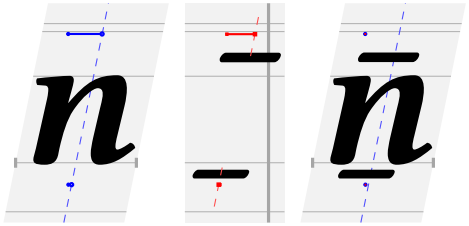


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline ($k = 170$) and the anchor of the macron is 141 units left of its axis ($k = 141$), so the macron lands 29 units left of the midline of n. The model's own positions, with $K = 150$, would be 20 units right of the filled circle and 9 units left of the filled square.

Mark anchor model

The anchor of a mark follows the same model in every style. Horizontally, it lies on the mark's own axis, $X = C + Y \tan\theta - K$ ($\theta = 0$ and $K = 0$ in the roman styles, $K = 0$ in italic, $K = 150$ above and -12 below in semibold italic). An asymmetric mark, such as the grave, acute, double acute or double grave, keeps the offset from its axis that it has in regular. Vertically, the anchor is at a fixed distance from the mark's outline: the anchor of an above-mark is a certain height above the bottom of the outline, and the anchor of a below-mark is a certain depth below the top of the outline. These distances differ from mark to mark (a short mark such as the macron sits a little higher than a tall one), but since the base anchors have the same Y coordinates in every style, a mark sits at the same height in every style only if its distance is the same as in regular.

Three kinds of evidence check a mark anchor: its X offset from its axis, compared with regular; the distance between its anchor and its outline, compared with regular; and the designer's precomposed glyphs, which HarfBuzz substitutes wherever they exist, so that a composed mark should match them. The precomposed glyphs are good evidence for Y, since every x-height base has its above anchor at 645. They are weaker evidence for X, because they also carry the offsets of the base anchors, and they are not always consistent between styles. In what follows, "lower" and "higher" and "left" and "right" describe where the mark lands on a base whose anchor conforms.

What the patch changes.

- Semibold's second above class `top_punkt` (dot above, diaeresis, ring, hook above, the commas above, …) is merged into the above class (stage 1), so its marks attach to every base with an above anchor.
- The marks a style lacks are added with regular's anchors (stage 3; semibold and semibold italic).
- The capital forms (`grave.cap`, `acute.cap`, …) are set from the designer's precomposed capitals (stage 4). This fixed semibold italic, whose capital forms lacked K, and italic, which had almost no capital-form anchors.
- The mark-axis rule (stage 4): every mirror-symmetric above and below mark, and the combining Latin letters, has its anchor X exactly on its own axis. The axis is the centre of the mark's ink bounds with the slant removed; for the rounder combining letters it is the centre of the ink area, for combining i the stem. For these marks the model decides X, not the precomposed glyphs. Asymmetric marks (grave, acute, double acute, double grave, hook, commas, half rings, …) keep the designer's offset.
- Curated anchors (stage 7), one mark at a time:
  - Vertical moves where a composed mark sits at the same distance from the precomposed glyphs on every base: the macron and double acute in several styles, the semibold tilde.
  - Consistency across styles, where the precomposed glyphs disagree between styles: the dot above, dot below and diaeresis below now sit at the same place in their mark zone in every style, as do the semibold diaeresis, the semibold italic ring and the semibold breve below.
  - The tall below marks (tacks, half rings, plus sign, ring, up arrow) are raised so that they clear descenders, and the tacks are placed consistently.
  - Single fixes, mostly in semibold and semibold italic: anchors that lacked K, marks that sat too low or too high against the other styles, the koronis, and the semibold italic macron below.

Moving every mark anchor onto the model at once was tested and rejected: as many marks got further from the precomposed glyphs as got closer.

Open decisions. None at present.

Decided: left as they are. Where the evidence does not support a move, the designer's anchor is kept: the above marks on the italic lowercase letters, which land a little right of the precomposed accents mainly on letters whose precomposed accents are themselves off centre (s, w); the italic macron and a few semibold and semibold italic marks whose precomposed glyphs disagree or do not support a move; and the semibold italic tilde, which the mark-axis rule moves further from its precomposed glyphs.

The detail behind all of this (figures, dates, test sheets and rejected alternatives) is in `MARK-ANCHOR-HISTORY.md` in the repository.

Mark outline zones

Figures 5, 6, 7 and 8 overlay the outlines of all the above and below marks of the patched fonts on a solid o, at three times the scale of the anchor model figures. Each mark is placed so that its anchor is on the anchor of the o for its class (the blue dots), which is where the shaper puts it. The common marks are black outlines. The other marks are gray with a light fill, so the shading builds up where many marks overlap. The grid is in font units of the o, with lines every 100 units. In semibold italic the above marks lie well to the right of the blue dot, because both the base and the mark anchor carry the shift K.

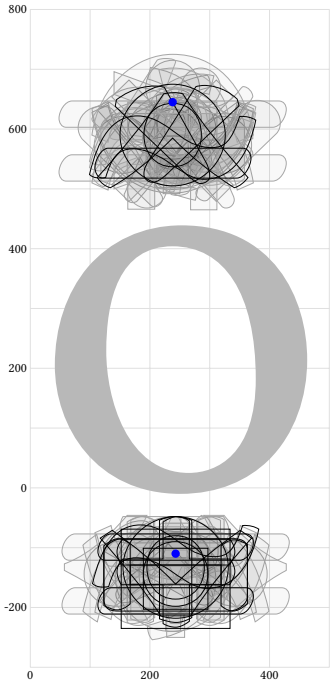


Figure 5: Mark zones, regular.

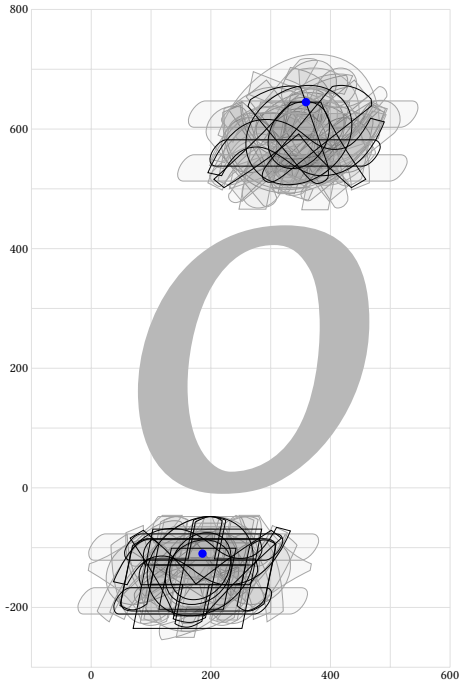


Figure 6: Mark zones, italic.

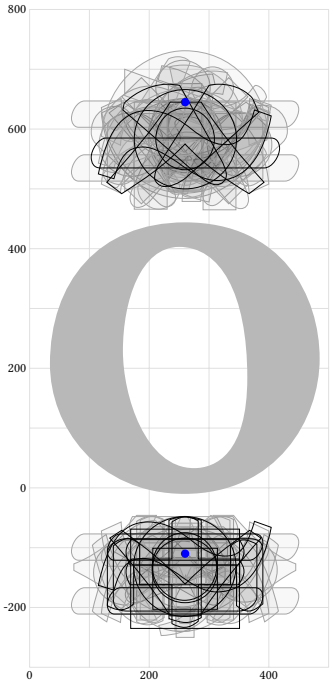


Figure 7: Mark zones, semibold.

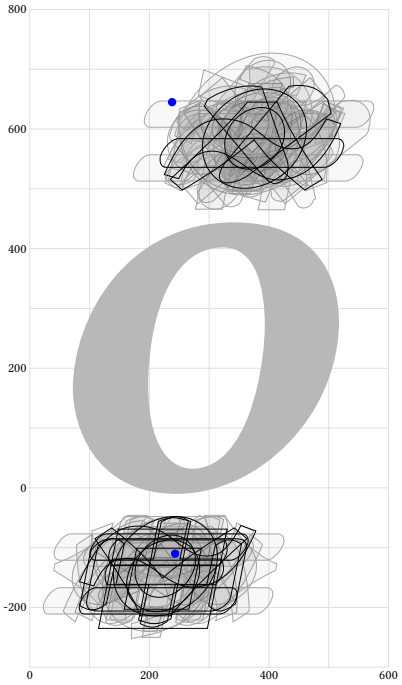


Figure 8: Mark zones, semibold italic.

Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in the following tables (Tables 1–12), and the mark anchors of above- and below-marks (Tables 13–16). The Latin and IPA tables leave out the letters whose above and below anchors in the patched fonts are all inherited from their base letter, in every style (derived letters, stage 8, and precomposed letters, stage 9: Đ Ø ø Đ đ Ħ Ł ł Ł ł Ŧ ŧ Ƀ Ć Đ Ɖ Ǵ Ɨ ł Θ Ƥ ƥ ţ T f Ţ Y Ԍ g ł ŧ Ⱥ Ȼ Ł Ŧ ɇ Ɉ R ɾ Ɏ ɏ ƀ ɗ ɟ ɩ ɨ ł ɵ ʃ ʃ ţ ɠ ʔ ʕ ə̓ f f y Ł ł Ł Ƥ ą Ḥ ḥ Ḿ ƥ ɇ): their anchors follow the base letter's, which is in the tables. A letter that keeps a designer, curated or heuristic anchor in any style is kept. The patch records the owner of every anchor in `data/anchor_provenance/`. Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for boldface styles.

Table row headings

hex: Unicode character codepoint in 4-hex.

cat: Unicode character category. L = letter. Lu = uppercase (capital) letter, Ll = lowercase (small) letter, Lo = other letter. Used to determine expected ideal bounding box. M = mark. Mn = nonspacing mark.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show anchor positioning (via combining grapheme joiner) when the base has an anchor; otherwise HarfBuzz default positioning applies.

patch: Use the patched Libertinus font to print the character alone, with combining dot above, and with combining macron below to show anchor positioning of the patched font (via combining grapheme joiner).

ideal: top and bottom ideal bounding box reference lines, with offset from Y_{max} or Y_{min} for non-full descender or non-full ascender. c = capline, x = x-height, a = ascender, b = baseline, d = descender.

a: X, Y coordinates of the above-anchor.

a?: X, Y coordinates of the above-anchor candidate.

aδ: X, Y coordinate differences between above-anchor and above-anchor candidate

b: X, Y coordinates of the below-anchor.

b?: X, Y coordinates of the below-anchor candidate.

bδ: X, Y coordinate differences between below-anchor and below-anchor candidate

a!, b!: the patched font's anchor, where the a/b and a?/b? cells do not already show it.

The owner of the patched anchor, the stage that last set it, is shown in bold:

- a or b in bold: the designer's anchor, kept by the patch (or snapped by stage 5 by up to 5 units to a reference line).
- The X of a or b and the Y of a? or b? in bold: the designer's X, with Y moved by stage 5 to the candidate's reference line (semibold's capitals to 850, and snaps within 5 units).
- a? or b? in bold: the candidate (stage 6).
- a! or b! in bold: a curated anchor (stage 7, and the stem-axis rule, stage 7b).
- a! or b! with a tag, not bold: i, inherited from the base letter (stage 8 or 9); n, semibold G's above anchor, from the `top_punkt` merge (stage 1) and moved to 850; h, the candidate of ʅ, which stage 2 redraws, so that the a? and b? cells (from the original outline) do not show it.

An empty column block means the patched font has no anchor in that class.

In the tables of marks, each column is a mark and the rows are as follows.

reg, it, sb, si, patch: the mark on n and on H (via combining grapheme joiner), in the original and patched fonts.

m: X, Y coordinates of the mark anchor in the original font. A class in parentheses is an anchor class other than above (0) or below (2): semibold puts the dot above, diaeresis, ring, hook above and comma above in class 1, whose base anchors are lower (n has 550 in class 1 and 646 in class 0). This is the designer's anchor class `top_punkt`; it places the marks at the same height as class 0, but only 24 bases have it. The patched semibold merges it into class 0, with the mark anchors raised by 96.

mp: X, Y coordinates of the mark anchor in the patched font.

mδ: patched minus original, X, Y; empty where the anchor is unchanged, "new" where the original font has no anchor (a mark added by the patch). For semibold's class 1 it is taken from the anchor raised by 96, so it shows how far the mark moves.

Table 1: Font metrics for A–Z 1/2

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠG	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
patch	AȦẠ	BḂḄ	CĊÇ	DḊḌ	EĖẸ	FḞF	GĠG	HḢḤ	IİỊ	JJ̇J	KḴḲ	LĿḶ	MṀṂ
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+60	c, b	c, b	c, b
a	354, 850	**267, 850**	**340, 850**	**306, 850**	**281, 850**	**284, 850**	**365, 850**	**361, 850**	150, 850	**182, 850**	**336, 850**	**151, 850**	**431, 850**
a?	346, 850	280, 850	324, 850	336, 850	271, 850	244, 850	352, 850	362, 850	148, 850	120, 850	323, 850	260, 850	418, 850
aδ	8, 0	-14, 0	16, 0	-30, 0	10, 0	40, 0	14, 0	-1, 0	2, 0	62, 0	13, 0	-110, 0	13, 0
a!	**361, 850**								**148, 850**				
b	308, -110	**268, -110**	**363, -110**	**309, -110**	**287, -110**	**209, -110**	**367, -110**	**362, -110**	147, -110	109, -319	**342, -110**	**262, -110**	**393, -110**
b?	346, -110	280, -110	324, -110	336, -110	271, -110	244, -110	352, -110	362, -110	148, -110	**120, -257**	323, -110	260, -110	418, -110
bδ	-38, 0	-12, 0	39, 0	-27, 0	16, 0	-36, 0	16, 0	0, 0	-1, 0	-11, -62	19, 0	2, 0	-25, 0
b!	**339, -110**								**148, -110**				
it	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
patch	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*JJ̇J*	*KḴḲ*	*LĿḶ*	*MṀṂ*
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+32, d+32	c, b	c, b	c, b
a	**505, 850**	**447, 850**	**500, 850**	396, 850	**413, 850**	**406, 850**	**513, 850**	481, 850	316, 850	**392, 850**	**466, 850**	389, 850	**557, 850**
a?	514, 850	459, 850	489, 850	514, 850	444, 850	409, 850	513, 850	517, 850	321, 850	388, 850	499, 850	440, 850	583, 850
aδ	-9, 0	-12, 0	11, 0	-118, 0	-31, 0	-3, 0	0, 0	-36, 0	-5, 0	4, 0	-33, 0	-51, 0	-26, 0
a!				**471, 850**				**504, 850**	**311, 850**			**314, 850**	
b	**308**, -108	**238**, -108	**333, -110**	**279**, -107	269, -107	**131**, -111	**345, -110**	**320**, -107	127, -109	109, -319	**292, -110**	**252, -110**	**363, -110**
b?	310, **-110**	255, **-110**	285, -110	310, **-110**	240, -110	205, **-110**	309, -110	313, **-110**	117, -110	**146, -285**	295, -110	236, -110	379, -110
bδ	-2, 2	-17, 2	48, 0	-31, 3	29, 3	-74, -1	36, 0	7, 3	10, 1	-37, -34	-3, 0	16, 0	-16, 0
b!					**246, -110**				**115, -110**				
sb	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠG**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
patch	**AȦẠ**	**BḂḄ**	**CĊÇ**	**DḊḌ**	**EĖẸ**	**FḞF**	**GĠG**	**HḢḤ**	**IİỊ**	**JJ̇J**	**KḴḲ**	**LĿḶ**	**MṀṂ**
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, d+50	c, b	c, b	c, b
a										**203**, 806			
a?	**350, 850**	300, 850	**332, 850**	**352, 850**	**283, 850**	**268, 850**	368, 850	**362, 850**	**160, 850**	131, **850**	**340, 850**	267, 850	**428, 850**
aδ										72, -44			
a!		**283, 850**					385, 850[n]					**161, 850**	
b								**362, -110**		109, -319			
b?	**350, -110**	**300, -110**	**332, -110**	**352, -110**	**283, -110**	**268, -110**	**368, -110**	362, -110	**160, -110**	**131, -267**	**340, -110**	**267, -110**	**428, -110**
bδ								0, 0		-22, -52			
b!													
si	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢḤ***	***IİỊ***	***JJ̇J***	***KḴḲ***	***LĿḶ***	***MṀṂ***
patch	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢH***	***IİỊ***	***JJ̇J***	***KḴK***	***LĿḶ***	***MṀM***
ideal	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c+34, d+68	c, b	c, b	c, b
a	**354, 850**	**267, 850**	**340, 850**	**306, 850**	**273, 850**	**276, 850**	**365, 850**	361, 850	150, 850	432, 850	336, 850	149, 850	**431, 850**
a?	382, 850	319, 850	341, 850	367, 850	292, 850	269, 850	371, 850	**396, 850**	194, 850	**249, 850**	**369, 850**	310, 850	441, 850
aδ	-28, 0	-52, 0	-1, 0	-61, 0	-19, 0	7, 0	-6, 0	-35, 0	-44, 0	183, 0	-33, 0	-161, 0	-10, 0
a!									**196, 850**			**209, 850**	
b	**308, -110**	**268, -110**	**363, -110**	**279, -110**	**279, -110**	**201, -110**	**367, -110**	**365, -110**	147, -110	109, -319	**342, -110**	**260, -110**	**393, -110**
b?	349, -110	286, -110	308, -110	334, -110	259, -110	236, -110	338, -110	363, -110	161, -110	**187, -249**	336, -110	277, -110	408, -110
bδ	-41, 0	-18, 0	55, 0	-55, 0	20, 0	-35, 0	29, 0	2, 0	-14, 0	-78, -70	6, 0	-17, 0	-15, 0
b!									**174, -110**				

Table 2: Font metrics for A–Z 2/2

hex	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
reg	NN̈N̲	OO̊O̲	PP̊P̲	QQ̊Q̲	RR̊R̲	SS̊S̲	TT̈T̲	UŮU̲	VV̇V̲	WẄW̲	XẌX̲	YẎY̲	ZŻZ̲
patch	NN̈N̲	OO̊O̲	PP̊P̲	QQ̊Q̲	RR̊R̲	SS̊S̲	TT̈T̲	UŮU̲	VV̇V̲	WẄW̲	XẌX̲	YẎY̲	ZŻZ̲
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	**346, 850**	**335, 850**	**277, 850**	**348, 850**	**269, 850**	**232, 850**	307, 850	**346, 850**	**336, 850**	**450, 850**	**332, 850**	**295, 850**	**320, 850**
a?	353, 850	352, 850	260, 850	359, 850	301, 850	236, 850	297, 850	331, 850	326, 850	474, 850	323, 850	288, 850	310, 850
aδ	-7, 0	-16, 0	18, 0	-11, 0	-32, 0	-4, 0	10, 0	15, 0	10, 0	-24, 0	9, 0	8, 0	10, 0
a!							**308, 850**						
b	**348, -110**	**350, -110**	**240, -110**		258, -110	**234, -110**	300, -110	**328, -110**	**318, -110**	**479, -110**	**307, -110**	**295, -110**	**287, -110**
b?	353, -110	352, -110	260, -110	**359, -294**	**301, -110**	236, -110	297, -110	331, -110	326, -110	474, -110	323, -110	288, -110	310, -110
bδ	-5, 0	-2, 0	-20, 0		-43, 0	-2, 0	3, 0	-3, 0	-8, 0	5, 0	-16, 0	8, 0	-22, 0
b!							**308, -110**						
it	*NN̈N̲*	*OO̊O̲*	*PP̊P̲*	*QQ̊Q̲*	*RR̊R̲*	*SS̊S̲*	*TT̈T̲*	*UŮU̲*	*VV̇V̲*	*WẄW̲*	*XẌX̲*	*YẎY̲*	*ZŻZ̲*
patch	*NN̈N̲*	*OO̊O̲*	*PP̊P̲*	*QQ̊Q̲*	*RR̊R̲*	*SS̊S̲*	*TT̈T̲*	*UŮU̲*	*VV̇V̲*	*WẄW̲*	*XẌX̲*	*YẎY̲*	*ZŻZ̲*
ideal	c, b	c, b	c, b	c, d+23	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b	c, b
a	**476, 850**	515, 850	**426, 850**	**498, 850**	**419, 850**	**402, 850**	457, 850	486, 850	481, 850	**610, 850**	**482, 850**	435, 850	**450, 850**
a?	514, 850	515, 850	430, 850	515, 850	458, 850	408, 850	453, 850	498, 850	479, 850	635, 850	495, 850	457, 850	470, 850
aδ	-38, 0	0, 0	-4, 0	-17, 0	-39, 0	-6, 0	4, 0	-12, 0	2, 0	-25, 0	-13, 0	-22, 0	-20, 0
a!		**493, 850**					**456, 850**	**508, 850**	**458, 850**			**441, 850**	
b	**328**, -109	310, -113	**140, -110**		**248**, -111	**204**, -111	277, -109	**308**, -109	**263, -110**	**439**, -107	**287**, -107	**270**, -109	**267**, -107
b?	310, **-110**	311, -110	226, -110	**272, -294**	254, **-110**	204, **-110**	249, -110	294, **-110**	275, -110	431, **-110**	291, **-110**	253, **-110**	266, **-110**
bδ	18, 1	-1, -3	-86, 0		-6, -1	0, -1	28, 1	14, 1	-12, 0	8, 3	-4, 3	17, 1	1, 3
b!		**310, -110**					**261, -110**						
sb	**NNN̲**	**OOQ**	**PPP̲**	**QQ̊Q̲**	**RR̊R̲**	**SS̊S̲**	**TT̈T̲**	**UUU̲**	**VV̇V̲**	**WẄW̲**	**XẌX̲**	**YẎY̲**	**ZŻZ̲**
patch	**NN̈N̲**	**OO̊O̲**	**PP̊P̲**	**QQ̊Q̲**	**RR̊R̲**	**SS̊S̲**	**TT̈T̲**	**UŮU̲**	**VV̇V̲**	**WẄW̲**	**XẌX̲**	**YẎY̲**	**ZŻZ̲**
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a		363, 802		**363**, 802	**304**, 808	252, 802	298, 802		**331**, 805	**493**, 805	**364**, 804	**316**, 805	321, 803
a?	**354, 850**	366, 850	**278, 850**	366, **850**	320, **850**	**238, 850**	299, 850	**339, 850**	318, **850**	489, **850**	354, **850**	308, **850**	309, 850
aδ		-2, -48		-2, -48	-16, -42	14, -48	-1, -48		12, -45	4, -45	10, -46	8, -45	12, -47
a!		**363, 850**					**299, 850**						**321, 850**
b					**292**, -111	**234**, -111	301, -107		**303**, -108	**470**, -107	**318**, -107	**316**, -109	**308**, -107
b?	**354, -110**	**366, -110**	**278, -110**	**366, -319**	320, **-110**	238, **-110**	299, -110	**339, -110**	318, **-110**	489, **-110**	354, **-110**	308, **-110**	309, **-110**
bδ					-28, -1	-4, -1	2, 3		-16, 2	-19, 3	-36, 3	8, 1	-1, 3
b!							**299, -110**						
si	***NN̈N̲***	***OO̊O̲***	***PP̊P̲***	***QQ̊Q̲***	***RR̊R̲***	***SS̊S̲***	***TT̈T̲***	***UŮU̲***	***VV̇V̲***	***WẄW̲***	***XẌX̲***	***YẎY̲***	***ZŻZ̲***
patch	***NN̈N̲***	***OO̊O̲***	***PP̊P̲***	***QQ̊Q̲***	***RR̊R̲***	***SS̊S̲***	***TT̈T̲***	***UŮU̲***	***VV̇V̲***	***WẄW̲***	***XẌX̲***	***YẎY̲***	***ZŻZ̲***
ideal	c, b	c, b	c, b	c, d	c, b	c, b	c+28, b	c, b	c, b	c, b	c, b	c, b	c, b
a	346, 850	335, 850	**316, 850**	**348, 850**	**269, 850**	**232, 850**	307, 850	346, 850	336, 850	**450, 850**	**332, 850**	295, 850	**320, 850**
a?	**394, 850**	**390, 850**	284, 850	390, 850	315, 850	250, 850	309, 850	364, 850	328, 850	501, 850	362, 850	319, 850	322, 850
aδ	-48, 0	-55, 0	32, 0	-42, 0	-46, 0	-18, 0	-2, 0	-18, 0	8, 0	-51, 0	-30, 0	-24, 0	-2, 0
a!							**295, 850**	**376, 850**	**313, 850**			**291, 850**	
b	**348, -110**	**350, -110**	**240, -110**		258, -110	**234, -110**	300, -110	**328, -110**	**318, -110**	**479, -110**	**307, -110**	295, -110	**287, -110**
b?	361, -110	357, -110	251, -110	**314, -319**	**282, -110**	217, -110	276, -110	331, -110	295, -110	468, -110	329, -110	286, -110	289, -110
bδ	-13, 0	-7, 0	-11, 0		-24, 0	17, 0	24, 0	-3, 0	23, 0	11, 0	-22, 0	9, 0	-2, 0
b!							**274, -110**					**277, -110**	

Table 3: Font metrics for a–z 1/2

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	aȧą	bḃḅ	cċç	dḋḍ	eėę	fḟf	gġg	hḣḥ	iįi	jȷj	kķk	lļl	mṁṃ
patch	aȧą	bḃḅ	cċç	dḋḍ	eėę	fḟf	gġg	hḣḥ	iįi	jȷj	kķk	lļl	mṁṃ
ideal	x, b	a, b	x, b	a, b	x, b	a, b	x+33, d	a, b	a-51, b	a-51, d+9	a, b	a, b	x, b
a	**209, 645**		**242, 645**	218, 645	**243, 645**		**231, 645**	299, 645	137, 795		306, 784	135, 885	392, 645
a?	246, 645	232, 885	218, 645	270, 885	222, 645	208, 885	256, 676	270, 885	142, 834	76, 834	261, 885	134, 885	402, 645
aδ	-36, 0		24, 0	-52, -240	21, 0		-26, -31	28, -240	-5, -39		45, -101	1, 0	-10, 0
a!		**279, 885**		**235, 885**		**273, 885**		**282, 885**	**142, 834**	**152, 834**	**272, 885**	**134, 885**	**404, 645**
b	**196, -110**	**234, -110**	236, -110	221, -110	**234, -110**	132, -110	**229, -319**	**272, -110**	136, -110	**109, -319**	290, -214	135, -110	408, -110
b?	246, -110	232, -110	**218, -110**	270, -110	222, -110	208, -110	256, -319	270, -110	**142, -110**	76, -308	**261, -110**	134, -110	402, -110
bδ	-50, 0	2, 0	18, 0	-50, 0	12, 0	-76, 0	-28, 0	2, 0	-6, 0	34, -11	29, -104	1, 0	6, 0
b!				**235, -110**		**136, -110**						**134, -110**	**404, -110**
it	*aȧą*	*bḃḅ*	*cċç*	*dḋḍ*	*eėę*	*fḟf*	*gġg*	*hḣḥ*	*iįi*	*jȷj*	*kķk*	*lļl*	*mṁṃ*
patch	*aȧą*	*bḃḅ*	*cċç*	*dḋḍ*	*eėę*	*fḟf*	*gġg*	*hḣḥ*	*iįi*	*jȷj*	*kķk*	*lļl*	*mṁṃ*
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-80, b	a-80, d+36	a, b	a, b	x, b
a	**341**, 647		**343, 645**	218, 645	**332, 645**		**360, 645**	409, 645			405, 646	292, 890	**522, 645**
a?	380, **645**	**428, 890**	332, 645	**434, 890**	338, 645	**346, 890**	376, 645	**449, 890**	310, 810	302, 810	**432, 890**	322, 890	529, 645
aδ	-39, 2		11, 0	-216, -245	-6, 0		-16, 0	-40, -245			-27, -244	-30, 0	-7, 0
a!									**300, 810**	**305, 810**		**313, 890**	
b	174, -108	**214**, -108	**163**, -113	**221**, -107	**158**, -108	222, -111	**197, -319**	**229**, -111	136, -107	**0, -319**	**223**, -107	115, -107	382, -105
b?	**220, -110**	216, **-110**	171, **-110**	221, **-110**	177, **-110**	89, -319	171, -319	236, **-110**	**115, -110**	70, -281	220, **-110**	110, -110	368, -110
bδ	-46, 2	-2, 2	-8, -3	0, 3	-19, 2	133, 208	26, 0	-7, -1	21, 3	-70, -38	3, 3	5, 3	14, 5
b!						**30, -319**						**99, -110**	**373, -110**
sb	**aȧą**	**bḃḅ**	**cċç**	**dḋḍ**	**eėę**	**fḟf**	**gġg**	**hḣḥ**	**iįi**	**jȷj**	**kķk**	**lļl**	**mṁṃ**
patch	**aȧą**	**bḃḅ**	**cċç**	**dḋḍ**	**eėę**	**fḟf**	**gġg**	**hḣḥ**	**iįi**	**jȷj**	**kķk**	**lļl**	**mṁṃ**
ideal	x, b	a, b-13	x, b	a, b	x, b	a, b	x+33, d	a, b	a-48, b	a-48, d	a, b	a, b	x, b
a							**231, 645**		137, 795			142, 885	396, 645
a?	255, 645	**236, 885**	**224, 645**	**274, 885**	232, 645	**226, 885**	264, 676	**275, 885**	156, 837	78, 837	**262, 885**	148, 885	408, 645
aδ							-32, -31		-18, -42			-6, 0	-12, 0
a!	**219, 645**				**249, 645**				**156, 837**	**162, 837**		**150, 885**	**407, 645**
b							**229, -319**	**277, -110**	136, -107	**109, -319**		135, -110	396, -105
b?	**255, -110**	**236, -121**	**224, -110**	**274, -110**	**232, -110**	**226, -110**	264, -319	275, -110	**156, -110**	78, -319	**262, -110**	148, -110	408, -110
bδ							-34, 0	2, 0	-20, 3	31, 0		-14, 0	-12, 5
b!												**150, -110**	**407, -110**
si	***aȧą***	***bḃḅ***	***cċç***	***dḋḍ***	***eėę***	***fḟf***	***gġg***	***hḣḥ***	***iįi***	***jȷj***	***kķk***	***lļl***	***mṁṃ***
patch	***aȧą***	***bḃḅ***	***cċç***	***dḋḍ***	***eėę***	***fḟf***	***gġg***	***hḣḥ***	***iįi***	***jȷj***	***kķk***	***lļl***	***mṁṃ***
ideal	x, b	a, b	x, b	a, b	x, b	a, d	x, d	a, b	a-67, b	a-67, d+25	a, b	a, b	x, b
a	379, 645		**242, 645**	218, 645	**243, 645**		291, 645	469, 645	137, 795		306, 784	174, 890	**392, 645**
a?	**251, 645**	**290, 890**	203, 645	**309, 890**	218, 645	**216, 890**	**220, 645**	**331, 890**	180, 823	192, 823	**316, 890**	194, 890	407, 645
aδ	128, 0		39, 0	-91, -245	25, 0		71, 0	138, -245	-43, -28		-10, -106	-20, 0	-15, 0
a!									**169, 823**	**186, 823**		**184, 890**	
b	196, -110	234, -110	**236, -110**	**221, -110**	**234, -110**	132, -110	**199, -319**	**272, -110**	136, -110	**67, -319**	290, -214	135, -110	400, -110
b?	**260, -110**	**248, -110**	212, -110	268, -110	227, -110	132, -319	186, -319	290, -110	**153, -110**	127, -292	**275, -110**	153, -110	415, -110
bδ	-64, 0	-14, 0	24, 0	-47, 0	7, 0	0, 209	13, 0	-18, 0	-17, 0	-60, -27	15, -104	-18, 0	-15, 0
b!						**62, -319**						**138, -110**	**406, -110**

Table 4: Font metrics for a–z 2/2

hex	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	nn̈n̲	oo̱o̱	pp̈p̲	qq̈q̲	rr̈r̲	ss̈s̲	tẗt̲	uu̱u̲	vv̱v̲	ww̱w̲	xx̱x̲	yy̱y̲	zz̈z̲
patch	nn̈n̲	oo̱o̱	pp̈p̲	qq̈q̲	rr̈r̲	ss̈s̲	tẗt̲	uu̱u̲	vv̱v̲	ww̱w̲	xx̱x̲	yy̱y̲	zz̈z̲
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-116, b	x, b	x, b	x, b	x, b	x, d	x, b
a	**264**, 646	**238, 645**	**271**, 646	**238, 645**	**191**, 646	**206, 645**		**255, 645**	**256**, 646	345, 646	**256**, 646	**262, 645**	**220**, 646
a?	276, **645**	252, 645	242, **645**	268, 645	190, **645**	199, 645	163, 769	271, 645	249, **645**	**373, 645**	245, **645**	260, 645	220, **645**
aδ	-12, 1	-14, 0	28, 1	-30, 0	1, 1	7, 0		-16, 0	7, 1	-28, 1	11, 1	2, 0	0, 1
a!							**139, 769**						
b	**265, -110**	**243, -110**	179, -319	348, -319	135, -110	**196, -110**	**163, -110**	**239, -110**	**241, -110**	352, -110	**224, -110**	**242, -319**	**207, -110**
b?	276, -110	252, -110	242, -319	268, -319	190, -110	199, -110	163, -110	271, -110	249, -110	**373, -110**	245, -110	260, -319	220, -110
bδ	-10, 0	-9, 0	-64, 0	80, 0	-55, 0	-3, 0	0, 0	-32, 0	-8, 0	-21, 0	-21, 0	-18, 0	-14, 0
b!			**120, -319**	**386, -319**	**140, -110**								
it	*nn̈n̲*	*oo̱o̱*	*pp̈p̲*	*qq̈q̲*	*rr̈r̲*	*ss̈s̲*	*tẗt̲*	*uu̱u̲*	*vv̱v̲*	*ww̱w̲*	*xx̱x̲*	*yy̱y̲*	*zz̈z̲*
patch	*nn̈n̲*	*oo̱o̱*	*pp̈p̲*	*qq̈q̲*	*rr̈r̲*	*ss̈s̲*	*tẗt̲*	*uu̱u̲*	*vv̱v̲*	*ww̱w̲*	*xx̱x̲*	*yy̱y̲*	*zz̈z̲*
ideal	x, b	x, b	x+62, d	x, d	x, b	x, b	x+132, b	x, b	x, b	x, b	x, b	x, d	x, b
a	**394**, 646	**359, 645**	**411, 646**	**369, 645**	**312**, 646	**327, 645**		**386**, 646		348, 646	**366**, 646	**367**, 644	**360**, 646
a?	396, **645**	361, 645	394, 705	383, 645	316, **645**	314, 645	**318, 775**	398, **645**	**373, 645**	**481, 645**	375, **645**	389, **645**	355, **645**
aδ	-2, 1	-2, 0	17, -59	-14, 0	-4, 1	13, 0		-12, 1		-133, 1	-9, 1	-22, -1	5, 1
a!													
b	**235**, -111	**186**, -107	248, -110	169, -110	**113, -110**	**169**, -111	**133**, -107	**226**, -111		**355, -110**	**224, -110**	368, -107	**187**, -111
b?	236, **-110**	200, **-110**	**177, -319**	178, -319	155, -110	153, **-110**	130, **-110**	237, **-110**	**213, -110**	321, -110	214, -110	184, -319	195, **-110**
bδ	-1, -1	-14, 3	71, 209	-9, 209	-42, 0	16, -1	3, 3	-11, -1		34, 0	10, 0	184, 212	-8, -1
b!				**300, -319**								**240, -319**	
sb	**nn̈n̲**	**oo̱o̱**	**pp̈p̲**	**qq̈q̲**	**rr̈r̲**	**ss̈s̲**	**tẗt̲**	**uu̱u̲**	**vv̱v̲**	**ww̱w̲**	**xx̱x̲**	**yy̱y̲**	**zz̈z̲**
patch	**nn̈n̲**	**oo̱o̱**	**pp̈p̲**	**qq̈q̲**	**rr̈r̲**	**ss̈s̲**	**tẗt̲**	**uu̱u̲**	**vv̱v̲**	**ww̱w̲**	**xx̱x̲**	**yy̱y̲**	**zz̈z̲**
ideal	x, b	x, b	x, d	x, d	x, b	x, b	a-103, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646							**264**, 646					
a?	**290, 645**	**260, 645**	**254, 645**	**273, 645**	**212, 645**	**201, 645**	**168, 782**	276, **645**	**262, 645**	**384, 645**	**260, 645**	**264, 645**	**222, 645**
aδ	-26, 1							-12, 1					
a!													
b	**265**, -111						**163**, -107	**248**, -111					
b?	290, **-110**	**260, -110**	**254, -319**	**273, -319**	212, -110	**201, -110**	168, **-110**	276, **-110**	**262, -110**	**384, -110**	**260, -110**	**264, -319**	**222, -110**
bδ	-24, -1						-5, 3	-28, -1					
b!					**152, -110**								
si	***nn̈n̲***	***oo̱o̱***	***pp̈p̲***	***qq̈q̲***	***rr̈r̲***	***ss̈s̲***	***tẗt̲***	***uu̱u̲***	***vv̱v̲***	***ww̱w̲***	***xx̱x̲***	***yy̱y̲***	***zz̈z̲***
patch	***nn̈n̲***	***oo̱o̱***	***pp̈p̲***	***qq̈q̲***	***rr̈r̲***	***ss̈s̲***	***tẗt̲***	***uu̱u̲***	***vv̱v̲***	***ww̱w̲***	***xx̱x̲***	***yy̱y̲***	***zz̈z̲***
ideal	x, b	x, b	x+36, d	x, d	x, b	x, b	a-125, b	x, b	x, b	x, b	x, b	x, d	x, b
a	264, 646	**238, 645**	**271, 646**	**238, 645**	**191**, 646	326, 645		**255, 645**	**256**, 646	**345**, 646		**262, 645**	**220**, 646
a?	**284, 645**	235, 645	257, 679	267, 645	203, **645**	**190, 645**	**189, 765**	262, 645	220, **645**	332, **645**	**240, 645**	236, 645	210, **645**
aδ	-20, 1	3, 0	14, -33	-29, 0	-12, 1	136, 0		-7, 0	36, 1	13, 1		26, 0	10, 1
a!													
b	265, -110	**243, -110**	**179, -319**	**348, -319**	**135, -110**	**196, -110**	**133**, -107	239, -110	**241, -110**	**352, -110**		**242, -319**	**207, -110**
b?	**293, -110**	243, -110	216, -319	233, -319	212, -110	198, -110	173, **-110**	**270, -110**	228, -110	341, -110	**248, -110**	202, -319	218, -110
bδ	-28, 0	0, 0	-37, 0	115, 0	-77, 0	-2, 0	-40, 3	-31, 0	13, 0	11, 0		40, 0	-11, 0
b!													

Table 5: Font metrics for Latin and IPA 1/8

hex	00C6	00DE	00DF	00E6	00F0	00FE	0127	0131	014A	014B	0152	0153	017F	0180	0182	0183
cat	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Ll	Lu	Ll
reg	ÆḂÆ	ÞḂÞ	ßßß	æḃæ	ððð	þþþ	ħħḥ	ıiı	ŊŊŊ	ŋŋŋ	ŒḂŒ	œḃœ	ſſſ	ƀƀƀ	ƁƁƁ	ɓɓɓ
patch	ÆḂÆ	ÞḂÞ	ßßß	æḃæ	ððð	þþþ	ħħḥ	ıiı	ŊŊŊ	ŋŋŋ	ŒḂŒ	œḃœ	ſſſ	ƀƀƀ	ƁƁƁ	ɓɓɓ
ideal	c, b	c, b	a, b	x, b	a, b	a, d	a, b	x, b	c, d+60	x, d+9	c, b	x, b	a, b	a, b	c, b	a, b
a	480, 850	**281, 850**		**350, 645**		278, 646		132, 645	**346**, 804	**264**, 646	**460, 850**	**405, 645**				
a?	426, 850	250, 850	**259, 885**	344, 645	**240, 885**	**236, 885**	270, 885	**142, 645**	316, **850**	240, **645**	436, 850	388, 645	**190, 885**	231, 885	**274, 850**	228, 885
aδ	54, 0	32, 0		6, 0		42, -239		-10, 0	30, -46	24, 1	24, 0	18, 0				
a!	**555, 850**						282, 885^{i}							279, 885^{i}		275, 885^{i}
b	**549**, -107			**327**, -109				139, -108	**353, -109**	265, -111						234, -109
b?	426, **-110**	**250, -110**	**259, -110**	344, **-110**	**240, -110**	**236, -319**	270, -110	**142, -110**	316, -257	240, -308	**436, -110**	388, -110	**190, -110**	231, -110	**274, -110**	228, -110
bδ	122, 3			-16, 1				-3, 2	37, 148	25, 197						6, 1
b!							272, -110^{i}			**265, -319**		**408, -110**		234, -110^{i}		230, -110^{i}
it	*ÆḂÆ*	*ÞḂÞ*	*ßßß*	*æḃæ*	*ððð*	*þþþ*	*ħħḥ*	*ıiı*	*ŊŊŊ*	*ŋŋŋ*	*ŒḂŒ*	*œḃœ*	*ſſſ*	*ƀƀƀ*	*ƁƁƁ*	*ɓɓɓ*
patch	*ÆḂÆ*	*ÞḂÞ*	*ßßß*	*æḃæ*	*ððð*	*þþþ*	*ħħḥ*	*ıiı*	*ŊŊŊ*	*ŋŋŋ*	*ŒḂŒ*	*œḃœ*	*ſſſ*	*ƀƀƀ*	*ƁƁƁ*	*ɓɓɓ*
ideal	c, b	c, b	a, d	x, b	a+6, b	a, d-7	a, b	x, b	c, d+32	x, d	c, b	x, b	a, d	a, b	c, b	a, b
a	615, 850			**480, 645**		278, 646		267, 645	346, 804	264, 646	**575, 850**	480, 645				
a?	595, 850	**431, 850**	**461, 890**	481, 645	**429, 896**	**439, 890**	**449, 890**	275, 645	**495, 850**	382, 645	587, 850	490, 645	**339, 890**	**428, 890**	**465, 850**	**413, 890**
aδ	20, 0			-1, 0		-161, -244		-8, 0	-149, -46	-118, 1	-12, 0	-10, 0				
a!	**674, 850**							**265, 645**		**411, 645**		**507, 645**				
b	**549**, -107			**327**, -109				139, -108	**364, -109**	265, -111						**230**, -109
b?	391, **-110**	**227, -110**	**204, -319**	320, **-110**	**216, -110**	**181, -324**	236, -110	**115, -110**	253, -285	**177, -319**	**383, -110**	**329, -110**	**82, -319**	**216, -110**	**261, -110**	201, **-110**
bδ	158, 3			7, 1				24, 2	111, 176	88, 208						29, 1
b!							229, -110^{i}									
sb	**ÆḂÆ**	**ÞḂÞ**	**ßßß**	**æḃæ**	**ððð**	**þþþ**	**ħħḥ**	**ıiı**	**ŊŊŊ**	**ŋŋŋ**	**ŒḂŒ**	**œḃœ**	**ſſſ**	**ƀƀƀ**	**ƁƁƁ**	**ɓɓɓ**
patch	**ÆḂÆ**	**ÞḂÞ**	**ßßß**	**æḃæ**	**ððð**	**þþþ**	**ħħḥ**	**ıiı**	**ŊŊŊ**	**ŋŋŋ**	**ŒḂŒ**	**œḃœ**	**ſſſ**	**ƀƀƀ**	**ƁƁƁ**	**ɓɓɓ**
ideal	c, b	c, b	a, b	x, b	a+19, b	a, d	a, b	x, b	c, d+50	x, d+36	c, b	x, b	a, b	a, b-13	c, b	a-15, b-16
a	510, 850			**355, 645**				151, 645			**450, 850**	**430, 645**				
a?	446, 850	**287, 850**	**284, 885**	362, 645	**256, 904**	**260, 885**	275, 885	**156, 645**	**328, 850**	**248, 645**	432, 850	402, 645	**217, 885**	236, 885	**268, 850**	223, 870
aδ	64, 0			-8, 0				-4, 0			18, 0	28, 0				
a!	**555, 850**						275, 885^{i}							236, 885^{i}		222, 885^{i}
b								135, -108								
b?	446, -110	**287, -110**	**284, -110**	**362, -110**	**256, -110**	**260, -319**	275, -110	**156, -110**	**328, -267**	**248, -281**	**432, -110**	**402, -110**	**217, -110**	236, -121	**268, -110**	223, -124
bδ								-20, 2								
b!	**510, -110**						277, -110^{i}							236, -121^{i}		222, -121^{i}
si	*ÆḂÆ*	*ÞḂÞ*	*ßßß*	*æḃæ*	*ððð*	*þþþ*	*ħħḥ*	*ıiı*	*ŊŊŊ*	*ŋŋŋ*	*ŒḂŒ*	*œḃœ*	*ſſſ*	*ƀƀƀ*	*ƁƁƁ*	*ɓɓɓ*
patch	*ÆḂÆ*	*ÞḂÞ*	*ßßß*	*æḃæ*	*ððð*	*þþþ*	*ħħḥ*	*ıiı*	*ŊŊŊ*	*ŋŋŋ*	*ŒḂŒ*	*œḃœ*	*ſſſ*	*ƀƀƀ*	*ƁƁƁ*	*ɓɓɓ*
ideal	c, b	c, b	a, d	x, b	a+19, b	a, d	a, b	x, b	c, d+49	x, d+16	c, b	x, b	a+13, d-9	a, b	c, b	a-15, b
a	500, 850	**281, 850**		**350, 645**		278, 646		132, 645	359, 804	**264**, 646	**480, 850**	**415, 645**				
a?	459, 850	303, 850	**325, 890**	339, 645	**291, 909**	**300, 890**	**331, 890**	144, 645	**390, 850**	253, **645**	450, 850	384, 645	**209, 903**	290, 890	**303, 850**	269, 875
aδ	41, 0	-22, 0		11, 0		-22, -244		-12, 0	-31, -46	11, 1	30, 0	31, 0				
a!	**545, 850**							**133, 645**						290, 890^{i}		274, 890^{i}
b	**549**, -107			**327**, -109				139, -108	**373, -109**	265, -111						234, -109
b?	426, **-110**	**270, -110**	**241, -319**	347, **-110**	**246, -110**	**216, -319**	290, -110	**153, -110**	325, -268	**222, -301**	**417, -110**	**393, -110**	**121, -326**	248, -110	**270, -110**	231, -110
bδ	123, 3			-20, 1				-14, 2	48, 159	43, 190						3, 1
b!							272, -110^{i}							248, -110^{i}		232, -110^{i}

Table 6: Font metrics for Latin and IPA 2/8

hex	0184	0186	0188	018B	018C	018E	018F	0190	0191	0192	0194	0196	0198	0199	019C	019D
cat	Lu	Lu	Ll	Lu	Ll	Lu	Lu	Lu	Lu	Ll	Lu	Lu	Lu	Ll	Lu	Lu
reg	Ƅƅ	ƆɔƆ	ƈƈ	Ƌƌ	ƌƍ	ƎǝƎ	ƏəƏ	ƐɛƐ	ƑƒF	*fff*	ƔɣƳ	ƖɩI	ƘƙĶ	ƙƙķ	ƜɯƜ	ƝɲƝ
patch	ƄƅƄ	ƆɔƆ	ƈƈ	ƋƌƋ	ƌƍ	ƎǝƎ	ƏəƏ	ƐɛƐ	ƑƒF	*fff*	ƔɣƳ	ƖɩI	ƘƙĶ	ƙƙķ	ƜɯƜ	ƝɲƝ
ideal	c, b	c, b	x+70, b	c, b	a, b	c, b	c, b	c, b	c, d+60	a, d	c, d	a, b	c, b	a, b	c, b	c, d+45
a	**267**, 805				214, 650						331, 805				431, 801	
a?	278, **850**	322, 850	264, 713	**292, 850**	275, 885	**280, 850**	326, 850	**256, 850**	183, 850	201, 885	**308, 850**	**152, 885**	**339, 850**	262, 885	**434, 850**	324, 850
aδ	-10, -45				-61, -235						22, -45				-2, -49	
a!		**290, 850**	243, 669^{i}		240, 885^{i}		**306, 850**		276, 850^{i}	**323, 885**				273, 885^{i}		345, 850^{i}
b	**268**, -108				217, -107									263, -107	450, -108	
b?	278, **-110**	322, -110	264, -110	**292, -110**	275, -110	**280, -110**	**326, -110**	**256, -110**	183, -257	201, -319	**308, -319**	**152, -110**	339, -110	262, -110	**434, -110**	324, -272
bδ	-10, 2				-58, 3									1, 3	16, 2	
b!		**294, -110**	218, -110^{i}		240, -110^{i}				202, -257^{i}	**84, -319**			342, -110^{i}	262, -110^{i}		347, -110^{i}
it	*ƄƅƄ*	*ƆɔƆ*	*ƈƈ*	*ƋƌƋ*	*ƌƍ*	*ƎǝƎ*	*ƏəƏ*	*ƐɛƐ*	*ƑƒF*	*fff*	*ƔɣƳ*	*ƖɩI*	*ƘƙĶ*	*ƙƙķ*	*ƜɯƜ*	*ƝɲƝ*
patch	*ƄƅƄ*	*ƆɔƆ*	*ƈƈ*	*ƋƌƋ*	*ƌƍ*	*ƎǝƎ*	*ƏəƏ*	*ƐɛƐ*	*ƑƒF*	*fff*	*ƔɣƳ*	*ƖɩI*	*ƘƙĶ*	*ƙƙķ*	*ƜɯƜ*	*ƝɲƝ*
ideal	c, b	c+14, b	x+70, b	c, b	a, b	c, b	c, b	c, b	c, d+60	a, d+60	c, d+20	a, b	c, b	a, b	c, b	c, d+45
a	267, 805				209, 650						331, 805				431, 801	
a?	**455, 850**	**503, 850**	**398, 713**	**450, 850**	**423, 890**	**465, 850**	**515, 850**	**439, 850**	419, 850	414, 890	**462, 850**	**322, 890**	**499, 850**	**429, 890**	**609, 850**	528, 850
aδ	-188, -45				-214, -240						-131, -45				-178, -49	
a!									422, 850^{i}	**525, 890**						483, 850^{i}
b	**268**, -108				**212**, -107									**263**, -107	450, -108	
b?	251, **-110**	**299, -110**	**223, -110**	**246, -110**	211, **-110**	**261, -110**	**311, -110**	**235, -110**	184, -257	170, -257	**218, -297**	**110, -110**	295, -110	216, **-110**	**405, -110**	290, -272
bδ	17, 2				1, 3									47, 3	45, 2	
b!									113, -257^{i}	**35, -257**			300, -110^{i}			337, -110^{i}
sb	**ƄƅƄ**	ƆɔƆ	**ƈƈ**	**ƋƌƋ**	**ƌƍ**	**ƎǝƎ**	**ƏəƏ**	**ƐɛƐ**	**ƑƒF**	***fff***	**ƔɣƳ**	**ƖɩI**	**ƘƙĶ**	**ƙƙķ**	**ƜɯƜ**	**ƝɲƝ**
patch	**ƄƅƄ**	ƆɔƆ	**ƈƈ**	**ƋƌƋ**	**ƌƍ**	**ƎǝƎ**	**ƏəƏ**	**ƐɛƐ**	**ƑƒF**	***fff***	**ƔɣƳ**	**ƖɩI**	**ƘƙĶ**	**ƙƙķ**	**ƜɯƜ**	**ƝɲƝ**
ideal	c, b	c, b	x+65, b	c, b	a-15, b	c, b	c, b	c, b	c, d+36	a, d+62	c, d+22	a, b	c, b	a, b	c, b	c, d+47
a		**275**, 833														
a?	**274, 850**	314, **850**	264, 708	**292, 850**	261, 870	**302, 850**	353, 850	**269, 850**	**236, 850**	221, 885	**294, 850**	**170, 885**	**338, 850**	**259, 885**	**444, 850**	**336, 850**
aδ		-38, -17														
a!			224, 645^{i}		255, 885^{i}		**331, 850**			**336, 885**						
b																
b?	**274, -110**	**314, -110**	264, -110	**292, -110**	261, -110	**302, -110**	**353, -110**	**269, -110**	236, -281	221, -255	**294, -295**	**170, -110**	338, -110	259, -110	**444, -110**	336, -270
bδ																
b!			224, -110^{i}		254, -110^{i}				236, -281^{i}	**104, -255**			342, -110^{i}	256, -110^{i}		336, -270^{i}
si	***ƄƅƄ***	*ƆɔƆ*	***ƈƈ***	***ƋƌƋ***	***ƌƍ***	***ƎǝƎ***	***ƏəƏ***	***ƐɛƐ***	***ƑƒF***	***fff***	***ƔɣƳ***	***ƖɩI***	***ƘƙĶ***	***ƙƙķ***	***ƜɯƜ***	***ƝɲƝ***
patch	***ƄƅƄ***	*ƆɔƆ*	***ƈƈ***	***ƋƌƋ***	***ƌƍ***	***ƎǝƎ***	***ƏəƏ***	***ƐɛƐ***	***ƑƒF***	***fff***	***ƔɣƳ***	***ƖɩI***	***ƘƙĶ***	***ƙƙķ***	***ƜɯƜ***	***ƝɲƝ***
ideal	c, b	c, b	x+65, b	c, b	a-15, b	c, b	c, b	c, b	c, d+36	a, d+62	c, d+22	a, b	c, b	a, b	c, b	c, d+47
a	267, 805				214, 650						331, 805				431, 801	
a?	**306, 850**	357, 850	253, 708	**304, 850**	275, 875	**320, 850**	369, 850	**293, 850**	**288, 850**	269, 890	**316, 850**	**194, 890**	**353, 850**	**283, 890**	**463, 850**	383, 850
aδ	-39, -45				-61, -225						15, -45				-32, -49	
a!		**313, 850**	270, 675^{i}		294, 890^{i}		**335, 850**			**386, 890**						366, 850^{i}
b	**268**, -108				**217**, -107									**263**, -107	450, -108	
b?	273, **-110**	**324, -110**	**249, -110**	**271, -110**	237, **-110**	**287, -110**	**336, -110**	**260, -110**	220, -281	198, -255	**245, -295**	**153, -110**	320, -110	241, **-110**	**430, -110**	317, -270
bδ	-5, 2				-20, 3									22, 3	20, 2	
b!									220, -281^{i}	**60, -255**			314, -110^{i}			288, -270^{i}

Table 7: Font metrics for Latin and IPA 3/8

hex	01A2	01A9	01B4	01B5	01B6	01B7	0221	0235	0237	023C	0243	0246	0249	024A	0250	0251
cat	Lu	Lu	Ll	Lu	Ll	Lu	Ll	Ll	Ll	Ll	Lu	Lu	Ll	Lu	Ll	Ll
reg	Ƣ	Ʃ	ƴ	Ƶ	ƶ	Ʒ	ȡ	ȵ	ȷ	ȼ	Ƀ	Ɇ	ɉ	Ɋ	ɐ	ɑ
patch	Ƣ	Ʃ	ƴ	Ƶ	ƶ	Ʒ	ȡ	ȵ	ȷ	ȼ	Ƀ	Ɇ	ɉ	Ɋ	ɐ	ɑ
ideal	c, d+52	c, b	x+128, d	c, b	x, b	c+22, d+21	a, d+107	x, b-93	x, d+9	x+25, b-28	c, b	c, b-13	a-51, d+9	c, d+40	x, b	x, b
a	**359**, 804		270, 644					261, 646	140, 645					**355**, 804	228, 645	**228, 645**
a?	469, **850**	277, 850	316, 771	310, 850	220, 645	**303, 850**	326, 885	333, 645	74, 645	216, 668	255, 850	250, 850	**116, 834**	380, **850**	**216, 645**	272, 645
aδ	-110, -46		-46, -127					-72, 1	66, 0					-26, -46	12, 0	-44, 0
a!		**268, 850**	257, 645^{i}	**320, 850**	220, 645^{i}		227, 885^{i}	260, 645^{i}	**152, 645**	243, 645^{i}	267, 850^{i}	281, 850^{i}				
b	**354, -113**					**272, -319**		262, -111						**350, -113**	229, -114	212, -114
b?	469, -265	277, -110	316, -319	310, -110	220, -110	303, -296	326, -210	333, -201	74, -308	216, -136	255, -110	**250, -121**	116, -308	380, -277	**216, -110**	272, -110
bδ	-115, 152					-31, -23		-71, 90						-30, 164	12, -4	-60, -4
b!		**277, -110**	237, -319^{i}	**287, -110**	207, -110^{i}		227, -210^{i}	261, -201^{i}	**109, -319**	218, -110^{i}	268, -110^{i}		109, -319^{i}			**248, -110**
it	*Ƣ*	*Ʃ*	*ƴ*	*Ƶ*	*ƶ*	*Ʒ*	*ȡ*	*ȵ*	*ȷ*	*ȼ*	*Ƀ*	⌧	*ɉ*	⌧	*ɐ*	*ɑ*
patch	*Ƣ*	*Ʃ*	*ƴ*	*Ƶ*	*ƶ*	*Ʒ*	*ȡ*	*ȵ*	*ȷ*	*ȼ*	*Ƀ*	⌧	*ɉ*	⌧	*ɐ*	*ɑ*
ideal	c, d+89	c, b	x+111, d	c, b	x, b	c+22, d+21	a, d+112	x, b-89	x, d+36	x+25, b-29	c, b		a-80, d+36		x, b	x, b
a	355, 804		270, 644					261, 646	140, 645							
a?	**622, 850**	**477, 850**	471, 754	470, 850	343, 645	485, 850	**507, 890**	460, 645	267, 645	352, 668	477, 850		**315, 810**		**365, 645**	**380, 645**
aδ	-267, -46		-201, -110					-199, 1	-127, 0							
a!			363, 645^{i}	**489, 850**	360, 645^{i}	**430, 850**		394, 645^{i}	**270, 645**	365, 645^{i}	457, 850^{i}					
b	**350, -113**							262, -111							229, -114	212, -114
b?	393, -228	**273, -110**	243, -319	266, -110	182, -110	241, -296	274, -205	281, -197	**70, -281**	**181, -137**	**273, -110**		83, -281		**205, -110**	220, -110
bδ	-43, 115							-19, 86							24, -4	-8, -4
b!			236, -319^{i}	**286, -110**	187, -110^{i}	**236, -319**	274, -205^{i}	217, -197^{i}					0, -319^{i}			**197, -110**
sb	**Ƣ**	**Ʃ**	**ƴ**	**Ƶ**	**ƶ**	**Ʒ**	**ȡ**	**ȵ**	**ȷ**	**ȼ**	**Ƀ**	⌧	**ɉ**	⌧	**ɐ**	**ɑ**
patch	**Ƣ**	**Ʃ**	**ƴ**	**Ƶ**	**ƶ**	**Ʒ**	**ȡ**	**ȵ**	**ȷ**	**ȼ**	**Ƀ**	⌧	**ɉ**	⌧	**ɐ**	**ɑ**
ideal	c, d+90	c, b	x+125, d	c, b	x, b	c+29, d+24	a, d+101	x, b-99	x, d	x+27, b-21	c, b		a-48, d		x, b-14	x, b
a									163, 645							
a?	**454, 850**	293, 850	**328, 768**	309, 850	222, 645	**316, 850**	338, 885	345, 645	78, 645	200, 670	294, 850		**112, 837**		**207, 645**	276, 645
aδ									85, 0							
a!		**283, 850**		**316, 850**	222, 645^{i}		272, 885^{i}	278, 645^{i}	**162, 645**	224, 645^{i}	283, 850^{i}					**238, 645**
b						**272, -319**									228, -114	
b?	**454, -227**	293, -110	**328, -319**	309, -110	222, -110	316, -293	338, -216	345, -207	**78, -319**	200, -129	294, -110		112, -319		**207, -122**	276, -110
bδ						-44, -26									21, 8	
b!		**298, -110**		**316, -110**	222, -110^{i}		272, -216^{i}	254, -207^{i}		224, -110^{i}	300, -110^{i}		109, -319^{i}			**237, -110**
si	***Ƣ***	***Ʃ***	***ƴ***	***Ƶ***	***ƶ***	***Ʒ***	***ȡ***	***ȵ***	***ȷ***	***ȼ***	***Ƀ***	⌧	***ɉ***	⌧	***ɐ***	***ɑ***
patch	***Ƣ***	***Ʃ***	***ƴ***	***Ƶ***	***ƶ***	***Ʒ***	***ȡ***	***ȵ***	***ȷ***	***ȼ***	***Ƀ***	⌧	***ɉ***	⌧	***ɐ***	***ɑ***
ideal	c, d+90	c, b	x+125, d+20	c+25, b	x, b	c+28, d+24	a, d+101	x, b-99	x, d+25	x+27, b-21	c, b		a-67, d+25		x, b	x, b
a	359, 804		270, 644					**261**, 646	140, 645							**228, 645**
a?	**464, 850**	**319, 850**	317, 768	332, 850	**187, 645**	**329, 850**	362, 890	324, **645**	121, 645	204, 670	319, 850		**192, 823**		**221, 645**	247, 645
aδ	-105, -46		-47, -124					-63, 1	19, 0							-19, 0
a!			**152, 645**	**312, 850**			311, 890^{i}		**150, 645**	255, 645^{i}	279, 850^{i}					
b	**354, -113**					**236, -319**		262, -111					109, -319		**229**, -114	212, -114
b?	407, -227	**286, -110**	**262, -297**	299, -110	**195, -110**	258, -293	299, -216	313, -207	**93, -292**	203, -129	286, -110		127, -292		230, **-110**	256, -110
bδ	-53, 114					-22, -26		-51, 96					-18, -27		-1, -4	-44, -4
b!				**267, -110**			200, -216^{i}	313, -207^{i}		248, -110^{i}	280, -110^{i}		79, -319^{i}			**225, -110**

Table 8: Font metrics for Latin and IPA 4/8

hex	0252	0254	0255	0256	0258	0259	025A	025B	025C	025D	025E	0260	0261	0262	0263	0264
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɒ̤ɒ̤	ɔ̤ɔ̤	ɕ̤ɕ̤	ɖ̤ɖ̤	ɘ̤ɘ̤	ə̤ə̤	ɚ̤ɚ̤	ɛ̤ɛ̤	ɜ̤ɜ̤	ɝ̤ɝ̤	ɞ̤ɞ̤	ɠ̤ɠ̤	ɡ̤ɡ̤	ɢ̤ɢ̤	ɣ̤ɣ̤	ɤ̤ɤ̤
patch	ɒ̤ɒ̤	ɔ̤ɔ̤	ɕ̤ɕ̤	ɖ̤ɖ̤	ɘ̤ɘ̤	ə̤ə̤	ɚ̤ɚ̤	ɛ̤ɛ̤	ɜ̤ɜ̤	ɝ̤ɝ̤	ɞ̤ɞ̤	ɠ̤ɠ̤	ɡ̤ɡ̤	ɢ̤ɢ̤	ɣ̤ɣ̤	ɤ̤ɤ̤
ideal	x, b	x, b	x, b-106	a, d+44	x, b	x, b	x+45, b	x, b	x, b	a-90, b	x, b	a, d	x, d	x, b	x, d+30	x, b
a	264, 645	**216, 645**	209, 645	218, 645	**204, 645**	195, 645	**229, 645**	209, 645	**194, 645**	209, 645	**239, 645**	209, 645	**239, 645**	**265, 645**	**259, 645**	**248, 645**
a?	**247, 645**	216, 645	222, 645	314, 885	220, 645	232, 645	306, 688	**202, 645**	196, 645	**234, 795**	242, 645	320, 885	228, 645	259, 645	258, 645	221, 645
aδ	17, 0	0, 0	-13, 0	-96, -240	-16, 0	-37, 0	-78, -43	8, 0	-2, 0	-26, -150	-2, 0	-112, -240	11, 0	6, 0	1, 0	27, 0
a!			243, 645^{i}	232, 885^{i}		**219, 645**						239, 885^{i}				
b	259, -114	**177**, -114	212, -114	212, -114	212, -114	186, -114	212, -114	212, -114	**184**, -114	212, -114	**256**, -114	236, -110	**226**, -320	**248, -110**	251, -277	**216**, -114
b?	**247, -110**	216, **-110**	222, -214	314, -273	220, -110	232, -110	306, -110	**202, -110**	196, **-110**	234, -110	242, **-110**	320, -319	228, **-319**	259, -110	**258, -287**	221, **-110**
bδ	12, -4	-38, -4	-10, 100	-102, 159	-8, -4	-46, -4	-94, -4	10, -4	-12, -4	-22, -4	14, -4	-84, 209	-2, -1	-11, 0	-7, 10	-5, -4
b!			218, -214^{i}	232, -273^{i}	**183, -110**	**212, -110**	**184, -110**			184, -110^{i}		**226, -319**				
it	*ɒ̤ɒ̤*	*ɔ̤ɔ̤*	*ɕ̤ɕ̤*	*ɖ̤ɖ̤*	*ɘ̤ɘ̤*	*ə̤ə̤*	*ɚ̤ɚ̤*	*ɛ̤ɛ̤*	*ɜ̤ɜ̤*	*ɝ̤ɝ̤*	*ɞ̤ɞ̤*	*ɠ̤ɠ̤*	*ɡ̤ɡ̤*	*ɢ̤ɢ̤*	*ɣ̤ɣ̤*	*ɤ̤ɤ̤*
patch	*ɒ̤ɒ̤*	*ɔ̤ɔ̤*	*ɕ̤ɕ̤*	*ɖ̤ɖ̤*	*ɘ̤ɘ̤*	*ə̤ə̤*	*ɚ̤ɚ̤*	*ɛ̤ɛ̤*	*ɜ̤ɜ̤*	*ɝ̤ɝ̤*	*ɞ̤ɞ̤*	*ɠ̤ɠ̤*	*ɡ̤ɡ̤*	*ɢ̤ɢ̤*	*ɣ̤ɣ̤*	*ɤ̤ɤ̤*
ideal	x, b	x, b	x, b-100	a+9, d+44	x, b	x, b	x+44, b	x, b	x, b	x+132, b	x, b-25	a+6, d+31	x, d+31	x, b	x, d+45	x, b+2
a		322, 654		218, 645	204, 645			322, 654						265, 645		
a?	**391, 645**	**332, 645**	352, 645	465, 899	356, 645	**356, 645**	**418, 687**	**338, 645**	**331, 645**	**413, 775**	**375, 645**	431, 896	**378, 645**	**388, 645**	**387, 645**	**338, 645**
aδ		-10, 9		-247, -254	-152, 0			-16, 9						-123, 0		
a!			350, 645^{i}	**440, 899**	**356, 645**							430, 890^{i}				
b	243, -114	**172**, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114	184, -114	212, -114	**256, -114**			**248**, -114		**177, -114**
b?	**231, -110**	171, **-110**	171, -208	215, -273	195, -110	195, -110	248, -110	**178, -110**	170, -110	225, -110	209, -133	**180, -286**	**180, -286**	227, **-110**	**192, -272**	178, -106
bδ	12, -4	1, -4	41, 94	-3, 159	17, -4	17, -4	-36, -4	34, -4	14, -4	-13, -4	47, 19			21, -4		-1, -8
b!			142, -208^{i}	215, -273^{i}	**202, -110**	**169, -110**	**161, -110**		**143, -110**	143, -110^{i}						
sb	**ɒ̤ɒ̤**	**ɔ̤ɔ̤**	**ɕ̤ɕ̤**	**ɖ̤ɖ̤**	**ɘ̤ɘ̤**	**ə̤ə̤**	**ɚ̤ɚ̤**	**ɛ̤ɛ̤**	**ɜ̤ɜ̤**	**ɝ̤ɝ̤**	**ɞ̤ɞ̤**	**ɠ̤ɠ̤**	**ɡ̤ɡ̤**	**ɢ̤ɢ̤**	**ɣ̤ɣ̤**	**ɤ̤ɤ̤**
patch	**ɒ̤ɒ̤**	**ɔ̤ɔ̤**	**ɕ̤ɕ̤**	**ɖ̤ɖ̤**	**ɘ̤ɘ̤**	**ə̤ə̤**	**ɚ̤ɚ̤**	**ɛ̤ɛ̤**	**ɜ̤ɜ̤**	**ɝ̤ɝ̤**	**ɞ̤ɞ̤**	**ɠ̤ɠ̤**	**ɡ̤ɡ̤**	**ɢ̤ɢ̤**	**ɣ̤ɣ̤**	**ɤ̤ɤ̤**
ideal	x, b	x, b-14	x, b-101	a, d+46	x, b	x, b	x+50, b	x, b	x, b	a-128, b	x, b	a+6, d+33	x, d	x, b	x, d+32	x, b
a		**199, 645**						229, 645								
a?	**251, 645**	199, 645	232, 645	322, 885	**230, 645**	**228, 645**	**319, 693**	**212, 645**	**216, 645**	**276, 757**	**249, 645**	336, 891	**235, 645**	**266, 645**	**272, 645**	**202, 645**
aδ		0, 0						16, 0								
a!			226, 645^{i}	272, 885^{i}								235, 885^{i}				
b		**184, -114**				**212**, -114										
b?	**251, -110**	199, -122	232, -209	322, -271	230, -110	228, **-110**	319, -110	**212, -110**	216, -110	276, -110	**249, -110**	**336, -284**	**235, -319**	**266, -110**	**272, -285**	**202, -110**
bδ		-15, 8				-16, -4										
b!			220, -209^{i}	272, -271^{i}	**212, -110**		**199, -110**		**204, -110**	205, -110^{i}						
si	***ɒ̤ɒ̤***	***ɔ̤ɔ̤***	***ɕ̤ɕ̤***	***ɖ̤ɖ̤***	***ɘ̤ɘ̤***	***ə̤ə̤***	***ɚ̤ɚ̤***	***ɛ̤ɛ̤***	***ɜ̤ɜ̤***	***ɝ̤ɝ̤***	***ɞ̤ɞ̤***	***ɠ̤ɠ̤***	***ɡ̤ɡ̤***	***ɢ̤ɢ̤***	***ɣ̤ɣ̤***	***ɤ̤ɤ̤***
patch	***ɒ̤ɒ̤***	***ɔ̤ɔ̤***	***ɕ̤ɕ̤***	***ɖ̤ɖ̤***	***ɘ̤ɘ̤***	***ə̤ə̤***	***ɚ̤ɚ̤***	***ɛ̤ɛ̤***	***ɜ̤ɜ̤***	***ɝ̤ɝ̤***	***ɞ̤ɞ̤***	***ɠ̤ɠ̤***	***ɡ̤ɡ̤***	***ɢ̤ɢ̤***	***ɣ̤ɣ̤***	***ɤ̤ɤ̤***
ideal	x, b	x, b	x, b-101	a, d+46	x, b	x, b	x+50, b	x, b	x, b	a-128, b	x, b	a, d+33	x, d	x, b	x, d+32	x, b
a	264, 645	**160, 645**		218, 645	**204, 645**			209, 645	194, 645					**265, 645**		
a?	**247, 645**	199, 645	**204, 645**	302, 890	207, 645	**218, 645**	**299, 693**	**194, 645**	**194, 645**	**261, 762**	**232, 645**	309, 890	**238, 645**	244, 645	**255, 645**	**185, 645**
aδ	17, 0	-39, 0		-84, -245	-3, 0			15, 0	0, 0					21, 0		
a!				311, 890^{i}								290, 890^{i}				
b	**259**, -114	**177**, -114	212, -114	212, -114	212, -114		212, -114	212, -114	184, -114	212, -114	**256**, -114			**248, -110**		**177**, -114
b?	256, **-110**	207, **-110**	192, -209	227, -271	216, -110	227, -110	298, -110	**203, -110**	203, -110	246, -110	241, **-110**	**232, -284**	**204, -319**	252, -110	**228, -285**	194, **-110**
bδ	3, -4	-30, -4	20, 95	-15, 157	-4, -4		-86, -4	9, -4	-19, -4	-34, -4	15, -4			-4, 0		-17, -4
b!			218, -209^{i}	189, -271^{i}	**201, -110**	**186, -110**	**188, -110**		**193, -110**	194, -110^{i}						

Table 9: Font metrics for Latin and IPA 5/8

hex	0265	0266	0267	0268	026A	026D	026E	026F	0270	0271	0272	0273	0274	0276	0277	0278
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɥ̰ɥ̰ɥ	ɦ̰ɦ̰ɦ	ɧ̰ɧ̰ɧ	ɨ̰ɨ̰ɨ	ɪ̰ɪ	ɭ̰ɭ	ɮ̰ɮ̰ɮ	ɯ̰ɯ̰ɯ	ɰ̰ɰ̰ɰ	ŋ̰ŋ̰ŋ	ɲ̰ɲ̰ɲ	ɳ̰ɳ̰ɳ	ɴ̰ɴ̰ɴ	ɶ̰ɶ̰ɶ	ɷ̰ɷ̰ɷ	ɸ̰ɸ̰ɸ
patch	ɥ̰ɥ̰ɥ	ɦ̰ɦ̰ɦ	ɧ̰ɧ̰ɧ	ɨ̰ɨ̰ɨ	ɪ̰ɪ	ɭ̰ɭ	ɮ̰ɮ̰ɮ	ɯ̰ɯ̰ɯ	ɰ̰ɰ̰ɰ	ŋ̰ŋ̰ŋ	ɲ̰ɲ̰ɲ	ɳ̰ɳ̰ɳ	ɴ̰ɴ̰ɴ	ɶ̰ɶ̰ɶ	ɷ̰ɷ̰ɷ	ɸ̰ɸ̰ɸ
ideal	x, d	a, b	a, d+11	a-51, b	x, b	a, d+44	a, d	x, b-13	x, d	x, d+11	x, d+11	x, d+11	x, b	x, b	x, b	a, d
a	**268, 645**	299, 565	299, 565	155, 795	133, 645	209, 645	209, 645	407, 645	209, 645	383, 645	264, 646	263, 645	209, 645	209, 645	209, 645	209, 645
a?	272, 645	262, 885	**228, 885**	150, 834	**136, 645**	182, 885	250, 885	**390, 645**	**379, 645**	366, 645	234, 645	296, 645	**278, 645**	340, 645	**357, 645**	**286, 885**
aδ	-4, 0	36, -320	72, -320	6, -39	-3, 0	27, -240	-42, -240	16, 0	-170, 0	18, 0	30, 1	-32, 0	-70, 0	-132, 0	-148, 0	-76, -240
a!		272, 885^{i}		**156, 834**		131, 885^{i}	134, 885^{i}			400, 645^{i}	253, 645^{i}	253, 645^{i}		**361, 645**		
b	219, -129	240, -114	240, -114	134, -114	139, -114	196, -110	196, -110	410, -114	294, -114	383, -105	265, -111	262, -112	**247**, -114	196, -110	**341**, -114	196, -110
b?	**272, -319**	262, -110	**228, -306**	150, -110	**136, -110**	182, -273	250, -319	**390, -121**	379, -319	366, -306	234, -306	296, -306	278, **-110**	340, -110	357, **-110**	**286, -319**
bδ	-52, 190	-22, -4	12, 192	-16, -4	3, -4	14, 163	-54, 209	20, 7	-85, 205	18, 201	30, 195	-34, 194	-32, -4	-144, 0	-16, -4	-90, 209
b!		262, -110^{i}		156, -110^{i}		131, -273^{i}	**275, -319**		**378, -121**	400, -306^{i}	254, -306^{i}	254, -306^{i}		**361, -110**		
it	*ɥ̰ɥ̰ɥ*	*ɦ̰ɦ̰ɦ*	*ɧ̰ɧ̰ɧ*	*ɨ̰ɨ̰ɨ*	*ɪ̰ɪ*	*ɭ̰ɭ*	*ɮ̰ɮ̰ɮ*	*ɯ̰ɯ̰ɯ*	*ɰ̰ɰ̰ɰ*	*ŋ̰ŋ̰ŋ*	*ɲ̰ɲ̰ɲ*	*ɳ̰ɳ̰ɳ*	*ɴ̰ɴ̰ɴ*	*ɶ̰ɶ̰ɶ*	*ɷ̰ɷ̰ɷ*	*ɸ̰ɸ̰ɸ*
patch	*ɥ̰ɥ̰ɥ*	*ɦ̰ɦ̰ɦ*	*ɧ̰ɧ̰ɧ*	*ɨ̰ɨ̰ɨ*	*ɪ̰ɪ*	*ɭ̰ɭ*	*ɮ̰ɮ̰ɮ*	*ɯ̰ɯ̰ɯ*	*ɰ̰ɰ̰ɰ*	*ŋ̰ŋ̰ŋ*	*ɲ̰ɲ̰ɲ*	*ɳ̰ɳ̰ɳ*	*ɴ̰ɴ̰ɴ*	*ɶ̰ɶ̰ɶ*	*ɷ̰ɷ̰ɷ*	*ɸ̰ɸ̰ɸ*
ideal	x, d	a, b	a, d	a-80, b	x, b	a, d+30	a, d+8	x, b-15	x, d	x, d	x, d	x, d	x, b	x, b	x, b-18	a, d+32
a	258, 645	299, 565			128, 645			407, 645		**383, 645**	264, 646	263, 645				
a?	**396, 645**	**449, 890**	**449, 890**	316, 810	281, 645	341, 890	447, 890	**517, 645**	**515, 645**	515, 645	396, 645	367, 645	**418, 645**	457, 645	**494, 645**	**481, 890**
aδ	-138, 0	-150, -325			-153, 0			-110, 0		-132, 0	-132, 1	-104, 0				
a!				**326, 810**	**280, 645**	**312, 890**	331, 890^{i}				414, 654^{i}	393, 645^{i}		**468, 645**		
b	219, -129	240, -114		134, -114	139, -114			410, -114	294, -114	383, -105	265, -111	262, -112	**247**, -114		**341, -114**	
b?	**191, -319**	236, -110	**192, -319**	121, -110	120, -110	91, -287	192, -309	**354, -123**	310, -319	310, -319	191, -319	162, -319	258, **-110**	296, -110	330, -126	**231, -285**
bδ	28, 190	4, -4		13, -4	19, -4			56, 9	-16, 205	73, 214	74, 208	100, 207	-11, -4		11, 12	
b!		229, -110^{i}		129, -110^{i}	**114, -110**	61, -287^{i}	**207, -319**		**362, -123**	310, -319^{i}	209, -319^{i}	190, -319^{i}		**308, -110**		
sb	**ɥ̰ɥ̰ɥ**	**ɦ̰ɦ̰ɦ**	**ɧ̰ɧ̰ɧ**	**ɨ̰ɨ̰ɨ**	**ɪ̰ɪ**	**ɭ̰ɭ**	**ɮ̰ɮ̰ɮ**	**ɯ̰ɯ̰ɯ**	**ɰ̰ɰ̰ɰ**	**ŋ̰ŋ̰ŋ**	**ɲ̰ɲ̰ɲ**	**ɳ̰ɳ̰ɳ**	**ɴ̰ɴ̰ɴ**	**ɶ̰ɶ̰ɶ**	**ɷ̰ɷ̰ɷ**	**ɸ̰ɸ̰ɸ**
patch	**ɥ̰ɥ̰ɥ**	**ɦ̰ɦ̰ɦ**	**ɧ̰ɧ̰ɧ**	**ɨ̰ɨ̰ɨ**	**ɪ̰ɪ**	**ɭ̰ɭ**	**ɮ̰ɮ̰ɮ**	**ɯ̰ɯ̰ɯ**	**ɰ̰ɰ̰ɰ**	**ŋ̰ŋ̰ŋ**	**ɲ̰ɲ̰ɲ**	**ɳ̰ɳ̰ɳ**	**ɴ̰ɴ̰ɴ**	**ɶ̰ɶ̰ɶ**	**ɷ̰ɷ̰ɷ**	**ɸ̰ɸ̰ɸ**
ideal	x, d	a, b	a, d	a-67, b	x, b	a, d	a, d	x-2, b-17	x, d	x, d	x, d	x, d	x, b	x, b	x, b	a, d+34
a					146, 645			377, 645								
a?	**278, 645**	274, 885	**243, 885**	152, 818	**152, 645**	194, 885	274, 885	**354, 641**	**388, 645**	363, 645	246, 645	306, 645	**290, 645**	334, 645	**369, 645**	**297, 885**
aδ					-6, 0			23, 4								
a!		273, 885^{i}		**156, 837**		150, 885^{i}	150, 885^{i}			396, 645^{i}	277, 645^{i}	275, 645^{i}		**338, 645**		
b				134, -114	156, -114			380, -114								
b?	**278, -319**	274, -110	**243, -319**	152, -110	**152, -110**	194, -319	274, -319	**354, -125**	388, -319	363, -319	246, -319	306, -319	**290, -110**	334, -110	**369, -110**	**297, -283**
bδ				-18, -4	4, -4			26, 11								
b!		274, -110^{i}		156, -110^{i}		150, -319^{i}	**283, -319**		**390, -125**	396, -319^{i}	253, -319^{i}	250, -319^{i}		**338, -110**		
si	***ɥ̰ɥ̰ɥ***	***ɦ̰ɦ̰ɦ***	***ɧ̰ɧ̰ɧ***	***ɨ̰ɨ̰ɨ***	***ɪ̰ɪ***	***ɭ̰ɭ***	***ɮ̰ɮ̰ɮ***	***ɯ̰ɯ̰ɯ***	***ɰ̰ɰ̰ɰ***	***ŋ̰ŋ̰ŋ***	***ɲ̰ɲ̰ɲ***	***ɳ̰ɳ̰ɳ***	***ɴ̰ɴ̰ɴ***	***ɶ̰ɶ̰ɶ***	***ɷ̰ɷ̰ɷ***	***ɸ̰ɸ̰ɸ***
patch	***ɥ̰ɥ̰ɥ***	***ɦ̰ɦ̰ɦ***	***ɧ̰ɧ̰ɧ***	***ɨ̰ɨ̰ɨ***	***ɪ̰ɪ***	***ɭ̰ɭ***	***ɮ̰ɮ̰ɮ***	***ɯ̰ɯ̰ɯ***	***ɰ̰ɰ̰ɰ***	***ŋ̰ŋ̰ŋ***	***ɲ̰ɲ̰ɲ***	***ɳ̰ɳ̰ɳ***	***ɴ̰ɴ̰ɴ***	***ɶ̰ɶ̰ɶ***	***ɷ̰ɷ̰ɷ***	***ɸ̰ɸ̰ɸ***
ideal	x, d	a, b	a, d	a-67, b	x, b	a, d	a, d	x-3, b	x, d	x, d	x, d	x, d	x, b	x-1, b	x, b	a, d+34
a	**258, 645**	299, 565	299, 565	137, 795	128, 645			407, 645		**383, 645**	**264**, 646	**263, 645**				
a?	258, 645	**302, 890**	**301, 890**	180, 823	137, 645	187, 890	301, 890	**372, 640**	**373, 645**	373, 645	254, **645**	254, 645	**275, 645**	313, 642	**350, 645**	**328, 890**
aδ	0, 0	-3, -325	-2, -325	-43, -28	-9, 0			35, 5		10, 0	10, 1	9, 0				
a!				**180, 823**	**139, 645**	**205, 890**	192, 890^{i}							**327, 645**		
b	219, -129	**240**, -114	240, -114	136, -110	139, -114			410, -114	294, -114	383, -105	265, -111	262, -112	**247**, -114		**341**, -114	
b?	**224, -319**	260, **-110**	**217, -319**	153, -110	145, -110	103, -319	217, -319	**382, -110**	339, -319	339, -319	220, -319	220, -319	283, **-110**	322, -110	359, **-110**	**251, -283**
bδ	-5, 190	-20, -4	23, 205	-17, 0	-6, -4			28, -4	-45, 205	44, 214	45, 208	42, 207	-36, -4		-18, -4	
b!				155, -110^{i}	**139, -110**	120, -319^{i}	**231, -319**		**386, -110**	339, -319^{i}	220, -319^{i}	220, -319^{i}		**336, -110**		

Table 10: Font metrics for Latin and IPA 6/8

hex	0279	027A	027B	027C	027D	027E	027F	0280	0281	0282	0283	0285	0287	0289	028A	028B
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
reg	ɹ	ɺ	ɾ	ɽ	ɻ	ɾ	ʀ	ʁ	ʕ	ɸ	β	θ	ʃ	ɧ	ʊ	ʋ
patch	ɹ	ɺ	ɾ	ɽ	ɻ	ɾ	ʀ	ʁ	ʕ	ɸ	β	θ	ʃ	ɧ	ʊ	ʋ
ideal	x, b-13	a-51, b-13	x, d+58	x, d	x, d	x, b	x, b	x, b	x, b	x, d+56	a, d+18	x+121, d+18	x, d+79	x, b	x, b	x, b
a	209, 645	209, 645		222, 703	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	**259, 645**
a?	178, 645	180, 834	226, 645	186, 645	186, 645	**174, 645**	**170, 645**	238, 645	**238, 645**	199, 645	**140, 885**	170, 764	**150, 645**	271, 645	**254, 645**	235, 645
aδ	32, 0	30, -189		36, 58	22, 0	34, 0	38, 0	-30, 0	-30, 0	10, 0	68, -240	38, -119	59, 0	-62, 0	-44, 0	24, 0
a!	**227, 645**	**232, 885**	**248, 645**	183, 645^{i}	175, 645^{i}			**226, 645**		206, 645^{i}		170, 645^{i}		255, 645^{i}		
b	190, -114	203, -114	196, -110	127, -305	196, -110	196, -110	**196, -110**	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	258, -114	**251**, -114
b?	**178, -121**	**180, -121**	226, -259	186, -319	186, -319	174, -110	170, -110	**238, -110**	**238, -110**	199, -261	140, -299	170, -299	**150, -238**	271, -110	**254, -110**	235, **-110**
bδ	12, 7	24, 7	-30, 149	-60, 14	10, 209	22, 0	26, 0	-42, 0	-42, 0	-3, 151	56, 189	26, 189	46, 128	-75, 0	4, -4	16, -4
b!			198, -259^{i}	**134, -319**	124, -319^{i}	**126, -110**				196, -261^{i}	**129, -319**	202, -306^{h}		239, -110^{i}		
it	*ɹ*	*ɺ*	*ɾ*	*ɽ*	*ɻ*	*ɾ*	*ʀ*	*ʁ*	*ʕ*	*ɸ*	*β*	*θ*	*ʃ*	*ɧ*	*ʊ*	*ʋ*
patch	*ɹ*	*ɺ*	*ɾ*	*ɽ*	*ɻ*	*ɾ*	*ʀ*	*ʁ*	*ʕ*	*ɸ*	*β*	*θ*	*ʃ*	*ɧ*	*ʊ*	*ʋ*
ideal	x, b	a-71, b-19	x, d+58	x, d+34	x, d+38	x, b	x, b	x, b	x, b	x, d+52	a+10, d+18	a+10, d+18	x, d+78	x, b	x, b-13	x, b-16
a				222, 703												**259, 645**
a?	315, 645	362, 819	352, 645	325, 645	320, 645	**311, 645**	**307, 645**	358, 645	**349, 645**	**314, 645**	**343, 900**	343, 900	**288, 645**	398, 645	**392, 645**	382, 645
aδ				-103, 58												-123, 0
a!	**275, 645**	**456, 890**	**389, 645**	318, 645^{i}	**359, 645**			**346, 645**				307, 645^{i}		**402, 645**		
b	190, -114	203, -114		127, -305											258, -114	**251, -114**
b?	**155, -110**	**161, -127**	160, -259	128, -283	124, -279	151, -110	**147, -110**	**197, -110**	**189, -110**	121, -265	88, -299	88, -299	**100, -239**	238, -110	**229, -121**	218, -124
bδ	35, -4	42, 13		-1, -22											29, 7	33, 10
b!			160, -259^{i}	**53, -319**	124, -279^{i}	**104, -110**				121, -265^{i}	**61, -319**	88, -319^{h}		**241, -110**		
sb	**ɹ**	**ɺ**	**ɾ**	**ɽ**	**ɻ**	**ɾ**	**ʀ**	**ʁ**	**ʕ**	**ɸ**	**β**	**θ**	**ʃ**	**ɧ**	**ʊ**	**ʋ**
patch	**ɹ**	**ɺ**	**ɾ**	**ɽ**	**ɻ**	**ɾ**	**ʀ**	**ʁ**	**ʕ**	**ɸ**	**β**	**θ**	**ʃ**	**ɧ**	**ʊ**	**ʋ**
ideal	x-2, b-17	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b	x, b	x, b	x, d+53	a, d+21	a, d+21	x, d+67	x, b	x, b	x, b
a															**266, 645**	
a?	144, 641	205, 816	238, 645	206, 645	198, 645	**186, 645**	**182, 645**	236, 645	**220, 645**	201, 645	**152, 885**	152, 885	**157, 645**	274, 645	267, 645	**256, 645**
aδ															-1, 0	
a!	**205, 645**	**271, 885**	260, 645^{i}	200, 645^{i}	196, 645^{i}			**241, 645**		201, 645^{i}		182, 645^{i}		264, 645^{i}		
b	190, -114															
b?	**144, -125**	**205, -110**	238, -257	206, -281	198, -319	186, -110	**182, -110**	**236, -110**	**220, -110**	201, -264	152, -296	152, -296	**157, -250**	274, -110	**267, -110**	**256, -110**
bδ	46, 11															
b!			196, -257^{i}	**141, -319**	135, -319^{i}	**138, -110**				201, -264^{i}	**141, -319**	214, -319^{h}		248, -110^{i}		
si	***ɹ***	***ɺ***	***ɾ***	***ɽ***	***ɻ***	***ɾ***	***ʀ***	***ʁ***	***ʕ***	***ɸ***	***β***	***θ***	***ʃ***	***ɧ***	***ʊ***	***ʋ***
patch	***ɹ***	***ɺ***	***ɾ***	***ɽ***	***ɻ***	***ɾ***	***ʀ***	***ʁ***	***ʕ***	***ɸ***	***β***	***θ***	***ʃ***	***ɧ***	***ʊ***	***ʋ***
ideal	x-3, b	a-69, b	x, d+60	x, d+36	x, d	x, b	x, b	x-1, b	x, b	x, d+52	a, d+21	a, d+21	x, d+67	x, b	x-1, b	x, b
a				222, 703												**259, 645**
a?	171, 640	217, 821	194, 645	181, 645	177, 645	**167, 645**	**167, 645**	214, 642	**214, 645**	**178, 645**	**195, 890**	195, 890	**149, 645**	255, 645	**248, 642**	241, 645
aδ				41, 58												18, 0
a!	**211, 645**	**313, 890**	246, 645^{i}	189, 645^{i}	**218, 645**			**206, 645**				167, 645^{i}		**275, 645**		
b	190, -114	203, -114		127, -305											**258**, -114	**251**, -114
b?	**180, -110**	**190, -110**	172, -257	155, -281	143, -319	176, -110	**176, -110**	**223, -110**	**223, -110**	155, -265	116, -296	116, -296	**129, -250**	264, -110	257, **-110**	249, **-110**
bδ	10, -4	13, -4		-28, -24											1, -4	2, -4
b!			185, -257^{i}	**78, -319**	143, -319^{i}	**132, -110**				155, -265^{i}	**88, -319**	131, -319^{h}		**284, -110**		

Table 11: Font metrics for Latin and IPA 7/8

hex	028C	028D	028E	028F	0290	0291	0292	0293	0294	0295	0296	0297	0299	029C	029D	029F
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll
reg	ʌʌ̇ʌ̲	ɯɯ̇ɯ̲	ʎʎ̇ʎ̲	ʏʏ̇ʏ̲	ʐʐ̇ʐ̲	ʑʑ̇ʑ̲	ʒʒ̇ʒ̲	ʒʒ̇ʒ̲	ʔʔ̇ʔ	ʕʕ̇ʕ	ɧɧ̇ɧ	cc̈c	ʙʙ̇ʙ̲	ʜʜ̇ʜ̲	jj̇j̲	ɭɭ̇ɭ̲
patch	ʌʌ̇ʌ̲	ɯɯ̇ɯ̲	ʎʎ̇ʎ̲	ʏʏ̇ʏ̲	ʐʐ̇ʐ̲	ʑʑ̇ʑ̲	ʒʒ̇ʒ̲	ʒʒ̇ʒ̲	ʔʔ̇ʔ	ʕʕ̇ʕ	ɧɧ̇ɧ	cc̈c	ʙʙ̇ʙ̲	ʜʜ̇ʜ̲	jj̇j̲	ɭɭ̇ɭ̲
ideal	x, b	x, b	a-37, b	x, b	x, d+47	x+19, d+110	x, d	x, d	c, b	c, b	c, b	x, d+11	x, b	x, b	a-51, d+55	x, b
a	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	**218**, 646	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	**244, 645**	**367, 645**	248, 848	**236, 645**	262, 645	218, 662	216, **645**	220, 645	**211, 850**	**221, 850**	**208, 850**	221, 645	232, 645	**293, 645**	126, 834	207, 645
aδ	-35, 0	-158, 0	-39, -203	-27, 0	-54, 0	-8, -17	2, 1	-11, 0	-2, -205	-12, -205	1, -205	-12, 0	-24, 0	-84, 0	83, -189	2, 0
a!			**242, 885**		220, 645[i]	206, 645[i]		**222, 645**				**252, 645**	**241, 645**		162, 834[i]	**136, 645**
b	196, -110	196, -110	196, -110	**239**, -107	196, -110	196, -110	**185, -319**	196, -110	**202**, -107	**230**, -107	196, -110	196, -110	196, -110	**305, -110**	196, -110	196, -110
b?	244, -110	367, -110	**248, -110**	236, **-110**	262, -270	218, -207	216, -319	220, -319	211, **-110**	221, **-110**	**208, -110**	**221, -306**	**232, -110**	293, -110	**126, -262**	**207, -110**
bδ	-48, 0	-171, 0	-52, 0	3, 3	-66, 160	-22, 97	-32, 0	-24, 209	-9, 3	9, 3	-12, 0	-25, 196	-36, 0	12, 0	70, 152	-11, 0
b!	**229, -110**	**378, -110**			207, -270[i]	192, -207[i]		220, -319[i]								
it	*ʌʌ̇ʌ̲*	*ɯɯ̇ɯ̲*	*ʎʎ̇ʎ̲*	*ʏʏ̇ʏ̲*	*ʐʐ̇ʐ̲*	*ʑʑ̇ʑ̲*	*ʒʒ̇ʒ̲*	*ʒʒ̇ʒ̲*	*ʔʔ̇ʔ*	*ʕʕ̇ʕ*	*ɧɧ̇ɧ*	*cc̈c*	*ʙʙ̇ʙ̲*	*ʜʜ̇ʜ̲*	*jj̇j̲*	*ɭɭ̇ɭ̲*
patch	*ʌʌ̇ʌ̲*	*ɯɯ̇ɯ̲*	*ʎʎ̇ʎ̲*	*ʏʏ̇ʏ̲*	*ʐʐ̇ʐ̲*	*ʑʑ̇ʑ̲*	*ʒʒ̇ʒ̲*	*ʒʒ̇ʒ̲*	*ʔʔ̇ʔ*	*ʕʕ̇ʕ*	*ɧɧ̇ɧ*	*cc̈c*	*ʙʙ̇ʙ̲*	*ʜʜ̇ʜ̲*	*jj̇j̲*	*ɭɭ̇ɭ̲*
ideal	x, b	x, b	a-52, b	x, b	x, d+41	x, d+110	x, d	x, d-80	c+27, b	c+38, b	c, b-23	x, d	x, b	x, b	a-80, d+63	x, b
a							**328**, 646									
a?	**383, 645**	**506, 645**	433, 838	**372, 645**	343, 645	352, 645	350, **645**	**355, 645**	373, 850	373, 850	**373, 850**	**411, 645**	352, 645	**416, 645**	303, 810	355, 645
aδ							-22, 1									
a!			**442, 890**		**354, 645**	**354, 645**			**376, 867**	**379, 878**			**371, 645**		325, 810[i]	**277, 645**
b				**239**, -107			**130, -319**		**202**, -107	**230**, -107				**273, -110**		
b?	222, -110	346, -110	**231, -110**	212, **-110**	147, -276	171, -207	145, -319	133, -397	169, **-110**	169, **-110**	**164, -131**	**206, -319**	**192, -110**	256, -110	**77, -254**	**194, -110**
bδ				27, 3			-15, 0		33, 3	61, 3				17, 0		
b!	**210, -110**	**361, -110**			147, -276[i]	171, -207[i]		**133, -399**								
sb	**ʌʌ̇ʌ̲**	**ɯɯ̇ɯ̲**	**ʎʎ̇ʎ̲**	**ʏʏ̇ʏ̲**	**ʐʐ̇ʐ̲**	**ʑʑ̇ʑ̲**	**ʒʒ̇ʒ̲**	**ʒʒ̇ʒ̲**	**ʔʔ̇ʔ**	**ʕʕ̇ʕ**	**ɧɧ̇ɧ**	**cc̈c**	**ʙʙ̇ʙ̲**	**ʜʜ̇ʜ̲**	**jj̇j̲**	**ɭɭ̇ɭ̲**
patch	**ʌʌ̇ʌ̲**	**ɯɯ̇ɯ̲**	**ʎʎ̇ʎ̲**	**ʏʏ̇ʏ̲**	**ʐʐ̇ʐ̲**	**ʑʑ̇ʑ̲**	**ʒʒ̇ʒ̲**	**ʒʒ̇ʒ̲**	**ʔʔ̇ʔ**	**ʕʕ̇ʕ**	**ɧɧ̇ɧ**	**cc̈c**	**ʙʙ̇ʙ̲**	**ʜʜ̇ʜ̲**	**jj̇j̲**	**ɭɭ̇ɭ̲**
ideal	x, b	x, b	a-36, b	x, b	x, d+41	x, d+107	x, d	x, d-96	c, b	c, b	c, b	x, d	x, b	x, b	a-64, d+65	x, b
a							**234, 645**									
a?	**231, 645**	**356, 645**	243, 849	**244, 645**	254, 645	**230, 645**	227, 645	236, 645	**206, 850**	**206, 850**	**206, 850**	**273, 645**	**246, 645**	**316, 645**	132, 821	219, 645
aδ							7, 0									
a!			**230, 885**		222, 645[i]			**231, 645**							172, 837[i]	**148, 645**
b							**193, -319**							**351, -110**		
b?	231, -110	**356, -110**	**243, -110**	**244, -110**	254, -276	230, -210	227, -319	236, -413	**206, -110**	**206, -110**	**206, -110**	**273, -319**	**246, -110**	316, -110	**132, -252**	**219, -110**
bδ							-34, 0							35, 0		
b!	**209, -110**				222, -276[i]	230, -210[i]		**236, -415**								
si	***ʌʌ̇ʌ̲***	***ɯɯ̇ɯ̲***	***ʎʎ̇ʎ̲***	***ʏʏ̇ʏ̲***	***ʐʐ̇ʐ̲***	***ʑʑ̇ʑ̲***	***ʒʒ̇ʒ̲***	***ʒʒ̇ʒ̲***	***ʔʔ̇ʔ***	***ʕʕ̇ʕ***	***ɧɧ̇ɧ***	***cc̈c***	***ʙʙ̇ʙ̲***	***ʜʜ̇ʜ̲***	***jj̇j̲***	***ɭɭ̇ɭ̲***
patch	***ʌʌ̇ʌ̲***	***ɯɯ̇ɯ̲***	***ʎʎ̇ʎ̲***	***ʏʏ̇ʏ̲***	***ʐʐ̇ʐ̲***	***ʑʑ̇ʑ̲***	***ʒʒ̇ʒ̲***	***ʒʒ̇ʒ̲***	***ʔʔ̇ʔ***	***ʕʕ̇ʕ***	***ɧɧ̇ɧ***	***cc̈c***	***ʙʙ̇ʙ̲***	***ʜʜ̇ʜ̲***	***jj̇j̲***	***ɭɭ̇ɭ̲***
ideal	x, b	x, b	a-56, b	x, b	x, d+38	x, d+107	x, d	x, d-96	c, b	c, b	c, b	x, d+15	x, b	x+28, b	a-78, d+65	x, b
a							218, 646									
a?	**239, 645**	**363, 645**	286, 834	**228, 645**	202, 645	208, 645	206, 645	211, 645	**227, 850**	**227, 850**	**227, 850**	**268, 645**	**213, 645**	**342, 671**	158, 812	211, 645
aδ							12, 1									
a!			**286, 890**		**244, 645**	**215, 645**	**211, 645**	**213, 645**							186, 823[i]	**137, 645**
b				**239**, -107			**141, -319**		**202**, -107	**230**, -107				**327, -110**		
b?	247, -110	371, -110	**256, -110**	237, **-110**	176, -279	196, -210	172, -319	157, -413	194, **-110**	194, **-110**	**194, -110**	**237, -302**	**221, -110**	345, -110	**103, -252**	**219, -110**
bδ				2, 3			-31, 0		8, 3	36, 3				-18, 0		
b!	**209, -110**	**369, -110**			176, -279[i]	196, -210[i]		**157, -415**								

Table 12: Font metrics for Latin and IPA 8/8

hex	02A0	02AE	02AF	03B2	03B8	03C7	1EFA	2132	2C64	2C66	2C69	2C6A	2C6B	2C6C	AB52
cat	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Lu	Lu	Ll	Lu	Ll	Lu	Ll	Ll
reg	ɋɋ̧ɋ̧	ɥɥ̧ɯ̄	ɥɥ̧ɯ̄	ββ̧β	θθ̧θ	χχ̇χ	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	ⱤⱤ̧Ɽ	ⱦⱦ̧	ⱪⱪ̧Ⱪ	ⱪ̇ⱪ̧	ⱬⱬ̧Ⱬ	ⱬ̇ⱬ̧	ꝟꝟ̧ꝟ	ꝟ̇ꝟ̧
patch	ɋɋ̧ɋ̧	ɥɥ̧ɯ̧	ɥɥ̧ɯ̧	ββ̧β	θθ̧θ	χχ̇χ	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	ⱤⱤ̧Ɽ	ⱦⱦ̧	ⱪⱪ̧Ⱪ	ⱪ̇ⱪ̧	ⱬⱬ̧Ⱬ	ⱬ̇ⱬ̧	ꝟꝟ̧ꝟ	ꝟ̇ꝟ̧
ideal	a, d	x, d	x, d	a, d	a, b	x, d	c, b	c, b	c, d+60	a-116, b-62	c, d+91	a, d+104	c+22, d+91	x+22, d+104	x, b
a	238, 645	278, -214	278, -214				**276**, 804	201, -203			336, 798	275, 646	**320**, 803	**218**, **646**	**376**, **645**
a?	320, 885	**288**, **645**	**328**, **645**	**262**, **885**	**245**, **885**	**254**, **645**	388, **850**	**241**, **850**	301, 850	146, 769	326, 850	259, 885	322, **850**	206, 665	332, 645
aδ	-82, -240	-10, -859	-50, -859				-112, -46	-40, -1053			10, -52	16, -239	-2, -47	12, -19	44, 0
a!	236, 885^{i}								269, 850^{i}	139, 769^{i}	328, 850^{i}	274, 885^{i}			
b	196, -110	318, 542	318, 542	275, -114			**387**, **-110**	276, 757			342, -109	263, -107	**287**, **-107**	204, -111	360, -110
b?	320, -319	**288**, **-319**	**328**, **-319**	**262**, **-319**	**245**, **-110**	**254**, **-319**	388, -110	**241**, **-110**	301, -257	146, -170	326, -226	259, -213	322, -226	**206**, **-213**	332, -110
bδ	-124, 209	30, 861	-10, 861	14, 205			-1, 0	35, 867			16, 117	4, 106	-34, 119	-2, 102	28, 0
b!	381, -319^{i}								301, -257^{i}	163, -110^{i}	334, -110^{i}	263, -213^{i}			360, -110^{i}
it	*ɋɋ̧ɋ*	*ɥɥ̧ɯ̄*	*ɥɥ̧ɯ̄*	*ββ̧β*	*θθ̧θ*	*χχ̇χ*	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	*ⱤⱤ̧Ɽ*	*ⱦⱦ̧*	*ⱪⱪ̧Ⱪ*	*ⱪ̇ⱪ̧*	*ⱬⱬ̧Ⱬ*	*ⱬ̇ⱬ̧*	*ꝟꝟ̧ꝟ*	*ꝟ̇ꝟ̧*	ⵊⵊ̧ⵊ
patch	*ɋɋ̧ɋ*	*ɥɥ̧ɯ̧*	*ɥɥ̧ɯ̧*	*ββ̧β*	*θθ̧θ*	*χχ̇χ*	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	*ⱤⱤ̧Ɽ*	*ⱦⱦ̧*	*ⱪⱪ̧Ⱪ*	*ⱪ̇ⱪ̧*	*ⱬⱬ̧Ⱬ*	*ⱬ̇ⱬ̧*	*ꝟꝟ̧ꝟ*	*ꝟ̇ꝟ̧*	ⵊⵊ̧ⵊ
ideal	a-19, d	x, d+12	x, d-9	a, d	a, b	x, d			c, d+12	x+132, b-28	c, b-115	a, b-115	c+22, b-99	x+22, b-99	
a	238, 645	278, -214	278, -214								336, 798	275, 646	320, 803	218, 646	
a?	460, 871	**422**, **645**	**433**, **645**	**425**, **890**	**413**, **890**	**372**, **645**			**465**, **850**	364, 775	**516**, **850**	**453**, **890**	**490**, **850**	**347**, **665**	
aδ	-222, -226	-144, -859	-155, -859								-180, -52	-178, -244	-170, -47	-129, -19	
a!	421, 890^{i}									358, 775^{i}					
b		318, 542	318, 542	275, -114							**342**, **-109**	263, -107	**287**, **-107**	204, -111	
b?	**207**, **-319**	**220**, **-305**	**227**, **-326**	**168**, **-319**	**200**, **-110**	**167**, **-319**			220, -305	**170**, **-136**	288, -223	216, -223	265, -207	**162**, **-207**	
bδ		98, 847	91, 868	107, 205							54, 114	47, 116	22, 100	42, 96	
b!									220, -305^{i}			216, -223^{i}			
sb	**ɋɋ̧ɋ**	**ɥɥ̧ɯ**	**ɥɥ̧ɯ**	**ββ̧β**	**θθ̧θ**	**χχ̇χ**	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	**ⱤⱤ̧Ɽ**	**ⱦⱦ̧**	**ⱪⱪ̧Ⱪ**	**ⱪ̇ⱪ̧**	**ⱬⱬ̧Ⱬ**	**ⱬ̇ⱬ̧**	**ꝟꝟ̧ꝟ**	**ꝟ̇ꝟ̧**	ⵊⵊ̧ⵊ
patch	**ɋɋ̧ɋ**	**ɥɥ̧ɯ**	**ɥɥ̧ɯ**	**ββ̧β**	**θθ̧θ**	**χχ̇χ**	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	**ⱤⱤ̧Ɽ**	**ⱦⱦ̧**	**ⱪⱪ̧Ⱪ**	**ⱪ̇ⱪ̧**	**ⱬⱬ̧Ⱬ**	**ⱬ̇ⱬ̧**	**ꝟꝟ̧ꝟ**	**ꝟ̇ꝟ̧**	ⵊⵊ̧ⵊ
ideal	a, d	x, d	x, d	a, d	a-17, b	x, d+42			c, d	a-103, b-62	c, b-113	a, b-114	c+25, b-98	x+24, b-98	
a															
a?	337, 885	**300**, **645**	**340**, **645**	**290**, **885**	**249**, **868**	**279**, **645**			**292**, **850**	146, 782	**345**, **850**	**273**, **885**	**327**, **850**	**212**, **667**	
aδ															
a!	269, 885^{i}									168, 782^{i}					
b															
b?	337, -319	**300**, **-319**	**340**, **-319**	**290**, **-319**	**249**, **-110**	**279**, **-275**			292, -319	146, -170	**345**, **-221**	273, -222	**327**, **-206**	**212**, **-206**	
bδ															
b!	269, -319^{i}								292, -319^{i}	162, -110^{i}		273, -222^{i}			
si	***ɋɋ̧ɋ***	***ɥɥ̧ɯ***	***ɥɥ̧ɯ***	***ββ̧β***	***θθ̧θ***	***χχ̇χ***	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	***ⱤⱤ̧Ɽ***	***ⱦⱦ̧***	***ⱪⱪ̧Ⱪ***	***ⱪ̇ⱪ̧***	***ⱬⱬ̧Ⱬ***	***ⱬ̇ⱬ̧***	***ꝟꝟ̧ꝟ***	***ꝟ̇ꝟ̧***	ⵊⵊ̧ⵊ
patch	***ɋɋ̧ɋ***	***ɥɥ̧ɯ***	***ɥɥ̧ɯ***	***ββ̧β***	***θθ̧θ***	***χχ̇χ***	ⵊⵊ̧ⵊ	ⅎⅎ̇ⅎ	***ⱤⱤ̧Ɽ***	***ⱦⱦ̧***	***ⱪⱪ̧Ⱪ***	***ⱪ̇ⱪ̧***	***ⱬⱬ̧Ⱬ***	***ⱬ̇ⱬ̧***	***ꝟꝟ̧ꝟ***	***ꝟ̇ꝟ̧***	ⵊⵊ̧ⵊ
ideal	a, d	x, d	x, d	a, d	a-17, b	x, d+42			c, d+15	a-125, b-62	c, b-113	a, b-114	c+25, b-98	x, b-98	
a	238, 645	278, -214	278, -214								336, 798	275, 646	320, 803	**218**, 646	
a?	318, 890	**279**, **645**	**289**, **645**	**302**, **890**	**277**, **873**	**256**, **645**			**307**, **850**	187, 765	**358**, **850**	**295**, **890**	**332**, **850**	187, **645**	
aδ	-80, -245	-1, -859	-11, -859								-22, -52	-20, -244	-12, -47	31, 1	
a!	278, 890^{i}									201, 765^{i}					
b		318, 542	318, 542	275, -114							**342**, **-109**	263, -107	**287**, **-107**	204, -111	
b?	**234**, **-319**	**245**, **-319**	**255**, **-319**	**218**, **-319**	**239**, **-110**	**231**, **-275**			235, -302	159, -170	302, -221	230, -222	279, -206	**176**, **-206**	
bδ		73, 861	63, 861	57, 205							40, 112	33, 115	8, 99	28, 95	
b!									235, -302^{i}	137, -140^{i}		230, -222^{i}			

Table 13: Font metrics for above marks 1/2

hex	0300	0301	0302	0303	0304	0306	0307	0308	0309	030A	030B	030C	030D	030F	0310
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	ǹH̀	ńH́	n̂Ĥ	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̉Ḥ	n̊Ḫ	ő̃Ḫ	n̆Ḫ	ṅḢ	n̋Ḫ	n̊̃Ḫ
patch	ǹH̀	ńH́	n̂Ĥ	ñH̃	n̄H̄	n̆H̆	ṅḢ	n̈Ḧ	n̉Ḥ	n̊Ḫ	ő̃Ḫ	n̆Ḫ	ṅḢ	n̋Ḫ	n̊̃Ḫ
m	-142, 683	-191, 682	-139, 697	-149, 697	-209, 644	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
mp	-142, 683	-191, 682	-140, 697	-149, 697	-212, 662	-161, 681	-180, 636	-210, 645	-160, 765	-182, 708	-219, 725	-204, 676	-154, 688	-140, 758	-251, 675
mδ			-1, 0		-3, 18		0, -31			2, 0	0, 11	1, 0	1, 0		
it	*ǹH*	*ńH*	*n̂H*	*ñH̃*	*n̄H̄*	*n̆H*	*ṅḢ*	*n̈Ḧ*	*n̉Ḥ*	*n̊Ḫ*	*ő̃Ḫ*	*n̆Ḫ*	*ṅḢ*	*n̋Ḫ*	*n̊̃Ḫ*
patch	*ǹH̀*	*ńH́*	*n̂Ĥ*	*ñH̃*	*n̄H̄*	*n̆H̆*	*ṅḢ*	*n̈Ḧ*	*n̉Ḥ*	*n̊Ḫ*	*ő̃Ḫ*	*n̆Ḫ*	*ṅḢ*	*n̋Ḫ*	*n̊̃Ḫ*
m	-3, 683	-52, 682	3, 697	-17, 697	-78, 645	-22, 681	-40, 689	-79, 645	-4, 765	-40, 708	-74, 714	-67, 676	-15, 688	14, 758	-114, 675
mp	-3, 683	-52, 682	3, 697	-6, 697	-79, 645	-22, 681	-41, 668	-79, 645	-4, 765	-37, 708	-55, 725	-66, 676	-13, 688	14, 758	-114, 675
mδ				11, 0	-1, 0		-1, -21			3, 0	19, 11	1, 0	2, 0		
sb	**ǹH**	**ńH**	**n̂H**	**ñH**	**n̄H**	**n̆H**	**ṅḢ**	**n̈Ḧ**	**n̉Ḥ**	**n̊Ḫ**	**ő̃Ḫ**	**n̆Ḫ**	**ṅḢ**	**n̋Ḫ**	**n̊̃Ḫ**
patch	**ǹH̀**	**ńH́**	**n̂Ĥ**	**ñH̃**	**n̄H̄**	**n̆H̆**	**ṅḢ**	**n̈Ḧ**	**n̉Ḥ**	**n̊Ḫ**	**ő̃Ḫ**	**n̆Ḫ**	**ṅḢ**	**n̋Ḫ**	**n̊̃Ḫ**
m	-127, 713	-180, 713	-127, 700	-133, 679	-205, 625	-149, 684	-169, 603 (1)	-198, 557 (1)	-162, 682 (1)	-170, 646 (1)	-197, 733	-193, 676	-137, 719	-127, 758	-239, 675
mp	-127, 683	-180, 683	-128, 700	-136, 693	-200, 613	-149, 684	-168, 661	-186, 644	-162, 778	-170, 715	-197, 733	-192, 676	-137, 688	-127, 758	-235, 675
mδ	0, -30	0, -30	-1, 0	-3, 14	5, -12		1, -38	12, -9		0, -27		1, 0	0, -31		4, 0
si	***ǹH̀***	***ńH́***	***n̂Ĥ***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̉Ḥ***	***n̊Ḫ***	***ő̃Ḫ***	***n̆Ḫ***	***ṅḢ***	***n̋Ḫ***	***n̊̃Ḫ***
patch	***ǹH̀***	***ńH́***	***n̂Ĥ***	***ñH̃***	***n̄H̄***	***n̆H̆***	***ṅḢ***	***n̈Ḧ***	***n̉Ḥ***	***n̊Ḫ***	***ő̃Ḫ***	***n̆Ḫ***	***ṅḢ***	***n̋Ḫ***	***n̊̃Ḫ***
m	-142, 683	-191, 682	-139, 697	-134, 700	-209, 645	-161, 681	-180, 667	-210, 645	-160, 765	-184, 708	-219, 714	-205, 676	-155, 688	-140, 758	-251, 675
mp	-142, 683	-191, 682	-136, 697	-145, 700	-225, 613	-161, 681	-183, 662	-248, 645	-142, 765	-175, 714	-217, 725	-205, 676	-147, 688	-140, 758	-248, 675
mδ			3, 0	-11, 0	-16, -32		-3, -5	-38, 0	18, 0	9, 6	2, 11		8, 0		3, 0

Table 14: Font metrics for above marks 2/2

hex	0311	0312	033D	033E	033F	030E	0346	034A	034B	034C	0350	0351	0352	0357	035B
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̑H̑	ṅḢ̑	n̑H̑	ṅḢ̑	n̄H̄	n̋H̋	n̑H̑	n̑H̑	n̊H̊	n̄H̄	n̂H̑	ṅḢ̑	n̊H̊	ṅḢ̑	n̂H̑
patch	n̑H̑	ṅḢ̑	n̑H̑	ṅḢ̑	n̄H̄	n̋H̋	n̑H̑	n̑H̑	n̊H̊	n̄H̄	n̂H̑	ṅḢ̑	n̊H̊	ṅḢ̑	n̂H̑
m	-205, 645	-110, 605	-265, 605	-265, 605	-235, 611	-245, 719	-251, 621	-215, 645	-205, 645	-235, 645	-201, 645	-171, 645	-251, 645	-171, 645	-164, 645
mp	-208, 645	-110, 605	-260, 605	-247, 605	-240, 611	-244, 719	-255, 621	-217, 645	-217, 645	-238, 645	-201, 645	-171, 645	-251, 645	-171, 645	-174, 645
mδ	-3, 0		5, 0	18, 0	-5, 0	1, 0	-4, 0	-2, 0	-12, 0	-3, 0					-10, 0
it	*n̂H̑*	*ṅḢ̑*	*n̑H̑*	*ṅḢ̑*	*n̄H̄*	*n̋H̋*	*n̑H̑*	*n̑H̑*	*n̊H̊*	*n̄H̄*	*n̂H̑*	*ṅḢ̑*	*n̊H̊*	*ṅḢ̑*	*n̂H̑*
patch	*n̂H̑*	*ṅḢ̑*	*n̑H̑*	*ṅḢ̑*	*n̄H̄*	*n̋H̋*	*n̑H̑*	*n̑H̑*	*n̊H̊*	*n̄H̄*	*n̂H̑*	*ṅḢ̑*	*n̊H̊*	*ṅḢ̑*	*n̂H̑*
m	-74, 645	13, 605	-142, 605	-142, 605	-111, 611	-99, 719	-125, 621	-84, 645	-74, 645	-104, 645	-70, 645	-40, 645	-120, 645	-40, 645	-33, 645
mp	-75, 645	13, 605	-137, 605	-123, 605	-115, 611	-97, 719	-128, 621	-85, 645	-85, 645	-106, 645	-70, 645	-40, 645	-119, 645	-40, 645	-43, 645
mδ	-1, 0		5, 0	19, 0	-4, 0	2, 0	-3, 0	-1, 0	-11, 0	-2, 0			1, 0		-10, 0
sb	**n̂H̑**	**ṅḢ**	**n□H□**	**n□H□**	**n□H□**	**n̋Ḣ**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**ṅH‘**	**n□H□**	**ṅH’**	**n̆H**
patch	**n̂H̑**	**ṅḢ̑**	**n̑H̑**	**ṅḢ̑**	**n̄H̄**	**n̋H̋**	**n̑H̑**	**n̑H̑**	**n̊H̊**	**n̄H̄**	**n̂H̑**	**ṅḢ̑**	**n̊H̊**	**ṅḢ̑**	**n̂H̑**
m	-205, 645	-97, 504 (1)				-232, 719						100, 547 (1)		100, 549 (1)	-180, 639
mp	-220, 645	-97, 600	-248, 605	-235, 605	-228, 611	-232, 719	-243, 621	-205, 645	-204, 645	-226, 645	-189, 645	90, 623	-239, 645	82, 613	-162, 645
mδ	-15, 0		new	new	new		new	new	new	new	new	-10, -20	new	-18, -32	18, 6
si	***n̂H̑***	***ṅḢ̑***	***n□H□***	***n□H□***	***n□H□***	***n̋Ḣ***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***
patch	***n̂H̑***	***ṅḢ̑***	***n̑H̑***	***ṅḢ̑***	***n̄H̄***	***n̋H̋***	***n̑H̑***	***n̑H̑***	***n̊H̊***	***n̄H̄***	***n̂H̑***	***ṅḢ̑***	***n̊H̊***	***ṅḢ̑***	***n̂H̑***
m	-205, 645	-110, 605				-245, 719									
mp	-226, 645	-110, 605	-275, 605	-262, 605	-253, 611	-236, 719	-266, 621	-224, 645	-224, 645	-245, 645	-201, 645	-171, 645	-258, 645	-171, 645	-181, 645
mδ	-21, 0		new	new	new	9, 0	new	new	new	new	new	new	new	new	new

Table 15: Font metrics for below marks 1/2

hex	0316	0317	0318	0319	031C	0320	0323	0324	0325	0326	0329	032A	032B	032C	032D	032E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̖	n̗	n̘	n̙	n̜	n̝	n̞	n̟	n̠	n̡	n̢	n̪	n̫	n̬	ṋ	n̮
patch	n̖	n̗	n̘	n̙	n̜	n̝	n̞	n̟	n̠	n̡	n̢	n̪	n̫	n̬	ṋ	n̮
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
mp	-219, -133	-210, -132	-249, -118	-249, -117	-216, -114	-252, -148	-216, -126	-217, -126	-217, -94	-255, -82	-256, -122	-255, -113	-252, -108	-204, -116	-195, -133	-252, -110
mδ			-25, 30	36, 30	0, 32	1, 0	0, -13	1, -10	1, 22		1, 0	-2, 0	-1, 0			-1, 0
it	*n̖*	*n̗*	*n̘*	*n̙*	*n̜*	*n̝*	*n̞*	*n̟*	*n̠*	*n̡*	*n̢*	*n̪*	*n̫*	*n̬*	*ṋ*	*n̮*
patch	*n̖*	*n̗*	*n̘*	*n̙*	*n̜*	*n̝*	*n̞*	*n̟*	*n̠*	*n̡*	*n̢*	*n̪*	*n̫*	*n̬*	*ṋ*	*n̮*
m	-246, -133	-237, -132	-254, -148	-315, -147	-246, -146	-283, -148	-239, -113	-242, -116	-242, -116	-272, -82	-282, -122	-276, -113	-273, -108	-228, -116	-222, -133	-275, -110
mp	-246, -133	-237, -132	-273, -118	-273, -117	-246, -114	-281, -148	-232, -86	-233, -86	-237, -96	-272, -82	-281, -122	-278, -113	-273, -108	-227, -116	-221, -133	-268, -110
mδ			-19, 30	42, 30	0, 32	2, 0	7, 27	9, 30	5, 20		1, 0	-2, 0		1, 0	1, 0	7, 0
sb	**n̖**	**n̗**	**n̘**	**n̙**	**n̜**	**n̝**	**n̞**	**n̟**	**n̠**	**n̡**	**n̢**	**n̪**	**n̫**	**n̬**	**ṋ**	**n̮**
patch	**n̖**	**n̗**	**n̘**	**n̙**	**n̜**	**n̝**	**n̞**	**n̟**	**n̠**	**n̡**	**n̢**	**n̪**	**n̫**	**n̬**	**ṋ**	**n̮**
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -149	-216, -113	-218, -116	-218, -116	-255, -73	-257, -124	-253, -113	-251, -108	-208, -116	-195, -133	-239, -110
mp	-219, -133	-210, -132	-249, -118	-249, -118	-216, -114	-251, -149	-216, -90	-218, -86	-216, -96	-255, -73	-256, -124	-255, -113	-252, -108	-204, -116	-195, -133	-240, -104
mδ			-25, 30	36, 29	0, 32	2, 0	0, 23	0, 30	2, 20		1, 0	-2, 0	-1, 0	4, 0		-1, 6
si	***n̖***	***n̗***	***n̘***	***n̙***	***n̜***	***n̝***	***n̞***	***n̟***	***n̠***	***n̡***	***n̢***	***n̪***	***n̫***	***n̬***	***ṋ***	***n̮***
patch	***n̖***	***n̗***	***n̘***	***n̙***	***n̜***	***n̝***	***n̞***	***n̟***	***n̠***	***n̡***	***n̢***	***n̪***	***n̫***	***n̬***	***ṋ***	***n̮***
m	-219, -133	-210, -132	-224, -148	-285, -147	-216, -146	-253, -148	-216, -113	-218, -116	-218, -116	-255, -82	-257, -122	-253, -113	-251, -108	-204, -116	-195, -133	-251, -110
mp	-219, -133	-210, -132	-249, -118	-261, -118	-216, -114	-257, -148	-207, -86	-210, -82	-212, -96	-255, -82	-257, -122	-254, -113	-249, -108	-204, -116	-198, -133	-244, -110
mδ			-25, 30	24, 29	0, 32	-4, 0	9, 27	8, 34	6, 20			-1, 0	2, 0		-3, 0	7, 0

Table 16: Font metrics for below marks 2/2

hex	032F	031D	031E	031F	0330	0331	0333	0339	033A	033B	033C	0347	0348	0349	034D	034E
cat	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn	Mn
reg	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̳H̳	n̯H̯	n̰H̰	n̤H̤	n̼H̼	n̿H̿	n̤H̤	n̤H̤	n̪H̪	n̤H̤
patch	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̯H̯	n̳H̳	n̯H̯	n̰H̰	n̤H̤	n̼H̼	n̿H̿	n̤H̤	n̤H̤	n̪H̪	n̤H̤
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110	-244, -110	-216, -146	-253, -113	-246, -113	-251, -108	-244, -110	-197, -110	-192, -110	-222, -110	-227, -110
mp	-251, -116	-249, -128	-247, -128	-252, -115	-205, -70	-240, -110	-240, -110	-216, -114	-255, -113	-252, -113	-252, -108	-240, -110	-195, -110	-192, -110	-228, -110	-228, -90
mδ	-1, 0	0, 15	0, -22	-3, 30	8, 0	4, 0	4, 0	0, 32	-2, 0	-6, 0	-1, 0	4, 0	2, 0		-6, 0	-1, 20
it	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̳H̳*	*n̯H̯*	*n̰H̰*	*n̤H̤*	*n̼H̼*	*n̿H̿*	*n̤H̤*	*n̤H̤*	*n̪H̪*	*n̤H̤*
patch	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̯H̯*	*n̳H̳*	*n̯H̯*	*n̰H̰*	*n̤H̤*	*n̼H̼*	*n̿H̿*	*n̤H̤*	*n̤H̤*	*n̪H̪*	*n̤H̤*
m	-274, -116	-278, -143	-269, -106	-278, -145	-227, -70	-266, -110	-266, -110	-246, -146	-276, -113	-269, -113	-273, -108	-266, -110	-219, -110	-214, -110	-244, -110	-249, -110
mp	-274, -116	-275, -128	-274, -128	-274, -115	-219, -70	-262, -110	-262, -110	-246, -114	-278, -113	-275, -113	-272, -108	-262, -110	-217, -110	-214, -110	-251, -110	-246, -90
mδ		3, 15	-5, -22	4, 30	8, 0	4, 0	4, 0	0, 32	-2, 0	-6, 0	1, 0	4, 0	2, 0		-7, 0	3, 20
sb	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**	**n□H□**
patch	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̯H̯**	**n̳H̳**	**n̯H̯**	**n̰H̰**	**n̤H̤**	**n̼H̼**	**n̿H̿**	**n̤H̤**	**n̤H̤**	**n̪H̪**	**n̤H̤**
m	-250, -116	-249, -143	-247, -106	-249, -146	-213, -70	-244, -58										
mp	-263, -116	-249, -129	-247, -128	-251, -116	-192, -70	-240, -58	-240, -110	-216, -114	-255, -113	-252, -113	-252, -108	-240, -110	-195, -110	-192, -110	-228, -110	-228, -90
mδ	-13, 0	0, 14	0, -22	-2, 30	21, 0	4, 0	new	new	new	new	new	new	new	new	new	new
si	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***	***n□H□***
patch	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̯H̯***	***n̳H̳***	***n̯H̯***	***n̰H̰***	***n̤H̤***	***n̼H̼***	***n̿H̿***	***n̤H̤***	***n̤H̤***	***n̪H̪***	***n̤H̤***
m	-250, -116	-249, -143	-247, -106	-249, -145	-213, -70	-244, -110										
mp	-262, -116	-248, -140	-249, -116	-251, -116	-183, -70	-228, -58	-238, -110	-216, -114	-254, -113	-251, -113	-249, -108	-238, -110	-194, -110	-192, -110	-227, -110	-223, -90
mδ	-12, 0	1, 3	-2, -10	-2, 29	30, 0	16, 52	new	new	new	new	new	new	new	new	new	new

Vertical references, anchor Y coordinates and reference, and clearance

A font has at least five vertical reference lines where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these Y coordinates are in the first five rows of Table 17. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as yra, yrc, yrx, $yr0$, yrd because they are Y values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

In the anchor model for Libertinus, the Y coordinate of an above-anchor (ay) or a below-anchor (by) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows a and b in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor Y coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 17. For example, all glyphs that sit on the baseline have below-anchor Y coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect Y value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor Y coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor Y coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the Y coordinate of the above- or below-anchor (ay or by) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender a, capital c, or x-height x. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where Y_{max} is the maximum Y coordinate of bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

Capitals are the exception: their top reference line is always the capline, and their above-anchor always has the capital reference Y ($ay = 850$), even where the outline rises above the capline. In humanist letterforms capitals do not have ascenders: an ascender is a stroke of the pen, not merely a high Y_{max}. What rises above the capline in Libertinus is away from the horizontal center of the glyph, where the mark sits: the top-left stroke of J (italic and semibold italic, 34 units above the capline), the serifs of T (semibold and semibold italic, 28), the slash of Ø (25), the top of Ʒ (up to 29), and the hooks of C̓ and G̓ (98). A mark at the capital reference Y does not collide with these. The one capital with a genuine ascender is the capital iota ɩ (U+0196), whose stem reaches the ascender line; it takes the ascender reference ($ay = 885$; italic styles 890). The glottal stop ʔ (U+0294) and pharyngeal ʕ (U+0295), counted here as capitals, rise above the capline only in italic (672 and 683); there they have curated anchors at $ay = 850 + (Y_{\text{max}} - 645) - 10$, with the bowl overshoot correction $\epsilon = 10$ instead of the default 2, which gives 867 and 878.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender d. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where Y_{min} in the minimum Y coordinate of the bounding box of the glyph, and ϵ is the correction for the vertical aspect range, usually for an overshoot.

The capital anchor Y reference and clearance of semibold are the patched values. The x-height bands marked * are widened above the designer's BlueValues (442, 447, 447) on 2026-11-07: several x-height letters of those styles rise 1–11 units above the band (ɐ ɔ ɡ ɵ ɽ ʂ ʐ ʑ in italic, ɢ ʑ in semibold, ʐ ʑ in semibold italic), which gave their candidates a Y of 657–666, above the base letters s, r, z, o at 645. Snapped to 645, no above-mark of the IPA set overlaps them (the nearest is 19 away); semibold italic ʜ (462) and p (470) stay outside.

The designer set semibold's capitals near 805 (clearance 160) and compensated only in the capital alternates of the marks, so a mark without a capital alternate crashed into the capital: in 225 of the 780 combinations of A–Z with 30 above marks, the bottom of the mark was below the top of the letter. The patch moves the capitals to 850 and adds 45 to the mark anchors of the capital alternates, which stay where they were. That leaves 4 such combinations, all on T, whose serif tips rise to 673 away from the marks; under the marks the gap is 21–26. A few capitals have the same problem in the other styles (Ŋ Ƅ Ƈ Ɋ Ħ Ɪ Ƙ Ƶ at 798–805), so since 2026-11-08 stage 5 moves every capital with an above-anchor in 795–835 to 850 in every style. In italic and semibold italic the designer's X of Ŋ Ƅ Ƈ Ƙ Ƶ was regular's, without the slant (italic Ƈ 355 against the candidate 622), so they are deleted in stage 1 and get the candidate; Ħ, outside the Latin/IPA scope, keeps regular's X there. These capitals are also in `@capitals`, so the twelve marks with a capital alternate take it on them.

Table 17: Vertical aspects, anchor Y references, and clearances of Libertinus

	v. aspect	regular	italic	semibold	semibold italic
yra	ascender	$698, [688, 698]$	$698, [688, 698]$	$698, [690, 698]$	$698, [696, 705]$
yrc	capital	$645, [645, 658]$	$645, [645, 658]$	$645, [645, 662]$	$645, [645, 662]$
yrx	x-height	$429, [429, 442]$	$429, [429, 450]^*$	$434, [433, 457]^*$	$434, [434, 458]^*$
$yr0$	baseline	$0, [-12, 0]$	$0, [-12, 1]$	$0, [-12, 1]$,	$0, [-20, 0]$
yrd	descender	$-232, [-238, -227]$	$-232, [-238, -227]$	$-232, [-238, -212]$	$-232, [-239, -219]$
$ayra$	ascender	885	890	885	890
$ayrc$	capital	850	850	850	850
$ayrx$	x-height	645	645	645	645
$byr0$	baseline	-110	-110	-110	-110
$byrd$	descender	-319	-319	-319	-319
$ay\delta a$	ascender	187	192	187	192
$ay\delta c$	capital	205	205	205	205
$ay\delta x$	x-height	216	216	211	211
$by\delta 0$	baseline	110	110	110	110
$by\delta d$	descender	87	87	87	87

Of the other designer anchors more than 5 from a reference (stage 5 lists them), most are legitimate: stacked marks on precomposed glyphs, superscripts at 805–815, small capitals at their own height, 685. Fixed on 2026-11-08: above and below swapped on ɥ ɥ̨ (regular, italic, semibold italic) and ƚ (regular), now deleted in stage 1 and given the candidates; and į i̇ ɨ, whose above-anchors were 9–59 units below i's, so that a mark keeping the dot crowded it, now curated at i's anchor.

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if y_{min} is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

Anchor survey, with patches

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show that some anchors are not set to good values. The base coverage gets worse for italic and semibold.

I will systematically survey above- and below anchors in the four styles, commenting on any features and patching any problems. Refer to Tables 1–4 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values — $a\delta$ and $b\delta$ — indicate possible problems, but these can often be explained by glyph symmetry.

Copied anchors

Many designer anchors are copies: within a style, clusters of letters share one above- or below-anchor exactly (a's $(209, 645)$ and $(196, -110)$ on some 40 IPA letters each. Some copies are harmless coincidences of similar letterforms; most are not.

`run_anchor_copy_report.py` lists the clusters (`tex/input/copied_anchors.tex`, no longer printed here). On 2026-11-07 every member of the regular clusters, and its counterparts in the other styles, was reviewed per character: each anchor is now the candidate, inherited from its base letter, or curated by a stated rule, the same way in all four styles.

A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For bx, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have ay regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, ay is at the x-height anchor reference, and therefore

wide marks collide with the ascender. For k, ay is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.
The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

Patch k regular, below-anchor

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.
Here is this positioning of k in a patched version of the font (with o for comparison only).
Original: k̠ k̠o ok̠
Patched: k̠ k̠o ok̠

Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set $ay = ayra = 885$.
For ax for b, d, f, and h, there are at least two strategies. First, set ax at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set ax centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. The ccmp feature swaps i and j for their dotless forms before every above-mark in its list, so only a mark outside the list attaches to i or j itself; among the IPA marks that is only the double overline U+033F. Such a mark must clear the dot. The designer's $(137, 795)$ on i (and the earlier curated j at $(126, 795)$) put the double overline only 13–16 above the dot in regular and semibold, against 69–74 above o. The patch (2026-11-07) curates i and j in every style on the stem's axis, at the candidate's Y, the dot top plus the ascender clearance (regular 834, italic 810, semibold 837, semibold italic 823), so the double overline sits 55–60 above the dot. The candidate's X is right for i but not for j, whose hook pulls the bounding-box midpoint left of the stem (by 84 in semibold).

Stem-axis rule (2026-11-12). Letters with a central stem and no curl have their anchors on the axis of the stem (`font_patching_stemaxis.py`, stage 7b, after the curated anchors: it sets X over the designer's and the curated anchors alike, and reports each curated X it changes; Y is not touched). The group is the basest such letters: i, ɪ, j, ɉ, l, m (its middle stem), I, T, ɪ, above and below (j and ɉ above only, their below anchor being under the hook). The letters derived from them (i̵ ɪ̵ ł ł̵ Ŧ T ɱ …) move with their base letter in stage 8; ɾ and ɹ are not in the group, because their hooked top puts the optical centre 40 to 55 units off the stem.

The stem's midline is the mean of its two straight edges, read from the control points (a pair of line segments on opposite sides of one ink interval, overlapping in height; where one side is a curve, as in the middle stem of semibold italic m, the straight edge moved by the median half-width). It is taken at the top of the straight part for the above anchor and at its bottom for the below anchor, and carried to the anchor's height at the style's declared slant (12°, 11.5°), less K in semibold italic.

The stems are not drawn at the declared slant: the lowercase stems are steeper (italic 12.1° to 13.5°, semibold italic 12.8° to 13.4°), the capitals I and T shallower (11°, 10°). Carrying the midline at the stem's own angle was tested and rejected (test sheet `tex/stem_angle_test_sheet.pdf`): it moves the above anchors of ɪ, ɉ and l 4 to 9 units right in italic and 9 to 12 in semibold italic, and the evidence points the other way: the dot of i and j, and the accents of the precomposed ì í î ĩ ī ĭ ï ỉ î ȷ̂ ȷ̌, sit left of the line at the declared slant, not right of it. Against the precomposed ṁ ṃ İ Į Ī Ṫ Ṭ Ṯ neither slant was consistently closer.

In the italic styles, curls change the optical centre, and the rule is adapted:

- Dotted i and j: the above anchor is over the centre of the dot, 11 to 13 units left of the stem's axis (only U+033F attaches there, right over the dot).
- Dotless ɪ and ɉ: the above anchor is on the axis of the dot of i and j, so that an accent takes the dot's place. Their curved entry stroke at the top left pulls the optical centre left of the stem, and the precomposed accents agree: composed minus precomposed (median), on the stem's axis and on the dot's: italic ɪ $+20$, $+8$, ɉ $+28$, $+15$; semibold italic ɪ $+14$, $+1$, ɉ $+10$, -14.

- The below anchors of i, ɪ and l (curled foot) and the above anchor of m (arches) are left out, to be curated (“To do”).

In regular and semibold the rule applies to the whole group, above and below. The accents of the precomposed accented i are 16 to 20 units left of ɪ’s stem there too, but the designer puts accents about 9 left of centre on lowercase letters generally, and the rule looked better on the test sheets.

Moves, in units: regular and semibold m $+11$ to $+12$, T below $+8$ (regular), others 0 to 2; italic ɪ above -10, ɟ -13, I above -5 and below -12, T below -16, m below -9, i and j above (dot) -12, -13; semibold italic ɪ above -11, ɟ -21, T above -12 and below -26, i and j above (dot) -11; the derived and precomposed letters move with them.

Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize ay to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ầ $\hat{A}$ $\mathbf{\hat{A}}$ $\boldsymbol{\hat{A}}$
O circumflex Ô Ồ $\hat{O}$ $\mathbf{\hat{O}}$ $\boldsymbol{\hat{O}}$
AE ligature Æ Ǽ $Æ$ $\mathbf{Æ}$ $\boldsymbol{Æ}$
Ṡ Ṫ V̇ $\dot{S}$ $\dot{T}$ $\dot{V}$ $\mathbf{\dot{S}}$ $\mathbf{\dot{T}}$ $\mathbf{\dot{V}}$ $\boldsymbol{\dot{S}}$ $\boldsymbol{\dot{T}}$ $\boldsymbol{\dot{V}}$
Ṡ Ṫ V̇ $\dot{S}$ $\dot{T}$ $\dot{V}$ $\mathbf{\dot{S}}$ $\mathbf{\dot{T}}$ $\mathbf{\dot{V}}$ $\boldsymbol{\dot{S}}$ $\boldsymbol{\dot{T}}$ $\boldsymbol{\dot{V}}$

Regular, above-anchor, capitals, X coordinate

Figure 9 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad X coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor X coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor X coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 9 for an illustration of increasingly bad (spoofed) X coordinates of N’s above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

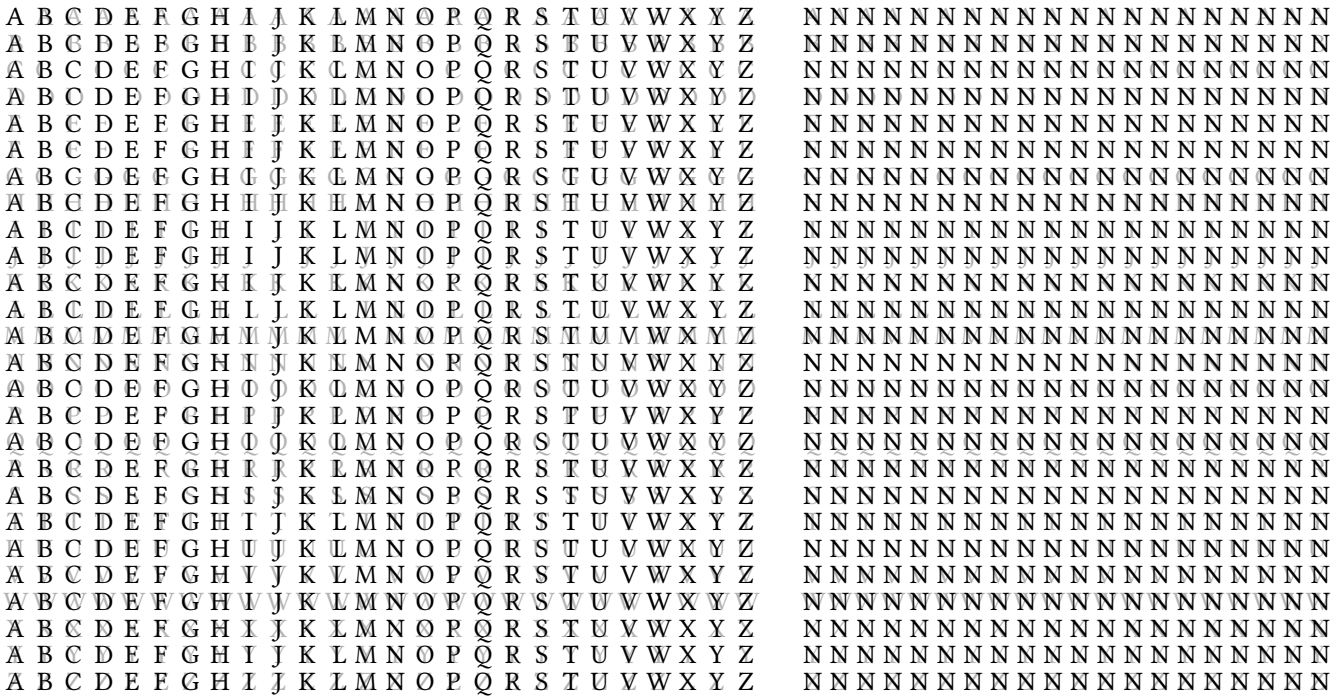


Figure 9: Above-anchor X coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad X coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the X coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) X coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̬

`\textsc{\char"013E}` ʟ̬

Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` Ç *Ç* **Ç** ***Ç***	`\char"00E7` ç *ç* **ç** ***ç***
`\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ***	`\char"1E08` ḉ *ḉ* **ḉ** ***ḉ***
`\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ***	`\char"1E11` ḑ *ḑ* **ḑ** ***ḑ***
`\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ***	`\char"0229` ȩ *ȩ* **ȩ** ***ȩ***
`\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ***	`\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ***
`\char"0122` Ģ *Ģ* **Ģ** ***Ģ***	`\char"0123` ģ *ģ* **ģ** ***ģ***
`\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ***	`\char"1E29` ḩ *ḩ* **ḩ** ***ḩ***
`\char"0136` Ķ *Ķ* **Ķ** ***Ķ***	`\char"0137` ķ *ķ* **ķ** ***ķ***
`\char"013B` Ļ *Ļ* **Ļ** ***Ļ***	`\char"013C` ļ *ļ* **ļ** ***ļ***
`\char"0145` Ņ *Ņ* **Ņ** ***Ņ***	`\char"0146` ņ *ņ* **ņ** ***ņ***
`\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ***	`\char"0157` ŗ *ŗ* **ŗ** ***ŗ***
`\char"015E` Ş *Ş* **Ş** ***Ş***	`\char"015F` ş *ş* **ş** ***ş***
`\char"0162` Ţ *Ţ* **Ţ** ***Ţ***	`\char"0163` ţ *ţ* **ţ** ***ţ***

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

`c\char"0327` ç *ç* **ç** ***ç***
`d\char"0327` ḑ *ḑ* **ḑ** ***ḑ***
`e\char"0327` ȩ *ȩ* **ȩ** ***ȩ***
`g\char"0327` ģ *ģ* **ģ** ***ģ***
`h\char"0327` ḩ *ḩ* **ḩ** ***ḩ***
`k\char"0327` ķ *ķ* **ķ** ***ķ***
`l\char"0327` ļ *ļ* **ļ** ***ļ***
`n\char"0327` ņ *ņ* **ņ** ***ņ***
`r\char"0327` ŗ *ŗ* **ŗ** ***ŗ***
`s\char"0327` ş *ş* **ş** ***ş***
`t\char"0327` ţ *ţ* **ţ** ***ţ***

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`. This behavior is tolerable for the other bases – c, d, e, h, s, t.

Greek koronis U+0343

Purpose. In polytonic (ancient) Greek the koronis marks crasis, two words run together with their vowels contracted: κἀγώ for καὶ ἐγώ. It has exactly the shape of the smooth breathing (psili), and Unicode writes both with one character, U+0313 COMBINING COMMA ABOVE. U+0343 COMBINING GREEK KORONIS was encoded for compatibility with older standards: its canonical decomposition is the single character U+0313, so any normalization (NFC or NFD) replaces it with U+0313, and no precomposed letter decomposes to it (ά is α + U+0313). The spacing forms U+1FBD GREEK KORONIS and U+1FBF GREEK PSILI are both compatibility equivalents of space + U+0313. Modern monotonic Greek uses neither. In every style and weight the koronis should look like the style's smooth breathing. Text should use U+0313 or the precomposed letters; nobody needs to type U+0343.

XeLaTeX and HarfBuzz. HarfBuzz applies the singleton decomposition: after a base letter (or a no-break space) U+0343 is shaped with the glyph of U+0313, and α + U+0343 composes to ά. The glyph `uni0343` is shaped only when the mark begins a word, as after a space, which XeTeX passes to HarfBuzz as glue, not as a character. Below, the first row is a + U+0313, the second a + U+0343 (the same glyphs), the third α + U+0343 (the precomposed ά), and the last U+0343 alone after a space (the glyph `uni0343`); each in the original fonts, then the patched fonts.

`a\char"0313` ả *ả* **ả** ***ả*** ả *ả* **ả** ***ả***
`a\char"0343` ả *ả* **ả** ***ả*** ả *ả* **ả** ***ả***
`\char"03B1\char"0343` ά *ά* **ά** ***ά*** ά *ά* **ά** ***ά***
`\ \char"0343` ’̓ ’̓ ’̓ ’̓

Libertinus. In regular and italic, `uni0343` is a reference to `uni0313`, with the same anchor: correct. In semibold it is a reference to the horn `uni031B`, moved 72 up and 72 right, and it had no mark anchor, so it attached to nothing. In semibold italic it is an outline of its own, the horn's size (188 by 178, against the comma's 134 by 180), drawn right of the origin with the comma above's anchor, so k was 371 against $K = 150$. The patch leaves both outlines and curates their anchors as regular's and italic's U+0313: 135 above the outline's bottom and 21 left of its ink centroid (along the slant, with K, in semibold italic): $(-92, 539)$ in semibold, $(128, 532)$ in semibold italic (2026-11-11). The glyph a reader actually sees on a letter is U+0313, which in semibold and semibold italic is drawn at the lighter style's weight (and in the original semibold attached only to the 24 bases with a `top_punkt` anchor; stage 1 merges that class). The precomposed Greek letters reference neither mark: ά έ Ἀ ῥ are built from the modifier apostrophe U+02BC (U+1FBD in semibold italic ά), and italic draws its own. The outlines are an item of the to-do list (commas and horns).

Æ: a letter the midpoint cannot centre

Why Æ has curated anchors in every style rather than the anchor model's candidate (see "Where candidates fail" in `ANCHOR-HEURISTICS.md`).

Over Æ, accents do not go at the midpoint of the glyph. The left leg of the A stops short of the top, so the top edge runs from the apex of the A to the end of the arm of the E, and its centre lies well to the right. Other fonts agree: Charis SIL, Gentium Plus, EB Garamond, Junicode, DejaVu Serif and Andika put the above-anchor at the macron of Ǣ, within 5 units, and below they use the common stroke or the same X. In Libertinus the designer's above-anchors missed the accents of Ǣ and Ǽ by 55–88 and 16–68 units, so the curated above-anchors lie between the two (555, 674, 555, 545 for regular, italic, semibold, semibold italic). In the italic styles the macron and the acute cannot both be met, and each lands about 30 units from its precomposed position. Below, semibold's anchor is curated under the common stroke, at 510. The designer's 549 of the other three styles stands: nothing collides there.

IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

IJ `\char"0132` U+0132 capital ligature IJ

ij `\char"0133` U+0133 small ligature ij

Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,

```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```

ĲĳĲ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute

ĳ́ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute

ĳ́ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute

In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).

ị̃ *ị̃* **ị̃** ***ị̃*** `\char"1E2D` i with tilde below (precomposed)

You cannot simply put a grave on it because it collides with the dot.

ị̃̀ *ị̃̀* **ị̃̀** ***ị̃̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave

Start over with i with grave (precomposed).

ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)

Try putting a grave below.

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input "i with grave (precomposed) with tilde below" into "i with grave with tilde below", then into canonical order "i with tilde below with grave", the recomposes it into "i with tilde below (precomposed) with grave", which reintroduces i's dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ı̰̀ *ı̰̀* **ı̰̀** ***ı̰̀*** `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ı̰̀ *ı̰̀* **ı̰̀** ***ı̰̀*** `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭ ì ḭ ì ḭ ḭ ì ḭ ì ḭ \char"1E2D \char"00EC \char"0131\char"0330\char"0300 i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ḭ ȉ ḭ ȉ \char"0131\char"0330\char"030F dotless i with tilde below with double grave

ḭ ́ ḭ ́ \char"0131\char"0330\char"0301 dotless i with tilde below with acute

ḭ ̋ ḭ ̋ \char"0131\char"0330\char"030B dotless i with tilde below with double acute

Lowercase a with double grave, followed by floating double grave, in italic

Superseded by "Floating tones" below: write `\floattone{a}{\char"030F}`.

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‟ `\char"2036`), which looks like a double grave, in italic.

ä‟ `\textit{ä\char"2036}}`

Then we lower it by a few points.

ä‟ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

ä‟ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

ë‟ `\textit{ë\raisebox{-0.05em}{\char"2036}}`

ë‟ `\textit{ë\raisebox{-0.045em}{\char"2036}}`

ɛ‟ `\textit{\char"025B\raisebox{-0.05em}{\char"2036}}`

Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` ḭ ́ ḭ ́ = ḭ ́ ḭ ́

`\char"0131\char"0330\char"0301` ı̰ ́ ḭ ́

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

`i\char"0301\char"0330` í ̰ ḭ ́ `i\char"0301` i ́ i ́ `i\char"0330` ḭ ́ ḭ ́

\char"00EC\char"0330 ḭ̀ *ḭ̀* ḭ̀ *ḭ̀*

\char"1E2D\char"0301 ḭ́ *ḭ́* ḭ́ *ḭ́* \char"1E2D ḭ *ḭ* ḭ *ḭ*

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, \char"034F, or \cgj if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ↄ a e i o u ɛ ɔ *A E I O U Ɛ Ↄ a e i o u ɛ ɔ* **A E I O U Ɛ Ↄ a e i o u ɛ ɔ** ***A E I O U Ɛ Ↄ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́* **Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́** ***Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́***

2 À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀* **À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀** ***À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀***

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***

4 Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́* **Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́** ***Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́***

5 Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* **Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀** ***Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀***

6 Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨***

7 Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

7c Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

8 Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***

9 Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ↄ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ↄ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ↄ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ̨̆ Ↄ̨̆ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ̨̆ ɔ̨̆***

10 Ą́̆ Ę́̆ Į́̆ Ǫ́̆ Ų́̆ Ɛ̨́̆ Ↄ̨́̆ ą́̆ ę́̆ į́̆ ǫ́̆ ų́̆ ɛ̨́̆ ɔ̨́̆ *Ą́̆ Ę́̆ Į́̆ Ǫ́̆ Ų́̆ Ɛ̨́̆ Ↄ̨́̆ ą́̆ ę́̆ į́̆ ǫ́̆ ų́̆ ɛ̨́̆ ɔ̨́̆* **Ą́̆ Ę́̆ Į́̆ Ǫ́̆ Ų́̆ Ɛ̨́̆ Ↄ̨́̆ ą́̆ ę́̆ į́̆ ǫ́̆ ų́̆ ɛ̨́̆ ɔ̨́̆** ***Ą́̆ Ę́̆ Į́̆ Ǫ́̆ Ų́̆ Ɛ̨́̆ Ↄ̨́̆ ą́̆ ę́̆ į́̆ ǫ́̆ ų́̆ ɛ̨́̆ ɔ̨́̆***

11 Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ̨̈̀ Ↄ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ̨̈̀ ɔ̨̈̀ *Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ̨̈̀ Ↄ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ̨̈̀ ɔ̨̈̀* **Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ̨̈̀ Ↄ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ̨̈̀ ɔ̨̈̀** ***Ą̈̀ Ę̈̀ Į̈̀ Ǫ̈̀ Ų̈̀ Ɛ̨̈̀ Ↄ̨̈̀ ą̈̀ ę̈̀ į̈̀ ǫ̈̀ ų̈̀ ɛ̨̈̀ ɔ̨̈̀***

Patched

A E I O U Ɛ Ↄ a e i o u ɛ ɔ *A E I O U Ɛ Ↄ a e i o u ɛ ɔ* **A E I O U Ɛ Ↄ a e i o u ɛ ɔ** ***A E I O U Ɛ Ↄ a e i o u ɛ ɔ***

Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́ *Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́* **Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́** ***Á É Í Ó Ú Ɛ́ Ↄ́ á é í ó ú ɛ́ ɔ́***

À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀ *À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀* **À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀** ***À È Ì Ò Ù Ɛ̀ Ↄ̀ à è ì ò ù ɛ̀ ɔ̀***

Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***

Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́* **Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́** ***Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́***

Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀ *Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀* **Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀** ***Ằ Ề Ỉ Ồ Ừ Ɛ̀ Ↄ̀ ằ ề ỉ ồ ừ ɛ̀ ɔ̀***

Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨***

cgj Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨ *Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨* **Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨** ***Ą Ę Į Ǫ Ų Ɛ̨ Ↄ̨ ą ę į ǫ ų ɛ̨ ɔ̨***

Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

cgj Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ̨́ Ↄ̨́ ą́ ę́ į́ ǫ́ ų́ ɛ̨́ ɔ̨́***

Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***

cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ̨̀ Ↄ̨̀ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ̨̀ ɔ̨̀***

Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***

cgj Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ̆ Ↄ̆ ă ĕ ĭ ŏ ŭ ɛ̆ ɔ̆***

Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́ *Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́* **Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́** ***Ắ Ế Ḯ Ố Ű Ɛ́ Ↄ́ ắ ế ḯ ố ű ɛ́ ɔ́***

cgj Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃
Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃
cgj Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃ Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ę̃ Ɔ̃ ą̏ ę̏ į̏ ǫ̏ ų̏ ę̃ ɔ̃
Patched not grave
À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃ *À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃* **À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃** ***À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃***
Patched grave
À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃ *À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃* **À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃** ***À ÈÈ Ì Ò Ù Ɛ̃ Ɔ̃ à è ì ò ù ɛ̃ ɔ̃***

Floating tones

A floating tone is written as a tone mark with no vowel of its own: after a vowel that bears the same tone (ȅ̗), or standing alone as a morpheme in a gloss. By Unicode convention, a mark that stands alone is applied to U+00A0 no-break space (the Unicode Standard prefers NBSP over U+0020 space, which allows a line break between the space and its mark). The patched fonts give U+00A0 an above and a below anchor at the lowercase references (645 above, −110 below), with X on each style's anchor model, so a mark on NBSP sits at exactly the height it has on a lowercase vowel, in all four styles.

How to write a floating tone in XeLaTeX

1. Load the patched fonts with `fontspec`, e.g.

```
\setmainfont{LibertinusSerif-Regular-patch.otf}[
    Path           = ../fonts/,
    ItalicFont     = LibertinusSerif-Italic-patch.otf,
    BoldFont       = LibertinusSerif-Semibold-patch.otf,
    BoldItalicFont = LibertinusSerif-SemiboldItalic-patch.otf]
```

2. Define three commands in the preamble (this report does):

```
\newcommand{\cgj}{\char"034F }                % combining grapheme joiner
\newcommand{\nbsp}{\char"00A0 }               % no-break space, base for a floating mark
\newcommand{\floattone}[2]{#1\cgj#2\nbsp#2}   % vowel + mark, then the same mark floating
```

3. Write a vowel followed by a floating tone of the same specification as `\floattone{`*vowel*`}{`*mark*`}`, e.g. `\floattone{e}{\char"030F}` gives ȅ̗. The mark may also be typed as the literal combining character between the braces. The vowel may carry other marks (`\floattone{a\char"0330}{\char"0300}` gives à̗), and may be any lowercase tone-bearing unit, such as a syllabic nasal (`\floattone{ŋ}{\char"0300}` gives ŋ̀).

4. Write a floating tone on its own as `\nbsp` followed by the mark: `-\nbsp\char"0300` gives -̀.

5. Punctuation may follow directly: `\floattone{e}{\char"0300}.` gives è̀. NBSP is a quarter em wide (250 units; 274 in semibold italic), so there is that much space before the punctuation.

Two things to avoid. Do not write the base as ~ or `\nobreakspace`: in TeX these are glue, not a character, so HarfBuzz sees no base and the mark is set on its own at its default position. Do not leave out the CGJ when the floating tone repeats the vowel's mark: without it, HarfBuzz substitutes the precomposed glyph where one exists (ȅ̗), and a precomposed glyph may

carry its mark at a different height from the anchor (the dot above U+0307 is 45 units higher in the precomposed glyphs of regular); with it, both marks are placed by anchors at the same height. `\floattone` inserts the CGJ. When typing `\nbsp` by hand before a literal combining mark, write `\nbsp{}`, since XeTeX reads a combining mark as a letter of the command name; `\floattone` has no such problem.

The anchors are lowercase ones, so a floating mark after a capital does not take its `.cap` form or the capital height. In fonts without NBSP anchors (original Libertinus, Noto Serif) the mark still appears, at its default position; Gentium Plus and Charis SIL have NBSP anchors of their own.

Comparison

Patched fonts, with `\floattone`: `-k\floattone{e}{\char"030F}`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched fonts, without CGJ: `-ke\char"030F\nbsp\char"030F`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Original Libertinus, with `\floattone`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Gentium Plus, with `\floattone`

-kȅ̏

Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Showcase

Each item is `\floattone{`*vowel*`}{`*mark*`}` for the vowels a e i o u ɛ ɔ ə ɪ ʊ ɨ ʉ ʌ æ and the syllabic nasals m n ŋ, in the patched fonts (regular, italic, semibold, semibold italic), at `\Large` font size. The floating mark matches the height of the vowel's mark throughout, since every x-height letter and NBSP share the same above reference (645) and below reference (−110).

U+0301 acute

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

U+0300 grave

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

U+030B double acute

a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

U+030F double grave

ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

U+0304 macron

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā ē ī ō ū ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

ā̄ ***ē̄*** ***ī̄*** ***ō̄*** ***ū̄*** ***ɛ̄̄*** ***ɔ̄*** ***ə̄*** ***ɪ̄*** ***ʊ̄*** ***ɨ̄*** ***ʉ̄*** ***ʌ̄*** ***ǣ*** ***m̄̄*** ***n̄*** ***ŋ̄***

U+0302 circumflex

âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ əˆ ɪ̂ ʊ̂ ɨ̂ ʉ̂ ʌ̂ æ̂ m̂ n̂ ŋ̂

âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ ə̂ ɪ̂ ʊ̂ ɨ̂ ʉ̂ ʌ̂ æ̂ m̂ n̂ ŋ̂

âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ əˆ ɪ̂ ʊ̂ ɨ̂ ʉ̂ ʌ̂ æ̂ m̂ n̂ ŋ̂

âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ ə̂ ɪ̂ ʊ̂ ɨ̂ ʉ̂ ʌ̂ æ̂ m̂ n̂ ŋ̂

U+030C caron

ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛ̌ ɔ̌ əˇ ǐˇ ʊ̌ ɨ̌ ʉ̌ ʌ̌ ǽ m̌ ň ŋ̌

ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛ̌ ɔ̌ ə̌ ǐˇ ʊ̌ ɨ̌ ʉ̌ ʌ̌ ǽ m̌ ň ŋ̌

ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛ̌ ɔ̌ əˇ ǐˇ ʊ̌ ɨ̌ ʉ̌ ʌ̌ ǽ m̌ ň ŋ̌

ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛ̌ ɔ̌ ə̌ ǐˇ ʊ̌ ɨ̌ ʉ̌ ʌ̌ ǽ m̌ ň ŋ̌

Squat reversed esh (Sinological apical vowels)

U+0285 LATIN SMALL LETTER SQUAT REVERSED ESH ʅ is the apical vowel of Sinological transcription after retroflex sibilants (Mandarin *shī* [ʂʅ^{55}], *zhī* [tʂʅ^{55}], *rì* [ʐʅ^{51}]). Its partner ɿ U+027F follows dental sibilants (*sī* [sɿ^{55}], *cí* [tsʰɿ^{35}]). ʅ is ɿ with a retroflex hook, the hook of ɳ ʈ ʂ ʐ ɭ: its top is ɿ's fishhook, at the x-height.

The designer's ʅ was 550 high in regular, 119 above the x-height, and in italic, semibold and semibold italic it is ʃ mirrored, rising to the ascender (semibold exactly so: every point $x \mapsto 305 - x$, hence the same bounding box as ʃ). In a survey of 22 fonts (2026-11-09) every font that draws the letter rather than mirroring ʃ tops it at the x-height, level with its own ɿ: Gentium Plus, Charis SIL, Andika, Noto Serif, STIX Two, Junicode, EB Garamond, DejaVu Serif, Tinos, Roboto. The ascender form occurs only as a mirrored ʃ (Gentium 1.x, which Gentium Plus replaced with the squat form; Lato) and in Libertinus and Langscitinus, which has Libertinus's glyph unchanged.

Stage 2 of the patch redraws ʅ in all four styles by control points: ɿ's fishhook and stem, with ɳ's retroflex hook in place of the foot serif. The hook reaches as deep as ɳ's (−221 in regular, −229 in the other styles), short of the descender (−234), like the other retroflex hooks. The left sidebearing is ɿ's, the right ɳ's at the hook. The above-anchor is ɿ's (stage 8, which treats ʅ as ɿ with a bottom tail); the below-anchor is the anchor model's candidate on the new outline (stage 6), at the descender reference. The designer's anchors, placed for the old outline, are deleted in stage 1, including regular's ogonek and cedilla anchors.

Original Libertinus	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	aʅa ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}**	**aʅa** ***ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}***	***aʅa***
Langscitinus	ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	aʅa ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}	*aʅa* **ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}**	**aʅa** ***ʂʅ^{55} tʂʰʅ^{55} ʐʅ^{51} sɿ^{55}***	***aʅa***
Patched	ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɿ^{55}	aɻa ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɿ^{55}	*aɻa* **ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɿ^{55}**	**aɻa** ***ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɿ^{55}***	***aɻa***
Gentium Plus	ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɻ^{55}	aɻa ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɻ^{55}	*aɻa* **ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɻ^{55}**	**aɻa** ***ʂɻ^{55} tʂʰɻ^{55} ʐɻ^{51} sɻ^{55}***	***aɻa***

With tone and other marks, patched: [ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [ʂɿ̰] [ɻ̩] *[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [ʂɿ̰] [ɻ̩]* **[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [ʂɿ̰] [ɻ̩]** ***[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [ʂɿ̰] [ɻ̩]***

Reversed r with fishhook

U+027F LATIN SMALL LETTER REVERSED R WITH FISHHOOK ɿ is the apical vowel after dental sibilants (see the previous section). Fonts draw it in two ways: on the baseline, as ɾ mirrored (Libertinus, Langscitinus, Gentium 1.x, Junicode, STIX Two, DejaVu Serif, EB Garamond), or with a straight descender (Gentium Plus, Charis SIL, Andika, Noto, Tinos, Roboto). SIL changed it from the baseline form in Gentium 1.x to the descending form in Gentium Plus, and the current SIL fonts offer no way back: they have one glyph for U+027F, no character variant or stylistic set for it, and no PUA alternate. Both forms are established, and the patch keeps Libertinus's baseline ɿ as U+027F.

For users who want the descending form, the patched fonts add it at a Private Use Area code point, U+E27F (glyph `uni027F.desc`), in all four styles. It is ɿ's fishhook and stem continued down to q's descender serif, with ɿ's advance width. The above-anchor is ɿ's (stage 8); the below-anchor is curated, under the stem at the descender reference (−319), as q's.

U+027F, patched	sɿ^{55} tsʰɿ^{35} [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɿ̰]	*sɿ^{55} tsʰɿ^{35} [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɿ̰]*	**sɿ^{55} tsʰɿ^{35} [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɿ̰]**	***sɿ^{55} tsʰɿ^{35} [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɿ̰]***
U+E27F, patched	sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]	*sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]*	**sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]**	***sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]***
Gentium Plus	sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]	*sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]*	**sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]**	***sɻ^{55} tsʰɻ^{35} [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̰]***

How to use the descending form

- In XeLaTeX or LuaLaTeX with the patched fonts, write `\char"E27F` or define a command, as this report does:

  ```
  \newcommand{\rfishdesc}{\char"E27F }   % reversed r with fishhook and descender
  ```

  `s\rfishdesc` gives sɻ. Before a literal combining mark write `\rfishdesc{}`, since XeTeX reads a combining mark as a letter of the command name (`s\rfishdesc{}` gives sɻ́).

- In other applications, enter the code point E27F with the application's Unicode input (e.g. type `e27f` then Alt+X in Word, or Ctrl+Shift+U e27f in GTK applications).

A Private Use code point has no meaning outside the font: in any other font U+E27F shows as a missing glyph, and searching, sorting, spell-checking and copied text see U+E27F, not U+027F. Use it for display where the descending form is wanted, and keep U+027F wherever the text must be interchanged or searched.

f with dot below

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

tiɣə́ ɾ̣fa

ɾ̣ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f̣ g̣ j̣ p̣ q̣ x̣

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.

For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original	Patched italics (f only)
tiɣə ṛfa	tiɣə ṛfa
tiɣə ṛfa	*tiɣə ṛfa*
tiɣə ṛfa	**tiɣə ṛfa**
tiɣə ṛfa	***tiɣə ṛfa***

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

Patched italics (f only)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

Ashkun

`a\char"0304\char"0303` ã *ã* **ã** ***ã***

`\char"0101\char"0303` ã *ã* **ã** ***ã***

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Patched

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`u\cgj\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\cgj\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ *ũ* **ũ** ***ũ***

Tacks

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ
a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ ὺ **ὺ ὺ**

ὺ ὺ **ὺ ὺ**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

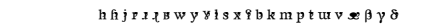
Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

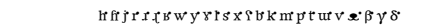
```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```

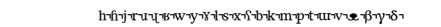
ʊ̀

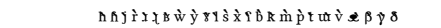
Patch superscript consonant anchors

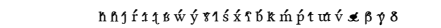
Libertinus serif regular

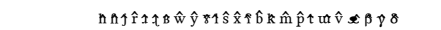




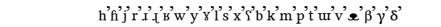


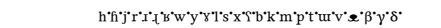


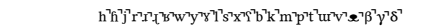


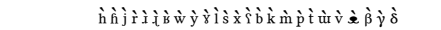


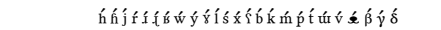
Libertinus serif regular, patched

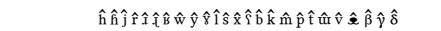






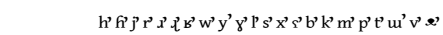


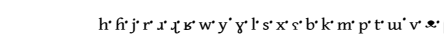


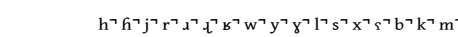


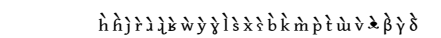


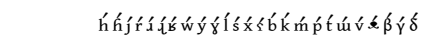
Gentium Plus

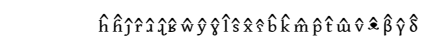














A B C D E F G H I J K L M N O P Q R S T U V W X Y Z a b c d e f g h i j k l m n o p q r s t u v w x y z Æ æ Œ œ Ø ø Đ ð Þ þ ß Ɽ Ð đ Ħ ħ ı ȷ Ł ł Ƀ ƀ
À Ḃ Ċ Ḋ È Ḟ Ġ Ḣ Ì J̀ Ǩ Ĺ Ṁ Ǹ Ò Ṗ Q̀ Ŕ Ṡ Ṫ Ù V̀ Ẁ Ẋ Ỳ Ż à ḃ ċ ḋ è ḟ ġ ḣ ì j̀ ǩ ĺ ṁ ǹ ò ṗ q̀ ŕ ṡ ṫ ù v̀ ẁ ẋ ỳ ż Æ̀ æ̀ Œ̀ œ̀ Ø̀ ø̀ Đ̀ ð̀ Þ̀ þ̀ ß̀ Ɽ̀ Đ̀ đ̀ Ħ̀ ħ̀ ì j̀ Ł̀ ł̀ Ƀ̀ ƀ̀

Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font's GSUB table.

Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣
x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣

Below marks – GPOS anchor 2

x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂
x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂
x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂
x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̴ x̵ x̶ x̷ x̸ x̹ x̺ x̻ x̼ x̽ x̾ x̿ x̀ x́ x͂

Left angle – GPOS anchor 3

x̚ x᳚ x᳛ x᳜

Below-right marks – GPOS anchor 4

x̙ x̜ x̞
x̙ x̜ x̞
x̙ x̜ x̞
x̙ x̜ x̞

Cedilla – GPOS anchor 5

x̡ x̧ x̨ x̢

Overlay marks – GPOS anchor 6

o̶ o̵ o̸ o̷ o̲
o̶ o̵ o̸ o̷ o̲
o̶ o̵ o̸ o̷ o̲
o̶ o̵ o̸ o̷ o̲

Double marks

x͡x x͡x x͞x x͞x x͠x x͠x x͡x
x͡x x͡x x͞x x͞x x͠x x͠x x͡x
x̋x x̋x x̏x x̂x x̌x x̆x ẍx
x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x

Greek marks

x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏ẍx̨
x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̋x̋x̏x̋x̨
x̀x́x̃x̄x̌ẋx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̃x̃ẍx̣
x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̃x̃ẍx̣
x̀x́x̃x̄x̌ẋẍx̣
x̍x́x̃x̨
x̂x̂x̀x̀x̃x̃ẍx̣

Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals. It adds `tildecomb.cap` to every style (2026-11-08): the tilde of the precomposed capitals, which is `tildecomb` itself in regular, italic and semibold, and in semibold italic the taller tilde of Ã (also of Ñ Õ Ĩ Ẽ Ỹ Ṽ and ã ñ õ), which differs from `tildecomb` there.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

		regular	semibold	italic	semibold italic	
U+0303	tilde (designer's, for comparison)	ãñũÕ	**ãñũÕ**	*ãñũÕ*	***ãñũÕ***	
U+030C	caron (designer's, for comparison)	ǎňǔǑ	**ǎňǔǑ**	*ǎňǔǑ*	***ǎňǔǑ***	
U+033D	x above	x̌ňǔǑ	**x̌ňǔǑ**	*x̌ňǔǑ*	***x̌ňǔǑ***	emboldened
U+033E	vertical tilde	̾ň̾ǔ̾Ǒ	**̾ň̾ǔ̾Ǒ**	*̾ň̾ǔ̾Ǒ*	***̾ň̾ǔ̾Ǒ***	emboldened
U+034A	not tilde above	ãn̈ũ̃Õ	**ãn̈ũ̃Õ**	*ãn̈ũ̃Õ*	***ãn̈ũ̃Õ***	emboldened
U+034B	homothetic above	ãn̈ũ̑Õ	**ãn̈ũ̑Õ**	*ãn̈ũ̑Õ*	***ãn̈ũ̑Õ***	own tilde; dots emboldened
U+034C	almost equal to above	ãn̈ũ̃̃Õ	**ãn̈ũ̃̃Õ**	*ãn̈ũ̃̃Õ*	***ãn̈ũ̃̃Õ***	emboldened
U+0350	right arrowhead above	ǎňǔǑ	**ǎňǔǑ**	*ǎňǔǑ*	***ǎňǔǑ***	emboldened
U+0351	left half ring above	á̜ǔ̜Ó	**á̜ǔ̜Ó**	*á̜ǔ̜Ó*	***á̜ǔ̜Ó***	emboldened (semibold italic only)
U+0352	fermata	ân̂ûÔ	**ân̂ûÔ**	*ân̂ûÔ*	***ân̂ûÔ***	own inverted breve; dot emboldened
U+0357	right half ring above	ả̜ǔ̜Ỏ	**ả̜ǔ̜Ỏ**	*ả̜ǔ̜Ỏ*	***ả̜ǔ̜Ỏ***	emboldened (semibold italic only)
U+035B	zigzag above	á̴ǔ̴Ó	**á̴ǔ̴Ó**	*á̴ǔ̴Ó*	***á̴ǔ̴Ó***	emboldened
U+0340	grave tone mark	àǹùÒ	**àǹùÒ**	*àǹùÒ*	***àǹùÒ***	own grave
U+0341	acute tone mark	áńúÓ	**áńúÓ**	*áńúÓ*	***áńúÓ***	own acute
U+0346	bridge above	ā͡n͡ū͡Ō	**ā͡n͡ū͡Ō**	*ā͡n͡ū͡Ō*	***ā͡n͡ū͡Ō***	own bridge below, raised

Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.



Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

Ơơ Ưư Ớớ Ờờ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̕wkt* [k̕^{wkx}t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’^{kx}ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is ^{wkx} and it is glottalized. In (2), the superscript consonant ^{x} is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: ̙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k˒ʷᵏˣ˒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k˒ʷᵏˣ’t]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k˒ʷᵏˣ˒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣt]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k˒ʷᵏˣ’t]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ɦ’ʲ’ʳ’ɹ’ɻ’ʁ’ʷ’ʸ’ʏ’ˡ’ˢ’ˣ’ʕ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ɯ’ᵛ’ᵜ’β’ᵞ’ᵟ’

ʰ‘ɦ‘ʲ‘ʳ‘ɹ‘ɻ‘ʁ‘ʷ‘ʸ‘ʏ‘ˡ‘ˢ‘ˣ‘ʕ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ɯ‘ᵛ‘ᵜ‘β‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣt]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ’ɦ’ʲ’ʳ’ɹ’ɻ’ʁ’ʷ’ʸ’ʏ’ˡ’ˢ’ˣ’ʕ’ᵇ’ᵏ’ᵐ’ᵖ’ᵗ’ɯ’ᵛ’ᵜ’β’ᵞ’ᵟ’

ʰ‘ɦ‘ʲ‘ʳ‘ɹ‘ɻ‘ʁ‘ʷ‘ʸ‘ʏ‘ˡ‘ˢ‘ˣ‘ʕ‘ᵇ‘ᵏ‘ᵐ‘ᵖ‘ᵗ‘ɯ‘ᵛ‘ᵜ‘β‘ᵞ‘ᵟ‘

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣ’t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

Bases

Here are some subgroups of bases.

Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ

IPA

ƅↃƇƉƊɑɋƎƏƐƑƒƓɣɦɬƙꟻƝɲƟɭʒƨ|‖ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪ̈ɫɬɮɭɰɱɲɳɴɵɶɷɸɹɺɻɽɾʀʁʂʃʄʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃʘʛʜʝʞʟʠ
ƅↃƇƉƊɑɋƎƏƐƑƒƓɣɦɬƙꟻƝɲƟɭʒƨ|‖ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪ̈ɫɬɮɭɰɱɲɳɴɵɶɷɸɹɺɻɽɾʀʁʂʃʄʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃʘʛʜʝʞʟʠ
ƅↃƇƉƊɑɋƎƏƐƑƒƓɣɦɬƙꟻƝɲƟɭʒƨ|‖ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪ̈ɫɬɮɭɰɱɲɳɴɵɶɷɸɹɺɻɽɾʀʁʂʃʄʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃʘʛʜʝʞʟʠ
ƅↃƇƉƊɑɋƎƏƐƑƒƓɣɦɬƙꟻƝɲƟɭʒƨ|‖ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɪɪ̈ɫɬɮɭɰɱɲɳɴɵɶɷɸɹɺɻɽɾʀʁʂʃʄʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖʗǀǁǂǃʘʛʜʝʞʟʠ

Superscript consonants

ʰʱʲʳʴʵʶʷʸˡˢˣˤˀᵇᵏᵐᵖᵗᵠᵛᵝᵞᵟ

ʰʱʲʳʴʵʶʷʸˡˢˣˤˀᵇᵏᵐᵖᵗᵠᵛ□□□□

ʰʱ ʲʳʴʵʶʷʸˡˢˣˤˀᵇᵏᵐᵖᵗᵠᵛ□□□□

ʰʱʲʳʴʵʶʷʸˡˢˣˤˀᵇᵏᵐᵖᵗᵠᵛ□□□□

Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω

Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ

Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẼẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẼẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẼẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẼẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐƠǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲƯǓǕǗǙǛỤỦỨỪỬỮỰ

Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻẽếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻẽếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűǔųưǖǘǚǜủụứừửữự

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Precomposed capital consonants

ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḨḪḴĶḰĶǨĹḶĽḺḼḸḾṀṂŃṆŇṊṄŊƠṔṖṕŖŘŗŕŖŚŜŞŠṢṠṤṦṨṬṪṰṮṬȚƯṼṾỤẂẀẄẂẊẌỲÝỸỶỴỲỸỾẎŹŻŽẔẐȤ

ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḨḪḴĶḰĶǨĹḶĽḺḼḸḾṀṂŃṆŇṊṄŊƠṔṖṕŖŘŗŕŖŚŜŞŠṢṠṤṦṨṬṪṰṮṬȚƯṼṾỤẂẀẄẂẊẌỲÝỸỶỴỲỸỾẎŹŻŽẔẐȤ

ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḨḪḴĶḰĶǨĹḶĽḺḼḸḾṀṂŃṆŇṊṄŊƠṔṖṕŖŘŗŕŖŚŜŞŠṢṠṤṦṨṬṪṰṮṬȚƯṼṾỤẂẀẄẂẊẌỲÝỸỶỴỲỸỾẎŹŻŽẔẐȤ

ḂḄÇḈĈĊČḊĎḐḒḌḎḐḞĜĞĠĢḠǦĤḢḤḦḨḪḴĶḰĶǨĹḶĽḺḼḸḾṀṂŃṆŇṊṄŊƠṔṖṕŖŘŗŕŖŚŜŞŠṢṠṤṦṨṬṪṰṮṬȚƯṼṾỤẂẀẄẂẊẌỲÝỸỶỴỲỸỾẎŹŻŽẔẐȤ

Precomposed small consonants

ƀƅƃƄƅƈċčḉćƈďđḓḍḏḑḋḡḟĝğġģḡǧǩĥħḫḥḧḣḩĵǰķḱǩḳƙĺḽľḻḹłḷḽḻḿṁṃṅṇňṋṉņṋŋɵɑɋṕƥƿŕṛřṙṟṝśŝşšṡṣṧṩẛţťṫṭṯṱƭưṽṿẁŵẃẅẋẍýÿŷỳẏỷỹẙźżžẓẕȥɜƨƽƿǯ

ƀƅƃƄƅƈċčḉćƈďđḓḍḏḑḋḡḟĝğġģḡǧǩĥħḫḥḧḣḩĵǰķḱǩḳƙĺḽľḻḹłḷḽḻḿṁṃṅṇňṋṉņṋŋɵɑɋṕƥƿŕṛřṙṟṝśŝşšṡṣṧṩẛţťṫṭṯṱƭưṽṿẁŵẃẅẋẍýÿŷỳẏỷỹẙźżžẓẕȥɜƨƽƿǯ

ƀƅƃƄƅƈċčḉćƈďđḓḍḏḑḋḡḟĝğġģḡǧǩĥħḫḥḧḣḩĵǰķḱǩḳƙĺḽľḻḹłḷḽḻḿṁṃṅṇňṋṉņṋŋɵɑɋṕƥƿŕṛřṙṟṝśŝşšṡṣṧṩẛţťṫṭṯṱƭưṽṿẁŵẃẅẋẍýÿŷỳẏỷỹẙźżžẓẕȥɜƨƽƿǯ

ƀƅƃƄƅƈċčḉćƈďđḓḍḏḑḋḡḟĝğġģḡǧǩĥħḫḥḧḣḩĵǰķḱǩḳƙĺḽľḻḹłḷḽḻḿṁṃṅṇňṋṉņṋŋɵɑɋṕƥƿŕṛřṙṟṝśŝşšṡṣṧṩẛţťṫṭṯṱƭưṽṿẁŵẃẅẋẍýÿŷỳẏỷỹẙźżžẓẕȥɜƨƽƿǯ

No marks: rare, digraph etc.

ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
ʗ ʤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ

Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ

Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͟ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

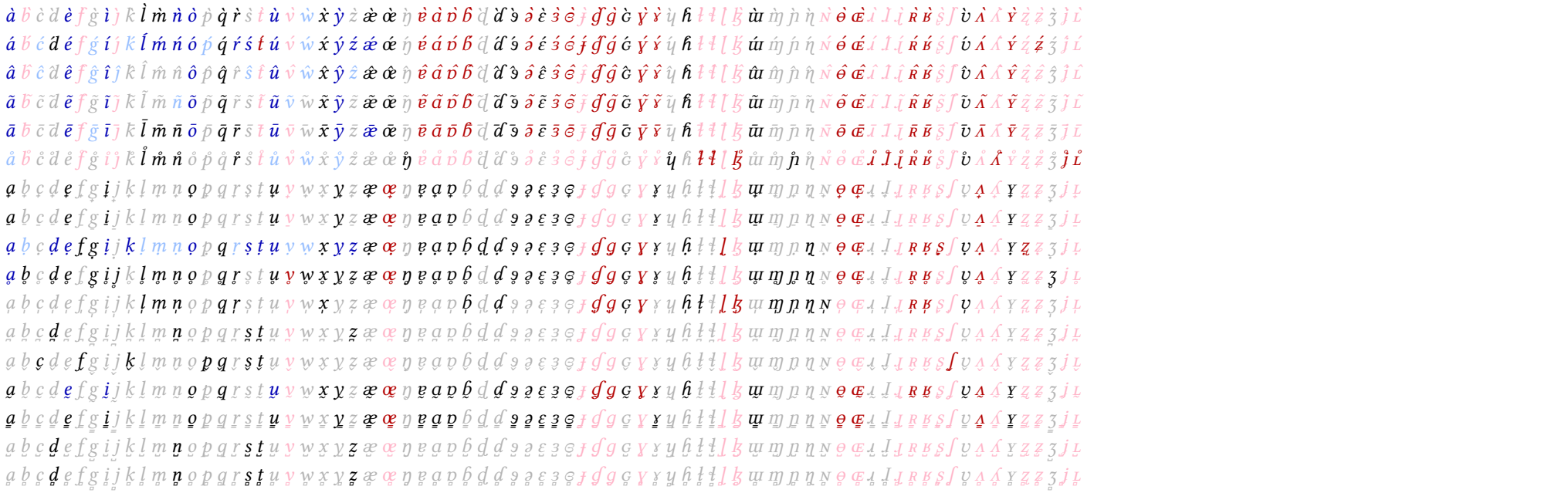
Regular, Bases requiring anchors for common marks, Common combining marks



Regular patched, Bases requiring anchors for common marks, Common combining marks



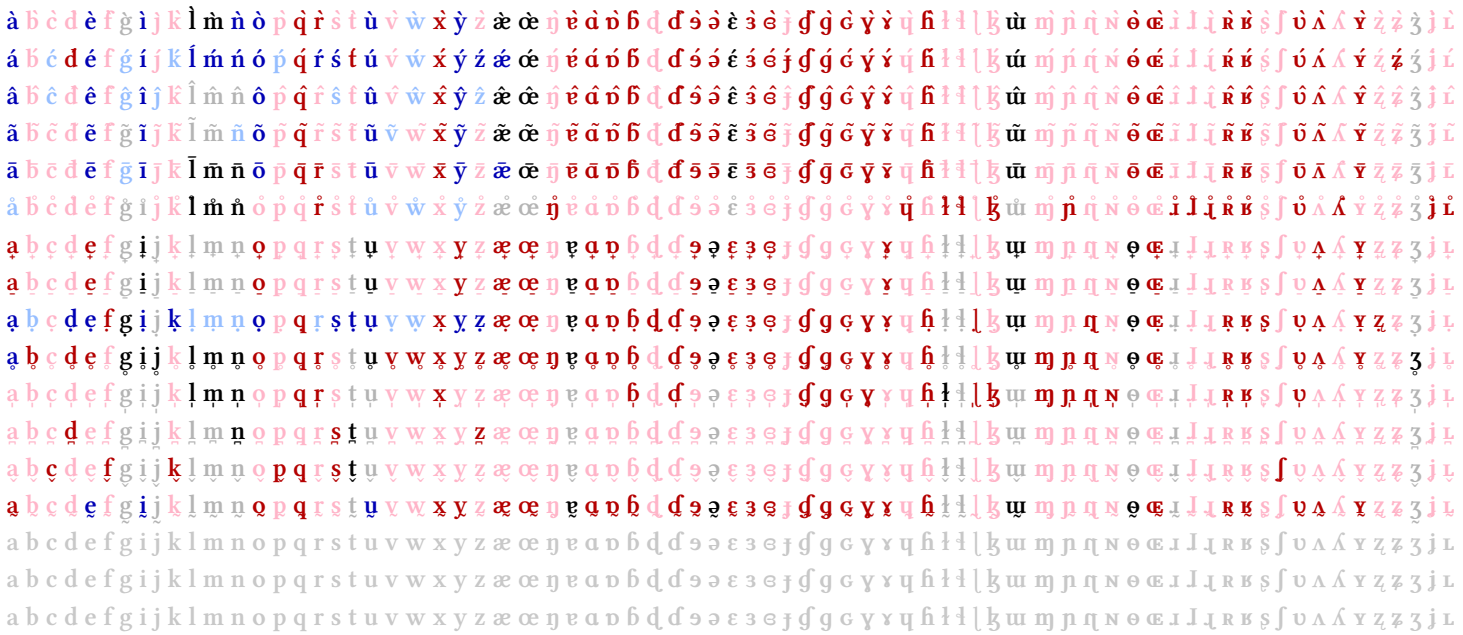
Italic, Bases requiring anchors for common marks, Common combining marks



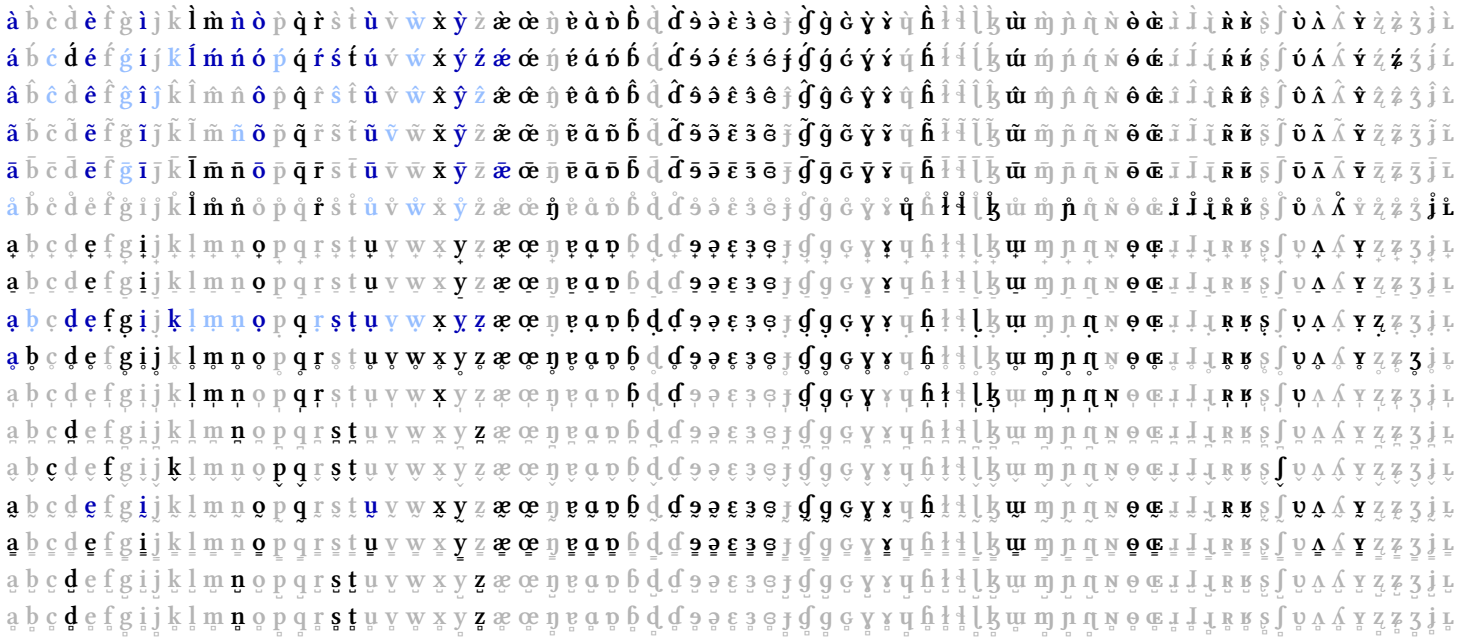
Italic patched, Bases requiring anchors for common marks, Common combining marks



Semibold, Bases requiring anchors for common marks, Common combining marks



Semibold patched, Bases requiring anchors for common marks, Common combining marks



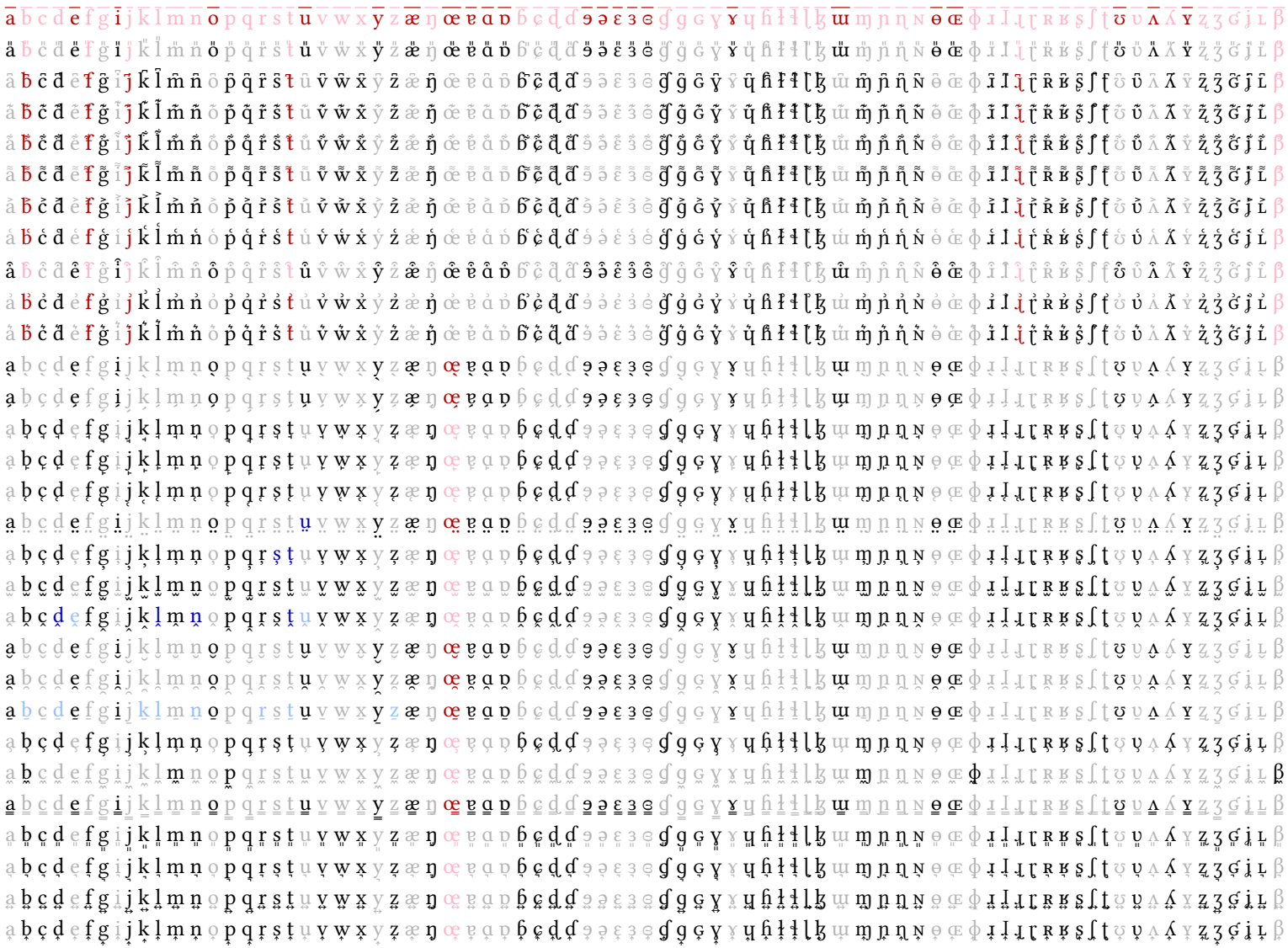
Semibold italic, Bases requiring anchors for common marks, Common combining marks



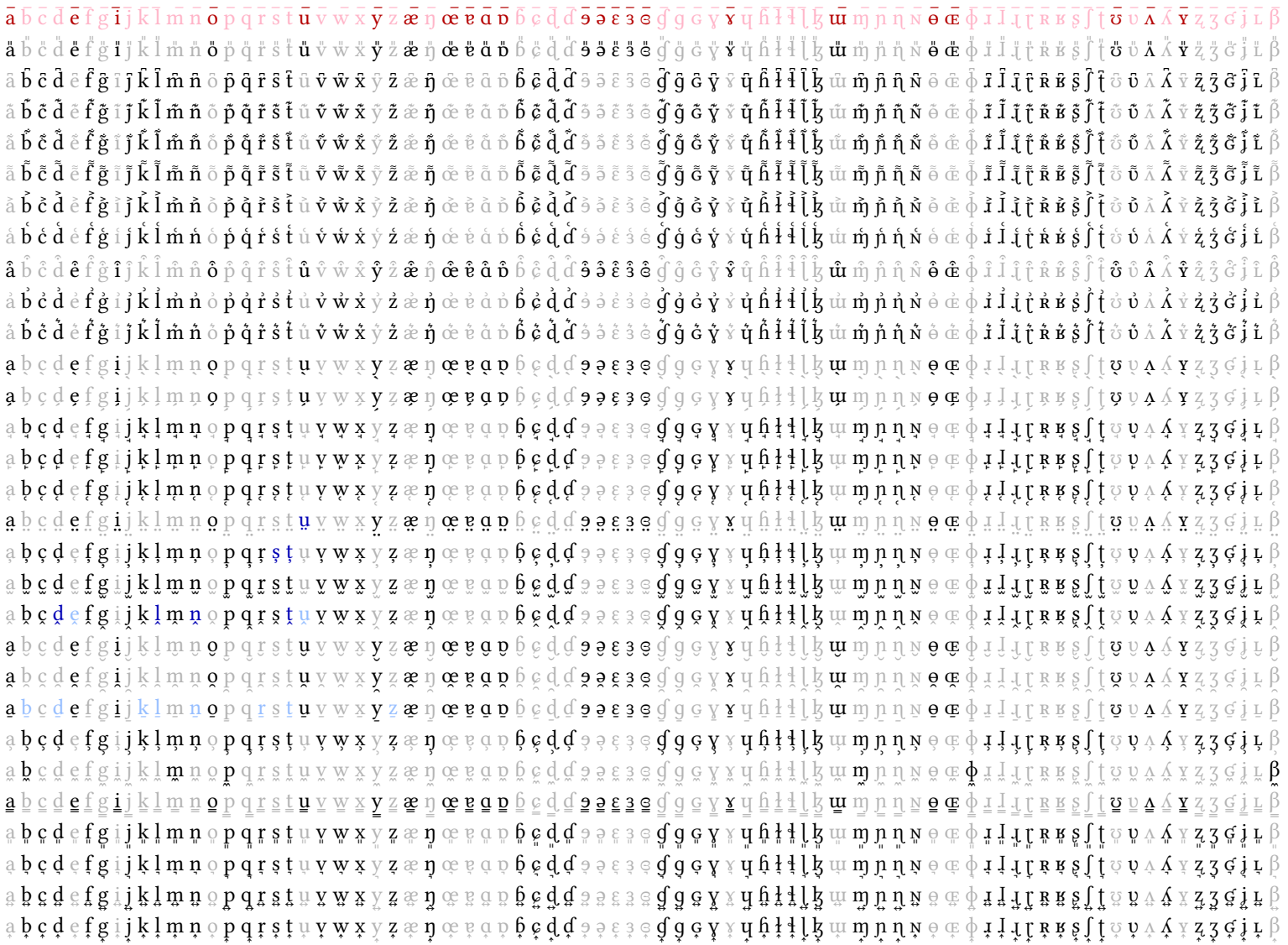
Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



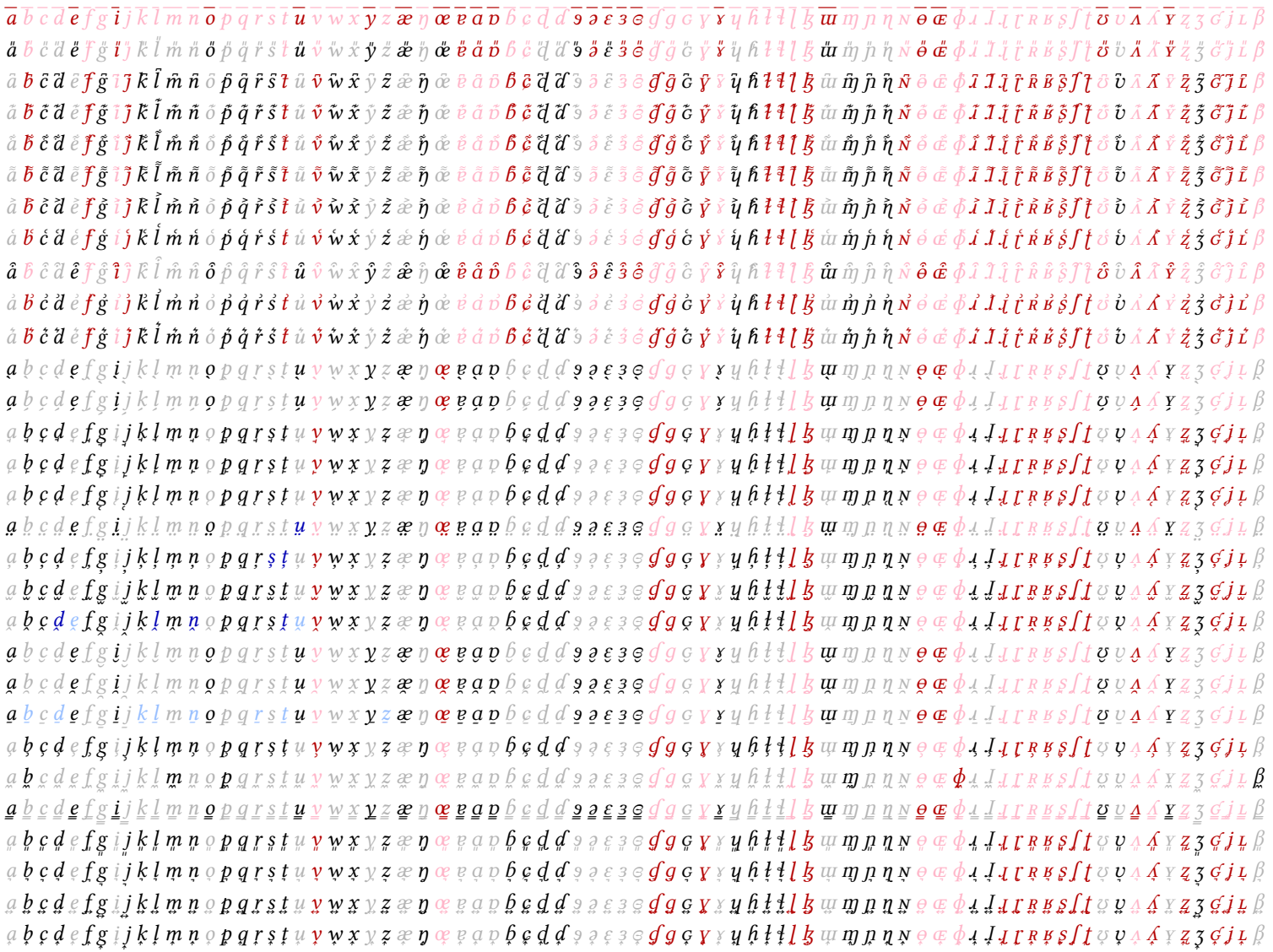
Regular, Bases requiring anchors for rare marks, Rare combining marks



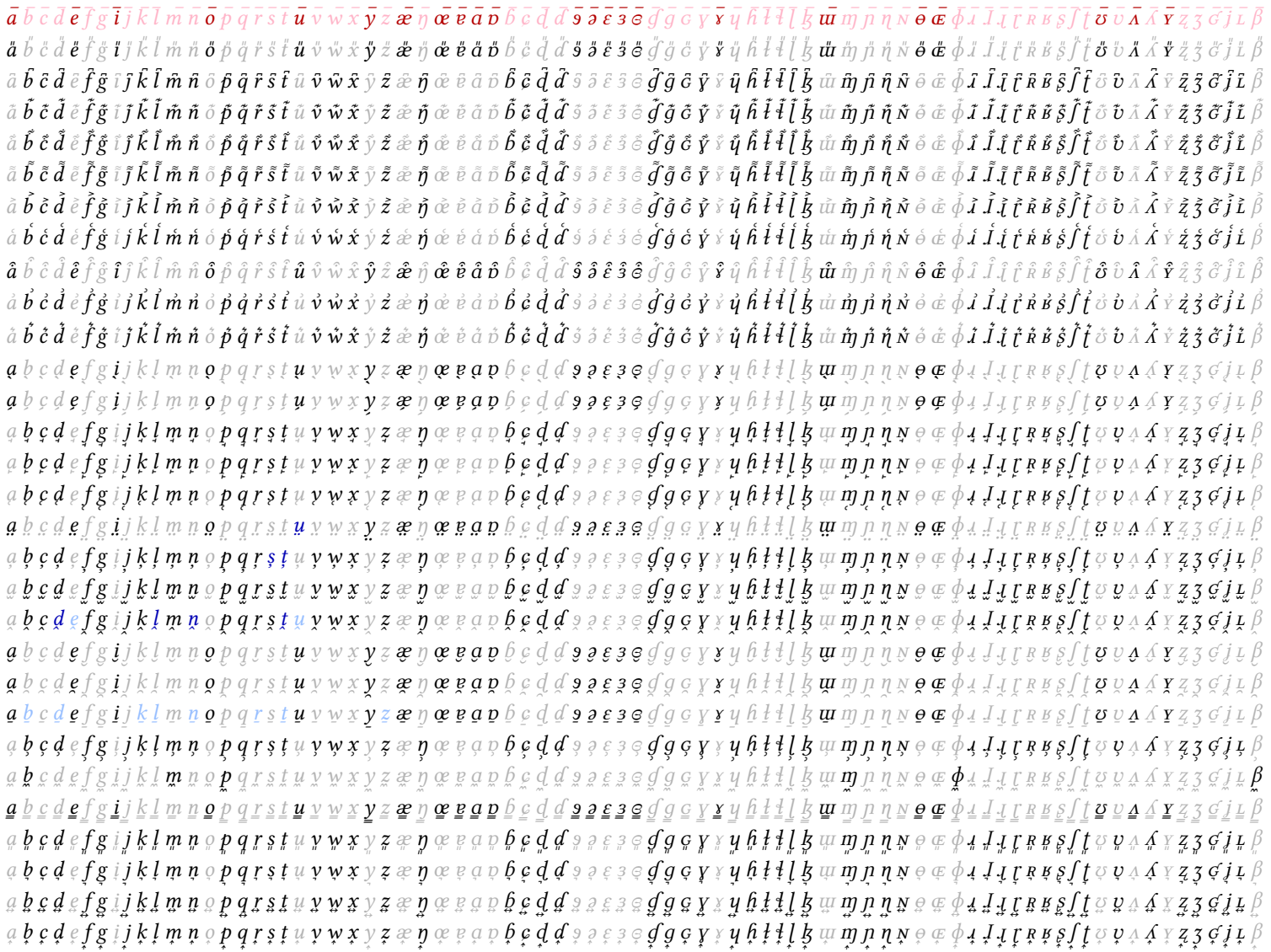
Regular patched, Bases requiring anchors for rare marks, Rare combining marks



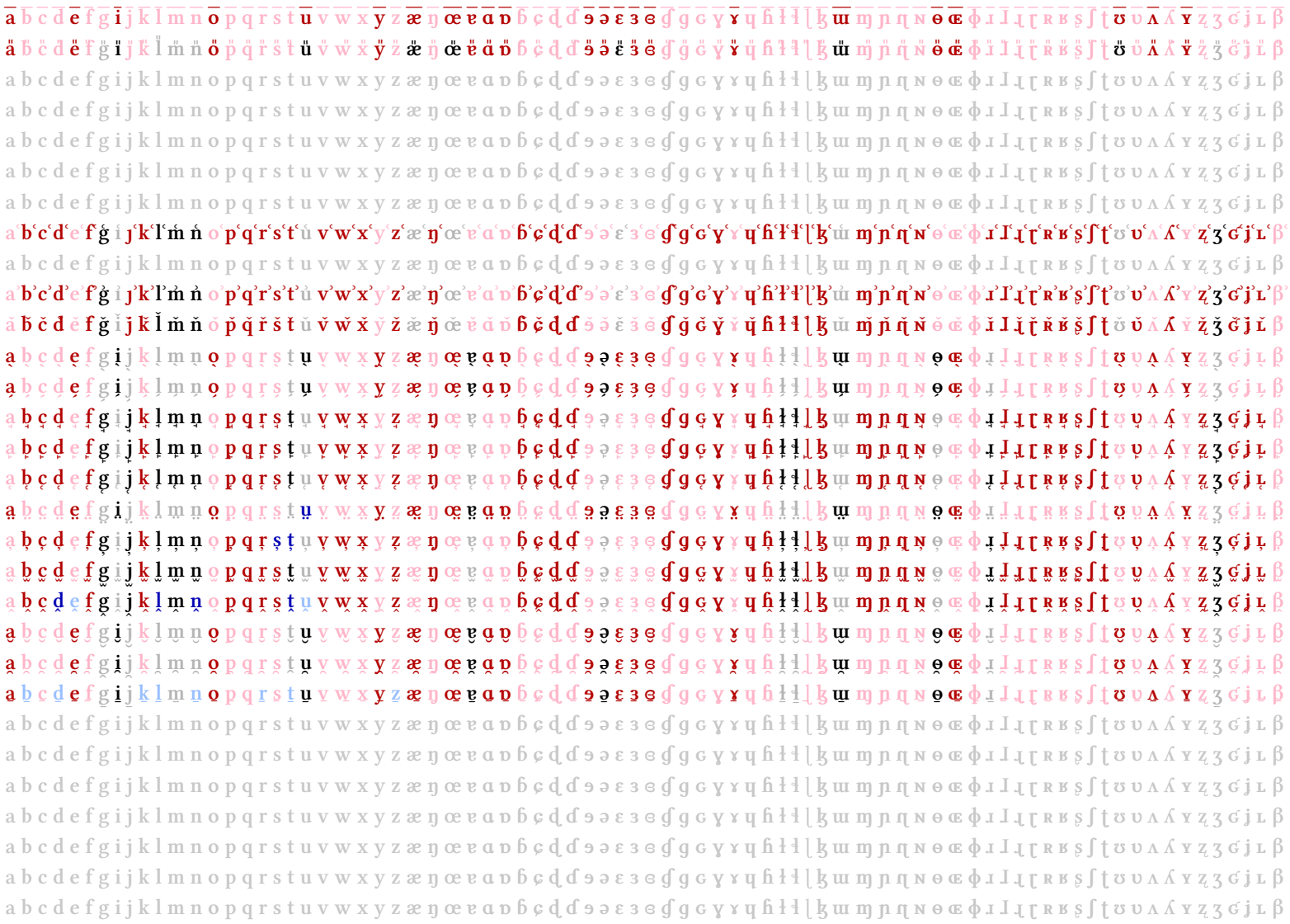
Italic, Bases requiring anchors for rare marks, Rare combining marks



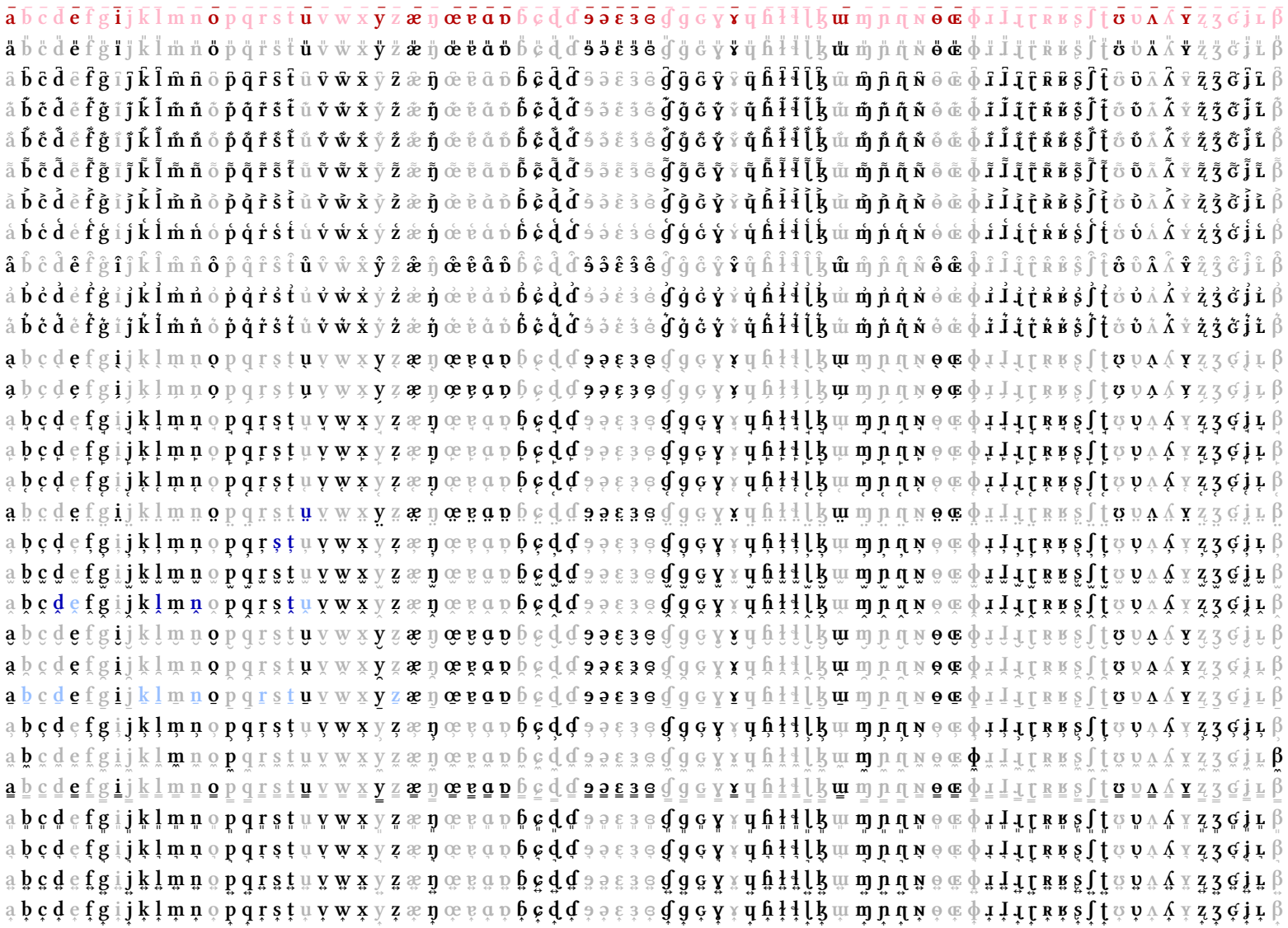
Italic patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold, Bases requiring anchors for rare marks, Rare combining marks



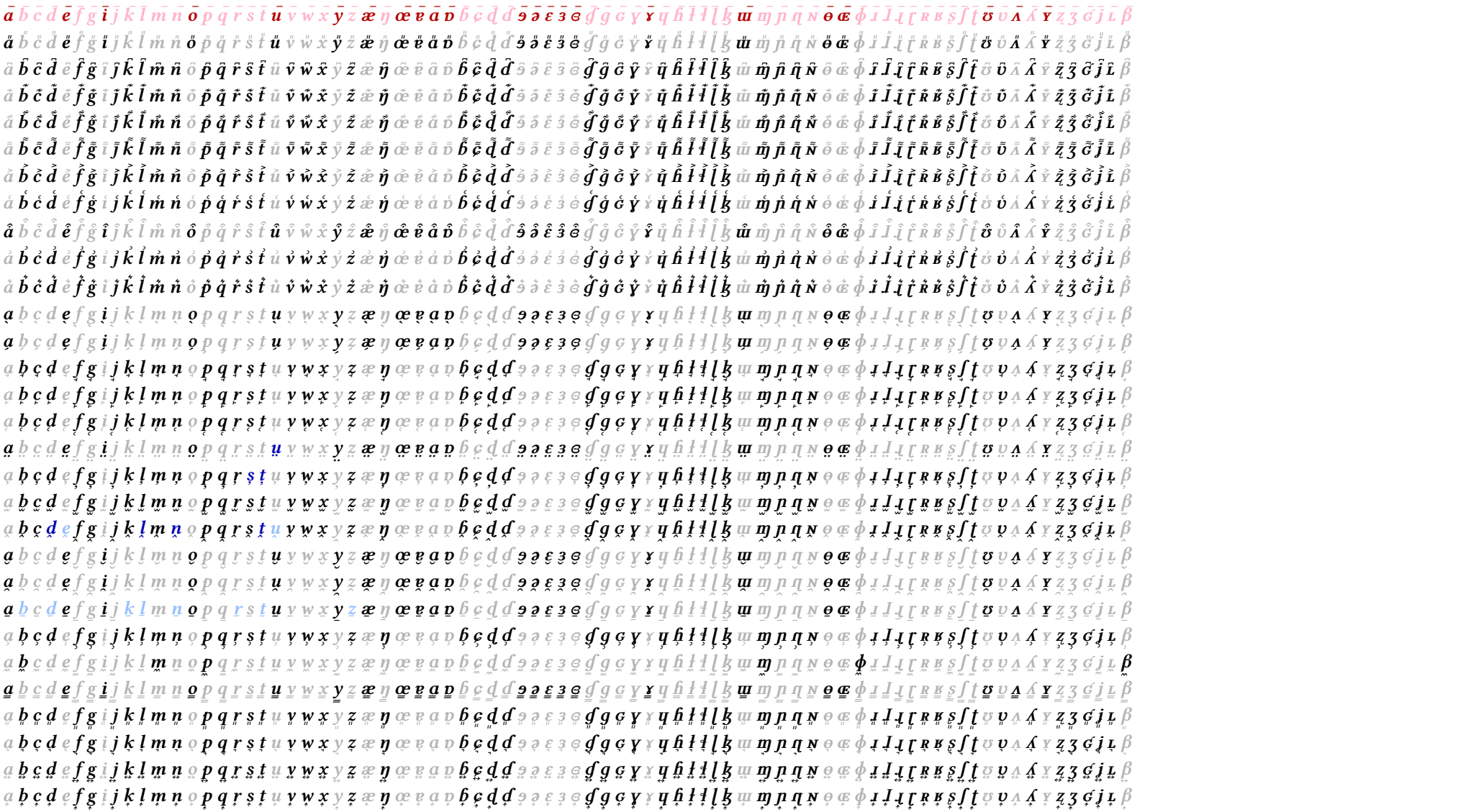
Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks. They leave out the letters whose anchors stage 8 or stage 9 sets, the derived and precomposed letters, which have their own grids after their base letters (sections “Derived letters” and “Precomposed characters”).

Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

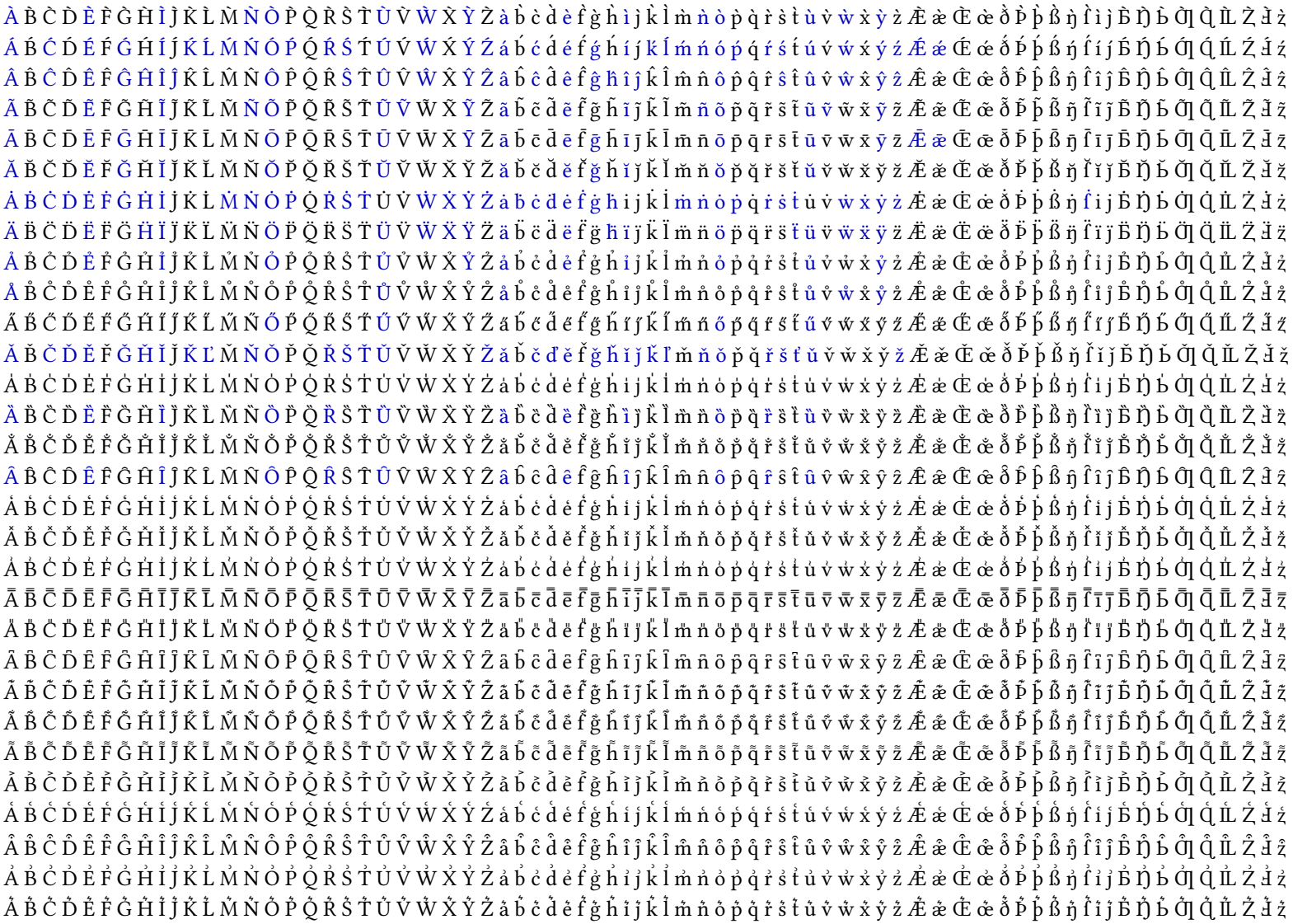
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

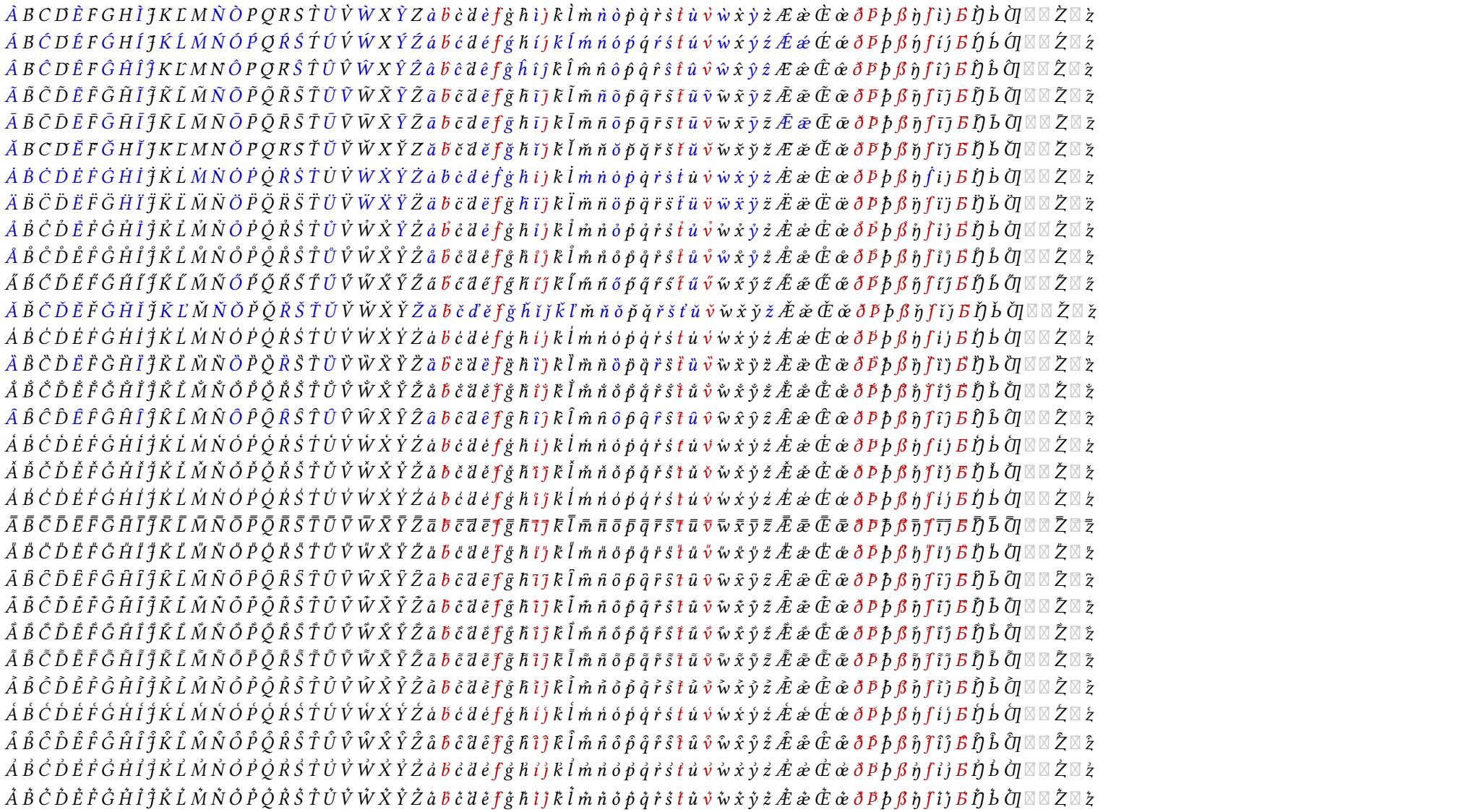
Regular, Latin, Above



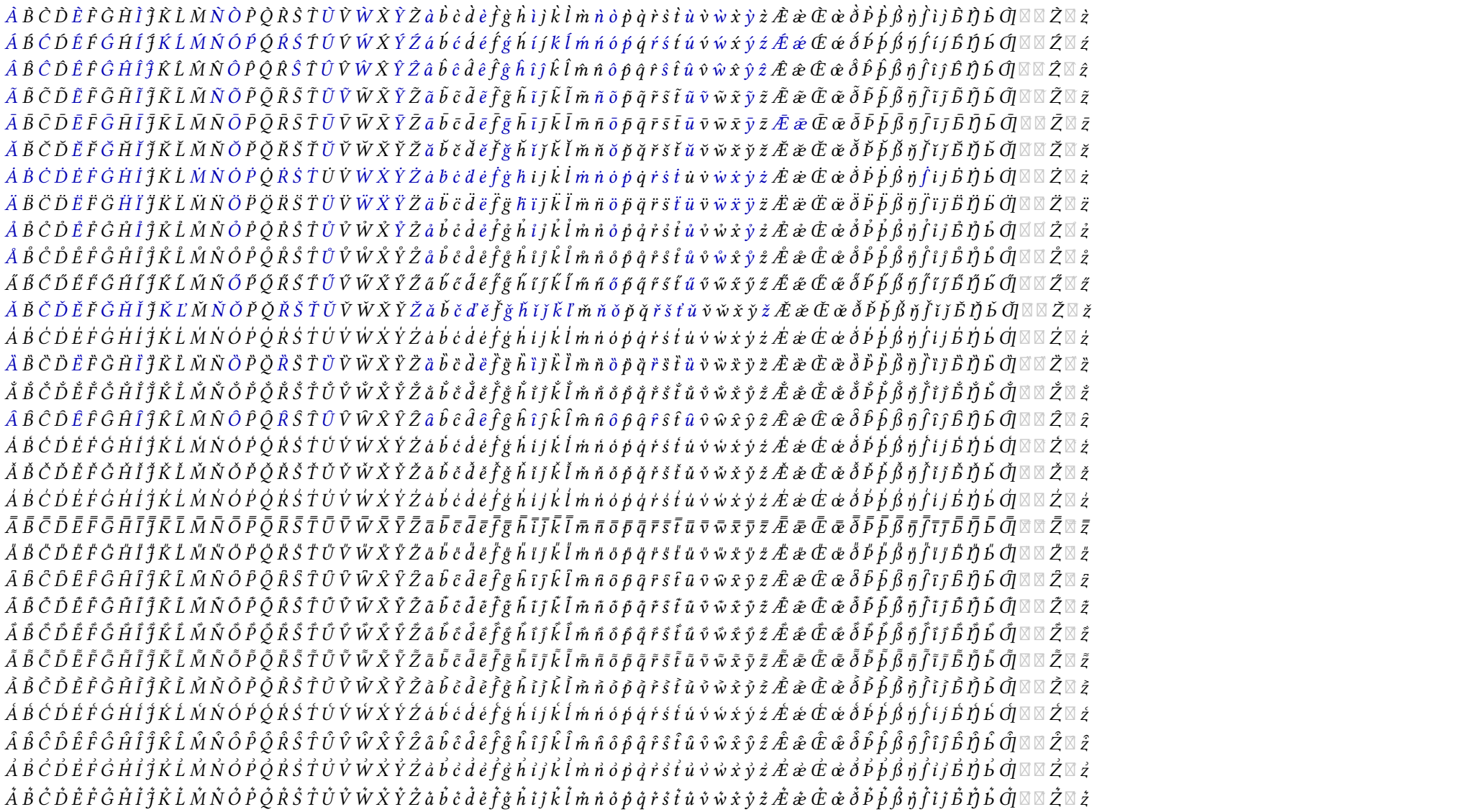
Regular patched, Latin, Above



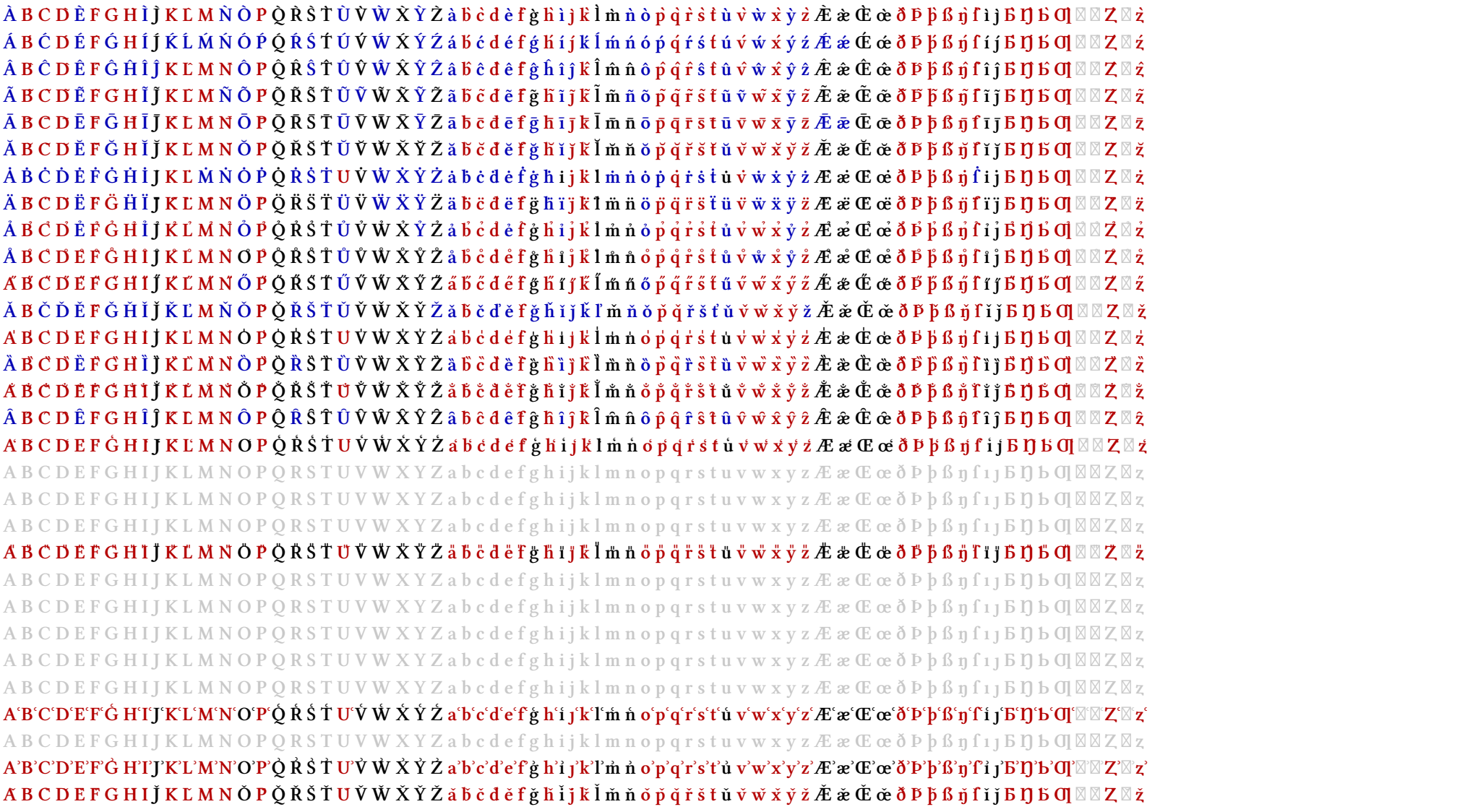
Italic, Latin, Above



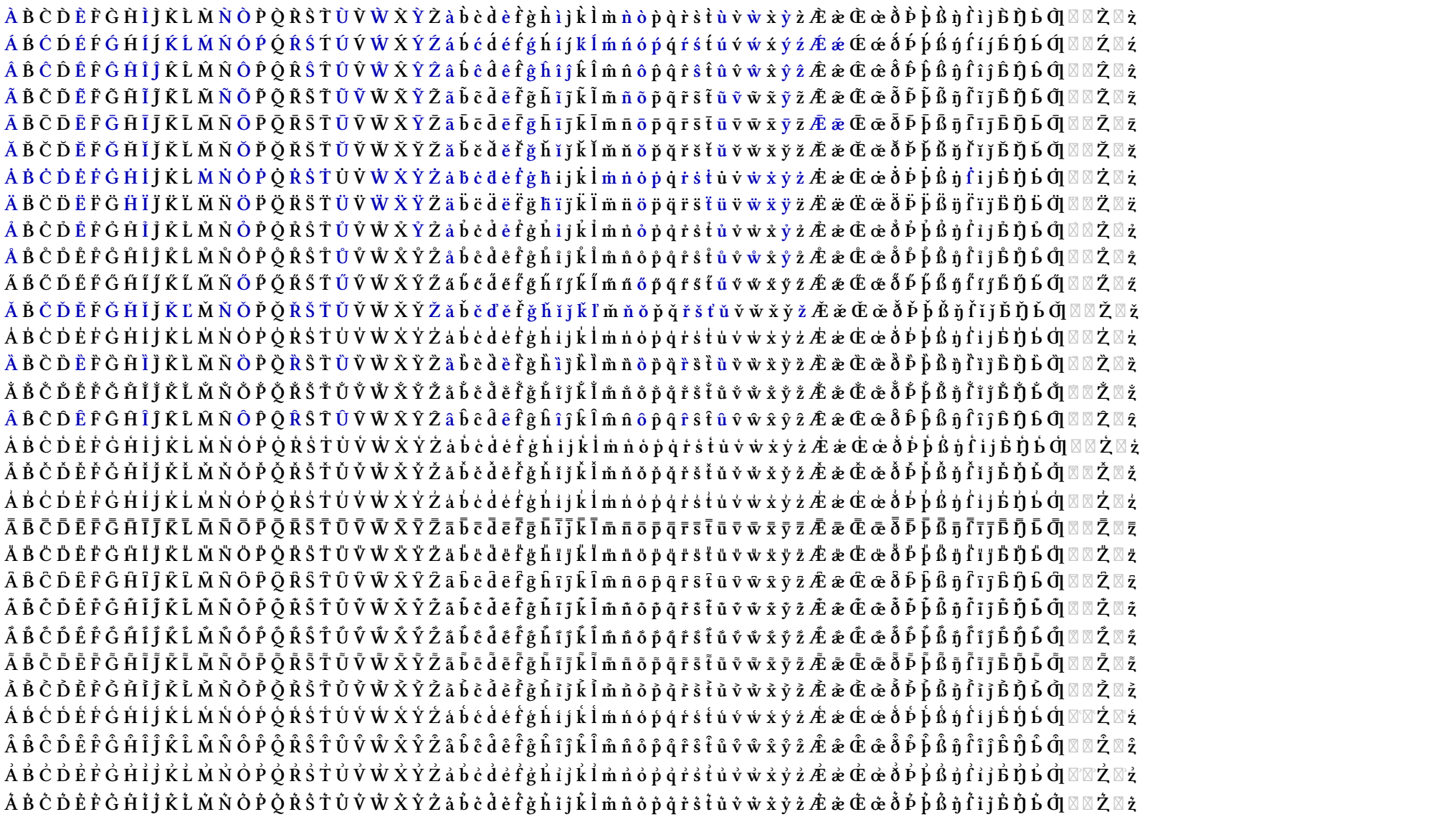
Italic patched, Latin, Above



Semibold, Latin, Above

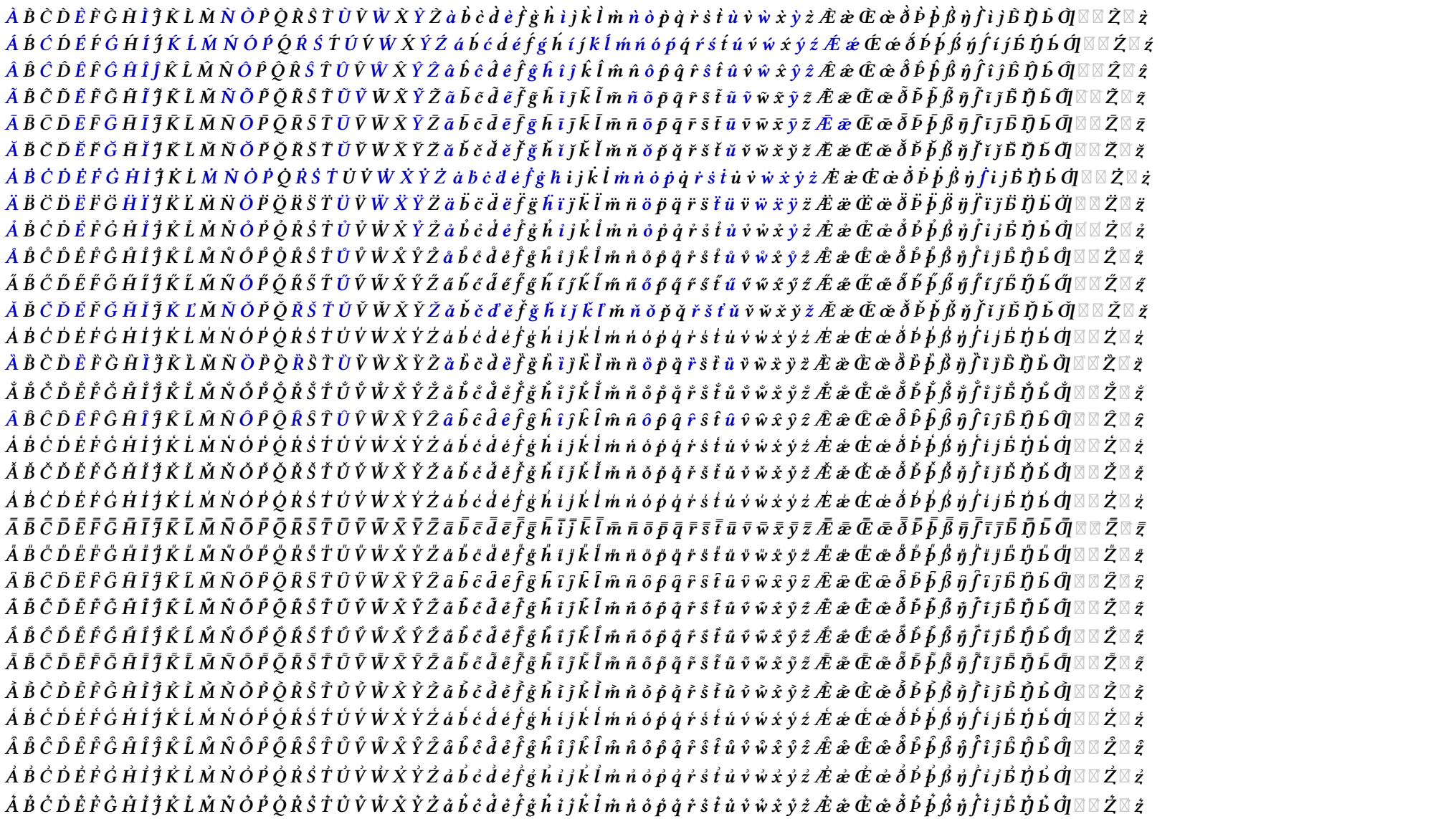


Semibold patched, Latin, Above

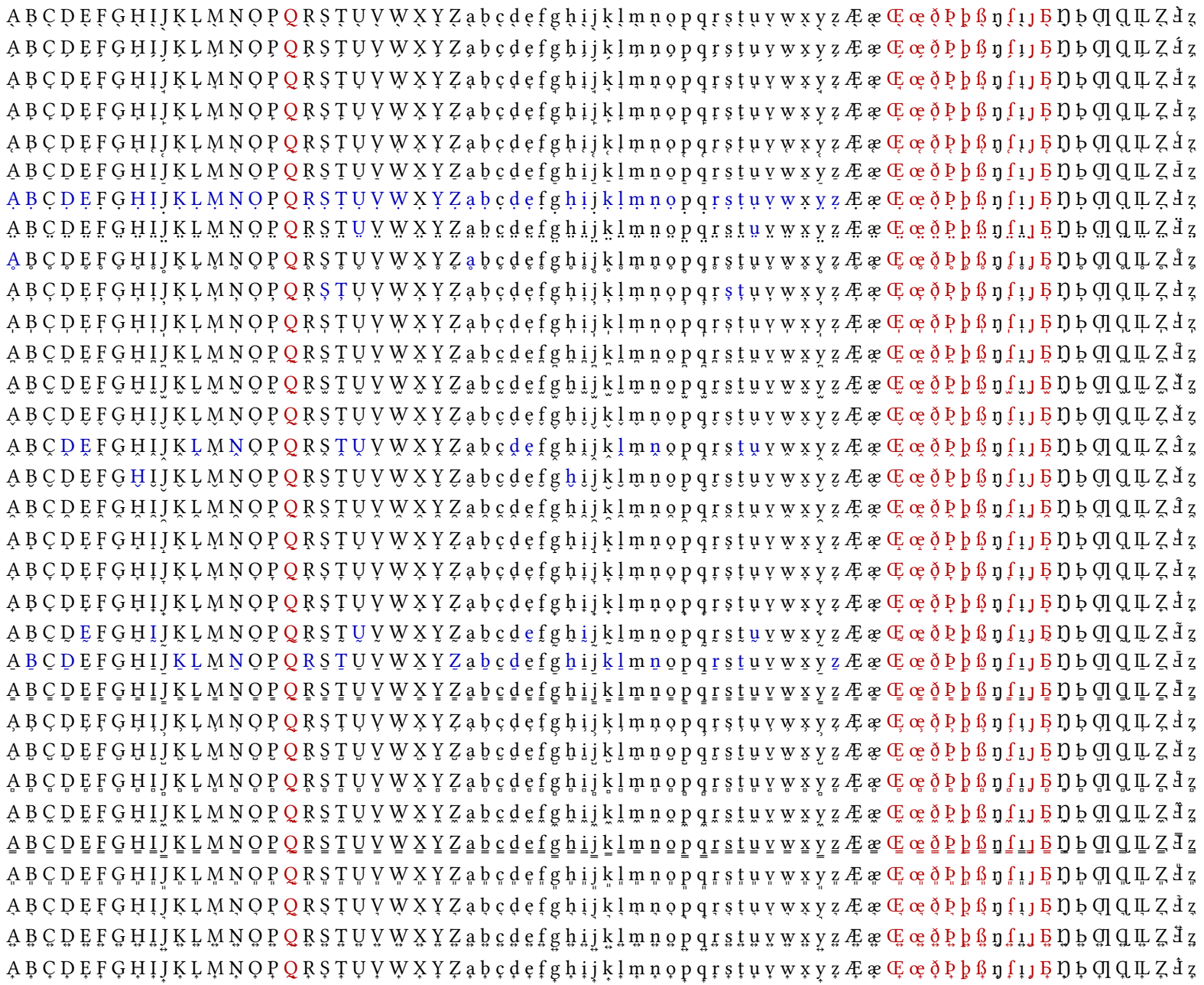


Semibold italic, Latin, Above

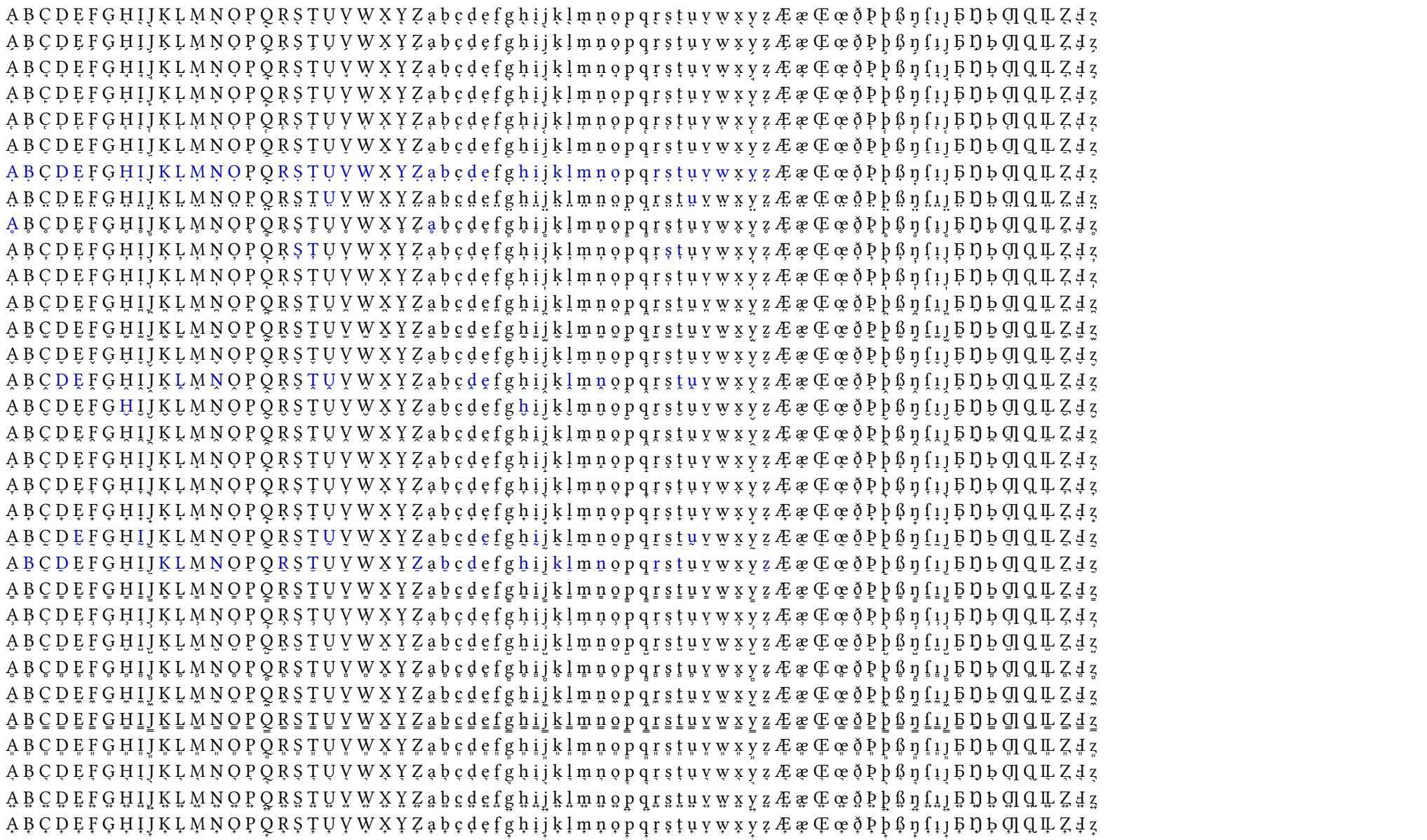
Semibold italic patched, Latin, Above



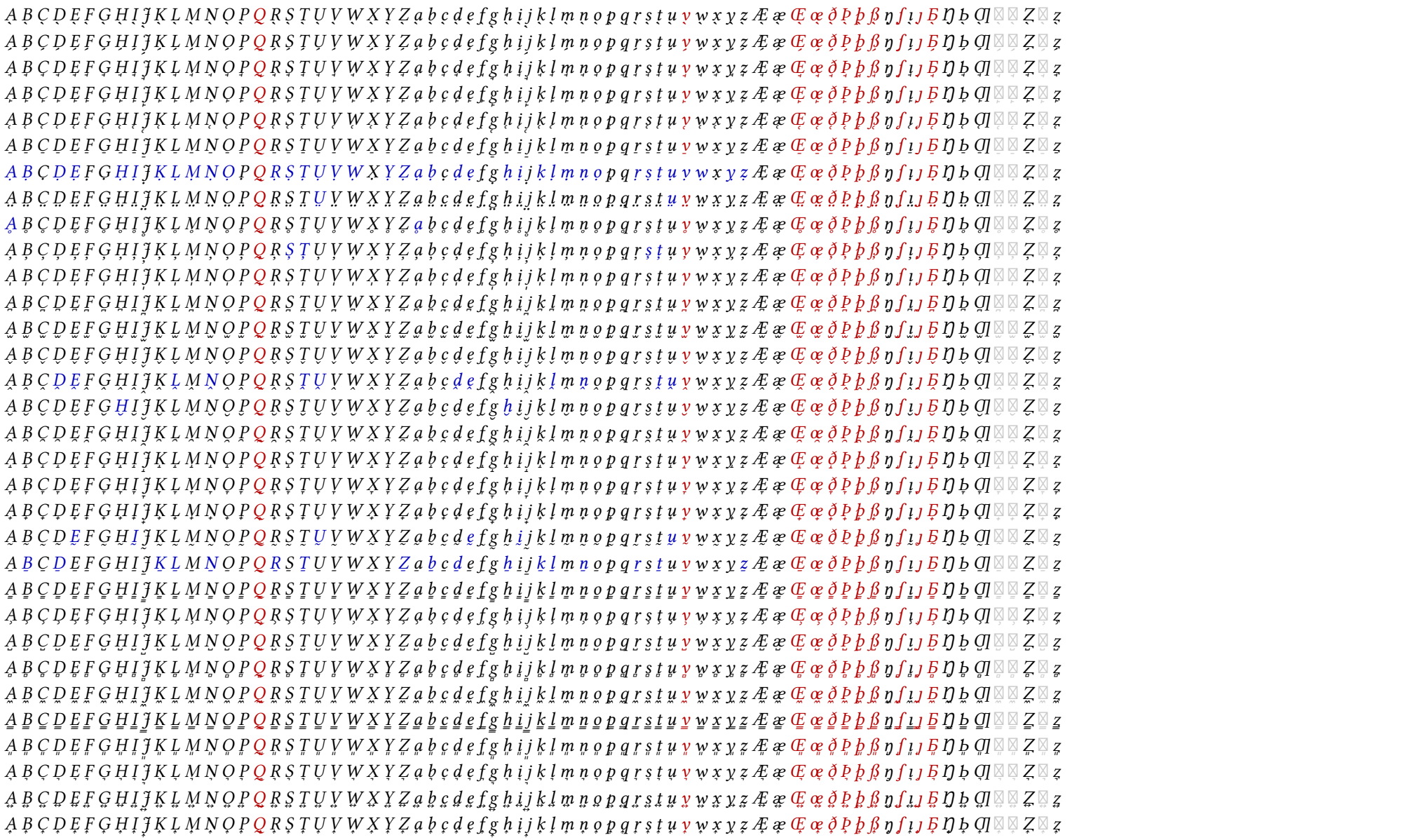
Regular, Latin, Below



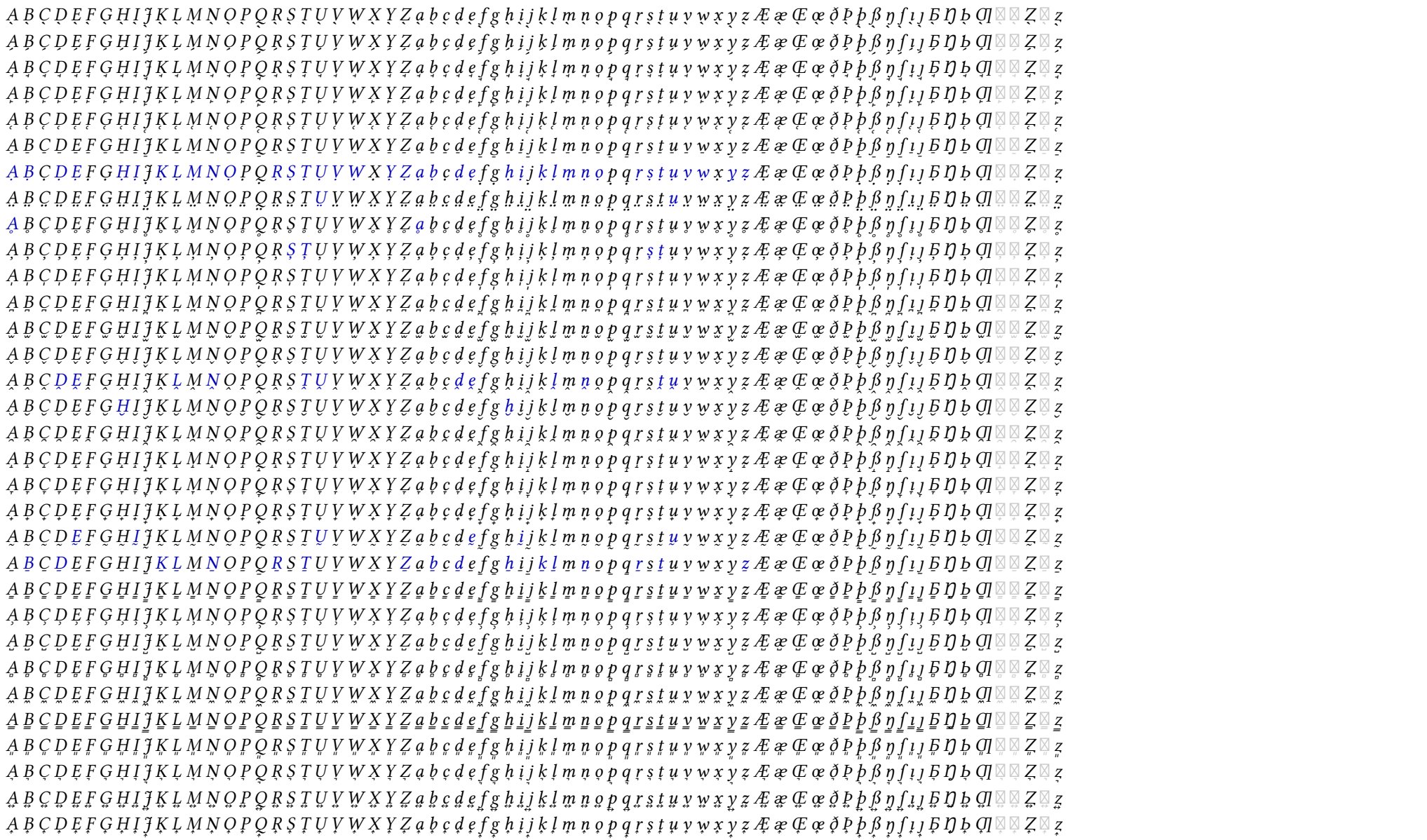
Regular patched, Latin, Below



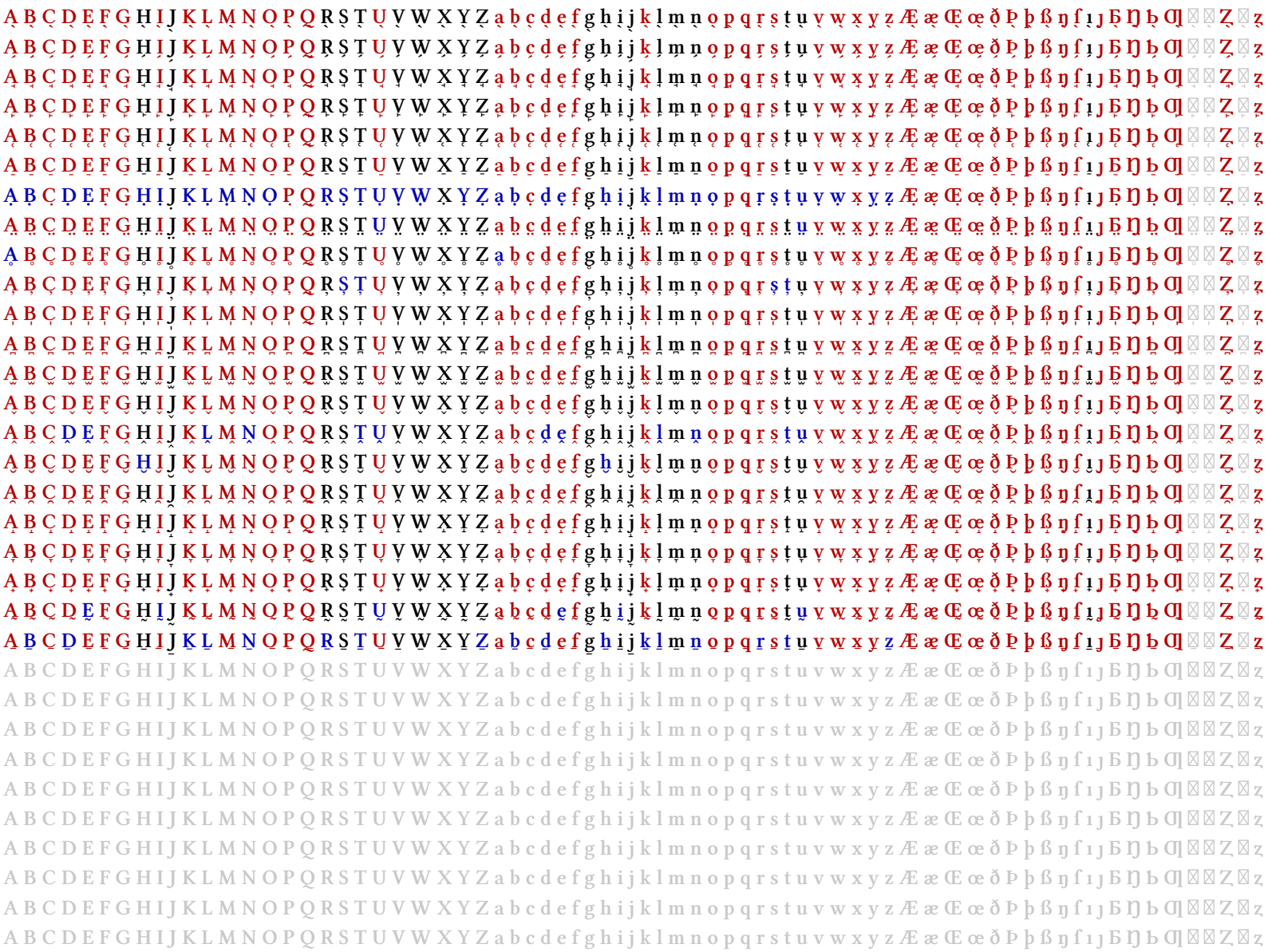
Italic, Latin, Below



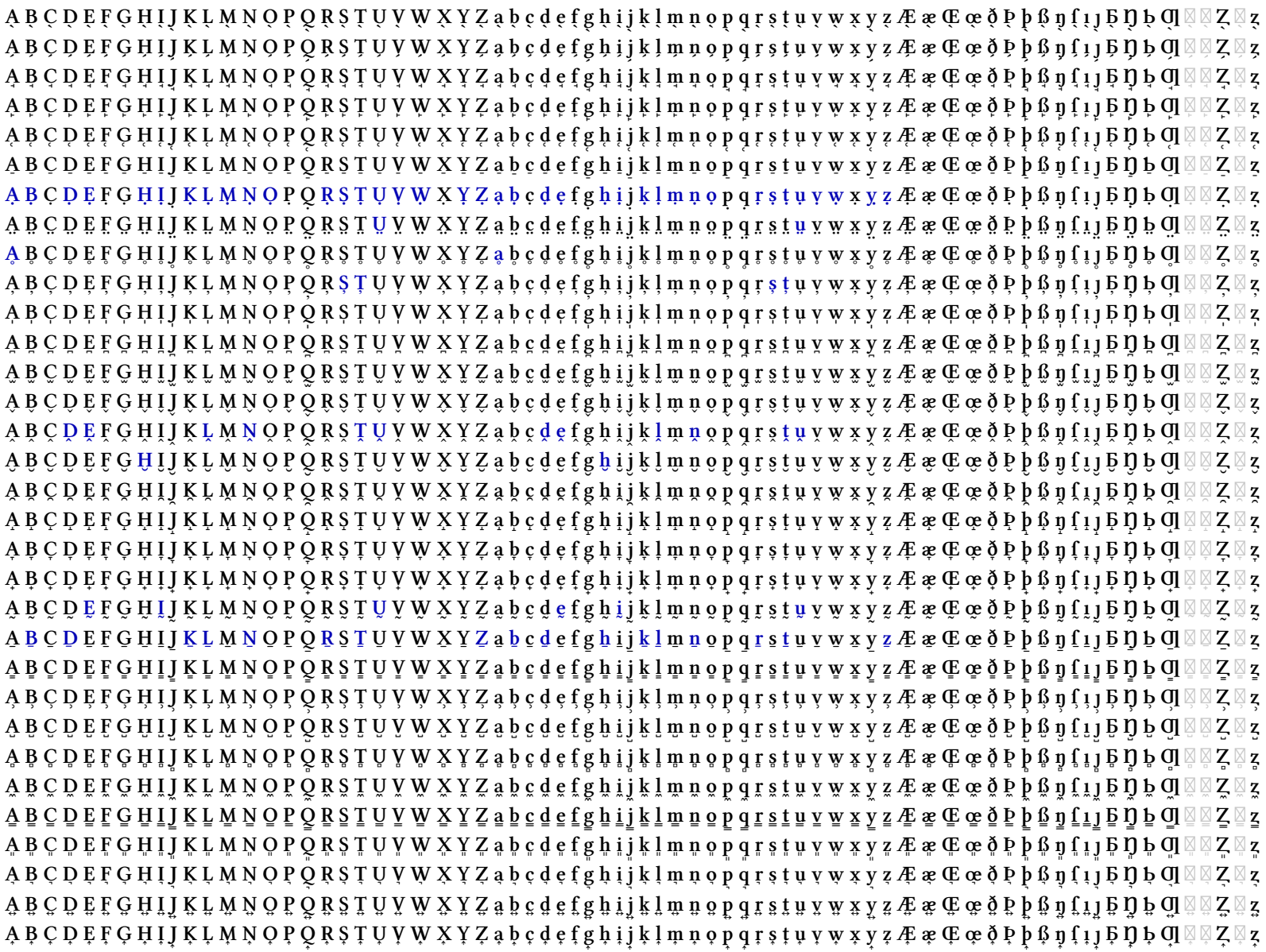
Italic patched, Latin, Below



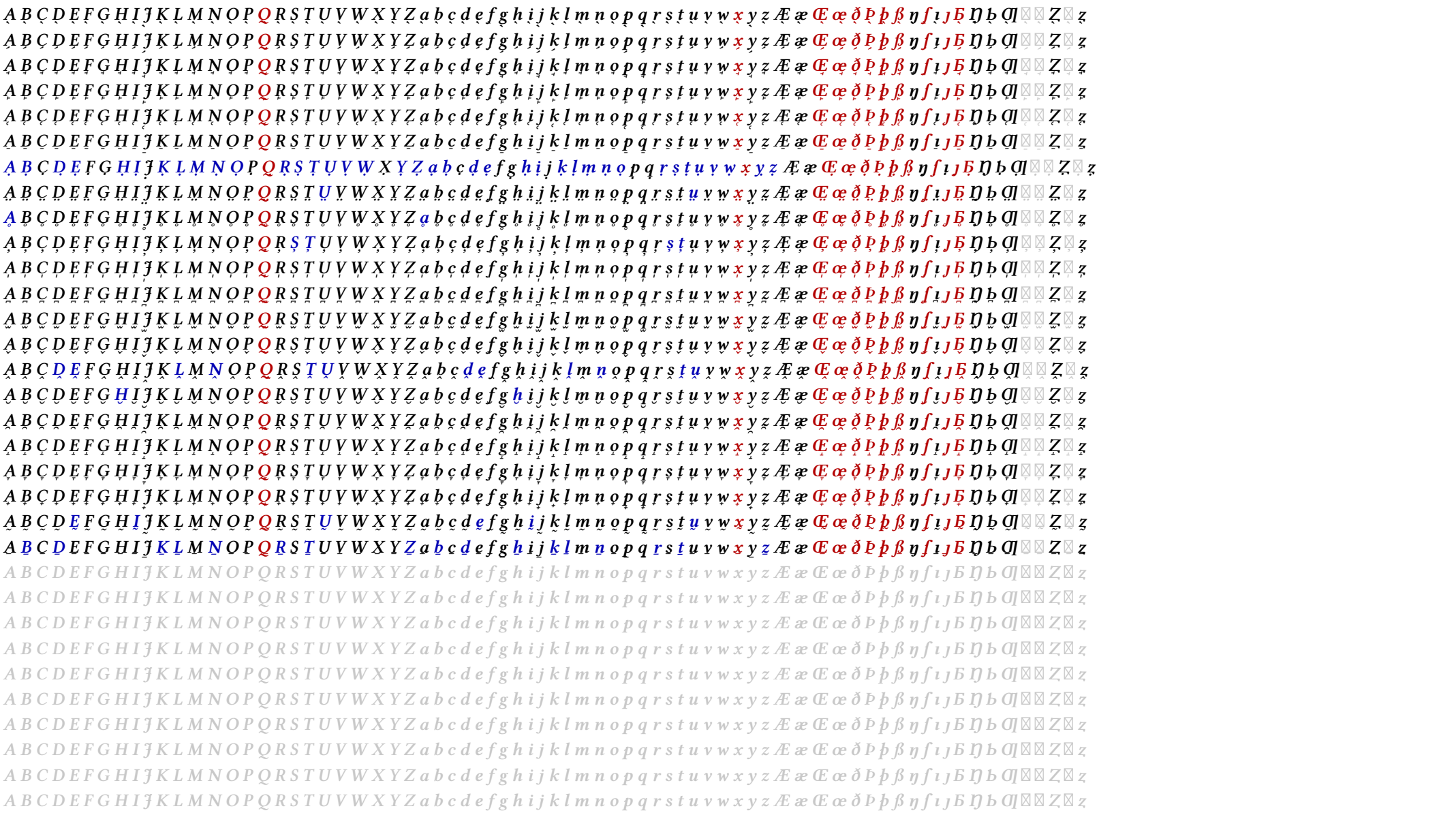
Semibold, Latin, Below



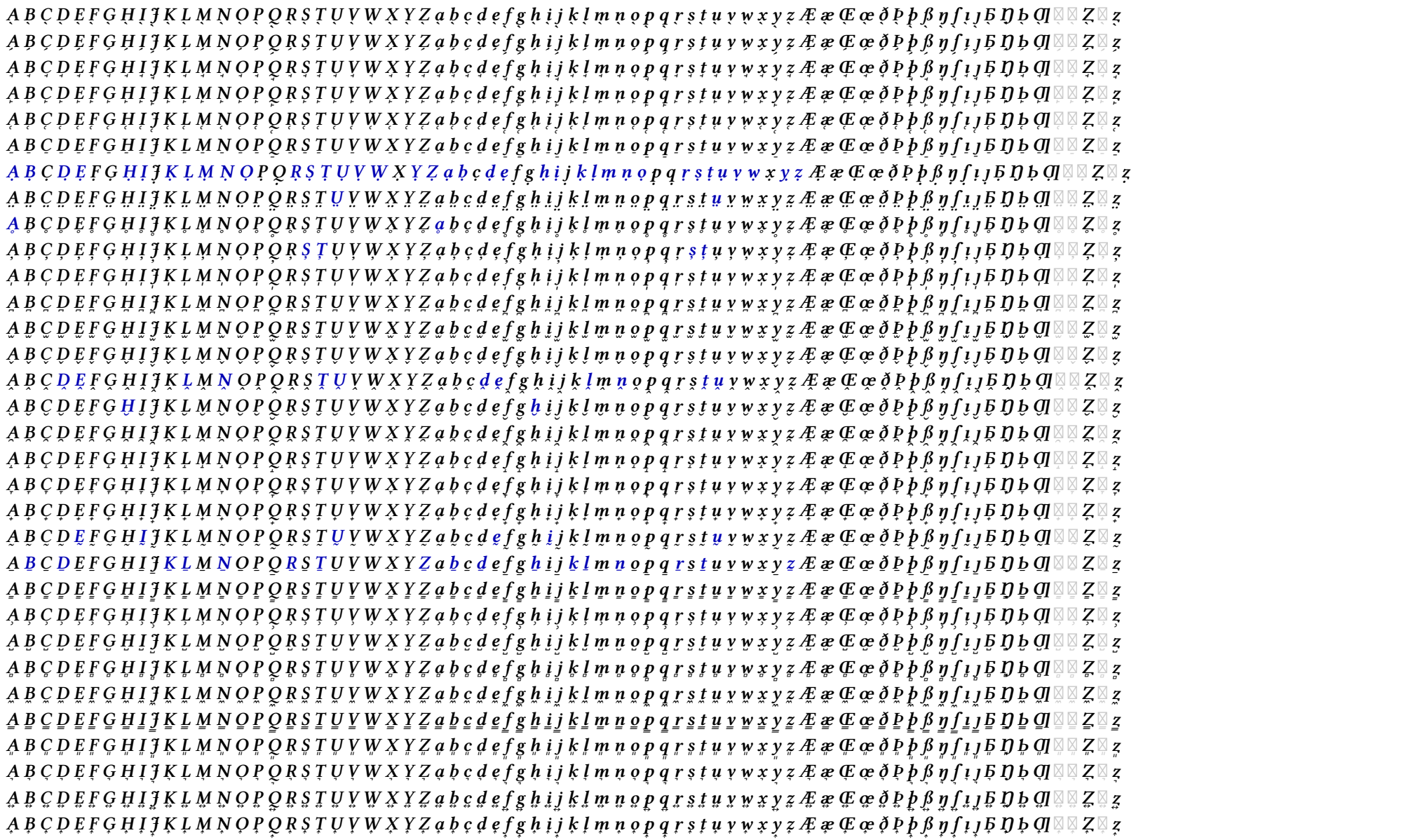
Semibold patched, Latin, Below



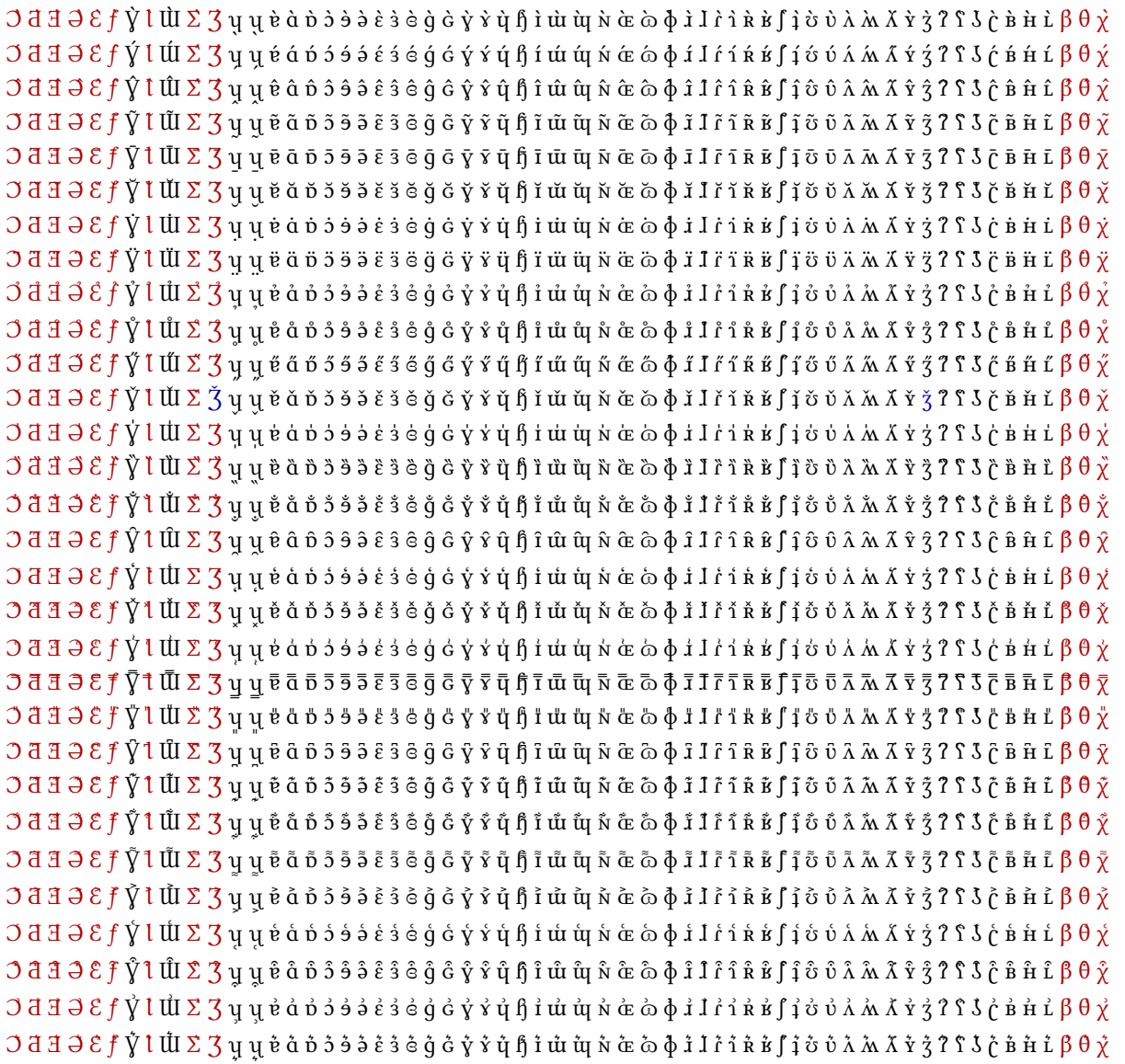
Semibold italic, Latin, Below



Semibold italic patched, Latin, Below



Regular, IPA, Above



Regular patched, IPA, Above

Ɔ ɑ ɐ ə ɛ ɦ ɣ ɩ ɯ ɤ ɜ ɥ ɲ ɵ ɸ ɹ ɻ ɽ ɾ ʁ ʃ ʇ ʊ ʋ ʌ ʍ ʎ ʏ ʒ ʔ ʕ ʖ ʗ ʙ ʜ ʟ �β ʚ ʛ
Ɔ́ ɑ́ ɐ́ ə́ ɛ́ ɦ́ ɣ́ ɩ́ ɯ́ ɤ́ ɜ́ ɥ́ ɲ́ ɵ́ ɸ́ ɹ́ ɻ́ ɽ́ ɾ́ ʁ́ ʃ́ ʇ́ ʊ́ ʋ́ ʌ́ ʍ́ ʎ́ ʏ́ ʒ́ ʔ́ ʕ́ ʖ́ ʗ́ ʙ́ ʜ́ ʟ́ �β́ ʚ́ ʛ́
Ɔ̂ ɑ̂ ɐ̂ ə̂ ɛ̂ ɦ̂ ɣ̂ ɩ̂ ɯ̂ ɤ̂ ɜ̂ ɥ̂ ɲ̂ ɵ̂ ɸ̂ ɹ̂ ɻ̂ ɽ̂ ɾ̂ ʁ̂ ʃ̂ ʇ̂ ʊ̂ ʋ̂ ʌ̂ ʍ̂ ʎ̂ ʏ̂ ʒ̂ ʔ̂ ʕ̂ ʖ̂ ʗ̂ ʙ̂ ʜ̂ ʟ̂ �β̂ ʚ̂ ʛ̂
Ɔ̃ ɑ̃ ɐ̃ ə̃ ɛ̃ ɦ̃ ɣ̃ ɩ̃ ɯ̃ ɤ̃ ɜ̃ ɥ̃ ɲ̃ ɵ̃ ɸ̃ ɹ̃ ɻ̃ ɽ̃ ɾ̃ ʁ̃ ʃ̃ ʇ̃ ʊ̃ ʋ̃ ʌ̃ ʍ̃ ʎ̃ ʏ̃ ʒ̃ ʔ̃ ʕ̃ ʖ̃ ʗ̃ ʙ̃ ʜ̃ ʟ̃ �β̃ ʚ̃ ʛ̃
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Ɔ̇ ɑ̇ ɐ̇ ə̇ ɛ̇ ɦ̇ ɣ̇ ɩ̇ ɯ̇ ɤ̇ ɜ̇ ɥ̇ ɲ̇ ɵ̇ ɸ̇ ɹ̇ ɻ̇ ɽ̇ ɾ̇ ʁ̇ ʃ̇ ʇ̇ ʊ̇ ʋ̇ ʌ̇ ʍ̇ ʎ̇ ʏ̇ ʒ̇ ʔ̇ ʕ̇ ʖ̇ ʗ̇ ʙ̇ ʜ̇ ʟ̇ ʚ̇ ʛ̇
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Ɔ̡ ɑ̡ ɐ̡ ə̡ ɛ̡ ɦ̡ ɣ̡ ɩ̡ ɯ̡ ɤ̡ ɜ̡ ɥ̡ ɲ̡ ɵ̡ ɸ̡ ɹ̡ ɻ̡ ɽ̡ ɾ̡ ʁ̡ ʃ̡ ʇ̡ ʊ̡ ʋ̡ ʌ̡ ʍ̡ ʎ̡ ʏ̡ ʒ̡ ʔ̡ ʕ̡ ʖ̡ ʗ̡ ʙ̡ ʜ̡ ʟ̡ ʚ̡ ʛ̡
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Ɔ̧ ɑ̧ ɐ̧ ə̧ ɛ̧ ɦ̧ ɣ̧ ɩ̧ ɯ̧ ɤ̧ ɜ̧ ɥ̧ ɲ̧ ɵ̧ ɸ̧ ɹ̧ ɻ̧ ɽ̧ ɾ̧ ʁ̧ ʃ̧ ʇ̧ ʊ̧ ʋ̧ ʌ̧ ʍ̧ ʎ̧ ʏ̧ ʒ̧ ʔ̧ ʕ̧ ʖ̧ ʗ̧ ʙ̧ ʜ̧ ʟ̧ ʚ̧ ʛ̧
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Ɔ̵ ɑ̵ ɐ̵ ə̵ ɛ̵ ɦ̵ ɣ̵ ɩ̵ ɯ̵ ɤ̵ ɜ̵ ɥ̵ ɲ̵ ɵ̵ ɸ̵ ɹ̵ ɻ̵ ɽ̵ ɾ̵ ʁ̵ ʃ̵ ʇ̵ ʊ̵ ʋ̵ ʌ̵ ʍ̵ ʎ̵ ʏ̵ ʒ̵ ʔ̵ ʕ̵ ʖ̵ ʗ̵ ʙ̵ ʜ̵ ʟ̵ ʚ̵ ʛ̵
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Italic, IPA, Above



Italic patched, IPA, Above



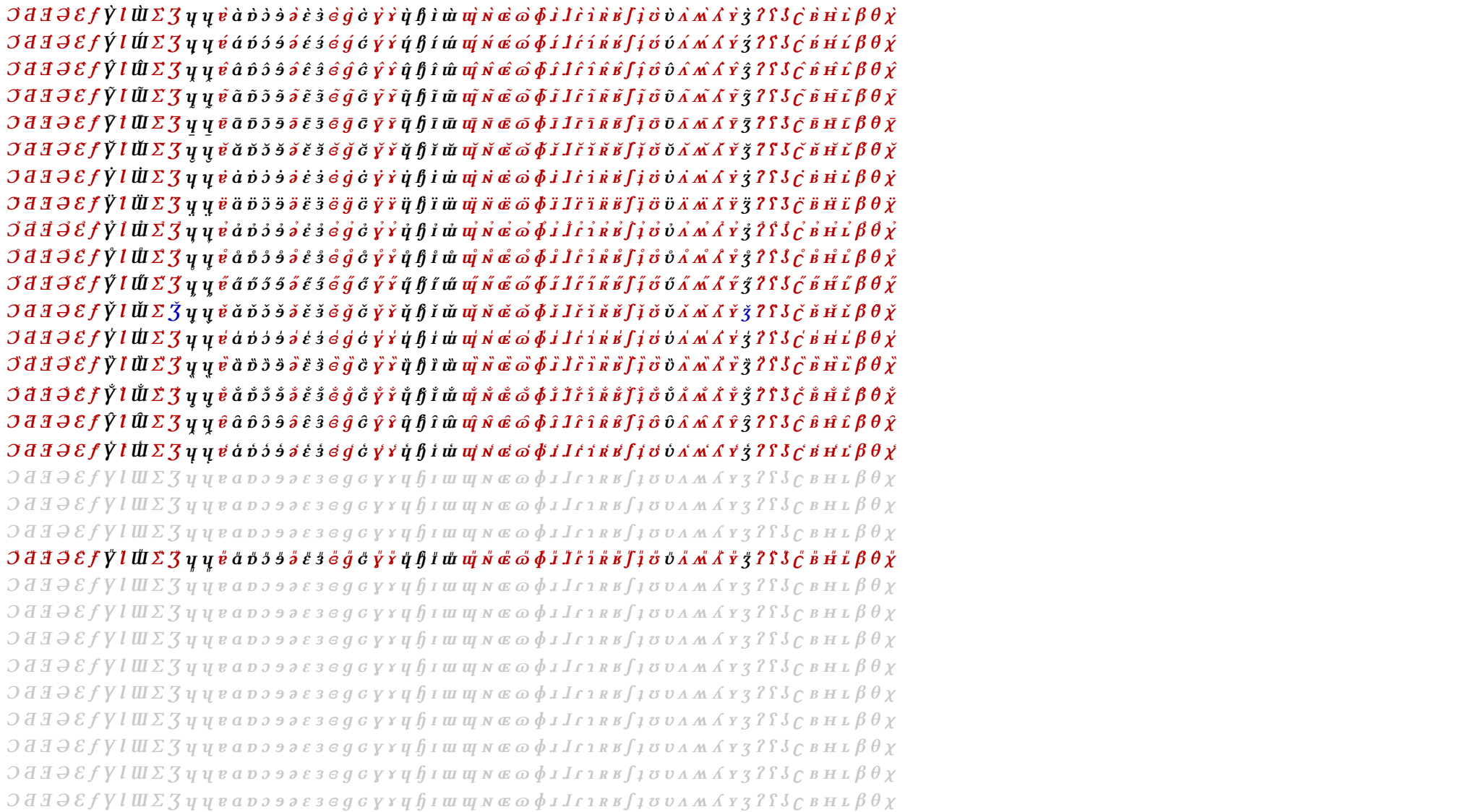
Semibold, IPA, Above



Semibold patched, IPA, Above

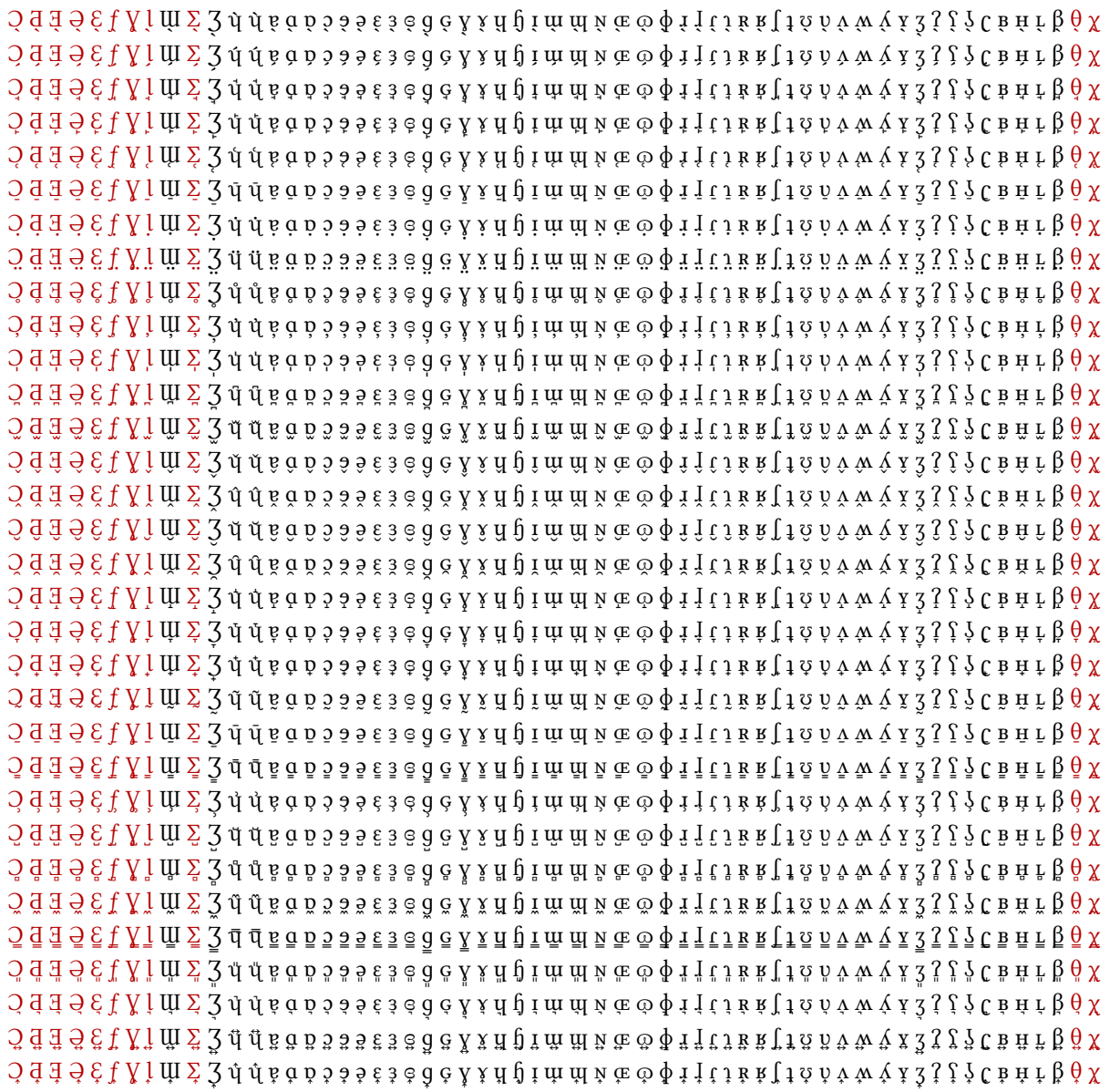


Semibold italic, IPA, Above

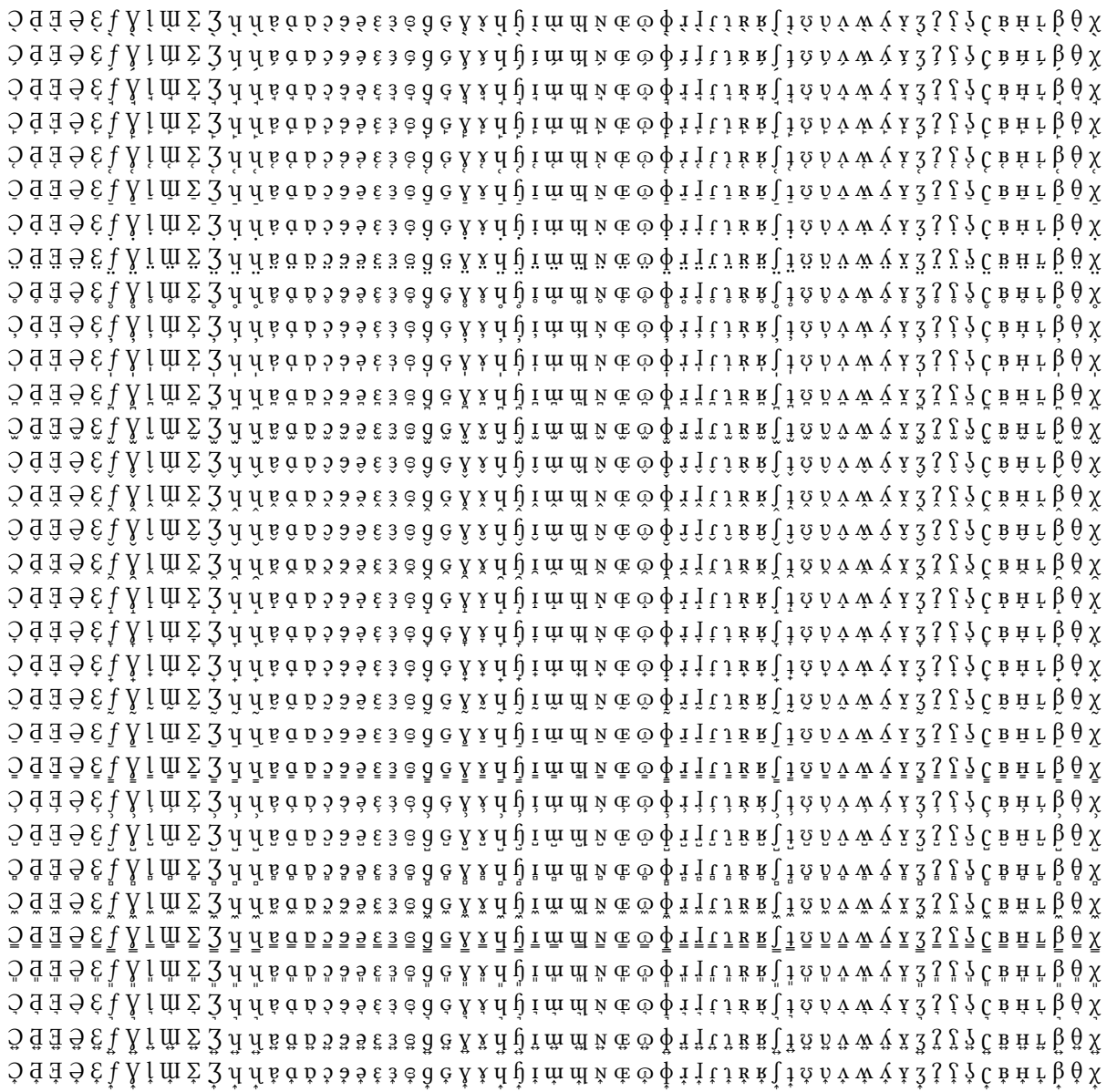


Semibold italic patched, IPA, Above

Regular, IPA, Below



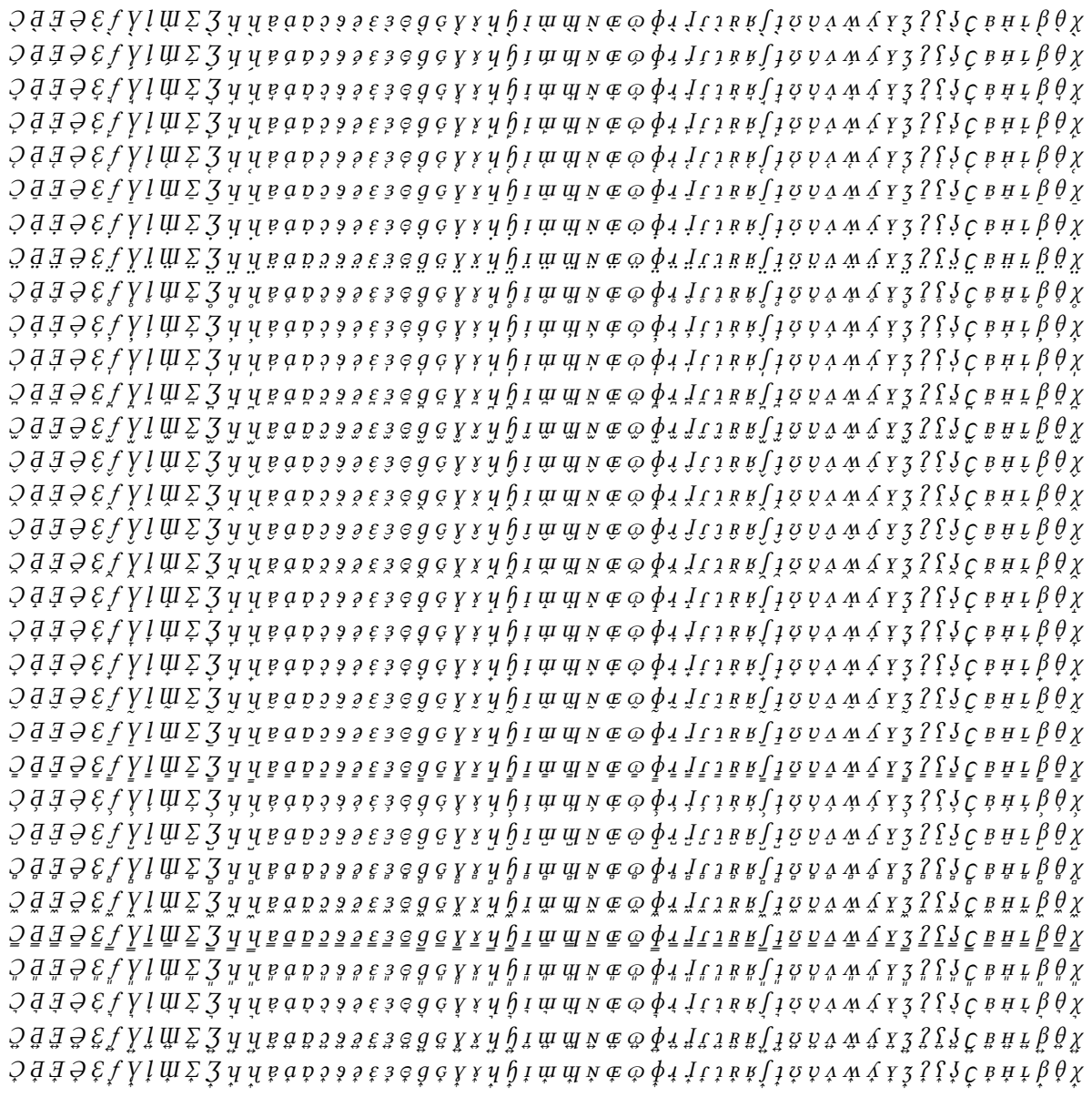
Regular patched, IPA, Below



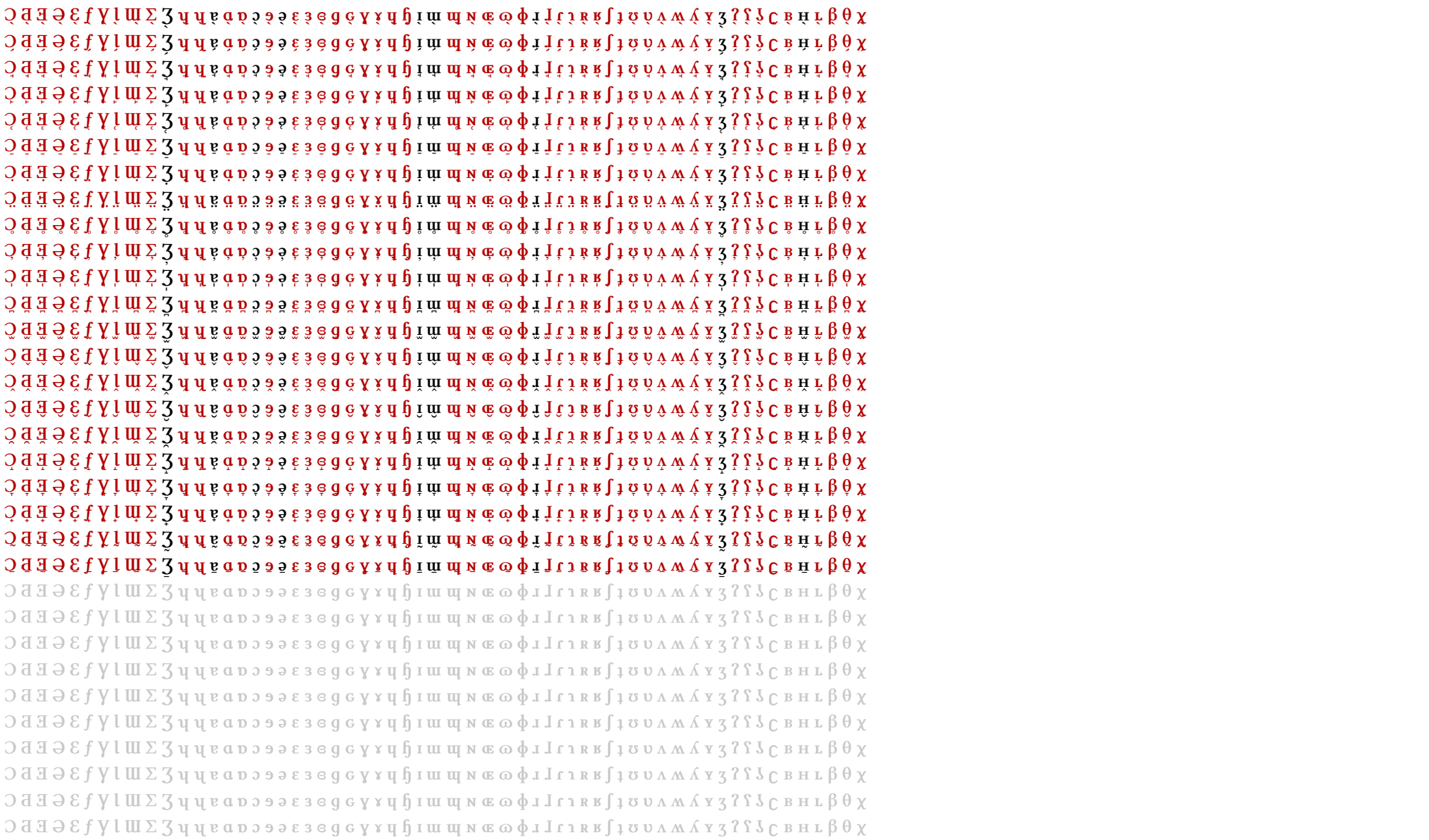
Italic, IPA, Below

ɔ̰ ɑ̰ ɘ̰ ɘ̰ ʃ ɣ̰ ɭ̰ ɻ̰ ɤ̰ ɯ̰ ɤ̰ ɯ̰ ɤ̰ ɯ

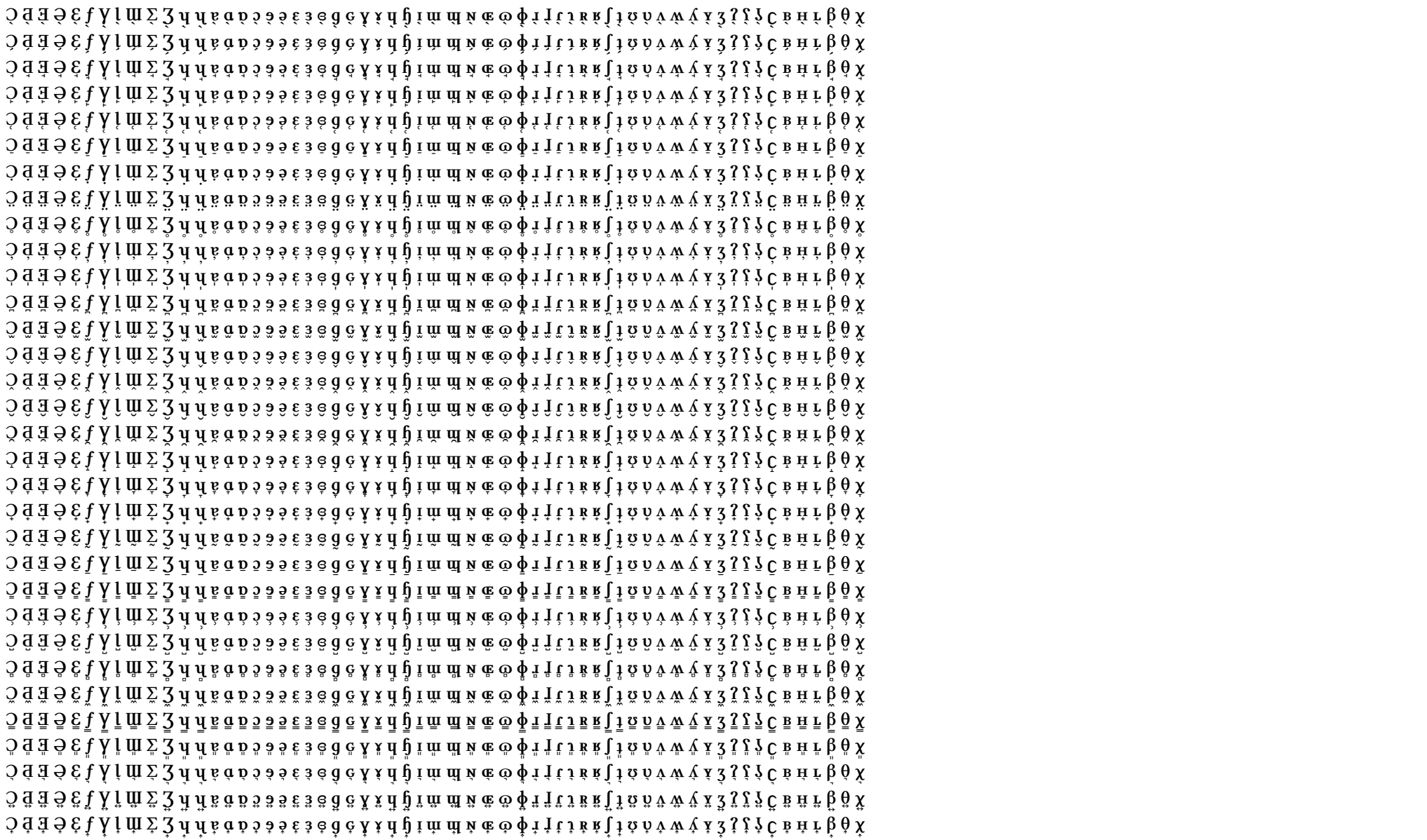
Italic patched, IPA, Below



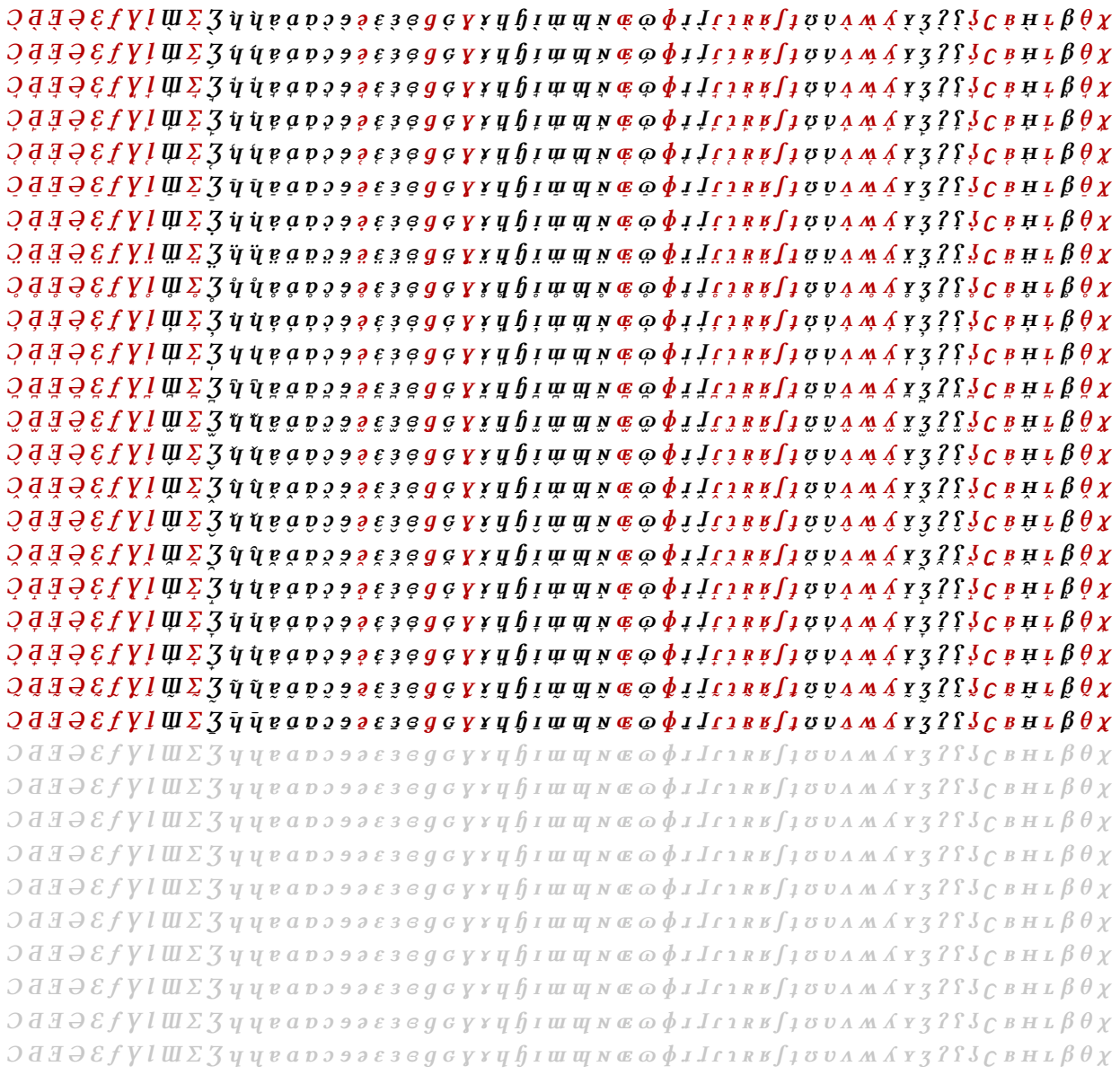
Semibold, IPA, Below



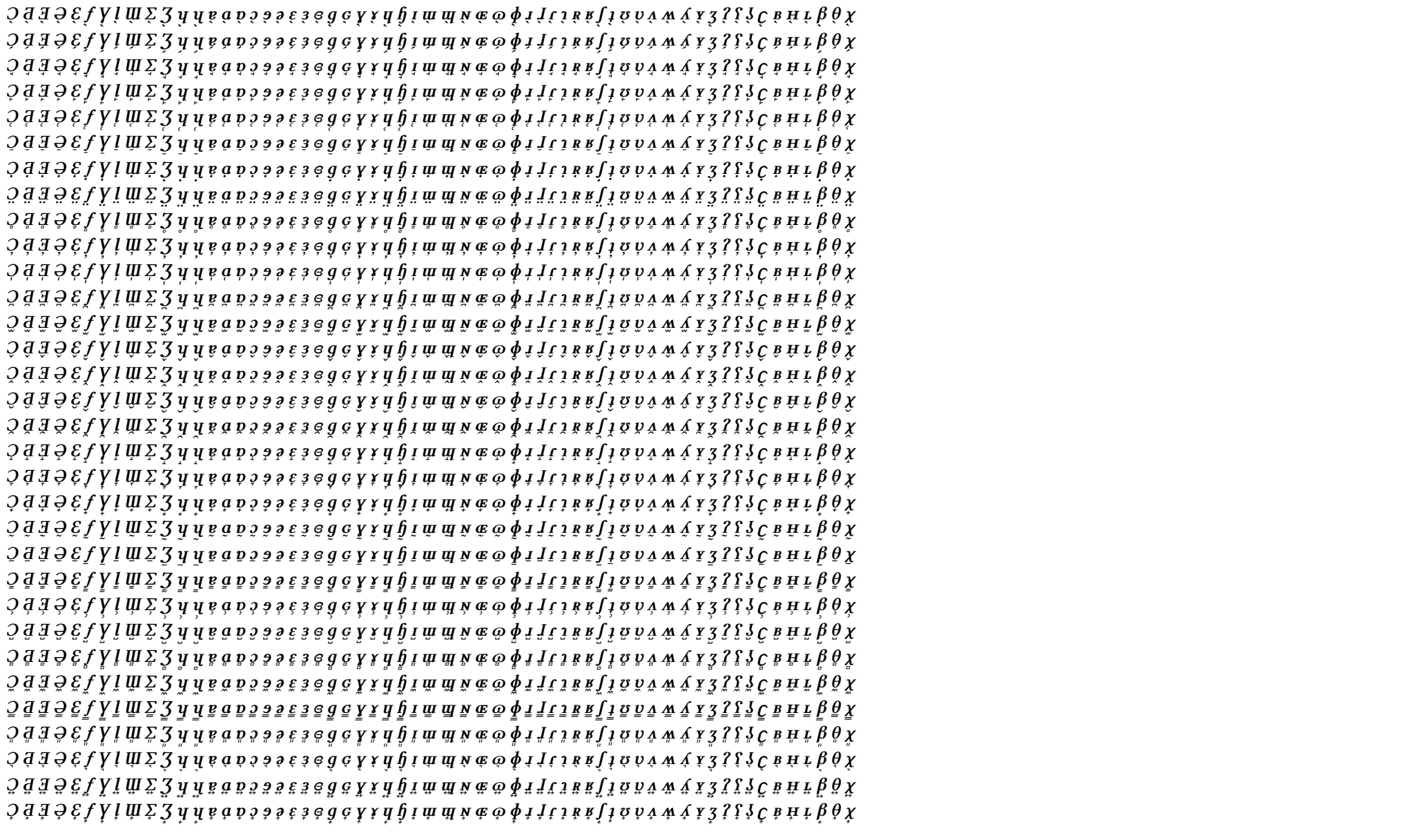
Semibold patched, IPA, Below



Semibold italic, IPA, Below



Semibold italic patched, IPA, Below



Capital alternates

A capital alternate (`grave.cap`, `acute.cap`, ...) is a form of an above-mark that sits lower over a capital, as the accents of the designer's precomposed capitals do. In the original fonts, `ccmp` substitutes only four of them, for the grave, acute, circumflex and breve, although the designer also drew the caron, double acute, double grave, inverted breve, dieresis and hook above (and, in regular and semibold, the dot above). The patched fonts substitute twelve, after every capital listed in `@capitals` of `data/fea/{style}/ccmp.fea` (stage 10): grave, acute, circumflex, tilde, breve, dot above, dieresis, hook above, double acute, caron, double grave and inverted breve. Stage 3 adds the two forms that were missing: `dotaccent.cap` in italic and semibold italic, the dot of the precomposed İ, and `tildecomb.cap` in every style (2026-11-08), the tilde of the precomposed tilded capitals (`tildecomb` itself, except in semibold italic, where the precomposed capitals use a taller tilde, that of Ã). The other above-marks have no capital form and attach with their own anchors. The ring above has none by decision: the precomposed Å sets the ring on the apex, and the combining ring keeps its clearance; nor do the vertical lines above U+030D and U+033E.

In stage 4 of the patched fonts (`font_patching_capmarks.py`), the mark anchor of each capital alternate is set from the designer's precomposed capitals A–Z. The mark's outline is found in each precomposed glyph that is a capital plus that mark, which gives how far the designer moved it there. The Y coordinate is the capital anchor reference (850) less the median height by which the mark is raised in the precomposed glyphs. The X coordinate of a symmetric mark is the centre of its outline on its own axis, $X = C + Y\tan\theta - K$. For the asymmetric grave, acute, double acute and double grave, it is the median position of the mark in the precomposed glyphs relative to the candidate above-anchor of the capital, $W/2 + 850\tan\theta - K$. K is 150 in semibold italic and 0 elsewhere. The precomposed placement is reproduced exactly where a capital's above-anchor is on its candidate; where it is not, the composed mark is displaced by the same amount. Human-curated anchors (stage 7) overrule stage 4.

The table compares each composed capital (base, capital alternate, anchors) with the designer's precomposed glyph, over the capitals that have one. Vertically every capital alternate matches: the median difference is 0 in every style, and no single capital differs by more than 12, except Ỏ, whose precomposed hook above is 24 units higher than the composed one in all styles but semibold italic. The numbers are the median horizontal distance; n is the number of precomposed capitals, and * marks a curated anchor.

mark	n	regular	italic	semibold	semibold italic
grave	8	17	30^*	4	10
acute	17	9	14^*	16	28
circumflex	12–13	4	9^*	14	16
tilde	7–8	5	10	10	25
breve	6	2	6^*	10	10
dot above	20	9	13	10	15
dieresis	7	2	6	8	8
hook above	6	4	12	12	16
double acute	2	0	2^*	18	6
caron	12–15	7	6^*	16	11
double grave	6	12	4^*	15	15
inverted breve	6	2	10^*	12	9

Notes by style:

- Regular. All scripted (2026-11-08). The curated grave $(-175, 723)$ was removed in favour of stage 4's $(-162, 724)$, as in semibold italic: the median distance from the precomposed graves goes from 18 to 17 units, and the median offset from 18 units right to 6, since the graves vary letter by letter. The other curated anchors are reproduced by the script and are commented out in `fontanchors_human`.
- Italic. Eight curated anchors (2026-11-07) overrule stage 4: grave -6 (stage 4: 17), acute -50 (-37), circumflex 15 (24), breve -14 (-1), caron -32 (-29), double acute -109 (-103), double grave -40 (-34), inverted breve -12 (-2). Each is fitted so that the accent lands, as a median over the capitals, where the designer drew it in the precomposed glyphs, with the capitals' actual above-anchors (so the signed median is 0). The dieresis, dot above, hook above and tilde are scripted.
- Semibold. All scripted. The designer's capital above-anchors were near 805, with the capital alternates drawn to compensate; stage 5 moves the capitals to 850, the reference stage 4 computes from.

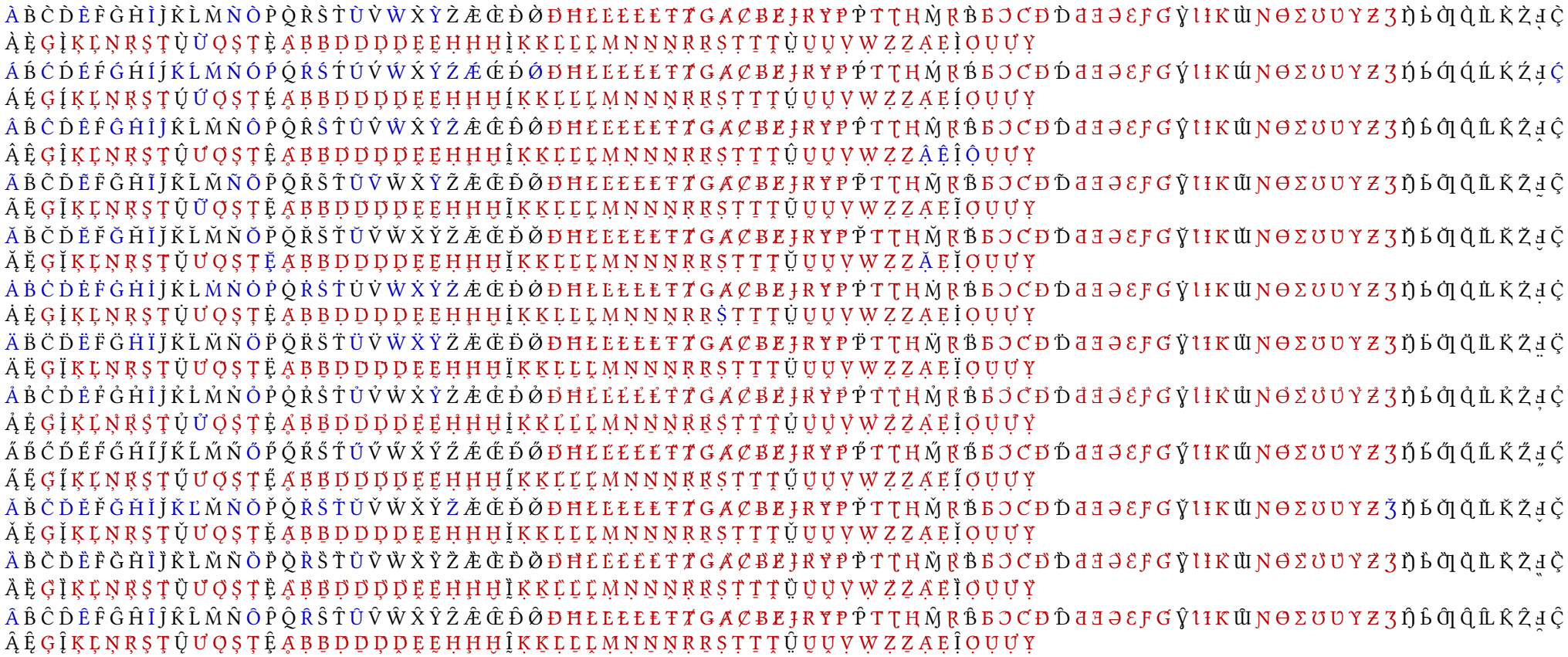
- Semibold italic. All scripted (2026-11-08). The acute, grave, double acute and double grave had been curated on 2026-11-06 at the mark-axis position, $X = C + Y \tan\theta + d - K$ with d from regular (-236, -194, -282, -214, against stage 4's -195, -168, -243, -167). Those reproduced the precomposed capitals only on the designer's capital anchors, which are left of the candidates (median k 173 against $K = 150$), because the designer drew the precomposed accents on those anchors. Stage 4 is preferred because the precomposed glyphs are used wherever they exist, so a composed capital without one should match them; and since H I J K L N O U V Y Æ were moved on 2026-11-07, it is also closer to them. The larger distances of the acute (28) and tilde (25) come from capitals whose above-anchors are off their candidates (A, B, D, Q, R, W, X; V, Y and E for the tilde), not from the mark anchors.

Notes by mark:

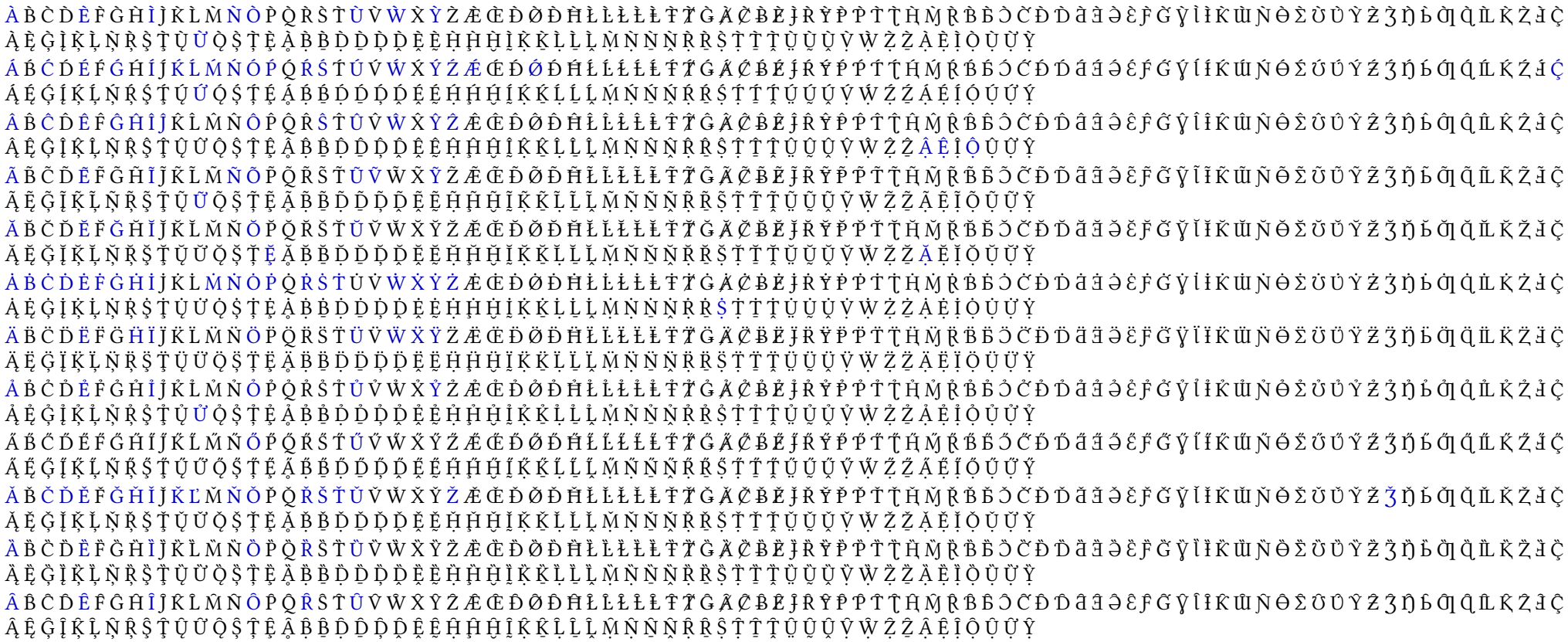
- The grave, acute and their doubles are placed by eye, letter by letter, in the precomposed glyphs, so the implied X varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- Tilde. Unicode has no precomposed G̃, which Guarani writes, so G + U+0303 is composed. With the combining tilde it sat 25 to 47 units higher than the tilde of the precomposed Ñ, Ã, Õ beside it; `tildecomb.cap` matches their height exactly. Gentium Plus does the same with its `tildecomb.LP`.
- Dot above. `dotaccent.cap` is new in italic and semibold italic; in regular, stage 4 raises its anchor by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), Ũ in semibold italic (tilde), and Ĵ in semibold italic.

The grids show these twelve marks on every Latin and IPA capital of `@capitals`: A–Z, Æ, Œ, Đ, Ø, Ɖ, Ħ, Ł, the IPA and African capitals, and the 59 precomposed capitals with a cedilla, ogonek, horn or below-mark that stage 9 gives an above-anchor (Ç Ą Ę Į Ų Ư Ḍ Ẹ Ọ …; added to `@capitals` 2026-11-08, so that, e.g., Lithuanian Ą́ and Yoruba Ẹ́ take the capital acute like Á and É). The Greek and Cyrillic capitals of `@capitals` are not shown: their combinations with these marks are all precomposed. Blue cells are precomposed glyphs, which the shaper substitutes for base + mark wherever the font has one, so they do not use the capital alternates. Black cells are composed: in the patched fonts, the base with the capital alternate; in the original fonts, the base with the alternate only for the grave, acute, circumflex and breve, and otherwise with the ordinary mark.

Regular, Capitals, Marks with capital alternates



Regular patched, Capitals, Marks with capital alternates



Italic, Capitals, Marks with capital alternates



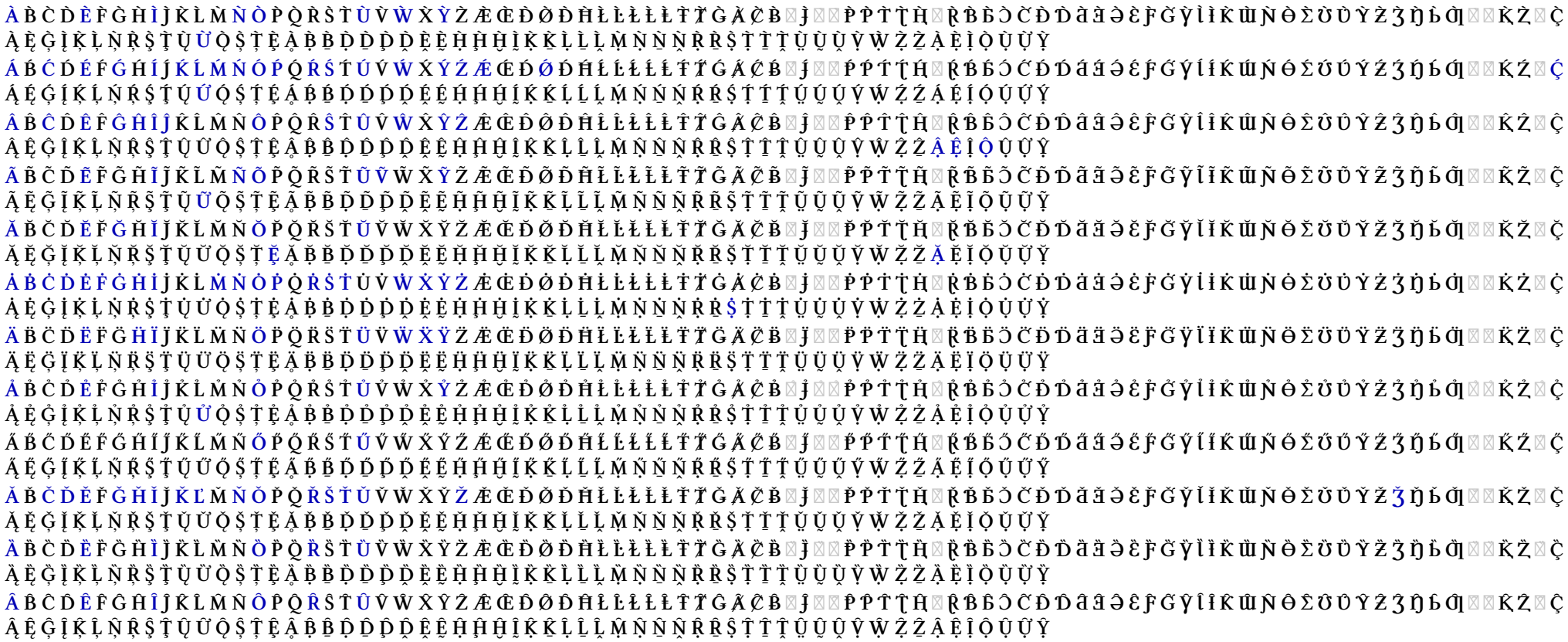
Italic patched, Capitals, Marks with capital alternates



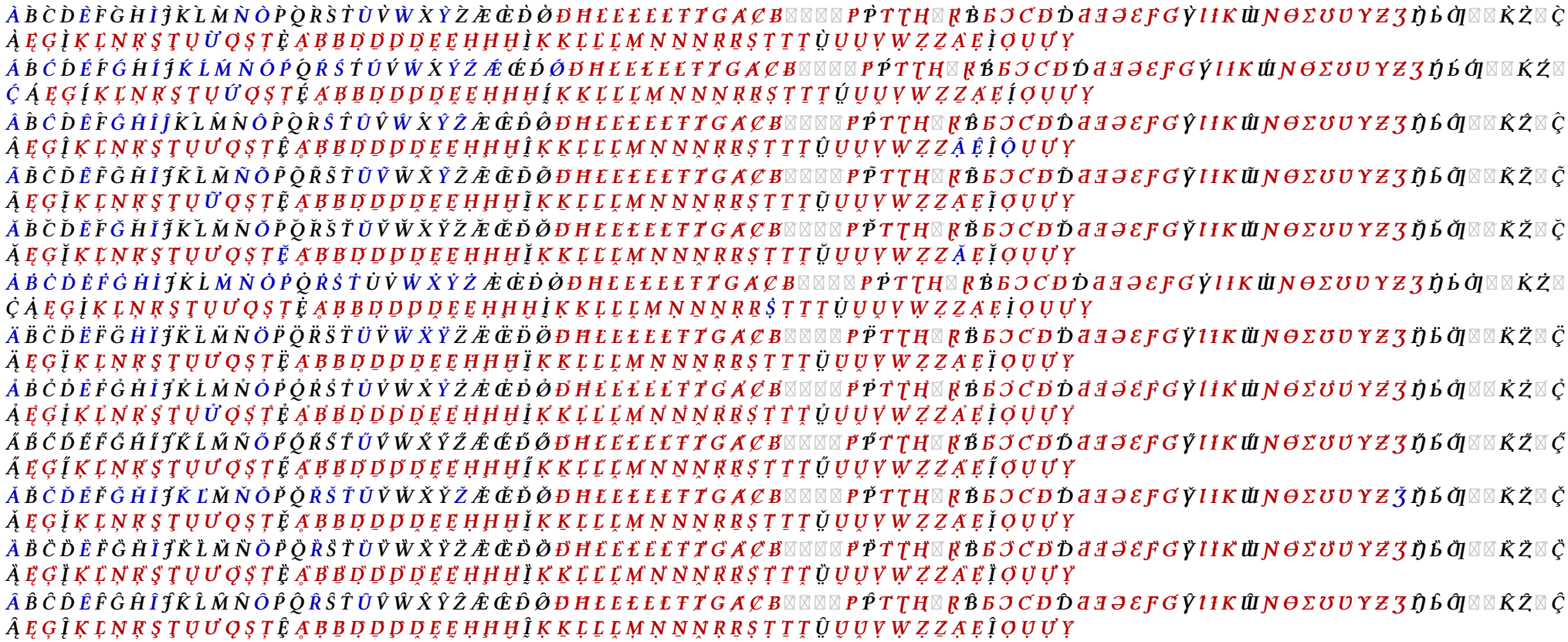
Semibold, Capitals, Marks with capital alternates



Semibold patched, Capitals, Marks with capital alternates



Semibold italic, Capitals, Marks with capital alternates



Semibold italic patched, Capitals, Marks with capital alternates

Derived letters

Many Latin and IPA letters are a base letter with a bar, stroke, belt, hook, tail or curl: Đ is D with a bar, Ɗ is D with a hook, ɖ is d with a tail. Unlike the precomposed letters of stage 9, they have no Unicode decomposition; the base letter is assigned to each one in `data/derived_letters.py`. Stage 8 (`src/libertinus_analysis/font_patching_derivedanchors.py`, letter lists in `data/derived_letters.py`) gives each such letter its base letter's final anchor, moved by where the base letter sits inside it, on each side where the added part stays clear of the marks. The heuristic candidate of stage 6 cannot do this: a hook or bar moves the bounding box it is centred on (regular Ƒ above: candidate 183, F's anchor moved into Ƒ 276), and the designer often drew the derived letter in a shifted frame (regular Ɗ is D moved 80 units right).

Stage 8 measures how well the derived letter matches its base letter (the share of the base's outline that fits, in percent) and where the base sits in it (a shift, and a scale), in one of four ways:

- **Fit** (most letters). On each inherited side, the base letter's ink is fitted to the derived letter's with a shift and a scale of up to 15%, as in stage 9; the anchor moves with the fit, so a letter drawn a few percent wider keeps the anchor at the same place in the shape (italic Đ is D drawn 4% wider). At least 80% of the base must fit; otherwise the letter keeps its stage 6 candidate (or a curated anchor) on that side.
- **Stems** (`stem_match`: a hook, tail or curl at the bottom, and ɗ). The fit uses only the stems above the tail, at least 70%. The below anchor goes to the descender, as the stage 6 candidate does, where the tail lies under the mark; otherwise it stays at the base letter's Y (Ɲ, Ⱨ).
- **Stem midline** (`stem_axis`: ɬ ɬ ɬ ɬ Ł ɪ ɟ, and T f Y 𝑇 𝑓 𝑌). The anchor moves by the distance between the two stems, whatever the fit, for letters whose bar, belt, tilde or hook spoils the fit but not the stem.
- **Fixed** Y (`derived_y`): ɓ ʝ ɠ ɾ ƥ ɋ f above, ɓ ɕ below take X from the base letter and a Y set by hand, usually because a hook rises to the ascender (ɠ ƥ ɋ f).

Where the added part reaches further than the base letter within 120 units of the mark's axis, Y moves away by the difference (ɛ above $+31$ to $+39$, italic 𝑓 $+96$). A derived letter is curated (stage 7) only where stage 8 cannot inherit a good anchor: the match is too low, or the inherited anchor would be wrong; stage 8 runs after stage 7 and leaves those anchors alone. Where stage 8 cannot inherit and nothing is curated, the letter keeps stage 6's candidate.

letters	inherit	note
Đ Ð Ɖ đ Ħ Ł ɬ Ø ø Θ ɵ Ɨ ɯ ɬ ɬ ɬ ɬ Ł Ł Ŧ ŧ 𝑇 𝑡 G g Ⱥ ⱥ Ȼ ɛ ɕ	both	bar, stroke, belt or middle dot
Ƀ ƀ Ɇ ɇ Ɉ Ʀ ɾ Ɏ ɏ Ᵽ Ƶ ƶ ſ f ƥ Ɬ ɬ ʔ ʕ		
Ɓ Ɗ Ƈ Ɠ Ƙ ɗ ƈ ɠ Ᵽ Ƴ ƴ ƙ ƈ Ƭ ɓ ɗ ƈ̓ ʊ	both	hook, topbar or ring clear of the marks
ħ ɦ	both (italics: below)	the bar or hook sits at the top of the italic ascender
ʒ ƨ ƥ ɋ f	below	the hook rises into the above mark (ƥ ɋ f above: fixed Y)
Ƒ Ɲ ɕ ɖ ɭ ɱ ɲ ɳ ɻ ɽ ʂ ʃ ʈ ʐ ƨ ʒ ʈ Ʈ ɗ ɬ ɳ ʈ Ʀ Ⱨ ɦ Ƙ ƙ Ɱ ƴ	both	tail at the bottom (stems)
ɨ ʝ	below	above: their dotless forms, from ɪ ɟ
ɮ ɻ U+E27F	above	below curated (ɮ) or the candidate
ɪ ɟ ſ	both	from ɪ, ɟ, and ſ (Libertinus draws ſ as ʃ with a bar)
ɓ ʝ ɠ ɾ	ɓ both, the others above	fixed Y, X from b, j, script g, r

Not derived, although the Unicode name says so: Ƀ ƀ (topbar), Ʊ ʊ ƒ ʒ ʂ Ƞ ƺ ƶ Ƴ ʋ ɾ, which match their base letters 5–50%, and Ƶ ƶ, whose base has no stems to match; they keep stage 6's candidate or a curated anchor. ħ Ħ take no marks in any orthography; ħ's copied anchors are deleted and both are in `BASE_NO_MARK`.

Fixes made on the way (2026-11-11): the copied x-height or baseline anchors of italic ɗ ƙ, semibold italic ƙ and ƶ (three styles) are deleted, so that stage 6 fills them; the curated Ƴ of regular and semibold, at Y's own X, was 63–78 units left of Ƴ's stem and is retired; semibold italic ƴ, redrawn narrower than y, is curated at $(152, 645)$; italic ɾ is rebuilt in stage 2 (its SFD glyph referred to IJ and two acutes as well) and now inherits r's anchors. Inheritance carries the base letter's anchor, right or wrong: Ƈ and Ȼ follow C's below anchor, which the designer put right of centre. The stem-midline check (Table 25) confirms that every anchor of an l-like or i-like letter lies within 5 units of its own stem, except ɮ below, under the ezh by design.

In the tables, each column is a derived letter, ordered by base letter, with its base letter's codepoint below it; per style: the letter with circumflex and macron below, original and patched (`reg`, `patch`), and its base letter patched (`base`); then for the above anchor the shift and fit used (a~; `s` stems, `|` stem midline, italics if below the threshold), the designer's anchor (a), the candidate (a?), the inherited anchor (a<, bold where the patched font has it) and a curated anchor (a!); the same below (b).

What the tables decide, per letter and side: a bold a< is the anchor the patched font has, inherited in stage 8, replacing the designer's anchor and the candidate (216 anchors in regular, 186 in semibold, 167 in italic, 165 in semibold italic); an a! is curated because inheritance fails or would be wrong there (5 anchors in regular and semibold, 12 in each italic style, mostly tail letters whose stems do not match: italic ɖ ɭ ɽ ʐ ƶ, ʒ), and the a< beside it, if any, is not used; an a~ in italics means the match is below the threshold (3 in regular, 11 in semibold, 23 in italic, 21 in semibold italic: italic redrawings such as ɓ ƃ ɗ ƙ ɛ̀, and tails such as Ⱪ ⱪ Ɽ), and the letter keeps what it had before stage 8, the candidate (a?) or a designer anchor (a) that stage 1 kept; an empty a< means the side is not inherited, because the added part is in the mark's way (ʒ̀ ɚ above, ħ ɦ above in the italics). The grids put the IPA marks on each base letter followed by its derived letters, patched fonts only, without `\cgj` (a cell shows the precomposed glyph in blue where one exists).

Turned letters

Turned letters are drawn from upright letters too, rotated by 180°: Ⅲ ɐ ɒ ɥ ɯ ɰ ɹ ɾ ɻ ʁ ɬ ʌ ʍ ʎ ʖ ɥ ɥ ɟ in the base groups. Stage 8 does not derive their anchors: not even swapping the upright letter's above and below anchors makes sense, since the turned letter has its own reference lines (h's ascender is ɥ's descender), and the anchor's Y and clearance come from those, not from the upright letter's. The designer's anchors on turned letters are bad: copied carelessly from other glyphs (a's (209, 645) and (196, −110) on ɰ ɹ ɾ ʁ ɬ ʌ ʍ ʎ ʖ in regular, and regular's numbers unslanted in the italics), or rotated along with the outline (ɥ ɥ ɟ, whose above anchor sat under the letter). So all of them are deleted in stage 1, except the few that happen to match the candidate within 4 units (regular and semibold italic ɥ above, semibold italic ɐ ɒ below), which stage 5 snaps to the reference line. The turned letters therefore get stage 6's candidate, or a curated anchor (stage 7) where the candidate is unsuitable.

- Where the letter is symmetric about its axis, both anchors are the candidate in every style: Ⅲ ɐ ɒ ɥ ɯ ʁ ɬ ʖ ɥ ɥ ɟ. ɯ's candidate lies on the central stem, within 2 to 7 units.
- Where a stem or a gap must be the centre and an arm pulls the candidate aside, the anchor is curated, in effect the upright letter's opposite anchor turned over: above, ɹ ɾ ɻ over the stem top, as r's below anchor sits on r's stem, and ʎ over the apex of the turned V, as y's below anchor; below, ʌ between the feet, ʍ under the middle foot (the candidate in semibold), and ɰ under the middle stem, at ɯ's candidate Y. Their other anchor is the candidate.
- ɰ is not derived from ɯ: it is narrower and its stems are spaced differently (outline match 21% in regular). ɻ is a derived letter of ɹ (a tail at the bottom) and inherits ɹ's above anchor in semibold and semibold italic; its above anchor is curated in regular and italic, and its below anchor follows stage 8's rule for a tail at the bottom (the candidate where the stems do not match, as in italic).

Table 18: Derived letters and inherited anchors 1/7

hex	023A	0181	0243	0187	023B	00D0	0110	0189	018A	0246	0191	0193	01E4	0126	2C67	0197
base	0041	0042	0042	0043	0043	0044	0044	0044	0044	0045	0046	0047	0047	0048	0048	0049
reg	Ⱥ	Ɓ	Ƀ	Ƈ	Ȼ	Ɗ	Đ	Ɖ	Ɗ	Ɇ	Ƒ	Ɠ	Ǥ	Ħ	Ⱨ	Ɨ
patch	Ⱥ	Ɓ	Ƀ	Ƈ	Ȼ	Ɗ	Đ	Ɖ	Ɗ	Ɇ	Ƒ	Ɠ	Ǥ	Ħ	Ⱨ	Ɨ
base	A	B	B	C	C	D	D	D	D	E	F	G	G	H	H	I
a	+0, 91%	+58, 98%	+0, 96%	+0, 100%	+0, 97%	+0, 96%	+0, 96%	+0, 96%	+79, 99%	+0, 81%	-8, 99% s	+0, 100%	+0, 95%	+0, 94%	-16, 100% s	+0, 94%
a		327, 805				341, 850			306, 805							
a?	346, 850	303, 850	255, 850	369, 850	324, 850	333, 850	336, 850	333, 850	368, 850	250, 850	183, 850	372, 850	366, 850	362, 850	345, 850	148, 850
a<	**361, 850**	**325, 850**	**267, 850**	**340, 850**	**340, 850**	**306, 850**	**306, 850**	**306, 850**	**385, 850**	**281, 850**	**276, 850**	**365, 850**	**365, 850**	**361, 850**	**345, 850**	**150, 850**
a!																
b	+0, 94%	+58, 96%	+0, 91%	+0, 100%	+0, 86%	+0, 100%	+0, 100%	+0, 100%	+79, 100%	*-2, 72%*	-8, 99% s	+0, 100%	+0, 92%	+0, 100%	-16, 100% s	+0, 100%
b		328, -108							369, -107							
b?	346, -218	303, -110	255, -110	369, -110	324, -218	333, -110	336, -110	333, -110	368, -110	250, -121	183, -257	372, -110	366, -110	362, -110	345, -226	148, -110
b<	**339, -110**	**326, -110**	**268, -110**	**363, -110**	**363, -110**	**309, -110**	**309, -110**	**309, -110**	**388, -110**	285, -110	**202, -257**	**367, -110**	**367, -110**	**362, -110**	**346, -110**	**147, -110**
b!																
it	*Ⱥ*	*Ɓ*	*Ƀ*	*Ƈ*	*Ȼ*	*Ɗ*	*Đ*	*Ɖ*	*Ɗ*	⌧	*Ƒ*	*Ɠ*	*Ǥ*	*Ħ*	*Ⱨ*	*Ɨ*
patch	*Ⱥ*	*Ɓ*	*Ƀ*	*Ƈ*	*Ȼ*	*Ɗ*	*Đ*	*Ɖ*	*Ɗ*	⌧	*Ƒ*	*Ɠ*	*Ǥ*	*Ħ*	*Ⱨ*	*Ɨ*
base	*A*	*B*	*B*	*C*	*C*	*D*	*D*	*D*	*D*	*E*	*F*	*G*	*G*	*H*	*H*	*I*
a	+5, 93%	+58, 99%	+10, 84%	+32, 96%	+6, 96%	+0, 96%	+0, 96%	+27, 89%	+78, 98%		+16, 100% s	+7, 99%	+7, 94%	+0, 94%	+10, 100% s	+0, 92%
a		327, 805							306, 805							
a?	528, 850	488, 850	477, 850	503, 850	503, 850	514, 850	514, 850	534, 850	553, 850		419, 850	541, 850	524, 850	517, 850	528, 850	321, 850
a<	**510, 850**	**505, 850**	**457, 850**	**532, 850**	**506, 850**	**471, 850**	**471, 850**	**498, 850**	**549, 850**		**422, 850**	**520, 850**	**520, 850**	**504, 850**	**514, 850**	**316, 850**
a!																
b	+10, 92%	+58, 100%	*+13, 74%*	+19, 94%	+13, 81%	+0, 100%	+0, 100%	+18, 98%	+78, 96%		+16, 100% s	+13, 98%	+13, 90%	+0, 100%	+10, 100% s	+0, 100%
b		328, -108							369, -107							
b?	313, -164	284, -110	273, -110	299, -110	287, -165	310, -110	310, -110	330, -110	349, -110		184, -257	337, -110	320, -110	313, -110	300, -223	117, -110
b<	**318, -110**	**296, -110**	251, -110	**352, -110**	**346, -110**	**279, -110**	**279, -110**	**297, -110**	**357, -110**		**113, -257**	**358, -110**	**358, -110**	**320, -110**	**331, -110**	**127, -110**
b!																
sb	Ⱥ	Ɓ	Ƀ	Ƈ	Ȼ	Ɗ	Đ	Ɖ	Ɗ		Ƒ	Ɠ	Ǥ	Ħ	Ⱨ	Ɨ
patch	Ⱥ	Ɓ	Ƀ	Ƈ	Ȼ	Ɗ	Đ	Ɖ	Ɗ	⌧	Ƒ	Ɠ	Ǥ	Ħ	Ⱨ	Ɨ
base	A	B	B	C	C	D	D	D	D	E	F	G	G	H	H	I
a	+0, 92%	+55, 93%	+0, 95%	+0, 99%	+0, 94%	+0, 95%	+0, 95%	+0, 95%	+82, 97%		*+18, 62% s*	+0, 100%	+0, 95%	+0, 92%	-17, 98% s	+0, 92%
a																
a?	350, 850	316, 850	294, 850	368, 850	332, 850	349, 850	349, 850	349, 850	381, 850		236, 850	374, 850	368, 850	362, 850	345, 850	158, 850
a<	**350, 850**	**338, 850**	**283, 850**	**332, 850**	**332, 850**	**352, 850**	**352, 850**	**352, 850**	**434, 850**			**385, 850**	**385, 850**	**362, 850**	**345, 850**	**160, 850**
a!																
b	+0, 94%	+58, 87%	+0, 89%	+0, 100%	+0, 87%	+0, 100%	+0, 100%	+0, 100%	+82, 98%		*+18, 62% s*	+0, 100%	+0, 91%	+0, 100%	-17, 98% s	+0, 100%
b																
b?	350, -218	316, -110	294, -110	368, -110	332, -218	349, -110	349, -110	349, -110	381, -110		236, -281	374, -110	368, -110	362, -110	345, -222	158, -110
b<	**350, -110**	**358, -110**	**300, -110**	**332, -110**	**332, -110**	**352, -110**	**352, -110**	**352, -110**	**434, -110**		**236, -281**	**368, -110**	**368, -110**	**362, -110**	**345, -110**	**160, -110**
b!																
si	***Ⱥ***	***Ɓ***	***Ƀ***	***Ƈ***	***Ȼ***	***Ɗ***	***Đ***	***Ɖ***	***Ɗ***	⌧	***Ƒ***	***Ɠ***	***Ǥ***	***Ħ***	***Ⱨ***	***Ɨ***
patch	***Ⱥ***	***Ɓ***	***Ƀ***	***Ƈ***	***Ȼ***	***Ɗ***	***Đ***	***Ɖ***	***Ɗ***	⌧	***Ƒ***	***Ɠ***	***Ǥ***	***Ħ***	***Ⱨ***	***Ɨ***
base	***A***	***B***	***B***	***C***	***C***	***D***	***D***	***D***	***D***	***E***	***F***	***G***	***G***	***H***	***H***	***I***
a	+12, 93%	+67, 93%	+12, 95%	+25, 90%	+12, 94%	+0, 95%	+0, 96%	+38, 95%	+94, 95%		*+15, 61% s*	+29, 85%	+12, 95%	+12, 92%	-44, 98% s	-17, 90%
a		327, 805				341, 850			306, 805							
a?	382, 850	359, 850	319, 850	359, 850	357, 850	367, 850	367, 850	388, 850	424, 850		288, 850	395, 850	388, 850	396, 850	370, 850	181, 850
a<	**366, 850**	**334, 850**	**279, 850**	**365, 850**	**352, 850**	**306, 850**	**306, 850**	**344, 850**	**400, 850**			**394, 850**	**377, 850**	**408, 850**	**353, 850**	**177, 850**
a!																
b	+12, 96%	+71, 87%	+12, 89%	+26, 92%	+12, 87%	+0, 100%	+0, 100%	+38, 96%	+94, 96%		*+15, 61% s*	+31, 98%	+12, 91%	+12, 100%	-44, 98% s	-17, 96%
b		328, -108							369, -107							
b?	327, -218	326, -110	286, -110	326, -110	302, -218	334, -110	334, -110	355, -110	391, -110		220, -281	362, -110	355, -110	363, -110	314, -222	148, -110
b<	**320, -110**	**339, -110**	**280, -110**	**389, -110**	**375, -110**	**279, -110**	**279, -110**	**317, -110**	**373, -110**		**220, -281**	**398, -110**	**379, -110**	**377, -110**	**322, -110**	**152, -110**
b!																

Table 19: Derived letters and inherited anchors 2/7

hex	0248	0198	2C69	013F	0141	023D	2C60	2C62	2C6E	019D	00D8	019F	01A4	2C63	024C	2C64
base	004A	004B	004B	004C	004C	004C	004C	004C	004D	004E	004F	004F	0050	0050	0052	0052
reg	Ɉ̲Ɉ̲	Ƙ̲Ƙ̲	Ⱪ̲Ⱪ̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ɱ̲Ɱ̲	Ɲ̲Ɲ̲	Ø̲Ø̲	Ɵ̲Ɵ̲	Ƥ̲Ƥ̲	Ƥ̲Ƥ̲	Ʀ̲Ʀ̲	Ʀ̲Ʀ̲
patch	Ɉ̲Ɉ̲	Ƙ̲Ƙ̲	Ⱪ̲Ⱪ̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ɱ̲Ɱ̲	Ɲ̲Ɲ̲	Ø̲Ø̲	Ɵ̲Ɵ̲	Ƥ̲Ƥ̲	Ƥ̲Ƥ̲	Ʀ̲Ʀ̲	Ʀ̲Ʀ̲
base	Ɉ̲Ɉ̲	Ƙ̲Ƙ̲	Ⱪ̲Ⱪ̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ł̲Ł̲	Ɱ̲Ɱ̲	Ɲ̲Ɲ̲	Ø̲Ø̲	Ɵ̲Ɵ̲	Ƥ̲Ƥ̲	Ƥ̲Ƥ̲	Ʀ̲Ʀ̲	Ʀ̲Ʀ̲
a	+50, 90%	*-6, 80%*	-8, 78% s	+0, 100%	+0, 87%	+0, 92%	+0, 85%	+0 |	+0, 100% s	-1, 96% s	+0, 95%	+0, 92%	+122, 84%	+0, 95%	+0, 97%	+0, 100% s
a			336, 798						431, 850		351, 850		400, 850			
a?	186, 850	339, 850	326, 850	260, 850	256, 850	248, 850	260, 850	235, 850	399, 850	324, 850	352, 850	352, 850	318, 850	260, 850	301, 850	301, 850
a<	**232, 850**	330, 850	**328, 850**	**151, 850**	**151, 850**	**151, 850**	**151, 850**	**151, 850**	**431, 850**	**345, 850**	**335, 850**	**335, 850**	**399, 850**	**277, 850**	**269, 850**	**269, 850**
a!																
b	+50, 96%	+0, 100%	-8, 78% s	+0, 100%	+0, 100%	+0, 100%	+0, 100%	+0 |	+0, 100% s	-1, 96% s	+0, 90%	+0, 100%	+123, 100%	+0, 100%	+0, 100%	+0, 100% s
b			342, -109						393, -110		350, -110	350, -110	363, -110			
b?	186, -257	339, -110	326, -226	260, -110	256, -110	248, -110	260, -110	235, -110	399, -257	324, -272	352, -121	352, -110	318, -110	260, -110	301, -110	301, -257
b<	**170, -257**	**342, -110**	**334, -110**	**262, -110**	**262, -110**	**262, -110**	**262, -110**	**262, -110**	**393, -257**	**347, -110**	**350, -110**	**350, -110**	**363, -110**	**240, -110**	**301, -110**	**301, -257**
b!																
it		*Ƙ̲Ƙ̲*	*Ⱪ̲Ⱪ̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*		*Ɲ̲Ɲ̲*	*Ø̲Ø̲*	*Ɵ̲Ɵ̲*	*Ƥ̲Ƥ̲*	*Ƥ̲Ƥ̲*		*Ʀ̲Ʀ̲*
patch	⌧	*Ƙ̲Ƙ̲*	*Ⱪ̲Ⱪ̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	⌧	*Ɲ̲Ɲ̲*	*Ø̲Ø̲*	*Ɵ̲Ɵ̲*	*Ƥ̲Ƥ̲*	*Ƥ̲Ƥ̲*	⌧	*Ʀ̲Ʀ̲*
base	*Ɉ̲Ɉ̲*	*Ƙ̲Ƙ̲*	*Ⱪ̲Ⱪ̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ł̲Ł̲*	*Ɱ̲Ɱ̲*	*Ɲ̲Ɲ̲*	*Ø̲Ø̲*	*Ɵ̲Ɵ̲*	*Ƥ̲Ƥ̲*	*Ƥ̲Ƥ̲*	*Ʀ̲Ʀ̲*	*Ʀ̲Ʀ̲*
a		*-10, 71%*	*+7, 52% s*	+0, 100%	+0, 87%	+0, 92%	+0, 85%	+1 |		+7, 99% s	+0, 96%	+7, 90%	+73, 95%	+0, 95%		*+3, 56% s*
a			336, 798													
a?		499, 850	516, 850	440, 850	440, 850	440, 850	444, 850	444, 850		528, 850	515, 850	515, 850	473, 850	430, 850		465, 850
a<		456, 850		**314, 850**	**314, 850**	**314, 850**	**314, 850**	**315, 850**		**483, 850**	**493, 850**	**500, 850**	**499, 850**	**426, 850**		
a!																
b		+8, 80%	*+7, 52% s*	+0, 100%	+0, 100%	+0, 100%	+0, 100%	+1 |		+7, 99% s	+0, 90%	+12, 98%	+71, 92%	+0, 100%		*+3, 56% s*
b			342, -109								310, -113					
b?		295, -110	288, -223	236, -110	236, -110	236, -110	240, -110	240, -110		290, -272	302, -151	311, -110	269, -110	226, -110		220, -305
b<		**300, -110**		**252, -110**	**252, -110**	**252, -110**	**252, -110**	**253, -110**		**337, -110**	**310, -110**	**322, -110**	**211, -110**	**140, -110**		**220, -305**
b!																
sb	**Ɉ̲Ɉ̲**	**Ƙ̲Ƙ̲**	**Ⱪ̲Ⱪ̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**		**Ɲ̲Ɲ̲**	**Ø̲Ø̲**	**Ɵ̲Ɵ̲**	**Ƥ̲Ƥ̲**	**Ƥ̲Ƥ̲**		**Ʀ̲Ʀ̲**
patch	**Ɉ̲Ɉ̲**	**Ƙ̲Ƙ̲**	**Ⱪ̲Ⱪ̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	⌧	**Ɲ̲Ɲ̲**	**Ø̲Ø̲**	**Ɵ̲Ɵ̲**	**Ƥ̲Ƥ̲**	**Ƥ̲Ƥ̲**	⌧	**Ʀ̲Ʀ̲**
base	**Ɉ̲Ɉ̲**	**Ƙ̲Ƙ̲**	**Ⱪ̲Ⱪ̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ł̲Ł̲**	**Ɱ̲Ɱ̲**	**Ɲ̲Ɲ̲**	**Ø̲Ø̲**	**Ɵ̲Ɵ̲**	**Ƥ̲Ƥ̲**	**Ƥ̲Ƥ̲**	**Ʀ̲Ʀ̲**	**Ʀ̲Ʀ̲**
a	+0, 92%	*-6, 66%*	*+2, 12% s*	+0, 100%	+0, 85%	+0, 91%	+0, 85%	+0 |		*+8, 37% s*	+0, 92%	+0, 90%	+67, 96%	+0, 95%		*-25, 68% s*
a	203, 806															
a?	131, 850	338, 850	345, 850	267, 850	258, 850	263, 850	266, 850	238, 850		336, 850	366, 850	366, 850	296, 850	276, 850		292, 850
a<	**203, 850**	334, 850		**161, 850**	**161, 850**	**161, 850**	**161, 850**	**161, 850**			**363, 850**	**363, 850**	**345, 850**	**278, 850**		
a!																
b	+0, 98%	+2, 96%	*+2, 12% s*	+0, 100%	+0, 100%	+0, 100%	+0, 100%	+0 |		*+8, 37% s*	+0, 87%	+0, 100%	+68, 96%	+0, 100%		*-25, 68% s*
b																
b?	131, -267	338, -110	345, -221	267, -110	258, -110	263, -110	266, -110	238, -110		336, -270	366, -153	366, -110	296, -110	276, -110		292, -319
b<	**131, -267**	**342, -110**		**267, -110**	**267, -110**	**267, -110**	**267, -110**	**267, -110**		**336, -270**	**366, -110**	**366, -110**	**346, -110**	**278, -110**		**292, -319**
b!																
si		***Ƙ̲Ƙ̲***	***Ⱪ̲Ⱪ̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***		***Ɲ̲Ɲ̲***	***Ø̲Ø̲***	***Ɵ̲Ɵ̲***	***Ƥ̲Ƥ̲***	***Ƥ̲Ƥ̲***		***Ʀ̲Ʀ̲***
patch	⌧	***Ƙ̲Ƙ̲***	***Ⱪ̲Ⱪ̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	⌧	***Ɲ̲Ɲ̲***	***Ø̲Ø̲***	***Ɵ̲Ɵ̲***	***Ƥ̲Ƥ̲***	***Ƥ̲Ƥ̲***	⌧	***Ʀ̲Ʀ̲***
base	***Ɉ̲Ɉ̲***	***Ƙ̲Ƙ̲***	***Ⱪ̲Ⱪ̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ł̲Ł̲***	***Ɱ̲Ɱ̲***	***Ɲ̲Ɲ̲***	***Ø̲Ø̲***	***Ɵ̲Ɵ̲***	***Ƥ̲Ƥ̲***	***Ƥ̲Ƥ̲***	***Ʀ̲Ʀ̲***	***Ʀ̲Ʀ̲***
a		*-36, 73%*	*-28, 11% s*	+12, 100%	+12, 81%	+12, 92%	+12, 85%	+14 |		-23, 99% s	+0, 93%	+1, 90%	+69, 96%	+12, 95%		*-11, 69% s*
a			336, 798								351, 850		400, 850			
a?		353, 850	358, 850	310, 850	300, 850	300, 850	300, 850	300, 850		383, 850	390, 850	387, 850	315, 850	296, 850		307, 850
a<		333, 850		**221, 850**	**221, 850**	**221, 850**	**221, 850**	**223, 850**		**366, 850**	**390, 850**	**391, 850**	**385, 850**	**328, 850**		
a!																
b		-28, 94%	*-28, 11% s*	+12, 100%	+12, 100%	+12, 100%	+12, 100%	+14 |		-23, 99% s	+0, 88%	+1, 100%	+70, 100%	+12, 100%		*-11, 69% s*
b			342, -109								350, -110		363, -110			
b?		320, -110	302, -221	277, -110	267, -110	267, -110	267, -110	267, -110		317, -270	349, -148	354, -110	282, -110	263, -110		235, -302
b<		**314, -110**		**272, -110**	**272, -110**	**272, -110**	**272, -110**	**274, -110**		**288, -270**	**350, -110**	**351, -110**	**310, -110**	**252, -110**		**235, -302**
b!																

Table 20: Derived letters and inherited anchors 3/7

hex	0166	01AC	01AE	023E	01B3	024E	01B5	1E9A	2C65	0180	0183	0188	023C	0255	A793	0111
base	0054	0054	0054	0054	0059	0059	005A	0061	0061	0062	0062	0063	0063	0063	0063	0064
reg	Ŧ̲	Ţ̲	Ţ̲	Ŧ̲	Ẏ̲	Ẏ̲	Ẓ̲	ả̲	ⱥ̲	ƀ̲	ƀ̲	c̲̓	ȼ̲	c̲̓	ȼ̲	đ̲
patch	T̲̂	T̲̂	T̲̂	T̲̂	Ŷ̲	Ŷ̲	Ẑ̲	ả̲	ⱥ̲	b̲̂	b̲̂	ĉ̲	ȼ̲	ĉ̲	ȼ̲	d̲̂
base	T̲̂	T̲̂	T̲̂	T̲̂	Ŷ̲	Ŷ̲	Ẑ̲	a̲	a̲	b̲̂	b̲̂	ĉ̲	ĉ̲	ĉ̲	ĉ̲	d̲̂
a	+0, 96%	+22 |	+0, 100% s	+0 |	+65 |	+0, 90%	+0, 94%	+0, 100%	+0, 84%	+0, 95%	-4, 93%	+1, 98%	+1, 85%	+1, 91% s	+85, 88%	+0, 95%
a														209, 645	327, 645	
a?	297, 850	308, 850	297, 850	297, 850	321, 850	288, 850	310, 850	272, 838	246, 690	231, 885	228, 885	264, 713	216, 668	222, 645	260, 645	270, 885
a<	**307, 850**	**329, 850**	**307, 850**	**307, 850**	**360, 850**	**295, 850**	320, 850	**209, 645**	**209, 645**	**279, 885**	**275, 885**	**243, 669**	**243, 645**	**243, 645**	**327, 645**	**235, 885**
a!							320, 850									
b	+0, 100%	+22 |	+0, 100% s	+0 |	+65 |	+0, 100%	+0, 100%	+0, 100%	+0, 96%	+0, 100%	-4, 100%	+0, 100%	+0, 85%		+85, 88%	+0, 100%
b											234, -109			212, -114	321, -110	
b?	297, -110	308, -110	297, -261	297, -165	321, -110	288, -110	310, -110	272, -110	246, -170	231, -110	228, -110	264, -110	216, -136	222, -214	260, -110	270, -110
b<	**300, -110**	**322, -110**	**300, -261**	**300, -110**	**360, -110**	**295, -110**	287, -110	**196, -110**	**196, -162**	**234, -110**	**230, -110**	**218, -110**	**218, -110**		**303, -110**	**235, -110**
b!							287, -110									
it	*Ŧ̲*	*Ţ̲*	*Ţ̲*	*Ŧ̲*	*Ẏ̲*		*Ẓ̲*	*ả̲*	*ⱥ̲*	*ƀ̲*	*ƀ̲*	*c̲̓*	*ȼ̲*	*c̲̓*		*đ̲*
patch	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	⌧	*Ẑ̲*	*ả̲*	*ⱥ̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ȼ̲*	*ĉ̲*	⌧	*d̲̂*
base	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	*Ŷ̲*	*Ẑ̲*	*a̲*	*a̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*d̲̂*
a	+0, 94%	+40 |	+0, 100% s	-8 |	+72 |		+0, 94%	+0, 100%	+1, 88%	*+4, 64%*	*-20, 64%*	*+18, 79%*	+22, 82%	+7, 83% s		+0, 95%
a																
a?	453, 850	488, 850	453, 850	453, 850	501, 850		470, 850	489, 843	385, 668	428, 890	413, 890	398, 713	352, 668	352, 645		434, 890
a<	**457, 850**	**497, 850**	**457, 850**	**469, 946**	**513, 850**		450, 850	**341, 645**	**342, 645**	432, 890	408, 890	361, 645	**365, 645**	**350, 645**		**434, 890**
a!							489, 850									
b	+0, 100%	+40 |	+0, 100% s	-8 |	+72 |		+0, 100%	+0, 100%	+0, 92%	*+15, 54%*	*-6, 54%*	*+5, 65%*	*+10, 44%*			+0, 100%
b											230, -109			212, -114		
b?	249, -110	284, -110	214, -274	237, -165	297, -110		266, -110	286, -110	214, -136	216, -110	201, -110	223, -110	181, -137	171, -208		221, -110
b<	**277, -110**	**317, -110**	**242, -274**	**269, -110**	**342, -110**		267, -110	**220, -110**	**220, -110**	229, -110	208, -110	168, -110	173, -110			**221, -110**
b!							286, -110									
sb	*Ŧ̲*	*Ţ̲*	*Ţ̲*	*Ŧ̲*	*Ẏ̲*		*Ẓ̲*	*ả̲*	*ⱥ̲*	*ƀ̲*	*ƀ̲*	*c̲̓*	*ȼ̲*	*c̲̓*		*đ̲*
patch	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	⌧	*Ẑ̲*	*ả̲*	*ⱥ̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ȼ̲*	*ĉ̲*	⌧	*d̲̂*
base	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	*Ŷ̲*	*Ẑ̲*	*a̲*	*a̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*d̲̂*
a	+0, 94%	+44 |	+13, 79% s	+0 |	+58 |		+0, 94%	+0, 100%	+0, 90%	+0, 95%	-14, 87%	+0, 95%	+0, 81%	+2, 75% s		+0, 95%
a																
a?	299, 850	320, 850	306, 850	299, 850	333, 850		309, 850	288, 840	255, 685	236, 885	223, 870	264, 708	200, 670	232, 645		274, 885
a<	**298, 850**	**342, 850**	**311, 850**	**298, 850**	**374, 850**		321, 850	**219, 645**	**219, 645**	**236, 885**	**222, 885**	**224, 645**	**224, 645**	**226, 645**		**274, 885**
a!							316, 850									
b	+0, 100%	+44 |	+13, 79% s	+0 |	+58 |		+0, 100%	+0, 100%	+0, 94%	+0, 100%	-14, 81%	+0, 100%	+0, 81%			+0, 100%
b																
b?	299, -110	320, -110	306, -281	299, -165	333, -110		309, -110	288, -110	255, -170	236, -121	223, -124	264, -110	200, -129	232, -209		274, -110
b<	**301, -110**	**345, -110**	**314, -281**	**301, -110**	**374, -110**		308, -110	**255, -110**	**255, -110**	**236, -121**	**222, -121**	**224, -110**	**224, -110**			**274, -110**
b!							316, -110									
si	*Ŧ̲*	*Ţ̲*	*Ţ̲*	*Ŧ̲*	*Ẏ̲*		*Ẓ̲*	*ả̲*	*ⱥ̲*	*ƀ̲*	*ƀ̲*	*c̲̓*	*ȼ̲*	*c̲̓*		*đ̲*
patch	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	⌧	*Ẑ̲*	*ả̲*	*ⱥ̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ȼ̲*	*ĉ̲*	⌧	*d̲̂*
base	*T̲̂*	*T̲̂*	*T̲̂*	*T̲̂*	*Ŷ̲*	*Ŷ̲*	*Ẑ̲*	*a̲*	*a̲*	*b̲̂*	*b̲̂*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*ĉ̲*	*d̲̂*
a	+12, 94%	+44 |	+9, 79% s	+11 |	+95 |		*+12, 62%*	+0, 100%	+11, 93%	+0, 94%	-16, 93%	+22, 93%	+13, 81%	*+1, 67% s*		+0, 94%
a																
a?	325, 850	330, 850	314, 850	325, 850	368, 850		332, 850	371, 844	234, 685	290, 890	269, 875	253, 708	204, 670	204, 645		309, 890
a<	**319, 850**	**351, 850**	**321, 850**	**318, 850**	**386, 850**		332, 850	**251, 645**	**262, 645**	**290, 890**	**274, 890**	**270, 675**	**255, 645**			**309, 890**
a!							312, 850									
b	+12, 100%	+44 |	+9, 79% s	+11 |	+95 |		*+14, 65%*	+0, 100%	+12, 95%	+0, 100%	-16, 81%	*+19, 67%*	+12, 85%			+0, 100%
b											234, -109			212, -114		
b?	292, -110	297, -110	246, -281	281, -165	335, -110		299, -110	339, -110	222, -170	248, -110	231, -110	249, -110	203, -129	192, -209		268, -110
b<	**312, -110**	**344, -110**	**280, -281**	**311, -110**	**372, -110**		301, -110	**260, -110**	**270, -122**	**248, -110**	**232, -110**	255, -110	**248, -110**			**221, -110**
b!							267, -110									

Table 21: Derived letters and inherited anchors 4/7

hex	018C	0221	0256	0257	0247	01E5	0127	0266	2C68	0268	0249	0199	2C6A	0140	0142	019A
base	0064	0064	0064	0064	0065	0067	0068	0068	0068	0069	006A	006B	006B	006C	006C	006C
reg	ƌ̲	ȡ̲	ɖ̲	ɗ̲	ɇ̲	ɠ̲	ħ̲	ɦ̲	ⱨ̲	ɨ̲	ɟ̲	ƙ̲	ⱪ̲	ŧ̲	ŧ̲	ŧ̲
patch	ƌ̲̂	ȡ̲̂	ɖ̲̂	ɗ̲̂	ɇ̲̂	ɠ̲̂	ħ̲̂	ɦ̲̂	ⱨ̲̂	ɨ̲̂	ɟ̲̂	ƙ̲̂	ⱪ̲̂	ŧ̲̂	ŧ̲̂	ŧ̲̂
base	ɖ̲̂	ɖ̲̂	ɖ̲̂	ɖ̲̂	ɇ̲	ɠ̲	ħ̲̂	ħ̲̂	ⱨ̲̂	ɨ̲	ɟ̲	ⱪ̲	ⱪ̲	l̲̂	l̲̂	l̲̂
a	+5, 94%	-8, 91% s	-3, 95% s	-2, 94% s	+0, 85%	+0, 100%	+0, 95%	-10, 82%	-11, 96% s	+14, 86%	+0, 90%	+1, 85%	+2, 97% s	+0, 100%	+0, 89%	+0 |
a	214, 650		218, 645	218, 645				299, 565	299, 645	139, 645	140, 645		275, 646			
a?	275, 885	326, 885	314, 885	322, 885	216, 668	256, 676	270, 885	262, 885	262, 885	150, 834	116, 834	262, 885	259, 885	163, 885	133, 885	132, 885
a<	**240, 885**	**227, 885**	**232, 885**	**233, 885**	**243, 645**	**231, 645**	**282, 885**	**272, 885**	**271, 885**	156, 645	**152, 645**	**273, 885**	**274, 885**	**135, 885**	**135, 885**	**135, 885**
a!										156, 834						
b	+5, 100%	-8, 91% s	-3, 95% s	-2, 94% s	+0, 88%	+0, 89%	+0, 100%	-10, 96%	-11, 96% s	+14, 92%	+0, 94%	+1, 91%	+2, 97% s	+0, 100%	+0, 100%	+0 |
b	217, -107		212, -114	212, -114				240, -114	259, -111	134, -114		263, -107	263, -107			
b?	275, -110	326, -210	314, -273	322, -110	216, -136	256, -319	270, -110	262, -110	262, -213	150, -110	116, -308	262, -110	259, -213	163, -110	133, -110	132, -110
b<	**240, -110**	**227, -210**	**232, -273**	**233, -110**	**234, -110**	**229, -319**	**272, -110**	**262, -110**	**262, -213**	**156, -110**	**109, -319**	**262, -110**	**263, -213**	**135, -110**	**135, -110**	**135, -110**
b!																
it	*ƌ̲*	*ȡ̲*	*ɖ̲*	*ɗ̲*	*ɇ̲*	*ɠ̲*	*ħ̲*	*ɦ̲*	*ⱨ̲*	*ɨ̲*	*ɟ̲*	*ƙ̲*	*ⱪ̲*	*ŧ̲*	*ŧ̲*	*ŧ̲*
patch	*ƌ̲̂*	*ȡ̲̂*	*ɖ̲̂*	*ɗ̲̂*	*ɇ̲̂*	*ɠ̲̂*	*ħ̲̂*	*ɦ̲̂*	*ⱨ̲̂*	*ɨ̲̂*	*ɟ̲̂*	*ƙ̲̂*	*ⱪ̲̂*	*ŧ̲̂*	*ŧ̲̂*	*ŧ̲̂*
base	*ɖ̲̂*	*ɖ̲̂*	*ɖ̲̂*	*ɖ̲̂*	*ɇ̲*	*ɠ̲*	*ħ̲̂*	*ħ̲̂*	*ⱨ̲̂*	*ɨ̲*	*ɟ̲*	*ⱪ̲*	*ⱪ̲*	*l̲̂*	*l̲̂*	*l̲̂*
a	*-7, 77%*	*+12, 4% s*	*+10, 4% s*	+0, 99% s	+0, 93%	+0, 100%			+12, 79% s	+14, 86%	+0, 86%	*+30, 48%*	*+24, 42% s*	+0, 100%	+0, 89%	+20 |
a	209, 650		218, 645	218, 645	332, 645			299, 565	299, 645	267, 645	254, 645		275, 646			
a?	423, 890	507, 890	465, 899	464, 890	348, 692	376, 645	449, 890	449, 890	448, 890	316, 810	315, 810	429, 890	453, 890	349, 890	322, 890	342, 890
a<	427, 890			**434, 890**	**340, 684**	**360, 645**			**461, 890**	289, 645	**283, 645**	462, 890		**312, 890**	**312, 890**	**332, 890**
a!			440, 899							326, 810						
b	*-2, 51%*	*+12, 4% s*	*+10, 4% s*	+0, 99% s	+0, 91%	+0, 92%	+0, 100%	+0, 100%	+12, 79% s	+14, 97%	+0, 92%	*+14, 60%*	*+24, 42% s*	+0, 100%	+0, 100%	+20 |
b	212, -107		212, -114	212, -114	158, -108			240, -114	259, -111	134, -114		263, -107	263, -107			
b?	211, -110	274, -205	215, -273	251, -110	170, -143	171, -319	236, -110	236, -110	211, -223	121, -110	83, -281	216, -110	216, -223	136, -110	110, -110	130, -110
b<	219, -110	**274, -205**	**215, -273**	**221, -110**	**153, -135**	**197, -319**	**229, -110**	**229, -110**	**216, -223**	**129, -110**	**0, -319**	237, -110	**216, -223**	**99, -110**	**99, -110**	**119, -110**
b!																
sb	**ƌ̲**	**ȡ̲**	**ɖ̲**	**ɗ̲**	**ɇ̲**	**ɠ̲**	**ħ̲**	**ɦ̲**	**ⱨ̲**	**ɨ̲**	**ɟ̲**	**ƙ̲**	**ⱪ̲**	**ŧ̲**	**ŧ̲**	**ŧ̲**
patch	**ƌ̲̂**	**ȡ̲̂**	**ɖ̲̂**	**ɗ̲̂**	**ɇ̲̂**	**ɠ̲̂**	**ħ̲̂**	**ɦ̲̂**	**ⱨ̲̂**	**ɨ̲̂**	**ɟ̲̂**	**ƙ̲̂**	**ⱪ̲̂**	**ŧ̲̂**	**ŧ̲̂**	**ŧ̲̂**
base	**ɖ̲̂**	**ɖ̲̂**	**ɖ̲̂**	**ɖ̲̂**	**ɇ̲**	**ɠ̲**	**ħ̲̂**	**ħ̲̂**	**ⱨ̲̂**	**ɨ̲**	**ɟ̲**	**ⱪ̲**	**ⱪ̲**	**l̲̂**	**l̲̂**	**l̲̂**
a	-19, 87%	-2, 92% s	-2, 93% s	-4, 92% s	+0, 83%	+0, 100%	+0, 95%	-2, 80%	-15, 97% s	+0, 87%	+0, 90%	*-5, 66%*	*-2, 2% s*	+0, 100%	+0, 85%	+0 |
a										151, 645						
a?	261, 870	338, 885	322, 885	324, 885	232, 686	258, 676	275, 885	274, 885	264, 885	152, 818	112, 837	259, 885	273, 885	212, 885	142, 885	138, 885
a<	**255, 885**	**272, 885**	**272, 885**	**270, 885**	**249, 678**	**231, 645**	**275, 885**	**273, 885**	**260, 885**	156, 645	**163, 645**	257, 885		**150, 885**	**150, 885**	**150, 885**
a!										156, 837						
b	-20, 83%	-2, 92% s	-2, 93% s	-4, 92% s	-1, 81%	+0, 92%	+0, 100%	-3, 96%	-15, 97% s	+0, 92%	+0, 94%	-6, 91%	*-2, 2% s*	+0, 100%	+0, 100%	+0 |
b										134, -114						
b?	261, -110	338, -216	322, -271	324, -110	232, -142	258, -319	275, -110	274, -110	264, -222	152, -110	112, -319	259, -110	273, -222	212, -110	142, -110	138, -110
b<	**254, -110**	**272, -216**	**272, -271**	**270, -110**	**231, -110**	**229, -319**	**277, -110**	**274, -110**	**262, -222**	**156, -110**	**109, -319**	**256, -110**	**273, -222**	**150, -110**	**150, -110**	**150, -110**
b!																
si	***ƌ̲***	***ȡ̲***	***ɖ̲***	***ɗ̲***	***ɇ̲***	***ɠ̲***	***ħ̲***	***ɦ̲***	***ⱨ̲***	***ɨ̲***	***ɟ̲***	***ƙ̲***	***ⱪ̲***	***ŧ̲***	***ŧ̲***	***ŧ̲***
patch	***ƌ̲̂***	***ȡ̲̂***	***ɖ̲̂***	***ɗ̲̂***	***ɇ̲̂***	***ɠ̲̂***	***ħ̲̂***	***ɦ̲̂***	***ⱨ̲̂***	***ɨ̲̂***	***ɟ̲̂***	***ƙ̲̂***	***ⱪ̲̂***	***ŧ̲̂***	***ŧ̲̂***	***ŧ̲̂***
base	***ɖ̲̂***	***ɖ̲̂***	***ɖ̲̂***	***ɖ̲̂***	***ɇ̲***	***ɠ̲***	***ħ̲̂***	***ħ̲̂***	***ⱨ̲̂***	***ɨ̲***	***ɟ̲***	***ⱪ̲***	***ⱪ̲***	***l̲̂***	***l̲̂***	***l̲̂***
a	-15, 89%	-2, 74% s	-2, 81% s	-5, 82% s	+0, 98%	+4, 100%			-13, 71% s	+0, 86%	+0, 86%	*-5, 45%*	*-11, 60% s*	+12, 100%	+17, 85%	+18 |
a	214, 650		218, 645	218, 645	243, 645			299, 565	299, 645	276, 645	290, 645		275, 646			
a?	275, 875	362, 890	302, 890	318, 890	226, 684	220, 645	331, 890	302, 890	290, 890	180, 823	192, 823	283, 890	295, 890	234, 890	190, 890	174, 890
a<	**294, 890**	**311, 890**	**311, 890**	**310, 890**	**249, 676**	**224, 645**			**315, 890**	144, 645	**171, 645**	311, 890		**192, 890**	**197, 890**	**198, 890**
a!										180, 823						
b	*-13, 57%*	-2, 74% s	-2, 81% s	-5, 82% s	+0, 91%	+4, 92%	+0, 100%	*-4, 71%*	-13, 71% s	+2, 93%	+12, 92%	*-17, 44%*	*-11, 60% s*	+12, 100%	+17, 100%	+18 |
b	217, -107		212, -114	212, -114	234, -110			240, -114	259, -111	136, -110	109, -319	263, -107	263, -107			
b?	237, -110	299, -216	227, -271	276, -110	221, -140	186, -319	290, -110	260, -110	225, -222	153, -110	127, -292	241, -110	230, -222	192, -110	148, -110	132, -110
b<	208, -110	**200, -216**	**189, -271**	**220, -110**	**230, -132**	**203, -319**	**272, -110**	268, -110	**233, -222**	**155, -110**	**79, -319**	258, -110	**230, -222**	**150, -110**	**155, -110**	**156, -110**
b!																

Table 22: Derived letters and inherited anchors 5/7

hex	0234	026B	026C	026D	026E	2C61	0271	0235	0272	0273	00F8	0275	01A5	02A0	024D	027D
base	006C	006C	006C	006C	006C	006C	006D	006E	006E	006E	006F	006F	0070	0071	0072	0072
reg	Ɫ̵	ɬ̵	ɮ̵	ɭ̵	ꞁ̵	ɫ̵	ɱ̵	ɲ̵	ɲ̵	ɳ̵	ø̵	ɵ̵	ƥ̵	ʠ̵	ɾ̵	ɽ̵
patch	Ɫ̵	ɬ̵	ɮ̵	ɭ̵	ꞁ̵	ɫ̵	ɱ̵	ɲ̵	ɲ̵	ɳ̵	ø̵	ɵ̵	ƥ̵	ʠ̵	ɾ̵	ɽ̵
base	Ɫ̵	ɬ̵	ɮ̵	ɭ̵	ꞁ̵	ɫ̵	ɱ̵	ɲ̵	ɲ̵	ɳ̵	ɵ̵	ɵ̵	ƥ̵	ʠ̵	ɾ̵	ɽ̵
a	+0, 100% s	+12 |	+90 |	-3, 100% s	+0, 91%	+0 |	-3, 97% s	-4, 99% s	-11, 96% s	-11, 96% s	+0, 91%	+0, 85%			+0, 93%	-16, 72% s
a		209, 645	209, 645	209, 645	209, 645		383, 645	261, 646	264, 646	263, 645		209, 645		238, 645		209, 645
a?	175, 885	144, 885	182, 885	182, 885	250, 885	132, 885	366, 645	333, 645	234, 645	296, 645	250, 668	252, 645	242, 885	320, 885	188, 645	186, 645
a<	**135, 885**	**147, 885**	**225, 885**	**132, 885**	**135, 885**	**135, 885**	**389, 645**	**260, 645**	**253, 645**	**253, 645**	**238, 645**	**238, 645**			**191, 645**	**175, 645**
a!																
b	+0, 100% s	+12 |	+90 |	-3, 100% s		+0 |	-3, 97% s	-4, 99% s	-11, 96% s	-11, 96% s	+0, 90%	+0, 83%	+0, 100%	-5, 97%	+0, 96%	-16, 72% s
b		146, -114	218, -120	196, -110	196, -110		383, -105	262, -111	265, -111	262, -112	243, -110	196, -110		196, -110		196, -110
b?	175, -216	144, -110	182, -110	182, -273	250, -319	132, -110	366, -306	333, -201	234, -306	296, -306	250, -136	252, -110	242, -319	320, -319	188, -110	186, -319
b<	**135, -216**	**147, -110**	**225, -110**	**132, -273**		**135, -110**	**398, -306**	**261, -201**	**254, -306**	**254, -306**	**243, -110**	**243, -110**	**120, -319**	**381, -319**	**140, -110**	**124, -319**
b!					275, -319											
it	*Ɫ̵*	*ɬ̵*	*ɮ̵*	*ɭ̵*	*ꞁ̵*	*ɫ̵*	*ɱ̵*	*ɲ̵*	*ɲ̵*	*ɳ̵*	*ø̵*	*ɵ̵*	*ƥ̵*	*ʠ̵*	*ɾ̵*	*ɽ̵*
patch	*Ɫ̵*	*ɬ̵*	*ɮ̵*	*ɭ̵*	*ꞁ̵*	*ɫ̵*	*ɱ̵*	*ɲ̵*	*ɲ̵*	*ɳ̵*	*ø̵*	*ɵ̵*	*ƥ̵*	*ʠ̵*	*ɾ̵*	*ɽ̵*
base	*Ɫ̵*	*ɬ̵*	*ɮ̵*	*ɭ̵*	*ꞁ̵*	*ɫ̵*	*ɱ̵*	*ɲ̵*	*ɲ̵*	*ɳ̵*	*ɵ̵*	*ɵ̵*	*ƥ̵*	*ʠ̵*	*ɾ̵*	*ɽ̵*
a	+14, 91% s	+12 |	+89 |	+0, 91% s	+18, 89%	+0 |	*-13, 70% s*	+0, 89% s	+18, 79% s	-1, 95% s	+0, 89%	+20, 84%			+0, 91%	*+4, 61% s*
a							383, 645	261, 646	264, 646	263, 645				238, 645	312, 646	
a?	389, 890	333, 890	368, 890	341, 890	447, 890	330, 890	515, 645	460, 645	396, 645	367, 645	370, 691	379, 645	429, 890	460, 871	515, 1008	320, 645
a<	**326, 890**	**324, 890**	**401, 890**	312, 890	**330, 890**	**312, 890**		**394, 645**	**414, 654**	**393, 645**	**359, 645**	**379, 645**			**312, 645**	
a!				312, 890												359, 645
b	+14, 91% s	+12 |	+89 |	+0, 91% s		+0 |	*-13, 70% s*	+0, 89% s	+18, 79% s	-1, 95% s	+0, 88%	+14, 83%	-17, 89%	*+4, 74%*	+0, 85%	*+4, 61% s*
b		146, -114	218, -120				383, -105	262, -111	265, -111	262, -112	186, -107				113, -110	
b?	155, -214	120, -110	156, -110	91, -287	192, -309	117, -110	310, -319	281, -197	191, -319	162, -319	190, -156	218, -110	172, -319	207, -319	246, -257	124, -279
b<	**91, -214**	**111, -110**	**188, -110**	**61, -287**		**99, -110**	**310, -319**	**217, -197**	**209, -319**	**190, -319**	**178, -148**	**200, -110**	**160, -319**	304, -319	**113, -110**	**124, -279**
b!					207, -319											
sb	**Ɫ̵**	**ɬ̵**	**ɮ̵**	**ɭ̵**	**ꞁ̵**	**ɫ̵**	**ɱ̵**	**ɲ̵**	**ɲ̵**	**ɳ̵**	**ø̵**	**ɵ̵**	**ƥ̵**	**ʠ̵**	**ɾ̵**	**ɽ̵**
patch	**Ɫ̵**	**ɬ̵**	**ɮ̵**	**ɭ̵**	**ꞁ̵**	**ɫ̵**	**ɱ̵**	**ɲ̵**	**ɲ̵**	**ɳ̵**	**ø̵**	**ɵ̵**	**ƥ̵**	**ʠ̵**	**ɾ̵**	**ɽ̵**
base	**Ɫ̵**	**ɬ̵**	**ɮ̵**	**ɭ̵**	**ꞁ̵**	**ɫ̵**	**ɱ̵**	**ɲ̵**	**ɲ̵**	**ɳ̵**	**ɵ̵**	**ɵ̵**	**ƥ̵**	**ʠ̵**	**ɾ̵**	**ɽ̵**
a	-6, 94% s	+12 |	+90 |	+0, 100% s	+0, 89%	+0 |	-10, 97% s	-12, 91% s	-13, 97% s	-15, 89% s	+1, 88%	+0, 89%			+0, 90%	-16, 92% s
a																
a?	212, 885	156, 885	190, 885	194, 885	274, 885	144, 885	363, 645	345, 645	246, 645	306, 645	260, 684	260, 645	236, 885	337, 885	210, 645	198, 645
a<	**144, 885**	**162, 885**	**240, 885**	**150, 885**	**150, 885**	**150, 885**	**386, 645**	**278, 645**	**277, 645**	**275, 645**	**261, 645**	**260, 645**			**212, 645**	**196, 645**
a!																
b	-6, 94% s	+12 |	+90 |	+0, 100% s		+0 |	-10, 97% s	-12, 91% s	-13, 97% s	-15, 89% s	+0, 85%	+0, 87%	-15, 93%	-4, 91%	+0, 100%	-16, 92% s
b		146, -114	218, -120									229, -114				
b?	212, -223	156, -110	190, -110	194, -319	274, -319	144, -110	363, -319	345, -207	246, -319	306, -319	260, -142	260, -110	236, -319	337, -319	210, -110	198, -319
b<	**144, -223**	**162, -110**	**240, -110**	**150, -319**		**150, -110**	**386, -319**	**254, -207**	**253, -319**	**250, -319**	**260, -110**	**260, -110**	**239, -319**	**269, -319**	**152, -110**	**135, -319**
b!					283, -319											
si	***Ɫ̵***	***ɬ̵***	***ɮ̵***	***ɭ̵***	***ꞁ̵***	***ɫ̵***	***ɱ̵***	***ɲ̵***	***ɲ̵***	***ɳ̵***	***ø̵***	***ɵ̵***	***ƥ̵***	***ʠ̵***	***ɾ̵***	***ɽ̵***
patch	***Ɫ̵***	***ɬ̵***	***ɮ̵***	***ɭ̵***	***ꞁ̵***	***ɫ̵***	***ɱ̵***	***ɲ̵***	***ɲ̵***	***ɳ̵***	***ø̵***	***ɵ̵***	***ƥ̵***	***ʠ̵***	***ɾ̵***	***ɽ̵***
base	***Ɫ̵***	***ɬ̵***	***ɮ̵***	***ɭ̵***	***ꞁ̵***	***ɫ̵***	***ɱ̵***	***ɲ̵***	***ɲ̵***	***ɳ̵***	***ɵ̵***	***ɵ̵***	***ƥ̵***	***ʠ̵***	***ɾ̵***	***ɽ̵***
a	+6, 89% s	+29 |	+106 |	+19, 93% s	+9, 87%	+11 |	*-7, 66% s*	*-2, 67% s*	*-4, 69% s*	*-5, 67% s*	+0, 89%	+32, 82%			+0, 92%	*-10, 50% s*
a							383, 645	261, 646	264, 646	263, 645				238, 645	191, 646	
a?	233, 890	187, 890	223, 890	187, 890	301, 890	184, 890	373, 645	324, 645	254, 645	254, 645	235, 645	229, 645	271, 890	318, 890	203, 645	177, 645
a<	**186, 890**	**209, 890**	**286, 890**	205, 890	**189, 890**	**191, 890**					**238, 645**	**270, 645**			**191, 645**	
a!				205, 890												218, 645
b	+6, 89% s	+29 |	+106 |	+19, 93% s		+11 |	*-7, 66% s*	*-2, 67% s*	*-4, 69% s*	*-5, 67% s*	+0, 89%	+25, 80%	-32, 88%	*-5, 65%*	+0, 85%	*-10, 50% s*
b		146, -114	218, -120				383, -105	262, -111	265, -111	262, -112	243, -110				135, -110	
b?	169, -223	145, -110	181, -110	103, -319	217, -319	142, -110	339, -319	313, -207	220, -319	220, -319	237, -138	238, -110	187, -319	234, -319	212, -110	143, -319
b<	**121, -223**	**167, -110**	**244, -110**	**120, -319**		**149, -110**	**339, -319**	**313, -207**	**220, -319**	**220, -319**	**243, -110**	**268, -110**	**147, -319**	343, -319	**135, -110**	**143, -319**
b!					231, -319											

Table 23: Derived letters and inherited anchors 6/7

hex	0282	0167	01AB	01AD	0236	0288	2C66	0289	AB52	01B4	024F	1EFF	01B6	0290	0291	A765
base	0073	0074	0074	0074	0074	0074	0074	0075	0075	0079	0079	0079	007A	007A	007A	00FE
reg																
patch																
base																
a	+0, 100% s	+0, 91%	+0, 100% s		-1, 94% s	-3, 86% s	+0, 89%	+0, 87%	*+119, 75%*	-5, 85%	+0, 90%	-4, 97% s	+0, 90%	+0, 100% s	-14, 79% s	+30, 93%
a	209, 645					209, 645		209, 645	376, 645	270, 644		270, 644		209, 645	209, 645	
a?	199, 645	160, 769	163, 769	186, 885	196, 770	158, 771	146, 769	271, 645	332, 645	316, 771	260, 645	256, 645	220, 645	262, 645	218, 662	240, 885
a<	**206, 645**	**139, 769**	**139, 769**		**138, 769**	**136, 769**	**139, 769**	**255, 645**	374, 645	**257, 645**	**262, 645**	**258, 645**	**220, 645**	**220, 645**	**206, 645**	**266, 885**
a!																
b	+0, 100% s	+0, 100%	+0, 100% s	+0 |	-1, 94% s	-3, 86% s	+0, 83%	+0, 96%	+121, 98%	-5, 92%	+0, 97%	-4, 97% s	+0, 89%	+0, 100% s	-14, 79% s	+30, 100%
b	196, -110			163, -107	163, -109	196, -110		196, -110	360, -110					196, -110	196, -110	
b?	199, -261	160, -110	163, -230	186, -110	196, -201	158, -306	146, -170	271, -110	332, -110	316, -319	260, -319	256, -319	220, -110	262, -270	218, -207	240, -319
b<	**196, -261**	**163, -110**	**163, -230**	**163, -110**	**162, -201**	**160, -306**	**163, -110**	**239, -110**	**360, -110**	**237, -319**	**242, -319**	**238, -319**	**207, -110**	**207, -270**	**192, -207**	**266, -319**
b!																

it																
patch									⌧		⌧	⌧				⌧
base																
a	*+11, 19% s*	+0, 90%	+3, 78% s		+3, 80% s	+2, 84% s	+40, 90%	*-3, 55%*		-4, 82%			+0, 89%	*-5, 22% s*	*-1, 15% s*	
a										270, 644						
a?	314, 645	318, 775	316, 776	341, 890	352, 776	314, 776	364, 775	398, 645		471, 754			343, 645	343, 645	352, 645	
a<		**318, 775**	**326, 798**		**326, 798**	**325, 798**	**358, 775**	383, 645		**363, 645**			**360, 645**			
a!								402, 645						354, 645	354, 645	
b	*+11, 19% s*	+0, 100%	+3, 78% s	+1 |	+3, 80% s	+2, 84% s	*+40, 74%*	*-1, 46%*		-4, 95%			+0, 90%	*-5, 22% s*	*-1, 15% s*	
b				163, -107	163, -109											
b?	121, -265	130, -110	102, -230	129, -110	144, -204	83, -311	170, -136	238, -110		243, -319			182, -110	147, -276	171, -207	
b<	**121, -265**	**133, -110**	**110, -230**	**134, -110**	**116, -204**	**92, -311**	173, -110	225, -110		**236, -319**			**187, -110**	**147, -276**	**171, -207**	
b!								241, -110								

sb																
patch									⌧		⌧	⌧				⌧
base																
a	+0, 100% s	+0, 91%	+0, 100% s		-6, 88% s	-6, 94% s	+0, 91%	+0, 86%		*+1, 52%*			+0, 90%	+0, 100% s	*-2, 65% s*	
a																
a?	201, 645	168, 782	168, 782	186, 885	206, 782	182, 782	146, 782	274, 645		328, 768			222, 645	254, 645	230, 645	
a<	**201, 645**	**168, 782**	**168, 782**		**162, 782**	**162, 782**	**168, 782**	**264, 645**		265, 645			**222, 645**	**222, 645**		
a!																
b	+0, 100% s	+0, 100%	+0, 100% s	-4 |	-6, 88% s	-6, 94% s	-1, 89%	+0, 96%		*+3, 36%*			+0, 89%	+0, 100% s	*-2, 65% s*	
b																
b?	201, -264	168, -110	168, -229	186, -110	206, -214	182, -319	146, -170	274, -110		328, -319			222, -110	254, -276	230, -210	
b<	**201, -264**	**163, -110**	**163, -229**	**159, -110**	**157, -214**	**157, -319**	**162, -110**	**248, -110**		267, -319			**222, -110**	**222, -276**	**230, -210**	
b!																

si																
patch									⌧		⌧	⌧				⌧
base																
a	*+7, 39% s*	+0, 91%	*-22, 86% s*		-9, 78% s	-9, 84% s	+12, 91%	*+15, 64%*		*-91, 20%*			*-233, 10%*	*-4, 11% s*	*+0, 4% s*	
a										270, 644						
a?	178, 645	175, 765	161, 787	183, 890	209, 787	172, 787	187, 765	255, 645		317, 768			187, 645	202, 645	208, 645	
a<		**189, 765**	**171, 787**		**184, 787**	**184, 787**	**201, 765**	270, 645		171, 645			-13, 645			
a!								275, 645		152, 645				244, 645	215, 645	
b	*+7, 39% s*	+0, 100%	*-22, 86% s*	*-23* |	-9, 78% s	-9, 84% s	+10, 90%	*+16, 52%*		*-37, 6%*			*+227, 11%*	*-4, 11% s*	*+0, 4% s*	
b				163, -107	163, -109											
b?	155, -265	159, -110	117, -228	142, -110	167, -214	109, -319	159, -170	264, -110		262, -297			195, -110	176, -279	196, -210	
b<	**155, -265**	**133, -110**	**87, -228**	**110, -110**	**103, -214**	**81, -319**	**137, -140**	286, -110		205, -319			434, -110	**176, -279**	**196, -210**	
b!								284, -110								

Table 24: Derived letters and inherited anchors 7/7

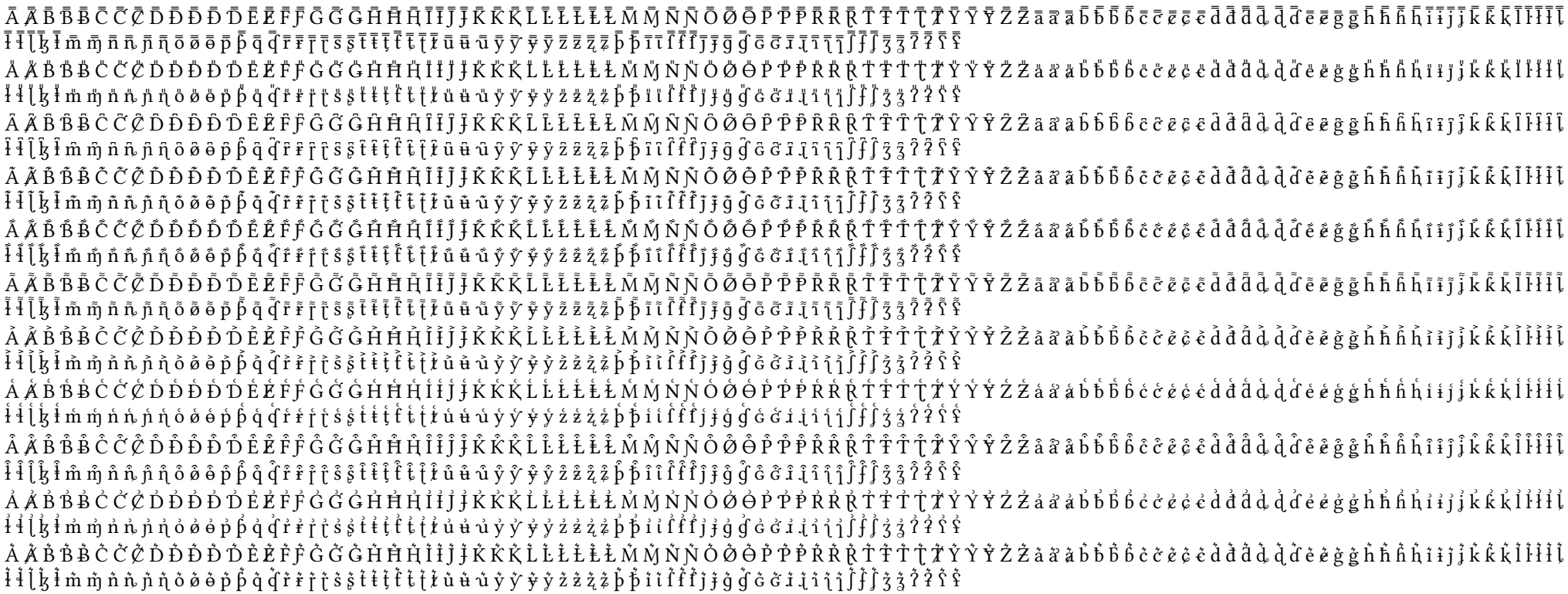
hex	0269	1E9C	1E9D	025F	025A	025D	029B	027B	0285	0284	0286	0293	02A1	02A2
base	0131	017F	017F	0237	0259	025C	0262	0279	027F	0283	0283	0292	0294	0295
reg	ɩ̥	ſ̊ſ̊	ſ̊ſ̊	ȷ̊ɟ̊	ə̊ɚ̊	ɜ̊ɝ̊	ɢ̊ʛ̊	ɻ̊ɺ̊	ʈ̊	ʃ̊ʃ̊	ʃ̊ʃ̊	ʒ̊ʒ̊	ʔ̊	ʕ̊
patch	ɩ̥	f̊f̊	f̊f̊	ɟ̊ɟ̊	ə̊ɚ̊	ɜ̊ɝ̊	ɢ̊ʛ̊	ɻ̊ɺ̊	ʈ̊	ʃ̊ʃ̊	ʃ̊ʃ̊	ʒ̊ʒ̊	ʔ̊ʔ̊	ʕ̊ʕ̊
base	ɩ̥	f̊f̊	f̊f̊	ɟ̊ɟ̊	ə̊ɚ̊	ɜ̊ɝ̊	ɢ̊ʛ̊	ɻ̊ɺ̊	ʈ̊	ʃ̊ʃ̊	ʃ̊ʃ̊	ʒ̊ʒ̊	ʔ̊ʔ̊	ʕ̊ʕ̊
a	-18 |	+0, 92%	+0, 94%	+16 |			+0, 100%	+20, 75% s	+0, 100% s	+0, 92%	+0, 100% s	*+4, 38% s*	+0, 92%	+0, 92%
a	123, 645			169, 645	229, 645	209, 645	209, 645		209, 645	209, 645	209, 645	209, 645	209, 645	209, 645
a?	170, 645	182, 885	190, 885	124, 645	306, 688	234, 795	294, 764	226, 645	170, 764	140, 885	150, 885	220, 645	211, 843	221, 843
a<	**124, 645**	**190, 885**	**190, 885**	**168, 645**			**265, 663**	247, 645	**170, 645**	**140, 885**	**140, 885**		**211, 850**	**221, 850**
a!								248, 645				222, 645		
b	-18 |	+0, 100%	+0, 100%	+16 |	-13, 81%	+0, 100%	+0, 100%	+20, 75% s		+0, 98%	+0, 100% s	*+4, 38% s*	+0, 92%	+0, 92%
b	153, -114			196, -110	212, -114	212, -114	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110	196, -110
b?	170, -110	182, -110	190, -110	124, -291	306, -110	234, -110	294, -110	226, -259	170, -299	140, -299	150, -310	220, -319	211, -110	221, -110
b<	**124, -119**	**190, -110**	**190, -110**	**125, -319**	199, -110	**184, -110**	**248, -110**	**198, -259**		**129, -319**	**129, -310**	**220, -319**	**202, -110**	**230, -110**
b!					184, -110									
it	*ɩ̥*	*ſ̊ſ̊*	*ſ̊ſ̊*	*ȷ̊ɟ̊*	*ə̊ɚ̊*	*ɜ̊ɝ̊*	*ɢ̊ʛ̊*	*ɻ̊ɺ̊*	*ʈ̊*	*ʃ̊ʃ̊*	*ʃ̊ʃ̊*	*ʒ̊ʒ̊*	*ʔ̊*	*ʕ̊*
patch	*ɩ̥*	*f̊f̊*	*f̊f̊*	*ɟ̊ɟ̊*	*ə̊ɚ̊*	*ɜ̊ɝ̊*	*ɢ̊ʛ̊*	*ɻ̊ɺ̊*	*ʈ̊*	*ʃ̊ʃ̊*	*ʃ̊ʃ̊*	*ʒ̊ʒ̊*	*ʔ̊ʔ̊*	*ʕ̊ʕ̊*
base	*ɩ̥*	*f̊f̊*	*f̊f̊*	*ɟ̊ɟ̊*	*ə̊ɚ̊*	*ɜ̊ɝ̊*	*ɢ̊ʛ̊*	*ɻ̊ɺ̊*	*ʈ̊*	*ʃ̊ʃ̊*	*ʃ̊ʃ̊*	*ʒ̊ʒ̊*	*ʔ̊ʔ̊*	*ʕ̊ʕ̊*
a	-19 |	+0, 90%	+0, 94%	+43 |			-9, 89%	*+111, 58% s*	+0, 100% s	+0, 92%	+0, 100% s	*+28, 26% s*	+0, 94%	+0, 94%
a	123, 645													
a?	266, 645	339, 890	339, 890	305, 645	418, 687	413, 775	410, 796	352, 645	343, 900	343, 900	343, 900	355, 645	376, 864	378, 875
a<	**256, 645**	**339, 890**	**339, 890**	**326, 645**			**379, 645**		**307, 645**	**343, 900**	**343, 900**		**376, 867**	**379, 878**
a!								389, 645						
b	-19 |	+0, 100%	+0, 100%	+43 |	*-10, 68%*	+0, 100%	-16, 89%	*+111, 58% s*		+0, 90%	+0, 100% s	*+28, 26% s*	+0, 85%	+0, 84%
b	153, -114				212, -114	212, -114								
b?	106, -110	82, -319	82, -319	100, -319	248, -110	225, -110	217, -110	160, -259	88, -299	88, -299	92, -283	133, -397	169, -110	169, -110
b<	**96, -110**	**82, -319**	**82, -319**	**105, -320**	159, -110	**143, -110**	**232, -110**	**160, -259**		**61, -319**	**69, -283**	133, -397	**202, -110**	**230, -110**
b!					161, -110							133, -399		
sb	**ɩ̥**			**ȷ̊ɟ̊**	**ə̊ɚ̊**	**ɜ̊ɝ̊**	**ɢ̊ʛ̊**	**ɻ̊ɺ̊**	**ʈ̊**	**ʃ̊ʃ̊**	**ʃ̊ʃ̊**	**ʒ̊ʒ̊**	**ʔ̊**	**ʕ̊**
patch	**ɩ̥**	⌧	⌧	**ɟ̊ɟ̊**	**ə̊ɚ̊**	**ɜ̊ɝ̊**	**ɢ̊ʛ̊**	**ɻ̊ɺ̊**	**ʈ̊**	**ʃ̊ʃ̊**	**ʃ̊ʃ̊**	**ʒ̊ʒ̊**	**ʔ̊ʔ̊**	**ʕ̊ʕ̊**
base	**ɩ̥**	**f̊f̊**	**f̊f̊**	**ɟ̊ɟ̊**	**ə̊ɚ̊**	**ɜ̊ɝ̊**	**ɢ̊ʛ̊**	**ɻ̊ɺ̊**	**ʈ̊**	**ʃ̊ʃ̊**	**ʃ̊ʃ̊**	**ʒ̊ʒ̊**	**ʔ̊ʔ̊**	**ʕ̊ʕ̊**
a	-16 |			+18 |			+0, 100%	+55, 97% s	+0, 100% s	+0, 92%	-1, 100% s	*+9, 49% s*	+0, 94%	+0, 94%
a														
a?	164, 645			144, 645	319, 693	276, 757	290, 764	238, 645	152, 885	152, 885	162, 885	236, 645	206, 845	206, 845
a<	**140, 645**			**181, 645**			**266, 645**	**260, 645**	**182, 645**	**152, 885**	**151, 885**		**206, 850**	**206, 850**
a!												231, 645		
b	-16 |			+18 |	+6, 81%	+1, 100%	+0, 100%	+55, 97% s		+0, 98%	-1, 100% s	*+9, 49% s*	+0, 96%	+0, 96%
b														
b?	164, -110			144, -319	319, -110	276, -110	290, -110	238, -257	152, -296	152, -296	162, -319	236, -413	206, -110	206, -110
b<	**140, -120**			**96, -319**	218, -110	**205, -110**	**266, -110**	**196, -257**		**141, -319**	**141, -319**	236, -413	**206, -110**	**206, -110**
b!					199, -110							236, -415		
si	***ɩ̥***			***ȷ̊ɟ̊***	***ə̊ɚ̊***	***ɜ̊ɝ̊***	***ɢ̊ʛ̊***	***ɻ̊ɺ̊***	***ʈ̊***	***ʃ̊ʃ̊***	***ʃ̊ʃ̊***	***ʒ̊ʒ̊***	***ʔ̊***	***ʕ̊***
patch	***ɩ̥***	⌧	⌧	***ɟ̊ɟ̊***	***ə̊ɚ̊***	***ɜ̊ɝ̊***	***ɢ̊ʛ̊***	***ɻ̊ɺ̊***	***ʈ̊***	***ʃ̊ʃ̊***	***ʃ̊ʃ̊***	***ʒ̊ʒ̊***	***ʔ̊ʔ̊***	***ʕ̊ʕ̊***
base	***ɩ̥***	***f̊f̊***	***f̊f̊***	***ɟ̊ɟ̊***	***ə̊ɚ̊***	***ɜ̊ɝ̊***	***ɢ̊ʛ̊***	***ɻ̊ɺ̊***	***ʈ̊***	***ʃ̊ʃ̊***	***ʃ̊ʃ̊***	***ʒ̊ʒ̊***	***ʔ̊ʔ̊***	***ʕ̊ʕ̊***
a	-15 |			-4 |			+12, 100%	+35, 86% s	+0, 100% s	+0, 92%	+1, 100% s	*+3, 56% s*	+12, 94%	+12, 94%
a	123, 645													
a?	138, 645			141, 645	299, 693	261, 762	278, 796	194, 645	195, 890	195, 890	189, 890	211, 645	227, 850	227, 850
a<	**131, 654**			**167, 645**			**284, 681**	**246, 645**	**167, 645**	**195, 890**	**195, 890**		**239, 850**	**239, 850**
a!												213, 645		
b	-15 |			-4 |	*-6, 60%*	+1, 100%	+12, 100%	+35, 86% s		+0, 98%	+1, 100% s	*+3, 56% s*	+13, 96%	+12, 96%
b	153, -114				212, -114	212, -114								
b?	147, -110			107, -319	298, -110	246, -110	256, -110	172, -257	116, -296	116, -296	105, -319	157, -413	194, -110	194, -110
b<	**138, -110**			**85, -314**	180, -110	**194, -110**	**260, -110**	**185, -257**		**88, -319**	**88, -319**	157, -413	**215, -110**	**242, -110**
b!					188, -110							157, -415		

Table 25: Stem-midline check: offset of the mark axis from the stem midline, patched fonts

	l	Ĺ	ļ	ľ	ŀ	ł	ƚ	ɬ	ɬ	ɭ	ɮ	ɬ	i	ɪ	i̇	i.dl	ɩ
hex	006C	013A	013C	013E	0140	0142	019A	026B	026C	026D	026E	2C61	0069	0131	0268	dotless	0269
reg a	+0		+0		+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0
b	+0	+0		+0	+0	+0	+0	+0	+0	+0	**+140**	+0	+0	+0	+0	+0	+0
it a	+2		+2		+2	+2	+2	+2	+2	+2	+0	+2	+2	+0	+2	+0	-1
b	+1	+1		+1	+2	+2	+1	+2	+1	+1	**+134**	+1	+0	+0	+0	+0	+0
sb a	+0		+0		+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0	+0
b	+0	+0		+0	+0	+0	+0	+0	+0	+0	**+133**	+0	+0	+0	+0	+0	+0
si a	+3		+3		+5	+4	+3	+3	+3	+3	+0	+3	+2	+0	+0	+0	+0
b	+2	+2		+2	+4	+4	+2	+2	+2	+2	**+126**	+3	+3	+1	+3	+1	+1

Regular patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above

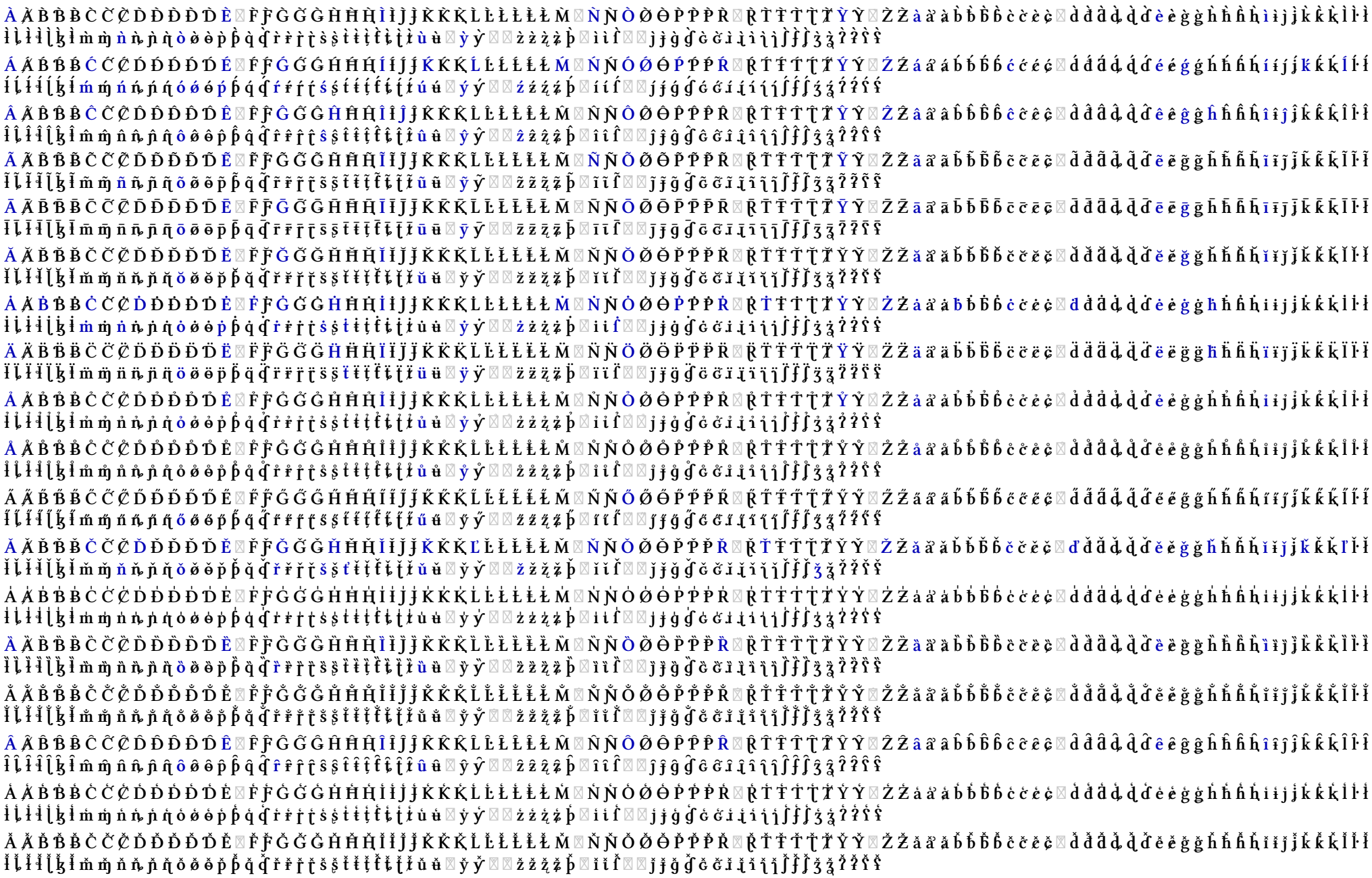


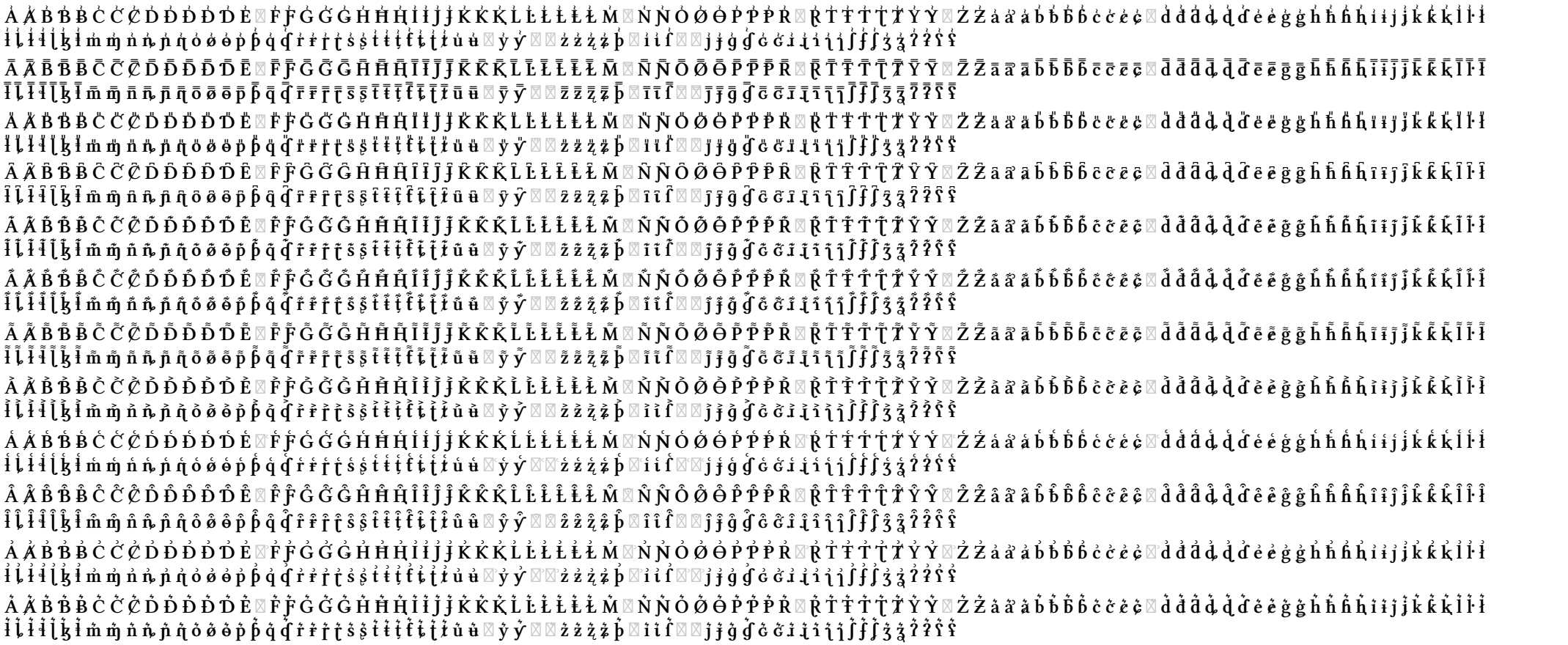


Italic patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above

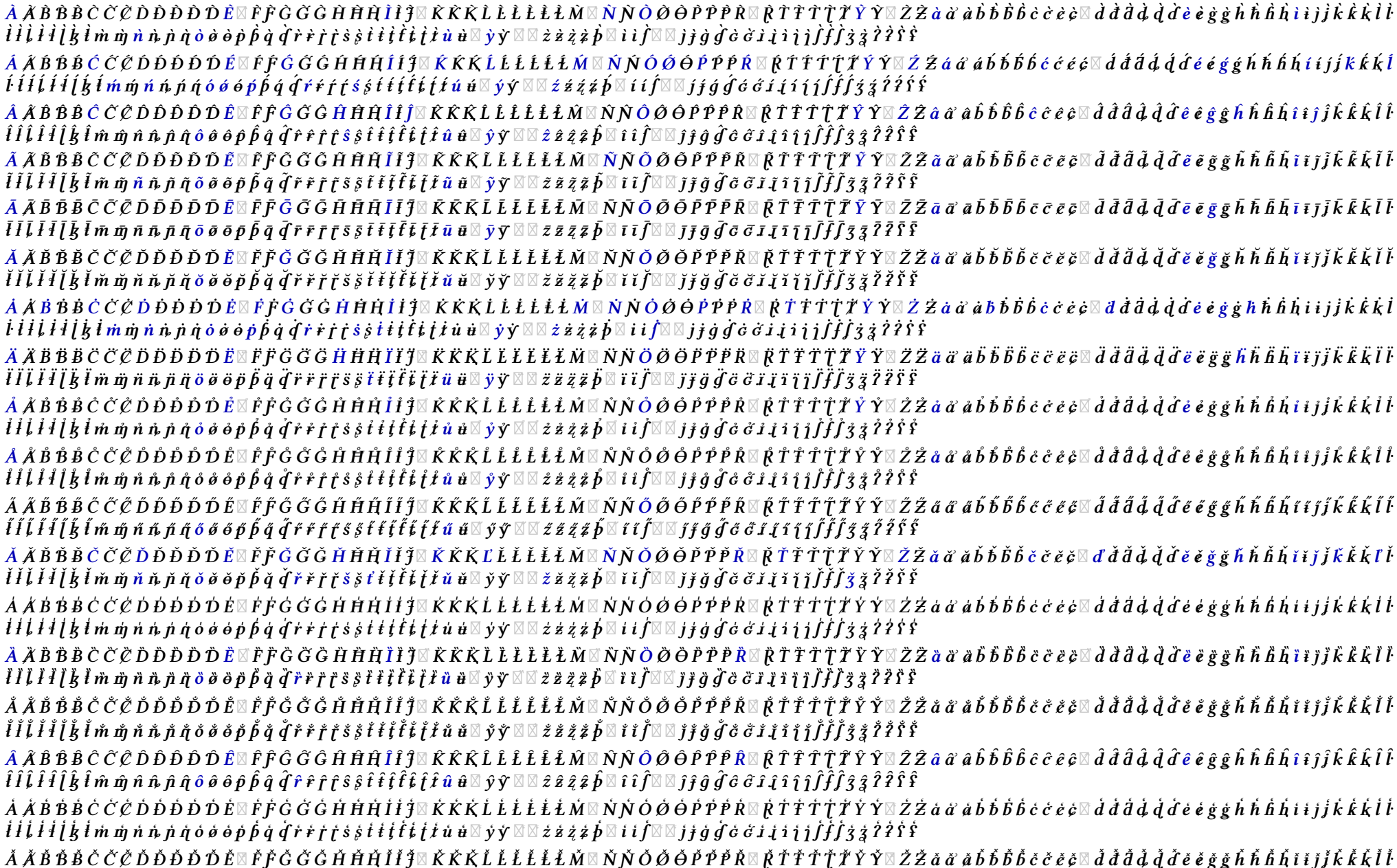


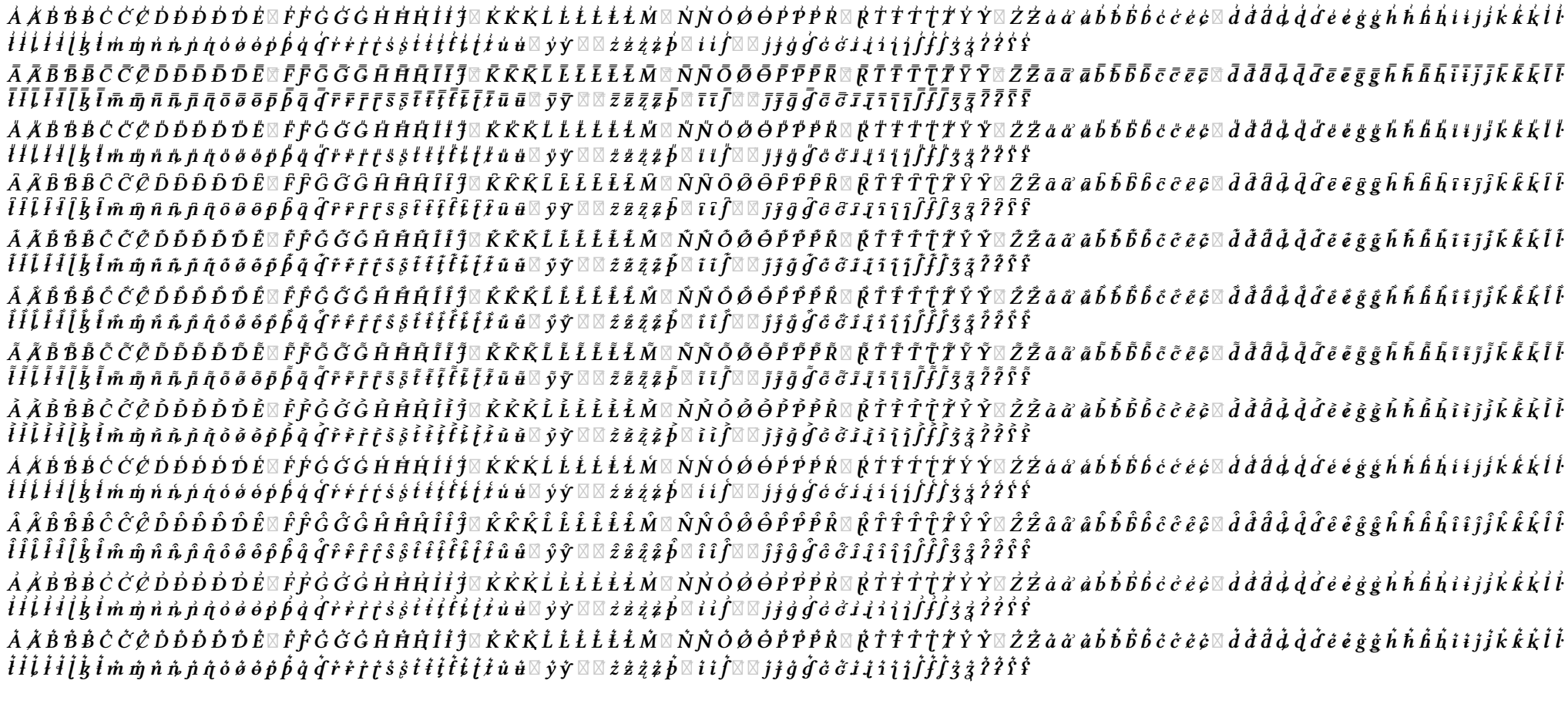
Semibold patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above





Semibold italic patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above



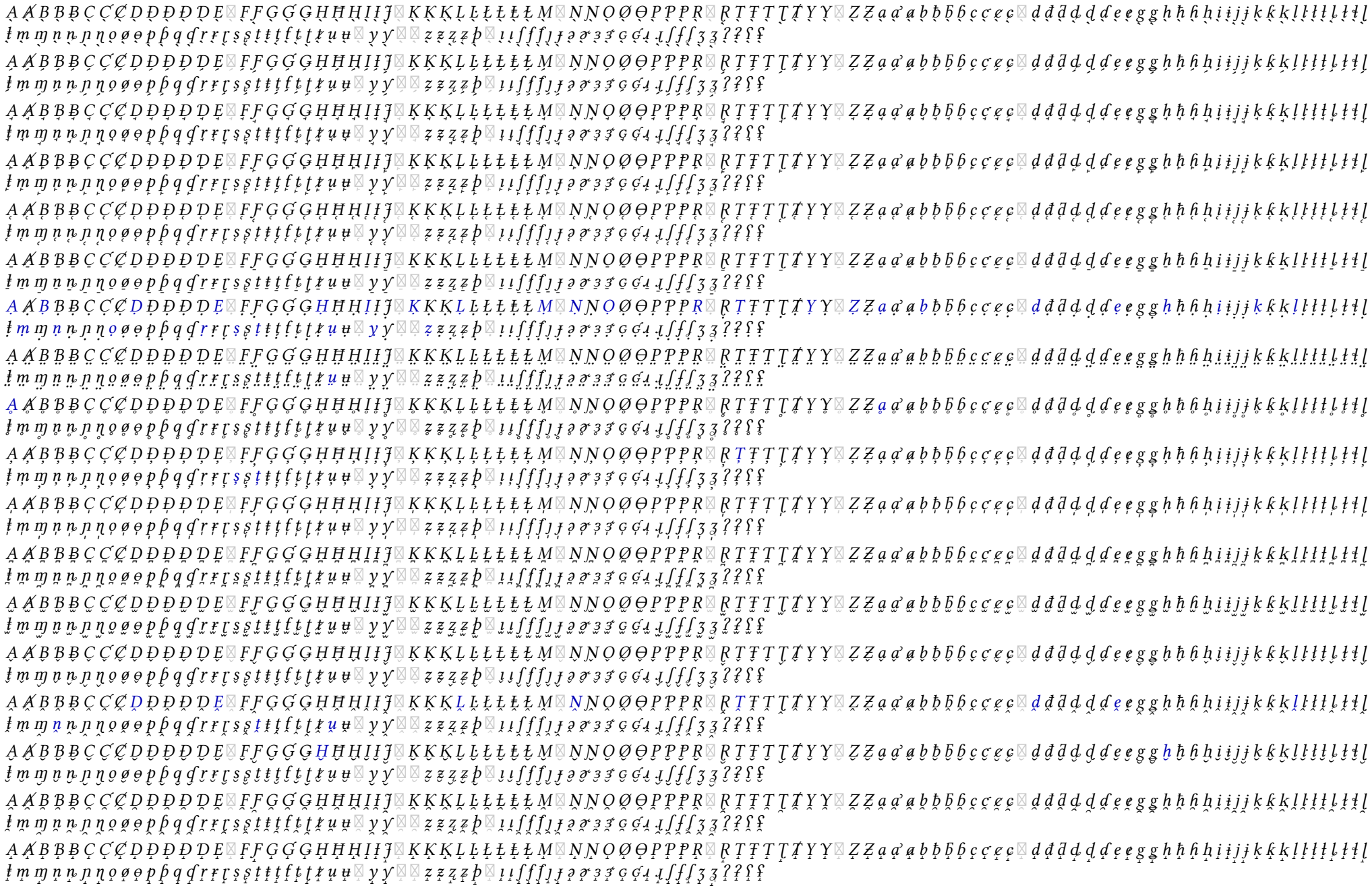


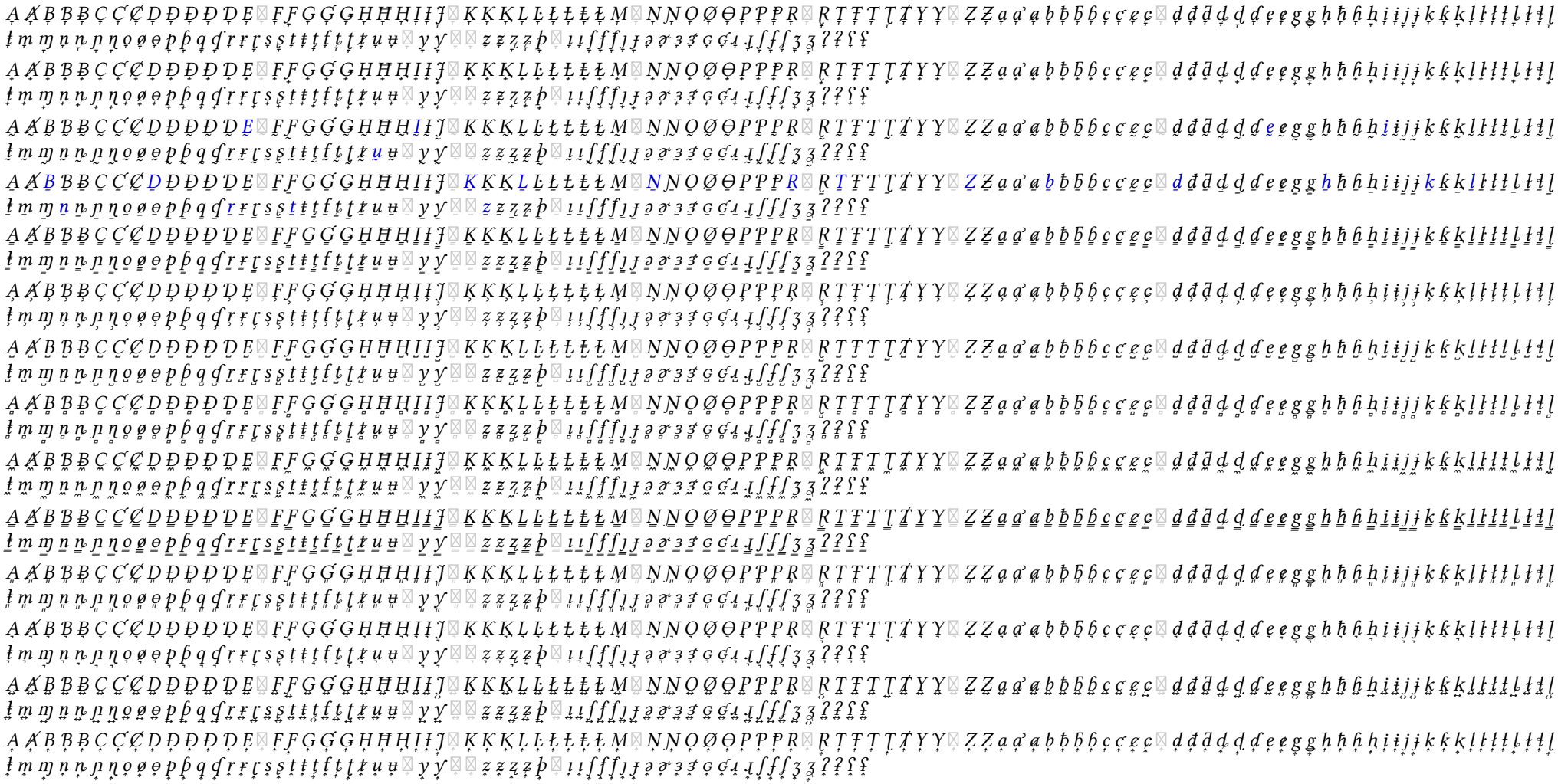
Regular patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below



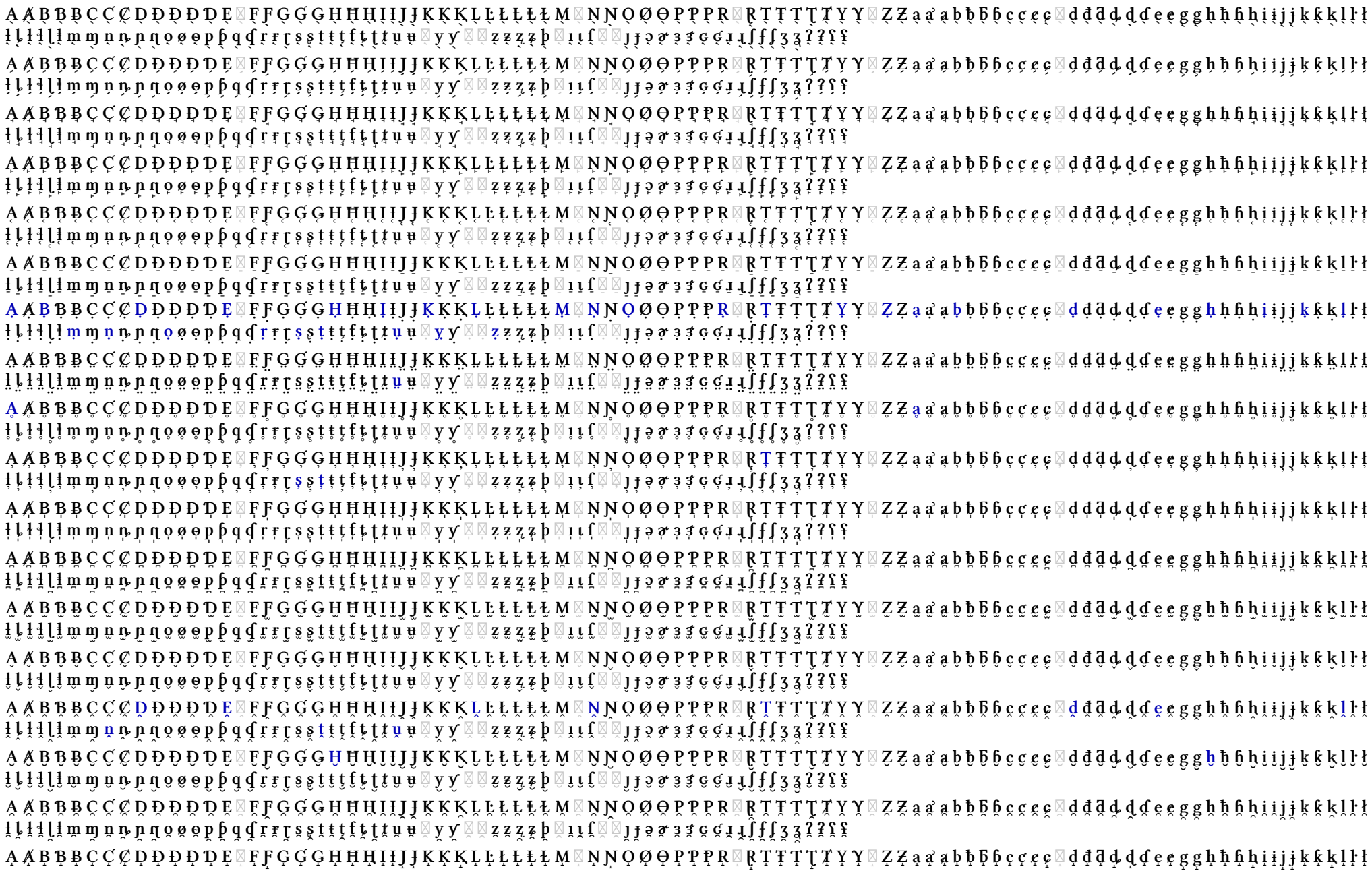
Ạ Ặ Ậ Ḅ Ḃ Ḇ Ç Ḉ Ḍ Ḏ Ḑ Ḓ Ḕ Ḝ Ḗ Ḟ Ḡ Ḣ Ḥ Ḧ Ḩ Ḫ Ḭ Ḯ Ḱ Ḳ Ḵ Ḷ Ḻ Ḽ Ḿ Ṁ Ṇ Ṉ Ṋ Ọ Ộ Ṑ Ṓ Ṕ Ṗ Ṙ Ṛ Ṝ Ṟ Ṯ Ṱ Ṳ Ṵ Ṷ Ṹ Ṻ Ỵ Ỷ Ỹ Ẓ Ẕ ạ ặ ậ ḅ ḃ ḇ ç ḉ ḍ ḏ ḑ ḓ ḕ ḝ ḗ ḟ ḡ ḣ ḥ ḧ ḩ ḫ ḭ ḯ ḱ ḳ ḵ ḷ ḻ ḽ ḿ ṁ ṇ ṉ ṋ ọ ộ ṑ ṓ ṕ ṗ ṙ ṛ ṝ ṟ ṯ ṱ ṳ ṵ ṷ ṹ ṻ ỵ ỷ ỹ ẓ ẕ ẖ ẗ ẘ ẙ ẚ ẛ ẜ ẝ ẞ

Italic patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below



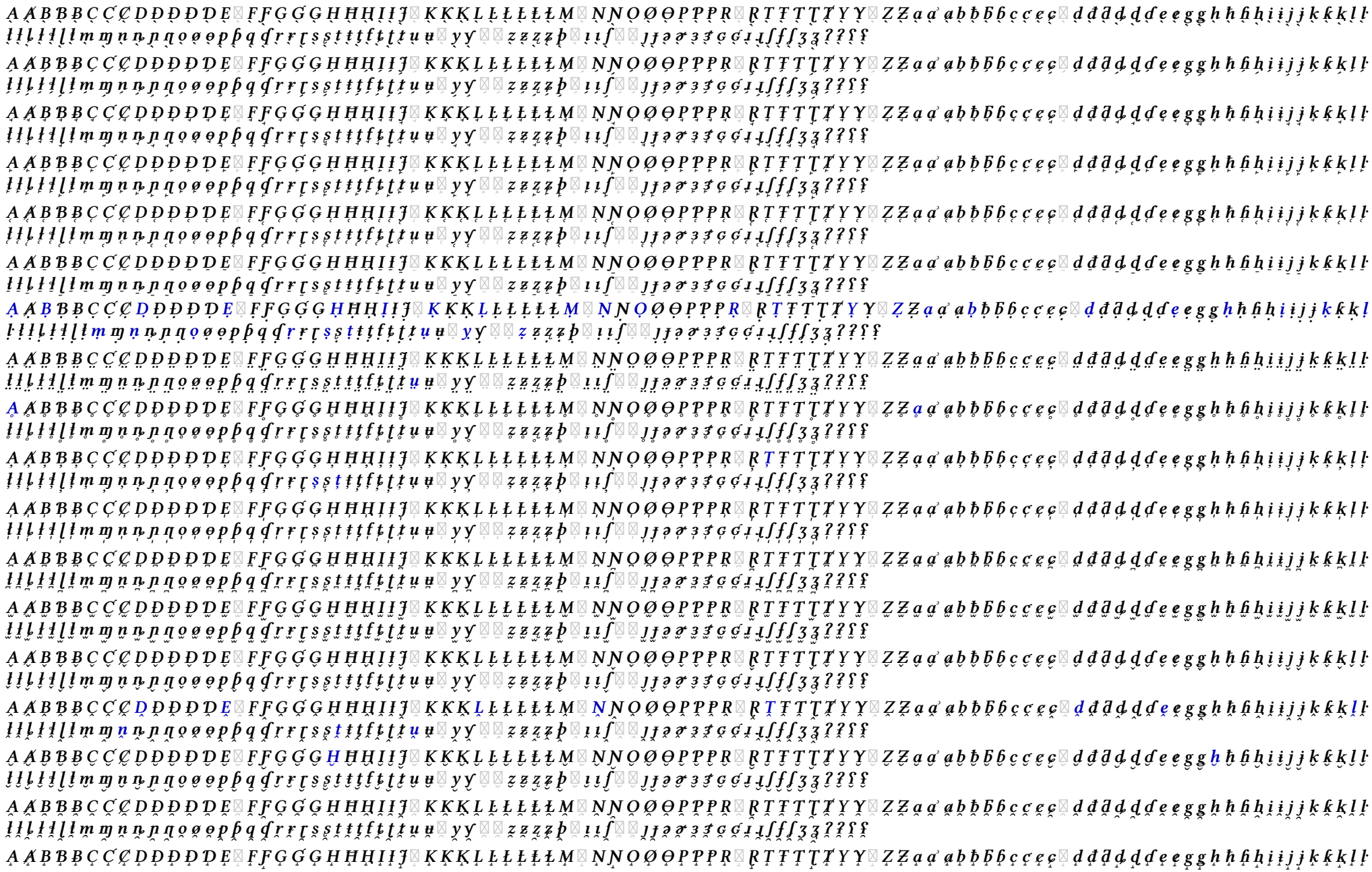


Semibold patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below





Semibold italic patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below





Precomposed characters

Stage 9 (`src/libertinus_analysis/font_patching_precomposed_anchors.py`, letter lists in `data/precomposed_letters.py`) gives a precomposed Latin letter the above- or below-anchor of its base letter where the other side is free: a letter with an above-mark (é, ñ) inherits a below-anchor, and a letter with a below-mark or attached below-mark (ẹ, ç, ą) inherits an above-anchor. (Letters with a bar or stroke, such as Ø, Ł, Đ, have no Unicode decomposition; they inherit in stage 8, previous section.) The stage also tests that the base letter's form is really there, since some precomposed letters were redrawn unlike their bases: it fits the base's ink on the inherited side to the precomposed glyph (a shift, and a scale of up to 15%), requires a fit of at least 80%, and moves the anchor by the fitted shift.

Precomposed letters inherit zealously. A designer anchor on a letter that can inherit is deleted in stage 1 (`"delete"` in `data/fontanchors_audit/overrides.py`) whenever it differs from the inherited anchor at all where the fit is 100%, or by more than 10 where the fit is lower; curated anchors that differed were commented out the same way. Most were copies of a base anchor that the patch has since changed, or copies that miss how the base sits in the precomposed glyph. Curated anchors (stage 7, `data/fontanchors_human/{style}.py`) are set on precomposed letters only where inheritance fails to find a good match. Every letter with a fit of 100% now has exactly the anchor it inherits (`run_precomposed_existing_report.py` checks this). The letters that do not simply inherit:

- ģ (U+0123) is formally g with a below-mark, a cedilla (U+0327), but is drawn with a comma above. It does inherit a below-anchor from g (fit 100% in every style), and gets no above-anchor: 30 IPA above-marks would collide with the comma.
- The horn letters Ơ ơ ư Ọ ọ ụ (U+01A0, U+01A1, U+01B0, U+1EE2, U+1EE3, U+1EF1) get no above-anchor in any style: 1–12 IPA above-marks per style would collide with the horn. Ơ ơ ư inherit a below-anchor (Ọ ọ ụ have a dot below), except semibold italic ư, whose below-anchor is curated (see the last item).
- **ą̄ ā** semibold ą (U+0105), above: the ogonek lowers the fit to 69%; curated at a's anchor plus the shift of ą's upper part (+12).
- *ĵ̲ ĵ̲* italic ĵ (U+0134), below: J moved 47 left with a shallower tail (-166, J -200), fit 65%; curated at the italic-model candidate from Ĵ's own metrics, (130, -251).
- ***ĵ̲ ĵ̲*** semibold italic ĵ, below: a flat top serif and a deeper tail (-196, J -164), fit 64%; curated at the candidate from Ĵ's own metrics, (125, -281).
- ***ư̲ ứ̲ ừ̲ ử̲ ữ̲ u̲*** semibold italic ư ứ ừ ử ữ (U+01B0, U+1EE9, U+1EEB, U+1EED, U+1EEF), below: the stems lie on u's but the bottom of the bowl is about 15 wider on each side, fit 75%; curated at u's anchor, (270, -110). The toned forms reference ư.

The grids show the IPA above-marks on the precomposed letters that inherit above and the below-marks on those that inherit below, in the patched fonts only. For completeness they include the letters whose anchor is curated instead of inherited (ą, ĵ, ư ứ ừ ử ữ). Samples and grid cells put `\cgj` between the letter and the mark, so that the mark is positioned by the letter's anchor and the shaper does not decompose and recompose the letter. The exception is a letter and mark that compose to one of the 24 precomposed letters with both an above- and a below-mark (Ç + acute = Ḉ, ạ + circumflex = ậ, Ȯ + dot below = Ọ̇): text gets the precomposed glyph, so the cell shows it, in blue, without `\cgj`. They are Ǫ ǫ Ḉ ḉ Ḝ ḝ Ḷ Ḯ Ḵ ḵ Ṣ ṣ Ậ ậ Ặ ặ Ệ ệ Ộ ộ, and, with the horn, Ợ ợ Ự ự.

Regular patched, Precomposed letters that inherit an above anchor in stage 9, Above



Italic patched, Precomposed letters that inherit an above anchor in stage 9, Above



Semibold patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ṭ ṭ Ḛ ḛ Ạ ạ Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḓ ḓ Ḏ ḏ Ḑ ḑ Ḝ ḝ Ḕ ḕ Ḗ ḗ Ḥ ḥ Ḫ ḫ Ḩ ḩ Ḭ Ḳ ḳ Ḵ ḵ Ḻ Ḽ Ḷ Ḹ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṻ ṻ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ ẖ Ả ả Ẹ ẹ Ỉ Ọ ọ Ủ ủ Ử Ỳ ỳ

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḛ ḛ Ḝ ḝ Ḟ ḟ Ḡ ḡ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḩ ḩ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḵ ḵ Ḷ ḷ Ḹ ḹ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṡ ṡ Ṣ ṣ Ṫ ṫ Ṭ ṭ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṵ ṵ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ṿ ṿ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ẓ ẓ Ẕ ẕ H̱ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ

Semibold italic patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Į Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ư Ǫ ǫ Ş ş Ţ ţ Ę ę Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḕ ḕ Ḗ ḗ Ḥ ḥ Ḫ ḫ Ḯ Ḳ ḳ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṽ ṽ
Ẁ ẁ Ẓ ẓ Ẕ ẕ ẖ Ą ą Ę ę Į Ǫ ǫ Ų ų Ư Ẏ ẏ

Regular patched, Precomposed letters that inherit a below anchor in stage 9, Below

ÀÁÂÃÄÅÈÉÊËÌÍÎÏÑÒÓÔÕÖÙÚÛÜÝàáâãäåèéêëìíîïñòóôõöùúûüýÿĀāĂăĆćĈĉĊċČčĎďĒēĔĕĖėĚěĜĝĞğĠġĢģĤĥĨĩĪīĬĭĮįİĴĵĶķĹĺĻļĽľŃńŅņŇňŌōŎŏŐőŔŕŖŗŘřŚśŜŝŞşŠšŢţŤťŪūŬŭŮůŰűŴŵŶŷŸŹźŻżŽžƠơƯưǍǎǏǐǑǒǓǔǕǖǗǘǙǚǛǜǞǟǦǧǨǩǪǫǮǯǴǵǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȞȟȦȧȨȩȪȫȬȭȮȯȰȱȲȳ
ḀḁḂḃḄḅḆḇḈḉḊḋḌḍḎḏḐḑḒḓḔḕḖḗḘḙḚḛḜḝḞḟḠḡḢḣḤḥḦḧḨḩḪḫḬḭḮḯḰḱḲḳḴḵḶḷḸḹḺḻḼḽḾḿṀṁṂṃṄṅṆṇṈṉṊṋṌṍṎṏṐṑṒṓṔṕṖṗṘṙṚṛṜṝṞṟṠṡṢṣṤṥṦṧṨṩṪṫṬṭṮṯṰṱṲṳṴṵṶṷṸṹṺṻṼṽṾṿẀẁẂẃẄẅẆẇẈẉẊẋẌẍẎẏẐẑẒẓẔẕẖẗẘẙẚẛẜẝẞẟẠạẢảẤấẦầẨẩẪẫẬậẮắẰằẲẳẴẵẶặẸẹẺẻẼẽẾếỀềỂểỄễỆệỈỉỊịỌọỎỏỐốỒồỔổỖỗỘộỚớỜờỞởỠỡỢợỤụỦủỨứỪừỬửỮữỰựỲỳỴỵỶỷỸỹ

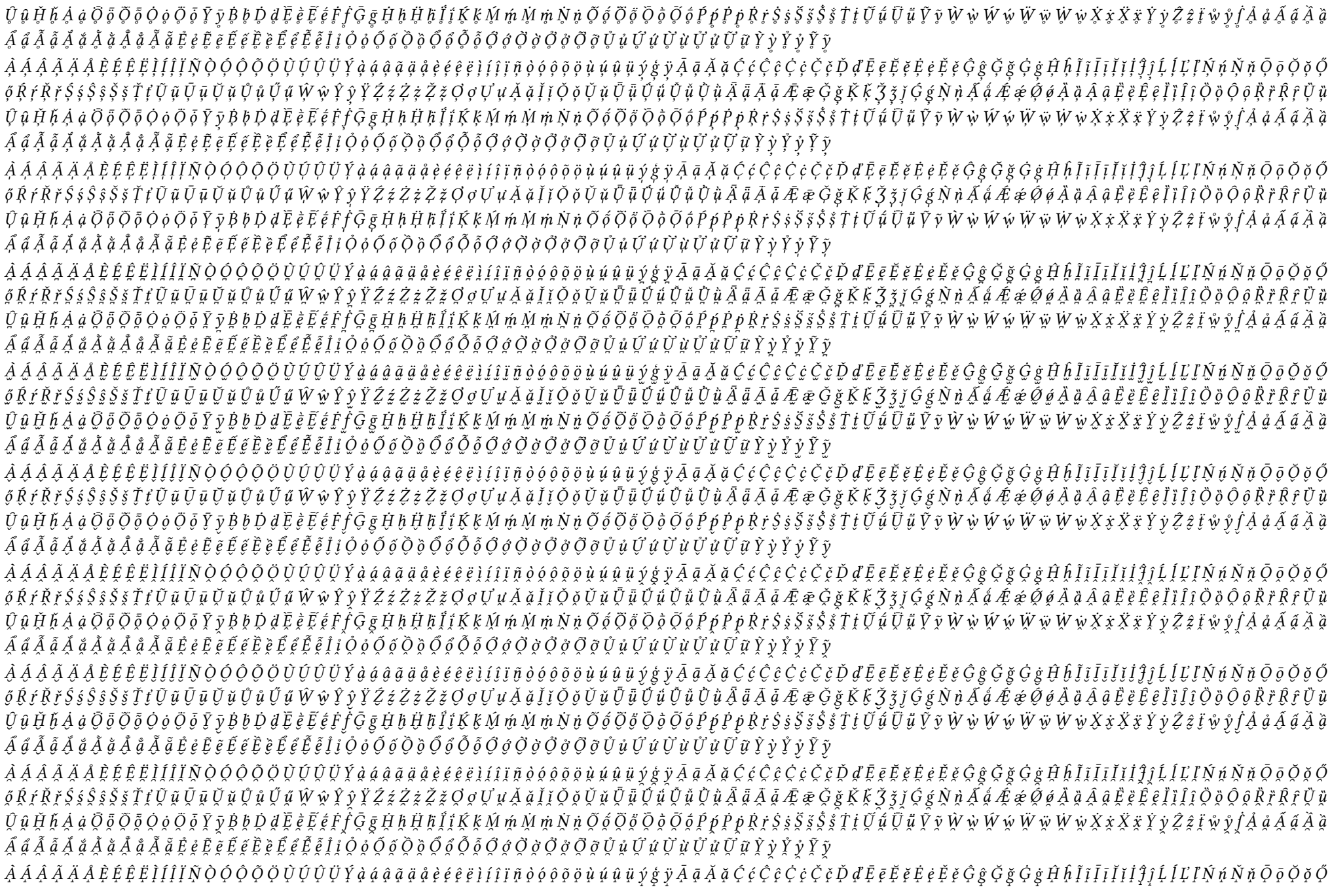
Ŗŗ Ř ř Ů ů Ű ű Ų ų Ẁ ẁ Ẃ ẃ Ẅ ẅ Ỳ ỳ Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỻ ỻ Ỽ ỽ Ỿ ỿ Ⱥ Ⱦ Ⱨ ⱨ Ⱪ ⱪ Ⱬ ⱬ Ⱳ ⱳ Ⱶ ⱶ Ȿ Ɀ Ɂ ɂ Ƀ Ʉ Ɇ ɇ Ɉ ɉ Ɍ ɍ Ɏ ɏ ɐ ɑ ɒ ɓ ɔ ɕ ɖ ɗ ɘ ə ɚ ɛ ɜ ɝ ɞ ɟ ɠ ɡ ɢ ɣ ɤ ɥ ɦ ɧ ɨ ɩ ɪ ɫ ɬ ɭ ɮ ɯ ɰ ɱ ɲ ɳ ɴ ɵ ɶ ɷ ɸ ɹ ɺ ɻ ɽ ɾ ʀ ʁ ʂ ʃ ʄ ʅ ʆ ʇ ʈ ʉ ʊ ʋ ʌ ʍ ʎ ʏ ʐ ʑ ʒ ʓ ʔ ʕ ʖ ʗ ʘ ʙ ʚ ʛ ʜ ʝ ʞ ʟ ʠ ʡ ʢ ʣ ʤ ʥ ʦ ʧ ʨ ʩ ʪ ʫ ʬ ʭ ʮ ʯ

Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ Ỏ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ụ Ủ Ứ Ừ Ử Ữ Ự Ỳ Ỵ Ỷ Ỹ

ǑǒǑǒǦǧǨǩǪǫǬǭǮǯǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏ
ȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽ
ȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɩɪɫɬɭ
ɮɯɰɱɲɳɴɵɶɷɸɹɺɻɼɽɾɿʀʁʂʃʄʅʆʇʈʉʊʋʌʍʎʏʐʑʒʓʔʕʖʗʘʙʚʛʜʝ
ʞʟʠʡʢʣʤʥʦʧʨʩʪʫʬʭʮʯʰʱʲʳʴʵʶʷʸʹʺʻʼʽʾʿˀˁ˂˃˄˅ˆˇˈˉˊˋˌˍˎˏ
ːˑ˒˓˔˕˖˗˘˙˚˛˜˝˞˟ˠˡˢˣˤ˥˦˧˨˩˪˫ˬ˭ˮ˯˰˱˲˳˴˵˶˷˸˹˺˻˼˽˾˿
̀́̂̃̄̅̆̇̈̉̊̋̌̍̎̏̐̑̒̓̔̕̚̕̚̕̚

Ő ő Ő ő Ř ř Ř ř Ś ś Ŝ ŝ Ş ş Š š Ť ť Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẃ ẃ Ŵ ŵ Ẅ ẅ Ỳ ỳ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ J̌ ǰ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ
Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ Ș ș Ț ț Ȟ ȟ Ȧ ȧ Ȩ ȩ Ȫ ȫ Ȭ ȭ Ȯ ȯ Ȱ ȱ Ȳ ȳ Ⱥ Ⱦ ɀ Ḃ ḃ Ḋ ḋ Ḟ ḟ Ġ ġ Ḣ ḣ Ḥ ḥ Ḧ ḧ Ḭ ḭ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṣ ṣ Ṥ ṥ Ṧ ṧ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẗ
ẘ ẙ ẛ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỵ ỵ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ğ ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ħ ħ Ĩ ĩ Ī ī Ĭ ĭ Į Ĵ ĵ Ľ ĺ Ļ ļ Ń ń Ň ň Ō ō
Ŏ ŏ

Italic patched, Precomposed letters that inherit a below anchor in stage 9, Below



ő Ř ř Ř ř Ş ş Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẁ ẁ Ŵ ŵ Ẅ ẅ Ỳ ỳ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ
Ụ ụ Ḫ ḫ Ḁ ḁ Ọ̃ ọ̃ Ọ̄ ọ̄ Ọ ọ Ö ö Ȫ ȫ Ȳ ȳ Ḅ ḅ Ḍ ḍ Ē ē É é Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṉ ṉ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṡ ṡ Ṫ ṫ Ṱ ṱ Ṹ ṹ Ṻ ṻ Ỹ ỹ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẇ ẇ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẑ ẑ Ẕ ẕ Ẩ ẩ Ầ ầ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ậ ậ Ẹ ẹ Ệ ệ Ễ ễ Ế ế Ề ề Ể ể Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỷ ỷ Ỷ ỷ Ỹ ỹ
Ậ ậ Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ä ä Å å Ḛ ḛ Ȅ ȅ Ȇ ȇ Ẻ ẻ Ẽ ẽ Ề ề Ế ế Ḕ ḕ Ḗ ḗ Ệ ệ Ỉ ỉ Ị ị Ḭ ḭ Ñ ñ Ò ò Ó ó Ô ô Õ õ Ö ö Ȭ ȭ Ù ù Ú ú Û û Ü ü Ỳ ỳ Ý ý à á â ã ä å è é ê ẽ ì í î ĩ ï ñ ò ó ô õ ö ù ú û ũ ü ý ğ ÿ Ā ā Ă ă Ç ç Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ į Ĵ ĵ Ļ ļ Ľ ľ Ń ń Ň ň Ō ō Ŏ ŏ Ő

Semibold patched, Precomposed letters that inherit a below anchor in stage 9, Below



Ồ ộ Ổ ổ Ỗ ỗ Ộ ộ Ổ ổ Ỗ ỗ Ộ ộ Ổ ổ Ỗ ỗ Ộ

Ō ō Ŏ ŏ Ő ő Ř ř Ŗ ŗ Ś ś Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ǫ ǫ Ư ư Ą ą Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ā ā Ą̄ ą̄ Æ æ Ḡ ḡ Ḳ ḳ Ǯ ǯ Ǧ ǧ Ǥ ǥ Ń ń Ň ň Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â È è Ê ê Ì ì Î î
Ȍ ȍ Ȏ ȏ Ȓ ȓ Ȑ ȑ Ȕ ȕ Ȗ ȗ Ṍ ṍ Ȯ ȯ Ȱ ȱ Ỳ ỳ Ḅ ḅ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṩ ṩ Ṧ ṧ Ṯ ṯ Ṳ ṳ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẅ ẅ Ẃ ẃ Ẅ ẅ Ẅ ẅ Ẋ ẋ
Ẍ ẍ Ỵ ỵ Ẓ ẓ Ẕ ẕ W̊ ẘ Ỷ ỷ Ḁ ḁ Ấ ấ Ậ ậ Ằ ằ Ẩ ẩ Ẫ ẫ Ắ ắ Ặ ặ Ẳ ẳ Ẵ ẵ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǵ ÿ Ā ā Ă ă Ç ç Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ľ ľ Ń ń Ň ň
Ō ō Ŏ ŏ Ő ő Ř ŕ Ř ř Ś ś Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű

Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẁ ẁ Ẃ ẃ Ẅ ẅ Ỳ ỳ Ý ý Ÿ ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â È è Ê ê Ì ì Î î
Ò ò Ô ô Ř ř Ŗ ŗ Ù ù Û û Ḧ ḧ Ạ ạ Ọ ọ Ộ ộ Ỏ ỏ Ỗ ỗ Ỹ ỹ Ḅ ḅ Ḍ ḍ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḫ ḫ Ḯ ḯ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṏ ṏ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẇ ẇ Ẉ ẉ Ẍ ẍ
Ẍ ẍ Ỳ ỳ Ẓ ẓ Ẕ ẕ W̊ ẘ Ỷ ỷ Ỉ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẻ ẻ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǧ ÿ Ā ā Ă ă Ç ç Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ě ě Ġ ġ Ģ ģ Ġ ġ Ĥ ĥ Ĭ ĭ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ľ ľ Ń ń Ň ň

Semibold italic patched, Precomposed letters that inherit a below anchor in stage 9, Below



Ê ê Ì ì Î î Ò ò Ô ô Ř ř Ř ř Ù ù Û û Ḧ ḧ Ạ ạ Ǫ ǫ Ǭ ǭ Ọ ọ Ǫ ǫ Ȳ ȳ Ƀ ƀ Ḍ ḍ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṣ ṣ Ṩ ṩ Ṩ ṩ Ṫ ṫ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ
Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẓ ẓ ẗ ẘ ẙ Ả ả Ấ ấ Ậ ậ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỳ ỳ Ỷ ỷ Ỹ ỹ
Ạ Ả Ấ Ầ Ẩ Ẫ Ậ Ắ Ằ Ẳ Ẵ Ặ Ẹ Ẻ Ẽ Ế Ề Ể Ễ Ệ Ỉ Ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỵ ỵ

Ǹ ǹ Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ỳ ỳ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ J̌ ǰ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ
Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ Ș ș Ț ț Ȟ ȟ Ȧ ȧ Ȩ ȩ Ȫ ȫ Ȭ ȭ Ȯ ȯ Ȱ ȱ Ȳ ȳ Ẃ ẃ Ẅ ẅ Ẍ ẍ Ẏ ẏ Ẑ ẑ Ḃ ḃ Ḋ ḋ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḱ ḱ Ḵ ḵ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṡ ṡ Ṩ ṩ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ
Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ T̈ ẗ W̊ ẘ Y̊ ẙ Ṡ ẛ Ạ ạ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
Ȁ Ȁ Ȃ Ã Ä Ả Ẹ É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǧ ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ļ ļ Ń ń
Ň ň

À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǵ ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń
Ň ň Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Â ȃ Ȅ ȅ
Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ṳ ṳ Ṷ ṷ Ḫ ḫ Ạ ạ Ọ ọ Ộ ộ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ẁ ẁ Ẃ ẃ Ẅ ẅ
Ẇ ẇ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẑ ẑ T̈ ẗ W̊ ẘ Y̊ ẙ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ọ ọ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǵ ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń
Ň ň Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Â ȃ Ȅ ȅ
Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ṳ ṳ Ṷ ṷ Ḫ ḫ Ạ ạ Ọ ọ Ộ ộ Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ